

Coarse-Graining Theory for Elliptic Equations

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Chapter 0

Notation

Chapter 1

Function spaces and basic inequalities

This chapter collects the analytic background for the coarse-graining theory. The goal is not to develop a maximally general theory of Besov and Sobolev spaces, but to record the specific tools that recur throughout the later chapters:

1. scale-invariant norms on cubes;
2. triadic Besov seminorms with positive and negative regularity;
3. the duality pairing between weak and strong multiscale norms;
4. the lemmas that convert weak control of ∇u or $\mathbf{a}\nabla u$ into usable control of u and of cutoff test terms.

Classical inputs are stated efficiently and cited, while the multiscale lemmas tailored to the triadic coarse-graining geometry are proved in full.

1.1 Classical and scale-invariant function spaces

1.1.1 Standard spaces and normalization

We first fix the basic vocabulary used throughout the notes. For an open set $U \subseteq \mathbb{R}^d$, $k \in \mathbb{N}_0$, $\alpha \in (0, 1]$, $s \in \mathbb{R}$, and $p \in [1, \infty]$, we use the standard Hölder spaces $C^{k,\alpha}(U)$ and Sobolev spaces $W^{s,p}(U)$. When $p = 2$, we write $H^s(U) = W^{s,2}(U)$. The space $W_0^{1,p}(U)$ denotes the closure of $C_c^\infty(U)$ in $W^{1,p}(U)$, and $X_{\text{loc}}(U)$ denotes the space of functions belonging to $X(U \cap B_R)$ for every $R \geq 1$.

For a bounded measurable set $U \subseteq \mathbb{R}^d$ and $p \in [1, \infty)$, we define the volume-normalized average and L^p norm by

$$(f)_U := \int_U f(x) dx \quad \text{and} \quad \|f\|_{\underline{L}^p(U)} := \left(\int_U |f(x)|^p dx \right)^{1/p}. \quad (1.1)$$

For $p = \infty$, we set $\|f\|_{\underline{L}^\infty(U)} = \|f\|_{L^\infty(U)}$. We also use the averaged finite sum

$$\sum_{a \in A} F(a) := \frac{1}{|A|} \sum_{a \in A} F(a)$$

for functions F on a finite set A .

If U is bounded and Lipschitz and $p \in [1, \infty)$, then the normalized Sobolev norm is

$$\|f\|_{\underline{W}^{1,p}(U)} := \left(\|\nabla f\|_{\underline{L}^p(U)}^p + |U|^{-p/d} \|f\|_{\underline{L}^p(U)}^p \right)^{1/p}, \quad (1.2)$$

and the corresponding seminorm is

$$[f]_{\underline{W}^{1,p}(U)} := \|\nabla f\|_{\underline{L}^p(U)}.$$

At the endpoint $p = \infty$, we use the scale-invariant norm

$$\|f\|_{\underline{W}^{1,\infty}(U)} := \|\nabla f\|_{L^\infty(U)} + |U|^{-1/d} \|f\|_{L^\infty(U)}.$$

The factor $|U|^{-1/d}$ makes the zeroth-order term scale like one derivative, so the underlined norms are dimensionless under the dilations that appear throughout the coarse-graining arguments.

1.1.2 Negative Sobolev norms

Let $p \in (1, \infty)$ and let $p' = \frac{p}{p-1}$ be the dual exponent. For a bounded Lipschitz domain U , we define the dual negative Sobolev seminorms by

$$\begin{cases} [f]_{\underline{W}^{-1,p'}(U)} := \sup \left\{ \int_U fg : g \in C_c^\infty(U), [g]_{\underline{W}^{1,p}(U)} \leq 1 \right\}, \\ [f]_{\widehat{\underline{W}}^{-1,p'}(U)} := \sup \left\{ \int_U fg : [g]_{\underline{W}^{1,p}(U)} \leq 1, (g)_U = 0 \right\}. \end{cases} \quad (1.3)$$

The second seminorm is the zero-average variant. It is better adapted to Poincaré-type arguments, because constants are invisible to it by construction.

When $p = p' = 2$, we write $\underline{H}^{-1}(U)$ in place of $\underline{W}^{-1,2}(U)$ and $\widehat{H}^{-1}(U)$ in place of $\widehat{\underline{W}}^{-1,2}(U)$. The guiding heuristic is simple: negative norms do not measure a field pointwise, but only after averaging against slowly varying test functions. They are therefore the natural language for coarse-grained information.

1.1.3 Hölder seminorms and the fractional viewpoint

For $\alpha \in (0, 1]$ and a bounded set U , we define

$$[f]_{C^{0,\alpha}(U)} := \sup_{x,y \in U, x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha} \quad \text{and} \quad \|f\|_{\underline{C}^{0,\alpha}(U)} := |U|^{-\alpha/d} \|f\|_{L^\infty(U)} + [f]_{C^{0,\alpha}(U)}. \quad (1.4)$$

Higher-order spaces $C^{k,\alpha}(U)$ are defined in the usual way by requiring $\partial^\beta f \in C^{0,\alpha}(U)$ for every $|\beta| \leq k$.

On cubes, the Hölder seminorm $[\cdot]_{C^{0,s}}$ is equivalent, up to dimensional constants, to the fractional Sobolev seminorm $[\cdot]_{\underline{W}^{s,\infty}}$. We will use this equivalence informally later on, since it explains why the fractional scale is the correct one for both oscillation estimates and large-scale regularity.

1.1.4 Potential and solenoidal spaces

For each bounded Lipschitz domain $U \subseteq \mathbb{R}^d$, define

$$\begin{cases} L_{\text{pot}}^2(U) := \nabla H^1(U) = \{\nabla u : u \in H^1(U)\}, \\ L_{\text{pot},0}^2(U) := \nabla H_0^1(U) = \{\nabla u : u \in H_0^1(U)\}, \\ L_{\text{sol}}^2(U) := L_{\text{pot},0}^2(U)^\perp = \left\{ \mathbf{g} \in L^2(U; \mathbb{R}^d) : \int_U \mathbf{g} \cdot \nabla u = 0 \quad \forall u \in H_0^1(U) \right\}, \\ L_{\text{sol},0}^2(U) := L_{\text{pot}}^2(U)^\perp = \left\{ \mathbf{g} \in L^2(U; \mathbb{R}^d) : \int_U \mathbf{g} \cdot \nabla u = 0 \quad \forall u \in H^1(U) \right\}. \end{cases} \quad (1.5)$$

These are the natural spaces of potential and solenoidal vector fields. The inclusion $L_{\text{pot},0}^2(U) \subseteq L_{\text{pot}}^2(U)$ immediately gives $L_{\text{sol},0}^2(U) \subseteq L_{\text{sol}}^2(U)$. Moreover, every field in $L_{\text{sol},0}^2(U)$ has zero average: if $\mathbf{g} \in L_{\text{sol},0}^2(U)$ and $e \in \mathbb{R}^d$, then the affine function $x \mapsto e \cdot x$ belongs to $H^1(U)$, so

$$e \cdot \int_U \mathbf{g}(x) dx = \int_U \mathbf{g}(x) \cdot e dx = \int_U \mathbf{g}(x) \cdot \nabla(e \cdot x) dx = 0.$$

The main point, however, is variational: in Chapter 2, the spaces $L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U)$ and $L_{\text{pot}}^2(U) \times L_{\text{sol}}^2(U)$ will play exactly the same role for the doubled block formalism that $H_0^1(U)$ and $H^1(U)$ play in the usual Dirichlet theory.

1.2 Triadic cubes and multiscale Besov seminorms

1.2.1 Triadic cubes as the analytic geometry of the book

Before defining Besov seminorms, briefly recall the triadic cubes $\square_m$ and their translates $z + \square_m$. The analytic definitions below are built directly on this triadic decomposition, since it is the geometry used later by the coarse-graining scheme. All the cube-based seminorms and norms introduced below are extended to translated triadic cubes $z + \square_m$ by translation.

1.2.2 Fractional Sobolev seminorms

Define, for $s \in (0, 1)$ and $p \in [1, \infty)$,

$$[u]_{\underline{W}^{s,p}(\square_m)} := \left(\oint_{\square_m} \int_{\square_m} \frac{|u(x) - u(y)|^p}{|x - y|^{d+sp}} dx dy \right)^{1/p}.$$

This gives the familiar fractional scale in the same normalized language as the rest of the chapter.

1.2.3 Positive-order triadic Besov seminorms

For $s \in (0, 1)$, $p \in [1, \infty)$, $q \in [1, \infty)$ and $m \in \mathbb{Z}$, we define the volume-normalized positive-order triadic Besov seminorm by

$$[u]_{\underline{B}_{p,q}^s(\square_m)} := \left(\sum_{n=-\infty}^m 3^{-nsq} \left(\sum_{\substack{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m \\ z + \square_n \subseteq \square_m}} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^p \right)^{q/p} \right)^{1/q}. \quad (1.6)$$

For $q = \infty$, we replace the geometric sum by the supremum:

$$[u]_{\underline{B}_{p,\infty}^s(\square_m)} := \sup_{n \in (-\infty, m] \cap \mathbb{Z}} 3^{-ns} \left(\sum_{\substack{z \in 3^{n-1} \mathbb{Z}^d \cap \square_m \\ z + \square_n \subseteq \square_m}} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^p \right)^{1/p}.$$

At the endpoint $s = 1$, we will only need the case $q = \infty$, and we define $[u]_{\underline{B}_{p,\infty}^1(\square_m)}$ by the same supremum formula with 3^{-n} in place of 3^{-ns} . The corresponding normalized Besov norm is

$$\|u\|_{\underline{B}_{p,q}^s(\square_m)} := [u]_{\underline{B}_{p,q}^s(\square_m)} + 3^{-sm} |(u)_{\square_m}|.$$

This triadic definition is preferable here to a Fourier-analytic one because:

- it is local;
- it matches the geometry of the renormalization and coarse-graining steps;
- it makes the later localization arguments nearly tautological.

The comparison with the classical fractional Sobolev seminorm is recorded below in Lemma 1.3.

1.2.4 Dual and circ negative-order Besov seminorms

There are two negative-order Besov objects in the background. The first is the genuine dual seminorm, obtained by testing against positive-order Besov functions. The second is the concrete triadic seminorm built from block averages. The latter is stronger, but it is the one that interacts most transparently with the coarse-graining decompositions.

Let $s \in (0, 1]$, $p \in (1, \infty)$, and $q \in [1, \infty]$, with $s = 1$ allowed only when $q = 1$. Let p' and q' denote the Hölder conjugates of p and q , respectively. We define the dual negative-order Besov seminorms by

$$\begin{cases} [f]_{\underline{B}_{p,q}^{-s}(\square_m)} := \sup \left\{ \left| \int_{\square_m} fg \right| : \|g\|_{\underline{B}_{p',q'}^s(\square_m)} \leq 1 \right\}, \\ [f]_{\hat{\underline{B}}_{p,q}^{-s}(\square_m)} := \sup \left\{ \left| \int_{\square_m} fg \right| : [g]_{\underline{B}_{p',q'}^s(\square_m)} \leq 1, (g)_{\square_m} = 0 \right\}. \end{cases} \quad (1.7)$$

The second seminorm is the direct Besov analogue of (1.3). It is the one that later appears naturally in testing and splitting arguments.

For $s \in (0, 1]$, $p \in [1, \infty)$, $q \in [1, \infty)$ and $m \in \mathbb{Z}$, with $s = 1$ allowed only when $q = 1$, we define the concrete circ negative-order Besov seminorm by

$$[f]_{\hat{\underline{B}}_{p,q}^{-s}(\square_m)} := \left(\sum_{n=-\infty}^m 3^{nsq} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(f)_{z+\square_n}|^p \right)^{q/p} \right)^{1/q}. \quad (1.8)$$

For $q = \infty$, we again replace the sum by the supremum:

$$[f]_{\hat{\underline{B}}_{p,\infty}^{-s}(\square_m)} := \sup_{n \in (-\infty, m] \cap \mathbb{Z}} 3^{ns} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(f)_{z+\square_n}|^p \right)^{1/p}.$$

The next lemma shows that the concrete circ seminorm dominates the true dual negative Besov seminorm. This is why the circ seminorm is so useful in practice: it can be inserted directly into multiscale splitting formulas while still controlling the genuinely dual quantity.

Lemma 1.1 (Circ domination of the dual negative Besov seminorm). *For every $m \in \mathbb{Z}$, $s \in (0, 1]$, $p \in (1, \infty)$, $q \in [1, \infty]$, with $s = 1$ allowed only when $q = 1$, and $f \in L^p(\square_m)$, we have*

$$[f]_{\dot{B}_{p,q}^{-s}(\square_m)} \leq 3^{d+s} [f]_{\dot{B}_{p,q}^{-s}(\square_m)}. \quad (1.9)$$

Moreover,

$$[f]_{B_{p,q}^{-s}(\square_m)} \leq 3^{d+s} [f]_{\dot{B}_{p,q}^{-s}(\square_m)} + 3^{sm} |(f)_{\square_m}|. \quad (1.10)$$

Proof. We begin with (1.9). Fix $g \in L^{p'}(\square_m)$ such that $(g)_{\square_m} = 0$ and $[g]_{B_{p',q'}^s(\square_m)} \leq 1$. By the Lebesgue differentiation theorem,

$$g = \sum_{n=-\infty}^m \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{z' \in 3^{n-1} \mathbb{Z}^d \cap (z + \square_n)} ((g)_{z' + \square_{n-1}} - (g)_{z + \square_n}) \mathbb{1}_{z' + \square_{n-1}}$$

almost everywhere in $\square_m$. Since $(g)_{\square_m} = 0$, this yields

$$\int_{\square_m} fg = \sum_{n=-\infty}^m \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{z' \in 3^{n-1} \mathbb{Z}^d \cap (z + \square_n)} ((g)_{z' + \square_{n-1}} - (g)_{z + \square_n}) (f)_{z' + \square_{n-1}}.$$

Therefore,

$$\begin{aligned} \left| \int_{\square_m} fg \right| &\leq \sum_{n=-\infty}^m \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{z' \in 3^{n-1} \mathbb{Z}^d \cap (z + \square_n)} |(g)_{z' + \square_{n-1}} - (g)_{z + \square_n}| |(f)_{z' + \square_{n-1}}| \\ &\leq 3^d \sum_{n=-\infty}^m \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \int_{z + \square_n} |g - (g)_{z + \square_n}|^{p'} \right)^{1/p'} \left(\sum_{z \in 3^{n-1} \mathbb{Z}^d \cap \square_m} |(f)_{z + \square_{n-1}}|^p \right)^{1/p}. \end{aligned}$$

Applying Hölder's inequality to the sum over n , we obtain

$$\begin{aligned} \left| \int_{\square_m} fg \right| &\leq 3^d \left(\sum_{n=-\infty}^m 3^{-snq'} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \int_{z + \square_n} |g - (g)_{z + \square_n}|^{p'} \right)^{q'/p'} \right)^{1/q'} \\ &\quad \times \left(\sum_{n=-\infty}^m 3^{snq} \left(\sum_{z \in 3^{n-1} \mathbb{Z}^d \cap \square_m} |(f)_{z + \square_{n-1}}|^p \right)^{q/p} \right)^{1/q} \\ &\leq 3^{d+s} [g]_{B_{p',q'}^s(\square_m)} [f]_{\dot{B}_{p,q}^{-s}(\square_m)} \\ &\leq 3^{d+s} [f]_{\dot{B}_{p,q}^{-s}(\square_m)}. \end{aligned}$$

The endpoint cases $q = 1$ and $q = \infty$ are handled in exactly the same way, replacing Hölder's inequality in the scale variable by the ℓ^∞ - ℓ^1 pairing. Taking the supremum over such g gives (1.9).

To prove (1.10), let $g \in L^{p'}(\square_m)$ satisfy $\|g\|_{B_{p',q'}^s(\square_m)} \leq 1$. Decompose

$$g = (g - (g)_{\square_m}) + (g)_{\square_m}.$$

Since $\|g\|_{B_{p',q'}^s(\square_m)} \leq 1$, we have $[g]_{B_{p',q'}^s(\square_m)} \leq 1$ and $|(g)_{\square_m}| \leq 3^{sm}$. Therefore,

$$\left| \int_{\square_m} fg \right| \leq \left| \int_{\square_m} f(g - (g)_{\square_m}) \right| + |(f)_{\square_m}| |(g)_{\square_m}| \leq 3^{d+s} [f]_{\dot{B}_{p,q}^{-s}(\square_m)} + 3^{sm} |(f)_{\square_m}|.$$

Taking the supremum over admissible g yields (1.10). $\square$

Lemma 1.2 (Reverse comparison after losing exponent). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $0 < t < s < 1$, and every $f \in L^2(\square_m)$,*

$$[f]_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C(1 - 3^{-2t})^{-1/2} (1 - 3^{-2(s-t)})^{-1/2} 3^{(s-t)m} [f]_{\dot{B}_{2,2}^{-t}(\square_m)}. \quad (1.11)$$

Equivalently, in scale-normalized form,

$$3^{-sm} [f]_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C(1 - 3^{-2t})^{-1/2} (1 - 3^{-2(s-t)})^{-1/2} 3^{-tm} [f]_{\dot{B}_{2,2}^{-t}(\square_m)}. \quad (1.12)$$

In particular, taking $t = s/2$ gives the useful estimate

$$3^{-sm} [f]_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-1} 3^{-\frac{s}{2}m} [f]_{\dot{B}_{2,2}^{-\frac{s}{2}}(\square_m)}. \quad (1.13)$$

The same estimates hold for vector fields componentwise, with the scalar dual norms combined in ℓ^2 over the components.

Proof. For $n \leq m$, set

$$A_n := \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(f)_{z+\square_n}|^2 \right)^{1/2}.$$

We first prove the depthwise estimate

$$3^{tn} A_n \leq C(1 - 3^{-2t})^{-1/2} [f]_{\dot{B}_{2,2}^{-t}(\square_m)}. \quad (1.14)$$

If $A_n = 0$, there is nothing to prove. Otherwise define the scale- n piecewise constant function

$$g_n := \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \frac{(f)_{z+\square_n}}{A_n} \mathbb{1}_{z+\square_n}.$$

Then $\|g_n\|_{\underline{L}^2(\square_m)} = 1$ and

$$\int_{\square_m} f g_n = \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (f)_{z+\square_n} \frac{(f)_{z+\square_n}}{A_n} = A_n.$$

It remains to estimate the positive Besov norm of g_n . Since g_n is constant on each scale- n cube, all oscillation terms at scales $\ell \leq n$ vanish. For $n < \ell \leq m$, the oscillation on every scale- ℓ cube is bounded by the global $\underline{L}^2(\square_m)$ norm of g_n , and hence by a dimensional constant. Therefore,

$$[g_n]_{\dot{B}_{2,2}^t(\square_m)} \leq C \left(\sum_{\ell=n+1}^m 3^{-2t\ell} \right)^{1/2} \leq C(1 - 3^{-2t})^{-1/2} 3^{-tn}.$$

The mean term satisfies

$$3^{-tm} |(g_n)_{\square_m}| \leq 3^{-tm} \|g_n\|_{\underline{L}^2(\square_m)} \leq 3^{-tn},$$

since $n \leq m$. Thus

$$\|g_n\|_{\dot{B}_{2,2}^t(\square_m)} \leq C(1 - 3^{-2t})^{-1/2} 3^{-tn}.$$

Applying the definition of the dual norm to g_n gives (1.14).

Using (1.14), we estimate

$$\begin{aligned} [f]_{\underline{B}_{2,2}^{-s}(\square_m)}^2 &= \sum_{n=-\infty}^m 3^{2sn} A_n^2 \\ &\leq C(1 - 3^{-2t})^{-1} [f]_{\underline{B}_{2,2}^{-t}(\square_m)}^2 \sum_{n=-\infty}^m 3^{2(s-t)n} \\ &\leq C(1 - 3^{-2t})^{-1} (1 - 3^{-2(s-t)})^{-1} 3^{2(s-t)m} [f]_{\underline{B}_{2,2}^{-t}(\square_m)}^2. \end{aligned}$$

Taking square roots gives (1.11), and multiplying by 3^{-sm} gives (1.12). The special case (1.13) follows from $1 - 3^{-r} \geq cr$ for $r \in (0, 1]$.

For vector fields, apply the scalar estimate to each coordinate and sum the resulting depthwise bounds in ℓ^2 over the coordinate index. $\square$

1.2.5 Optional comparisons worth recording

The first comparison we will use repeatedly is that the triadic Besov seminorm with matching summability exponent $q = p$ is equivalent to the classical fractional Sobolev seminorm.

Lemma 1.3 (Fractional Sobolev versus Besov seminorms). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, $s \in (0, 1)$, $p \in [1, \infty)$, and $u \in W^{s,p}(\square_m)$,*

$$C^{-1}[u]_{\underline{W}^{s,p}(\square_m)} \leq [u]_{\underline{B}_{p,p}^s(\square_m)} \leq C[u]_{\underline{W}^{s,p}(\square_m)}. \quad (1.15)$$

Proof. By translation and scaling it is enough to consider the unit cube. The estimate is proved in [1, Lemma A.4]. $\square$

1.3 Classical embeddings and inequalities to cite

1.3.1 Fractional Sobolev conjugate exponents

The classical exponents that appear in fractional Sobolev embeddings are

$$\begin{cases} p^*(s, d) := \frac{pd}{d - sp}, & p \in \left[1, \frac{d}{s}\right), \\ p_*(s, d) := \frac{pd}{d + sp}, & p \in [1, \infty). \end{cases} \quad (1.16)$$

These satisfy the duality relation

$$(p^*(s, d))' = (p')_*(s, d).$$

We will use this notation repeatedly in the statements of fractional Sobolev, Besov-duality, and negative-regularity estimates.

1.3.2 Fractional Poincaré-Sobolev and Morrey inequalities

The only classical embeddings we need later are the cube versions of the fractional Poincaré-Sobolev and Morrey inequalities.

Proposition 1.4 (Fractional Poincaré-Sobolev and Morrey inequalities on cubes). *Let $m \in \mathbb{Z}$.*

1. Let $s \in (0, 1]$ and $p \in [1, \frac{d}{s})$. Then, for every $f \in W^{s,p}(\square_m)$,

$$\|f - (f)_{\square_m}\|_{\underline{L}^{p^*(s,d)}(\square_m)} \leq C(s, p, d) 3^{sm} [f]_{\underline{W}^{s,p}(\square_m)}. \quad (1.17)$$

2. Let $s \in (0, 1)$ and $p \in (\frac{d}{s}, \infty]$. Then, for every $f \in W^{s,p}(\square_m)$,

$$\|f - (f)_{\square_m}\|_{L^\infty(\square_m)} \leq C(s, p, d) 3^{sm} [f]_{\underline{W}^{s,p}(\square_m)}. \quad (1.18)$$

Proof. By translation and scaling it is enough to consider the unit cube. The first estimate is the fractional Poincaré-Sobolev inequality [2, Theorem 6.7], and the second is the fractional Morrey inequality [2, Theorem 8.2]. $\square$

1.3.3 Calderón-Zygmund input

The proofs of the negative-Besov embeddings later in the chapter use only the most basic Calderón-Zygmund estimates for the Poisson equation on cubes. We record the exact form we need.

Proposition 1.5 (Calderón-Zygmund estimates for Poisson problems on cubes). *Let $m \in \mathbb{Z}$ and $q \in (1, \infty)$. There exists $C(q, d) < \infty$ such that:*

1. for every $F \in L^q(\square_m)$, if $w \in W^{2,q}(\square_m) \cap H_0^1(\square_m)$ solves

$$\begin{cases} -\Delta w = F & \text{in } \square_m, \\ w = 0 & \text{on } \partial\square_m, \end{cases}$$

then

$$\|\nabla^2 w\|_{\underline{L}^q(\square_m)} \leq C(q, d) \|F\|_{\underline{L}^q(\square_m)}; \quad (1.19)$$

2. for every $F \in L^q(\square_m)$, if $w \in W^{2,q}(\square_m)$ is the mean-zero solution of

$$\begin{cases} -\Delta w = F - (F)_{\square_m} & \text{in } \square_m, \\ \mathbf{n} \cdot \nabla w = 0 & \text{on } \partial\square_m, \\ (w)_{\square_m} = 0, \end{cases}$$

then

$$\|\nabla^2 w\|_{\underline{L}^q(\square_m)} \leq C(q, d) \|F - (F)_{\square_m}\|_{\underline{L}^q(\square_m)}. \quad (1.20)$$

Proof. By translation and scaling it is enough to consider the unit cube. The Dirichlet estimate follows by odd reflection across the faces of $\square_0$, while the Neumann estimate follows by even reflection. In both cases one reduces to the classical Calderón-Zygmund estimate for the Laplacian in $\mathbb{R}^d$; see, for instance, [3, Chapter V]. $\square$

1.3.4 A constant-coefficient Dirichlet regularity input

The duality argument in Chapter 3 also needs a fractional regularity estimate for the constant-coefficient Dirichlet problem with divergence-form right-hand side. We record the specific Besov estimate used there. In this subsection, when a Sobolev or Besov norm is applied to an $\mathbb{R}^d$ -valued field, we mean the natural componentwise extension with the Euclidean structure on the target.

Lemma 1.6 (Fractional Dirichlet regularity for constant-coefficient equations). *Let $m \in \mathbb{Z}$ and $s \in (0, 1)$. There exists $C(s, d) < \infty$ such that, for every vector field $\mathbf{h} \in H^s(\square_m; \mathbb{R}^d)$, if $w \in H_0^1(\square_m)$ is the solution of*

$$\begin{cases} -\Delta w = \nabla \cdot \mathbf{h} & \text{in } \square_m, \\ w = 0 & \text{on } \partial \square_m, \end{cases}$$

then

$$\|\nabla w\|_{\underline{B}_{2,2}^s(\square_m)} \leq C \|\mathbf{h}\|_{\underline{B}_{2,2}^s(\square_m)}. \quad (1.21)$$

Proof. By translation and scaling it is enough to consider the unit cube. For each $\mathbf{h} \in L^2(\square_0; \mathbb{R}^d)$, let $T(\mathbf{h}) := \nabla w$, where $w \in H_0^1(\square_0)$ solves

$$-\Delta w = \nabla \cdot \mathbf{h} \quad \text{in } \square_0.$$

Testing this equation with w gives

$$\|T(\mathbf{h})\|_{\underline{L}^2(\square_0)} \leq \|\mathbf{h}\|_{\underline{L}^2(\square_0)}.$$

If now $\mathbf{h} \in H^1(\square_0; \mathbb{R}^d)$, then the same odd-reflection argument used in Proposition 1.5 yields the constant-coefficient Dirichlet H^2 estimate

$$\|\nabla^2 w\|_{\underline{L}^2(\square_0)} \leq C \|\nabla \cdot \mathbf{h}\|_{\underline{L}^2(\square_0)} \leq C \|\nabla \mathbf{h}\|_{\underline{L}^2(\square_0)}.$$

We next use these two endpoint estimates through the elementary K -functional. For a vector field $F \in L^2(\square_0; \mathbb{R}^d)$ and $t \in (0, 1]$, set

$$K(t, F) := \inf_{G \in H^1(\square_0; \mathbb{R}^d)} (\|F - G\|_{\underline{L}^2(\square_0)}^2 + t^2 \|\nabla G\|_{\underline{L}^2(\square_0)}^2)^{1/2}.$$

We claim that

$$K(t, T(\mathbf{h})) \leq CK(t, \mathbf{h}).$$

Indeed, the map T is linear by uniqueness for the weak Dirichlet problem, and the H^2 endpoint estimate shows that $T(G) \in H^1(\square_0; \mathbb{R}^d)$ whenever $G \in H^1(\square_0; \mathbb{R}^d)$. Thus, for every such G , the endpoint estimates applied to $\mathbf{h} - G$ and to G give

$$\begin{aligned} K(t, T(\mathbf{h})) &\leq (\|T(\mathbf{h} - G)\|_{\underline{L}^2(\square_0)}^2 + t^2 \|\nabla T(G)\|_{\underline{L}^2(\square_0)}^2)^{1/2} \\ &\leq C(\|\mathbf{h} - G\|_{\underline{L}^2(\square_0)}^2 + t^2 \|\nabla G\|_{\underline{L}^2(\square_0)}^2)^{1/2}. \end{aligned}$$

Taking the infimum over G proves the claim. Therefore, with

$$|F|_{K,s} := \left(\int_0^1 t^{-2s} K(t, F)^2 \frac{dt}{t} \right)^{1/2},$$

we have

$$|T(\mathbf{h})|_{K,s} \leq C|\mathbf{h}|_{K,s}.$$

Together with the energy estimate, this gives

$$\|T(\mathbf{h})\|_{\underline{L}^2(\square_0)} + |T(\mathbf{h})|_{K,s} \leq C(\|\mathbf{h}\|_{\underline{L}^2(\square_0)} + |\mathbf{h}|_{K,s}).$$

Finally, we use the standard equivalence between the above K -functional norm and the fractional Sobolev norm on a cube, followed by Lemma 1.3. The fractional Poincaré-Sobolev inequality (1.17) with $p = 2$ then replaces the $\underline{L}^2$ term by the mean plus the triadic Besov seminorm. This yields

$$[\nabla w]_{\underline{B}_{2,2}^s(\square_0)} \leq C(\|T(\mathbf{h})\|_{\underline{L}^2(\square_0)} + |T(\mathbf{h})|_{K,s}) \leq C(\|\mathbf{h}\|_{\underline{L}^2(\square_0)} + |\mathbf{h}|_{K,s}) \leq C\|\mathbf{h}\|_{\underline{B}_{2,2}^s(\square_0)},$$

where the last inequality uses the same norm comparison in the reverse direction. For the mean term, we use

$$|(\nabla w)_{\square_0}| \leq \|\nabla w\|_{\underline{L}^2(\square_0)} \leq C\|\mathbf{h}\|_{\underline{L}^2(\square_0)} \leq C\|\mathbf{h}\|_{\underline{B}_{2,2}^s(\square_0)}.$$

Combining the seminorm and mean estimates gives (1.21) on $\square_0$. $\square$

1.4 Core multiscale lemmas

This is the genuinely book-specific core of the chapter. These are the statements that the later coarse-graining arguments repeatedly lean on.

The basic bookkeeping principle is simple: positive seminorms become stronger when one resolves a function on smaller cubes, while dual negative norms are controlled by summing appropriately rescaled cube-wise contributions. We first record these localization statements in the only case we will later need, namely the L^2 Besov scale. We then pass to the cube-wise Besov pairing that drives the coarse-grained Poincaré and Caccioppoli arguments in Chapter 3.

1.4.1 Localization across a triadic partition

The next two lemmas isolate the correct scaling when one passes between a cube and a partition into smaller triadic subcubes. The point is that the full normalized positive Besov norm does localize, but only after multiplying the local norms by the factor $3^{s(n-m)}$ dictated by the mean term.

Lemma 1.7 (Localization of the normalized positive Besov norm). *There exists $C(d) < \infty$ such that, for every $m, n \in \mathbb{Z}$ with $n \leq m$, every $s \in (0, 1)$, and every $u \in L^2(\square_m)$,*

$$\left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{s(n-m)} \|u\|_{\underline{B}_{2,2}^s(z + \square_n)} \right)^2 \right)^{1/2} \leq C \|u\|_{\underline{B}_{2,2}^s(\square_m)}. \quad (1.22)$$

Proof. By the triangle inequality, it is enough to estimate the seminorm part and the mean part separately.

For the seminorm part, expanding (1.6) gives

$$\begin{aligned} & \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{s(n-m)} [u]_{\underline{B}_{2,2}^s(z + \square_n)} \right)^2 \\ &= 3^{2s(n-m)} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{l=-\infty}^n 3^{-2sl} \sum_{\substack{y \in 3^{l-1} \mathbb{Z}^d \cap (z + \square_n) \\ y + \square_l \subseteq z + \square_n}} \|u - (u)_{y + \square_l}\|_{\underline{L}^2(y + \square_l)}^2. \end{aligned}$$

For each $l \leq n$, the families of scale- l subcubes of the cubes $\{z + \square_n\}_{z \in 3^n \mathbb{Z}^d \cap \square_m}$ partition the family

of all scale- l subcubes of $\square_m$. Therefore,

$$\begin{aligned}
& \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{s(n-m)} [u]_{\underline{B}_{2,2}^s(z+\square_n)} \right)^2 \\
&= 3^{2s(n-m)} \sum_{l=-\infty}^n 3^{-2sl} \sum_{\substack{y \in 3^{l-1} \mathbb{Z}^d \cap \square_m \\ y+\square_l \subseteq \square_m}} \|u - (u)_{y+\square_l}\|_{\underline{L}^2(y+\square_l)}^2 \\
&\leq \sum_{l=-\infty}^m 3^{-2sl} \sum_{\substack{y \in 3^{l-1} \mathbb{Z}^d \cap \square_m \\ y+\square_l \subseteq \square_m}} \|u - (u)_{y+\square_l}\|_{\underline{L}^2(y+\square_l)}^2 \\
&= [u]_{\underline{B}_{2,2}^s(\square_m)}^2.
\end{aligned}$$

For the mean part, we write

$$|(u)_{z+\square_n}| \leq |(u)_{z+\square_n} - (u)_{\square_m}| + |(u)_{\square_m}|.$$

Hence,

$$\begin{aligned}
& \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{s(n-m)} 3^{-sn} |(u)_{z+\square_n}| \right)^2 \right)^{1/2} \\
&= \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-sm} |(u)_{z+\square_n}| \right)^2 \right)^{1/2} \\
&\leq \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-sm} |(u)_{z+\square_n} - (u)_{\square_m}| \right)^2 \right)^{1/2} + 3^{-sm} |(u)_{\square_m}| \\
&\leq 3^{-sm} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \|u - (u)_{\square_m}\|_{\underline{L}^2(z+\square_n)}^2 \right)^{1/2} + 3^{-sm} |(u)_{\square_m}| \\
&= 3^{-sm} \|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} + 3^{-sm} |(u)_{\square_m}| \\
&\leq C[u]_{\underline{W}^{s,2}(\square_m)} + 3^{-sm} |(u)_{\square_m}| \\
&\leq C[u]_{\underline{B}_{2,2}^s(\square_m)} + 3^{-sm} |(u)_{\square_m}| \\
&\leq C\|u\|_{\underline{B}_{2,2}^s(\square_m)}.
\end{aligned}$$

In the penultimate step we used the fractional Poincaré-Sobolev inequality (1.17) with $p = 2$, followed by (1.15). Combining the seminorm and mean estimates yields (1.22). $\square$

Lemma 1.8 (Subadditivity of the dual negative Besov norm). *There exists $C(d) < \infty$ such that, for every $m, n \in \mathbb{Z}$ with $n \leq m$, every $s \in (0, 1)$, and every $f \in L^2(\square_m)$,*

$$[f]_{\underline{B}_{2,2}^{-s}(\square_m)} \leq C \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-s(n-m)} [f]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right)^2 \right)^{1/2}. \quad (1.23)$$

Proof. Fix $g \in L^2(\square_m)$ such that $\|g\|_{\underline{B}_{2,2}^s(\square_m)} \leq 1$. Then

$$\begin{aligned}
\left| \int_{\square_m} fg \right| &\leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \int_{z+\square_n} fg \right| \\
&\leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} [f]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \|g\|_{\underline{B}_{2,2}^s(z+\square_n)} \\
&= \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-s(n-m)} [f]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right) \left(3^{s(n-m)} \|g\|_{\underline{B}_{2,2}^s(z+\square_n)} \right) \\
&\leq \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-s(n-m)} [f]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right)^2 \right)^{1/2} \\
&\quad \times \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{s(n-m)} \|g\|_{\underline{B}_{2,2}^s(z+\square_n)} \right)^2 \right)^{1/2}.
\end{aligned}$$

Applying (1.22), we obtain

$$\left| \int_{\square_m} fg \right| \leq C \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-s(n-m)} [f]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right)^2 \right)^{1/2}.$$

Taking the supremum over such g yields (1.23). $\square$

1.4.2 Duality between negative and positive Besov seminorms

The concrete negative Besov seminorm is useful precisely because it pairs directly with positive Besov oscillation on the same triadic cube.

Lemma 1.9 (Cube-wise Besov pairing). *Let $m \in \mathbb{Z}$, $s \in (0, 1]$, $p \in (1, \infty)$, and $q \in [1, \infty]$, with $s = 1$ allowed only when $q = 1$. Then, for every $f \in L^p(\square_m)$ and every $g \in L^{p'}(\square_m)$,*

$$\left| \int_{\square_m} fg \right| \leq 3^{d+s} [f]_{\underline{B}_{p,q}^{-s}(\square_m)} \|g\|_{\underline{B}_{p',q'}^s(\square_m)}. \quad (1.24)$$

Proof. This is immediate from the definition of $[f]_{\underline{B}_{p,q}^{-s}(\square_m)}$ in (1.7) and the domination estimate (1.10). $\square$

1.4.3 Embedding from weak gradient control to L_q control of the function

For the next two subsections, the only case we need is the L^2 endpoint. It can be viewed as a multiscale Poincaré inequality measured by the negative Besov seminorm of the gradient.

Lemma 1.10 (Multiscale Poincaré from negative Besov control). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$ and $u \in H^1(\square_m)$,*

$$\|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} \leq C[\nabla u]_{\underline{B}_{2,1}^{-1}(\square_m)}. \quad (1.25)$$

Proof. Without loss of generality, assume that $(u)_{\square_m} = 0$. Let $w \in H^1(\square_m)$ be the mean-zero solution of the Neumann problem

$$\begin{cases} -\Delta w = u & \text{in } \square_m, \\ \mathbf{n} \cdot \nabla w = 0 & \text{on } \partial \square_m. \end{cases}$$

Classical Calderón-Zygmund theory gives

$$\|\nabla^2 w\|_{\underline{L}^2(\square_m)} \leq C \|u\|_{\underline{L}^2(\square_m)}.$$

Applying the usual Poincaré inequality on each triadic subcube $z + \square_k \subseteq \square_m$ and then taking the supremum over $k \leq m$, we obtain

$$[\nabla w]_{\underline{B}_{2,\infty}^1(\square_m)} \leq C \|\nabla^2 w\|_{\underline{L}^2(\square_m)} \leq C \|u\|_{\underline{L}^2(\square_m)}.$$

Since $(w)_{\square_m} = 0$, the ordinary Poincaré inequality also gives

$$3^{-m} \|\nabla w\|_{\underline{L}^2(\square_m)} \leq C \|\nabla^2 w\|_{\underline{L}^2(\square_m)} \leq C \|u\|_{\underline{L}^2(\square_m)}.$$

Consequently,

$$\|\nabla w\|_{\underline{B}_{2,\infty}^1(\square_m)} \leq C \|u\|_{\underline{L}^2(\square_m)}.$$

Testing the equation of w with u and using (1.24) with $f = \nabla u$, $g = \nabla w$, $s = 1$, $p = 2$, and $q = 1$, we find

$$\|u\|_{\underline{L}^2(\square_m)}^2 = \int_{\square_m} \nabla w \cdot \nabla u \leq C [\nabla u]_{\underline{B}_{2,1}^{-1}(\square_m)} \|\nabla w\|_{\underline{B}_{2,\infty}^1(\square_m)} \leq C [\nabla u]_{\underline{B}_{2,1}^{-1}(\square_m)} \|u\|_{\underline{L}^2(\square_m)}.$$

Dividing by $\|u\|_{\underline{L}^2(\square_m)}$ yields (1.25). $\square$

1.4.4 From negative Besov control of the gradient to positive Besov control of the function

The previous lemma can now be iterated across scales to turn weak information on the gradient into positive Besov control of the function itself.

Lemma 1.11 (From ∇u to u in Besov scale). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, $s \in (0, 1)$, and $u \in H^1(\square_m)$,*

$$[u]_{\underline{B}_{2,\infty}^s(\square_m)} \leq C s^{-1} [\nabla u]_{\underline{B}_{2,1}^{s-1}(\square_m)}. \quad (1.26)$$

Proof. For $n \leq m$, let

$$\mathcal{Z}_{n,m} := \{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m : z + \square_n \subseteq \square_m\}.$$

Applying (1.25) on each cube $z + \square_n$ gives

$$3^{-sn} \left(\sum_{z \in \mathcal{Z}_{n,m}} \|u - (u)_{z+\square_n}\|_{\underline{L}^2(z+\square_n)}^2 \right)^{1/2} \leq C 3^{-sn} \left(\sum_{z \in \mathcal{Z}_{n,m}} [\nabla u]_{\underline{B}_{2,1}^{-1}(z+\square_n)}^2 \right)^{1/2}.$$

For each $z \in \mathcal{Z}_{n,m}$, we have

$$[\nabla u]_{\underline{B}_{2,1}^{-1}(z+\square_n)} \leq \sum_{k=-\infty}^n 3^k \left(\sum_{y \in z+3^k\mathbb{Z}^d \cap \square_n} |(\nabla u)_{y+\square_k}|^2 \right)^{1/2}.$$

Using Minkowski's inequality in the $\ell^2(\mathcal{Z}_{n,m})$ variable, we obtain

$$3^{-sn} \left(\sum_{z \in \mathcal{Z}_{n,m}} [\nabla u]_{\underline{B}_{2,1}^{-1}(z+\square_n)}^2 \right)^{1/2} \leq C \sum_{k=-\infty}^n 3^{-s(n-k)} 3^{(1-s)k} \left(\sum_{y \in \mathcal{Z}_{k,m}} |(\nabla u)_{y+\square_k}|^2 \right)^{1/2}.$$

Taking the supremum over $n \leq m$ and using the discrete convolution estimate

$$\sup_{n \leq m} \sum_{k=-\infty}^n 3^{-s(n-k)} b_k \leq C s^{-1} \sup_{k \leq m} b_k \leq C s^{-1} \sum_{k=-\infty}^m b_k,$$

with

$$b_k := 3^{(1-s)k} \left(\sum_{y \in \mathcal{Z}_{k,m}} |(\nabla u)_{y+\square_k}|^2 \right)^{1/2},$$

we deduce (1.26). $\square$

1.4.5 Cutoff and product estimates

For the coarse-grained Caccioppoli argument, we only need a product estimate tailored to cutoff functions. It is convenient to record it directly in the form used later.

Lemma 1.12 (Cutoff/product estimate in positive Besov norm). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, $s \in (0, 1)$, $u \in H^1(\square_m)$, and $\varphi \in W^{2,\infty}(\square_m)$,*

$$\|(u - (u)_{\square_m}) \nabla \varphi\|_{\underline{B}_{2,\infty}^s(\square_m)} \leq C 3^m \|\nabla \varphi\|_{W^{1,\infty}(\square_m)} [\nabla u]_{\underline{B}_{2,1}^{s-1}(\square_m)}. \quad (1.27)$$

Proof. Set $v := u - (u)_{\square_m}$, so that $(v)_{\square_m} = 0$. For $k \leq m$, let

$$\mathcal{Z}_{k,m} := \{z \in 3^{k-1} \mathbb{Z}^d \cap \square_m : z + \square_k \subseteq \square_m\}.$$

For each $k \in (-\infty, m] \cap \mathbb{Z}$, we have

$$\begin{aligned} & 3^{-sk} \left(\sum_{z \in \mathcal{Z}_{k,m}} \|v \nabla \varphi - (v \nabla \varphi)_{z+\square_k}\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2} \\ & \leq 2 3^{-sk} \left(\sum_{z \in \mathcal{Z}_{k,m}} \|v(\nabla \varphi - (\nabla \varphi)_{z+\square_k})\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2} \\ & \quad + 2 \|\nabla \varphi\|_{L^\infty(\square_m)} 3^{-sk} \left(\sum_{z \in \mathcal{Z}_{k,m}} \|v - (v)_{z+\square_k}\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2}. \end{aligned}$$

Since

$$\|\nabla \varphi - (\nabla \varphi)_{z+\square_k}\|_{L^\infty(z+\square_k)} \leq C 3^k \|\nabla^2 \varphi\|_{L^\infty(\square_m)},$$

we obtain

$$\begin{aligned} & 3^{-sk} \left(\sum_{z \in \mathcal{Z}_{k,m}} \|v \nabla \varphi - (v \nabla \varphi)_{z+\square_k}\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2} \\ & \leq C 3^{(1-s)k} \|\nabla^2 \varphi\|_{L^\infty(\square_m)} \left(\sum_{z \in \mathcal{Z}_{k,m}} \|v\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2} \\ & \quad + C \|\nabla \varphi\|_{L^\infty(\square_m)} [v]_{\underline{B}_{2,\infty}^s(\square_m)} \\ & \leq C 3^{(1-s)m} \|\nabla^2 \varphi\|_{L^\infty(\square_m)} \|v\|_{\underline{L}^2(\square_m)} + C \|\nabla \varphi\|_{L^\infty(\square_m)} [v]_{\underline{B}_{2,\infty}^s(\square_m)}. \end{aligned}$$

Taking the supremum over $k \leq m$ yields

$$[v \nabla \varphi]_{\underline{B}_{2,\infty}^s(\square_m)} \leq C 3^{(1-s)m} \|\nabla^2 \varphi\|_{L^\infty(\square_m)} \|v\|_{\underline{L}^2(\square_m)} + C \|\nabla \varphi\|_{L^\infty(\square_m)} [v]_{\underline{B}_{2,\infty}^s(\square_m)}. \quad (1.28)$$

Since $(v)_{\square_m} = 0$, we also have

$$3^{-sm} |(v \nabla \varphi)_{\square_m}| \leq \|\nabla \varphi\|_{L^\infty(\square_m)} [v]_{\underline{B}_{2,\infty}^s(\square_m)}.$$

Combining this with (1.28) and then applying (1.26) gives

$$\begin{aligned} \|v \nabla \varphi\|_{\underline{B}_{2,\infty}^s(\square_m)} &\leq C (\|\nabla \varphi\|_{L^\infty(\square_m)} + 3^m \|\nabla^2 \varphi\|_{L^\infty(\square_m)}) [v]_{\underline{B}_{2,\infty}^s(\square_m)} \\ &\leq C 3^m \|\nabla \varphi\|_{\underline{W}^{1,\infty}(\square_m)} [\nabla u]_{\dot{\underline{B}}_{2,1}^{s-1}(\square_m)}. \end{aligned}$$

This is exactly (1.27). $\square$

1.4.6 A standard radius iteration lemma

The coarse-grained Caccioppoli argument ends with a one-dimensional iteration in the radius variable. Since the same pattern occurs repeatedly in elliptic regularity, it is useful to isolate it once and for all.

Lemma 1.13 (Standard radius iteration). *There exists $C < \infty$ such that, for every $\beta > 0$, $A \geq 0$, $r < R$, and every bounded function $F : [r, R] \rightarrow [0, \infty)$ satisfying*

$$F(\rho_1) \leq \frac{1}{2} F(\rho_2) + A(\rho_2 - \rho_1)^{-\beta} \quad \forall \rho_1, \rho_2 \in [r, R], \rho_1 < \rho_2, \quad (1.29)$$

we have

$$F(r) \leq (C(1 + \beta))^\beta A(R - r)^{-\beta}. \quad (1.30)$$

Proof. Set

$$\theta := 2^{-1/2\beta} \quad \text{and} \quad \rho_n := R - (R - r)\theta^n \quad \text{for } n \in \mathbb{N}_0.$$

Then $\rho_0 = r$, $\rho_n \uparrow R$, and

$$\rho_{n+1} - \rho_n = (R - r)(1 - \theta)\theta^n.$$

Iterating (1.29) along the sequence $\{\rho_n\}_{n \geq 0}$, we obtain, for every $N \in \mathbb{N}$,

$$F(r) \leq \sum_{n=0}^{N-1} 2^{-n} A(\rho_{n+1} - \rho_n)^{-\beta} + 2^{-N} F(\rho_N).$$

Since $2^{-1}\theta^{-\beta} = 2^{-1/2} < 1$, letting $N \rightarrow \infty$ yields

$$F(r) \leq C A(R - r)^{-\beta} (1 - \theta)^{-\beta}.$$

It remains to bound $(1 - \theta)^{-\beta}$. If $\beta \in (0, 1)$, then $\theta \leq 2^{-1/2}$, and therefore

$$(1 - \theta)^{-\beta} \leq (1 - 2^{-1/2})^{-1} \leq C.$$

If $\beta \geq 1$, then $\frac{\log 2}{2\beta} \leq 1$, so the elementary bound $1 - e^{-t} \geq ct$ for $t \in (0, 1]$ gives

$$1 - \theta = 1 - \exp\left(-\frac{\log 2}{2\beta}\right) \geq \frac{c}{\beta},$$

and hence

$$(1 - \theta)^{-\beta} \leq (C\beta)^\beta.$$

Combining the two cases gives

$$(1 - \theta)^{-\beta} \leq (C(1 + \beta))^\beta,$$

which is exactly (1.30). $\square$

1.5 Summary

Chapter 2 uses the underlined norms and the triadic Besov language to define coarse-grained matrices and multiscale ellipticity quantities. Chapter 3 uses the Besov duality pairing, the multiscale Poincaré lemma, the cutoff/product estimate, and the radius iteration lemma to prove the coarse-grained Poincaré, Caccioppoli, and homogenization estimates. Later, the same negative-to-positive embeddings convert weak control of gradients and fluxes into regularity information.

Chapter 2

Coarse-graining: basic definitions

This chapter should be the dictionary for the coarse-graining theory. The goal is to introduce the objects that will be used throughout the rest of the book, explain why they are natural, and collect the structural identities they satisfy. The main analytic estimates should come only in Chapter 3. In particular, the role of the present chapter is to define the coarse-grained response functionals and matrices, develop the block-matrix formalism adapted to nonsymmetric operators, and introduce the multiscale quantities that later serve as the state variables for both the deterministic and stochastic arguments.

2.1 Motivation and setup

2.1.1 The coarse-graining problem

The basic problem is to associate to a rough divergence-form operator a finite-dimensional object which captures its effective linear response on a bounded domain. Given an operator

$$-\nabla \cdot \mathbf{a}(x) \nabla,$$

we seek, for each bounded domain U , a coarse-scale description which retains the effect of gradients and fluxes while discarding microscopic detail. The guiding idea is that, even when the coefficient field is rough or highly oscillatory, the operator should admit a simpler coarse-scale description on each cube.

2.1.2 Coefficient fields and their symmetric and antisymmetric parts

Throughout, we consider coefficient fields $\mathbf{a}$ which are measurable functions from an open subset $U \subseteq \mathbb{R}^d$ to the set $\mathbb{R}^{d \times d}$ of real d -by- d matrices which are uniformly elliptic in the following sense: there exist constants $0 < \lambda \leq \Lambda < \infty$ such that

$$e \cdot \mathbf{a}(x)e \geq \lambda |e|^2 \quad \text{and} \quad e \cdot \mathbf{a}^{-1}(x)e \geq \Lambda^{-1} |e|^2 \quad \forall e \in \mathbb{R}^d, x \in U. \quad (2.1)$$

The second inequality in (2.1) implies an upper bound on $\mathbf{a}$. Indeed,

$$|\mathbf{a}(x)e|^2 = \mathbf{a}(x)e \cdot \mathbf{a}(x)e \leq \Lambda \mathbf{a}(x)e \cdot \mathbf{a}^{-1}(x)\mathbf{a}(x)e = \Lambda e \cdot \mathbf{a}(x)e \leq \Lambda |e| |\mathbf{a}(x)e|,$$

and therefore

$$|\mathbf{a}(x)e| \leq \Lambda |e| \quad \forall x \in U, e \in \mathbb{R}^d.$$

For symmetric matrices this is equivalent to the usual upper bound $\mathbf{a}(x) \leq \Lambda \mathbf{I}_d$; for nonsymmetric matrices it is the more natural assumption for ellipticity, and it is the assumption that later turns out to be stable under coarse-graining.

Given a coefficient field $\mathbf{a} : U \rightarrow \mathbb{R}^{d \times d}$, we denote the symmetric and antisymmetric parts of $\mathbf{a}$ by

$$\mathbf{s} := \frac{1}{2}(\mathbf{a} + \mathbf{a}^t) \quad \text{and} \quad \mathbf{k} := \frac{1}{2}(\mathbf{a} - \mathbf{a}^t)$$

Observe that $\mathbf{a} = \mathbf{s} + \mathbf{k}$. The uniform ellipticity condition in (2.1) can be stated equivalently as

$$\mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k} \leq \Lambda \mathbf{I}_d \quad \text{and} \quad \mathbf{s}^{-1} \leq \lambda^{-1} \mathbf{I}_d. \quad (2.2)$$

Indeed, since $e \cdot \mathbf{a} e = e \cdot \mathbf{s} e$, the first inequality in (2.1) is equivalent to $\mathbf{s} \geq \lambda \mathbf{I}_d$, that is, to $\mathbf{s}^{-1} \leq \lambda^{-1} \mathbf{I}_d$. Likewise,

$$e \cdot \mathbf{a}^{-1} e = e \cdot \frac{1}{2}(\mathbf{a}^{-1} + \mathbf{a}^{-t}) e, \quad \frac{1}{2}(\mathbf{a}^{-1} + \mathbf{a}^{-t}) = \mathbf{a}^{-t} \mathbf{s} \mathbf{a}^{-1}.$$

Therefore the second inequality in (2.1) is equivalent to

$$\mathbf{a}^{-t} \mathbf{s} \mathbf{a}^{-1} \geq \Lambda^{-1} \mathbf{I}_d.$$

Since

$$(\mathbf{a}^{-t} \mathbf{s} \mathbf{a}^{-1})^{-1} = \mathbf{a} \mathbf{s}^{-1} \mathbf{a}^t = (\mathbf{s} + \mathbf{k}) \mathbf{s}^{-1} (\mathbf{s} - \mathbf{k}) = \mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k},$$

this is exactly the first matrix inequality in (2.2).

The space of all uniformly elliptic coefficient fields on U with constants λ, Λ is denoted by

$$\Omega[\lambda, \Lambda, U] := \{ \mathbf{a} : U \rightarrow \mathbb{R}^{d \times d} \text{ satisfies (2.1) } \}.$$

The set of all uniformly elliptic coefficients on U is

$$\Omega(U) := \bigcup_{\varepsilon \in (0,1]} \Omega[\varepsilon, \varepsilon^{-1}, U].$$

We denote $\Omega[\lambda, \Lambda] := \Omega[\lambda, \Lambda, \mathbb{R}^d]$. The set of coefficient fields on $\mathbb{R}^d$ which are locally uniformly elliptic is denoted by

$$\Omega := \left\{ \mathbf{a} : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d} \text{ measurable} : \mathbf{a}|_{B_R} \in \Omega(B_R) \text{ for every } R \geq 1 \right\}.$$

Every $\mathbf{a} \in \Omega$ restricts to an element of $\Omega(U)$ on each bounded domain $U \subseteq \mathbb{R}^d$, and conversely every $\mathbf{a} \in \Omega(U)$ can be extended, for instance by $\mathbf{I}_d$ outside U , to an element of Ω .

We emphasize that in this book uniform ellipticity is used only *qualitatively*. It is mainly used to guarantee that, on bounded Lipschitz domains and without introducing other exotic function spaces, the standard Dirichlet and variational problems are well-posed in H^1 . This will be the only level of ellipticity theory needed in the present chapter.

2.1.3 Solution spaces

For a bounded domain $U \subseteq \mathbb{R}^d$ and $\mathbf{a} \in \Omega(U)$, we denote the space of solutions by

$$\mathcal{A}(U; \mathbf{a}) := \left\{ u \in H^1(U) : \int_U \mathbf{a} \nabla u \cdot \nabla \phi = 0 \text{ for every } \phi \in H_0^1(U) \right\}.$$

These are the natural ambient spaces for the coarse-grained variational problems.

2.2 The response functional and the coarse-grained matrices

2.2.1 Definition of the response functional

From now on, let $U \subseteq \mathbb{R}^d$ be a bounded Lipschitz domain¹ and $\mathbf{a} = \mathbf{s} + \mathbf{k} \in \Omega(U)$. We also write

$$\mathcal{A}^*(U; \mathbf{a}) := \mathcal{A}(U; \mathbf{a}^t)$$

for the space of solutions of the adjoint equation. In the nonsymmetric setting, the right scalar quantity to begin with is not a separate Dirichlet or Neumann energy, but a single response functional which records at once the effect of coarse gradients and coarse fluxes.

For every $p, q \in \mathbb{R}^d$, we define the *response functional*

$$J(U, p, q; \mathbf{a}) := \sup_{v \in \mathcal{A}(U; \mathbf{a})} \int_U \left(-\frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right). \quad (2.3)$$

The use of the symmetric part $\mathbf{s}$ in the quadratic term is essential: it keeps the functional strictly concave in the gradient variable while the admissible class remains the actual finite-energy solution space for the operator $-\nabla \cdot \mathbf{a} \nabla$. Since the functional in (2.3) depends only on ∇v , we may restrict the supremum to the slice

$$\mathcal{A}_0(U; \mathbf{a}) := \left\{ v \in \mathcal{A}(U; \mathbf{a}) : \int_U v = 0 \right\}.$$

On $\mathcal{A}_0(U; \mathbf{a})$, Poincaré's inequality and the bound $|\mathbf{a}(x)e| \leq \Lambda|e|$ give

$$\int_U \left(-\frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right) \leq -\frac{\lambda}{4} \int_U |\nabla v|^2 + C(U, \lambda, \Lambda)(|p|^2 + |q|^2),$$

so the functional is coercive on $\mathcal{A}_0(U; \mathbf{a})$. The set $\mathcal{A}_0(U; \mathbf{a})$ is weakly closed in $H^1(U)$, because $\mathbf{a} \in L^\infty(U)$ and the weak equation is stable under weak H^1 convergence. The direct method therefore yields a unique maximizer in $\mathcal{A}_0(U; \mathbf{a})$; viewed in $\mathcal{A}(U; \mathbf{a})$, it is unique up to additive constants. We denote it by

$$v(\cdot, U, p, q; \mathbf{a}).$$

Since constants belong to $\mathcal{A}(U; \mathbf{a})$, we immediately have $J(U, p, q; \mathbf{a}) \geq 0$.

Lemma 2.1 (First variation). *The maximizer $v(\cdot, U, p, q; \mathbf{a})$ satisfies, for every $w \in \mathcal{A}(U; \mathbf{a})$,*

$$q \cdot \int_U \nabla w - p \cdot \int_U \mathbf{a} \nabla w = \int_U \nabla w \cdot \mathbf{s} \nabla v(\cdot, U, p, q; \mathbf{a}). \quad (2.4)$$

Consequently, the map

$$(p, q) \mapsto \nabla v(\cdot, U, p, q; \mathbf{a}) \quad \text{is linear} \quad (2.5)$$

and therefore $(p, q) \mapsto J(U, p, q; \mathbf{a})$ is a quadratic form.

Proof. Since $\mathcal{A}(U; \mathbf{a})$ is a linear space, for every $t \in \mathbb{R}$ and every $w \in \mathcal{A}(U; \mathbf{a})$ the function $v + tw$ is an admissible competitor in (2.3). By maximality of v , the function

$$t \mapsto \int_U \left(-\frac{1}{2} \nabla(v + tw) \cdot \mathbf{s} \nabla(v + tw) - p \cdot \mathbf{a} \nabla(v + tw) + q \cdot \nabla(v + tw) \right)$$

¹In the Lean formalization, the deterministic domain interface in this chapter is restricted to bounded open convex domains, rather than general bounded Lipschitz domains, to keep the formal PDE and domain infrastructure simpler.

has a maximum at $t = 0$, and therefore its derivative at 0 vanishes. This is exactly (2.4).

To obtain linearity, fix $(p_1, q_1), (p_2, q_2) \in \mathbb{R}^d \times \mathbb{R}^d$ and write

$$v_i := v(\cdot, U, p_i, q_i; \mathbf{a}), \quad v_{12} := v(\cdot, U, p_1 + p_2, q_1 + q_2; \mathbf{a}).$$

Applying (2.4) to v_{12} and then to v_1 and v_2 , we find that, for every $w \in \mathcal{A}(U; \mathbf{a})$,

$$\int_U \nabla w \cdot \mathbf{s} \nabla v_{12} = \int_U \nabla w \cdot \mathbf{s} \nabla (v_1 + v_2).$$

Since

$$\delta := v_{12} - v_1 - v_2$$

belongs to $\mathcal{A}(U; \mathbf{a})$, we may choose $w = \delta$ and obtain

$$\int_U \nabla \delta \cdot \mathbf{s} \nabla \delta = 0.$$

The uniform ellipticity of $\mathbf{s}$ therefore implies $\nabla \delta = 0$, that is,

$$\nabla v_{12} = \nabla v_1 + \nabla v_2.$$

The same argument with (tp, tq) in place of (p, q) gives homogeneity in scalar multiples, hence the linearity of $(p, q) \mapsto \nabla v(\cdot, U, p, q; \mathbf{a})$. Substituting the maximizing gradient back into the variational formula (2.3) then shows that $(p, q) \mapsto J(U, p, q; \mathbf{a})$ is quadratic. $\square$

2.2.2 Definition of the coarse-grained matrices

By Lemma 2.1, the response functional J is quadratic in (p, q) . We may therefore extract the coarse-grained matrices from J one piece at a time. Moreover, the quadratic form $q \mapsto J(U, 0, q; \mathbf{a})$ is strictly positive away from the origin. Indeed, for each $e \in \mathbb{R}^d$, let $u_e \in e \cdot x + H_0^1(U)$ be the unique weak solution of $-\nabla \cdot \mathbf{a} \nabla u_e = 0$ in U . Since $u_e - e \cdot x \in H_0^1(U)$, we have

$$\int_U \nabla u_e = e.$$

Therefore, for $q \neq 0$ and $e := q|q|^{-1}$, the competitors tu_e give

$$J(U, 0, q; \mathbf{a}) \geq \sup_{t \in \mathbb{R}} \left(t|q| - \frac{t^2}{2} \int_U \nabla u_e \cdot \mathbf{s} \nabla u_e \right) > 0.$$

We first define $\mathbf{s}_*(U; \mathbf{a})$ to be the symmetric positive definite matrix satisfying

$$\frac{1}{2} q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) q = J(U, 0, q; \mathbf{a}) \quad \forall q \in \mathbb{R}^d. \quad (2.6)$$

Next we define the mixed matrix $\mathbf{k}(U; \mathbf{a})$ by

$$q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p = J(U, p, q; \mathbf{a}) - J(U, p, 0; \mathbf{a}) - J(U, 0, q; \mathbf{a}) + p \cdot q \quad \forall p, q \in \mathbb{R}^d. \quad (2.7)$$

We define $\mathbf{s}(U; \mathbf{a})$ to be the symmetric matrix satisfying

$$\frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p = J(U, p, 0; \mathbf{a}) - \frac{1}{2} p \cdot \mathbf{k}^t(U; \mathbf{a}) \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p \quad \forall p \in \mathbb{R}^d. \quad (2.8)$$

By construction, these matrices satisfy

$$J(U, p, q; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p + \frac{1}{2} (q + \mathbf{k}(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q + \mathbf{k}(U; \mathbf{a}) p) - p \cdot q \quad \forall p, q \in \mathbb{R}^d. \quad (2.9)$$

The matrices $\mathbf{s}(U; \mathbf{a})$, $\mathbf{s}_*(U; \mathbf{a})$ and $\mathbf{k}(U; \mathbf{a})$ are the primary objects in the coarse-grained description of the elliptic operator $-\nabla \cdot \mathbf{a} \nabla$. From them we define the derived matrices

$$\mathbf{b}(U; \mathbf{a}) := \mathbf{s}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}) \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) \quad (2.10)$$

and

$$\mathbf{a}(U; \mathbf{a}) := \mathbf{s}(U; \mathbf{a}) - \mathbf{k}^t(U; \mathbf{a}), \quad \mathbf{a}_*(U; \mathbf{a}) := \mathbf{s}_*(U; \mathbf{a}) - \mathbf{k}^t(U; \mathbf{a}). \quad (2.11)$$

The symmetric case, in which these objects reduce to the classical Dirichlet and Neumann finite-volume matrices, is discussed in Subsection 2.3.2. In the general nonsymmetric setting, the matrices $\mathbf{s}(U; \mathbf{a})$, $\mathbf{s}_*(U; \mathbf{a})$, and $\mathbf{k}(U; \mathbf{a})$ form the right finite-dimensional summary of the response of the operator on U , while $\mathbf{a}(U; \mathbf{a})$, $\mathbf{a}_*(U; \mathbf{a})$, and $\mathbf{b}(U; \mathbf{a})$ are the combinations that later appear naturally in formulas for average gradients, average fluxes, and ellipticity estimates.

2.3 Structural properties of the coarse-grained matrices

2.3.1 First algebraic and variational identities

The identities in this subsection explain what the coarse-grained matrices are actually measuring. They are deterministic statements attached to a fixed domain U and a fixed coefficient field $\mathbf{a}$.

Lemma 2.2 (Basic variational identities). *The following assertions hold for every $p, q \in \mathbb{R}^d$.*

1. *The matrices $\mathbf{s}(U; \mathbf{a})$ and $\mathbf{s}_*(U; \mathbf{a})$ are symmetric, and they satisfy the ellipticity bounds*

$$\left(\int_U \mathbf{s}^{-1}(x) dx \right)^{-1} \leq \mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a}) \leq \mathbf{b}(U; \mathbf{a}) \leq \int_U (\mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k})(x) dx. \quad (2.12)$$

2. *For every $w \in \mathcal{A}(U; \mathbf{a})$, we have the second variation identity*

$$\begin{aligned} J(U, p, q; \mathbf{a}) - \int_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{s} \nabla w - p \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) \\ = \int_U \frac{1}{2} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w) \cdot \mathbf{s} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w). \end{aligned} \quad (2.13)$$

3. *The response functional is exactly the energy of the maximizer:*

$$J(U, p, q; \mathbf{a}) = \int_U \frac{1}{2} \nabla v(\cdot, U, p, q; \mathbf{a}) \cdot \mathbf{s} \nabla v(\cdot, U, p, q; \mathbf{a}). \quad (2.14)$$

4. *The averages of the gradient and flux of the maximizer are given by*

$$\begin{cases} \int_U \nabla v(\cdot, U, p, q; \mathbf{a}) = -p + \mathbf{s}_*^{-1}(U; \mathbf{a}) (q + \mathbf{k}(U; \mathbf{a}) p), \\ \int_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) = (\mathbf{I}_d - \mathbf{k}^t(U; \mathbf{a}) \mathbf{s}_*^{-1}(U; \mathbf{a})) q - \mathbf{b}(U; \mathbf{a}) p. \end{cases} \quad (2.15)$$

Proof. The symmetry of $\mathbf{s}(U; \mathbf{a})$ and $\mathbf{s}_*(U; \mathbf{a})$ is built into their definitions. Moreover, the inequality $\mathbf{s}(U; \mathbf{a}) \leq \mathbf{b}(U; \mathbf{a})$ is immediate from (2.10), since the matrix

$$\mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a})\mathbf{k}(U; \mathbf{a})$$

is symmetric and nonnegative definite. The proof of the ordering $\mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a})$ is postponed; it will be proved below in (2.61), using the doubled block-matrix formalism. It therefore remains for the first item to prove the lower bound on $\mathbf{s}_*(U; \mathbf{a})$ and the upper bound on $\mathbf{b}(U; \mathbf{a})$.

For every $x \in U$ and $\xi, q \in \mathbb{R}^d$, completion of the square gives

$$-\frac{1}{2}\xi \cdot \mathbf{s}(x)\xi + q \cdot \xi = -\frac{1}{2}(\xi - \mathbf{s}^{-1}(x)q) \cdot \mathbf{s}(x)(\xi - \mathbf{s}^{-1}(x)q) + \frac{1}{2}q \cdot \mathbf{s}^{-1}(x)q \leq \frac{1}{2}q \cdot \mathbf{s}^{-1}(x)q.$$

By (2.6), for every $q \in \mathbb{R}^d$,

$$\frac{1}{2}q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})q = \sup_{u \in \mathcal{A}(U; \mathbf{a})} \int_U \left(-\frac{1}{2}\nabla u \cdot \mathbf{s}\nabla u + q \cdot \nabla u \right) \leq \int_U \frac{1}{2}q \cdot \mathbf{s}^{-1}(x)q \, dx.$$

Thus

$$\mathbf{s}_*^{-1}(U; \mathbf{a}) \leq \int_U \mathbf{s}^{-1}(x) \, dx.$$

Since both sides are symmetric positive definite, inversion reverses the Loewner order and yields

$$\left(\int_U \mathbf{s}^{-1}(x) \, dx \right)^{-1} \leq \mathbf{s}_*(U; \mathbf{a}).$$

Likewise, for every $x \in U$ and $\xi, p \in \mathbb{R}^d$,

$$\begin{aligned} -\frac{1}{2}\xi \cdot \mathbf{s}(x)\xi - p \cdot \mathbf{a}(x)\xi &= -\frac{1}{2}(\xi + \mathbf{s}^{-1}(x)\mathbf{a}^t(x)p) \cdot \mathbf{s}(x)(\xi + \mathbf{s}^{-1}(x)\mathbf{a}^t(x)p) \\ &\quad + \frac{1}{2}\mathbf{a}^t(x)p \cdot \mathbf{s}^{-1}(x)\mathbf{a}^t(x)p \\ &\leq \frac{1}{2}\mathbf{a}^t(x)p \cdot \mathbf{s}^{-1}(x)\mathbf{a}^t(x)p. \end{aligned}$$

Likewise, by (2.9) with $q = 0$ and the definition of $\mathbf{b}(U; \mathbf{a})$,

$$\begin{aligned} \frac{1}{2}p \cdot \mathbf{b}(U; \mathbf{a})p &= J(U, p, 0; \mathbf{a}) = \sup_{u \in \mathcal{A}(U; \mathbf{a})} \int_U \left(-\frac{1}{2}\nabla u \cdot \mathbf{s}\nabla u - p \cdot \mathbf{a}\nabla u \right) \\ &\leq \int_U \frac{1}{2}\mathbf{a}^t(x)p \cdot \mathbf{s}^{-1}(x)\mathbf{a}^t(x)p \, dx \\ &= \int_U \frac{1}{2}p \cdot (\mathbf{s} + \mathbf{k}^t\mathbf{s}^{-1}\mathbf{k})(x)p \, dx, \end{aligned}$$

which is the asserted upper bound on $\mathbf{b}(U; \mathbf{a})$. This completes the proof of (2.12), apart from the deferred inequality $\mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a})$.

We next prove (2.13). Denote $v := v(\cdot, U, p, q; \mathbf{a})$. Since both v and w belong to $\mathcal{A}(U; \mathbf{a})$, their difference belongs to $\mathcal{A}(U; \mathbf{a})$ as well. Writing $h := w - v$, we compute

$$\begin{aligned} J(U, p, q; \mathbf{a}) &- \int_U \left(-\frac{1}{2}\nabla w \cdot \mathbf{s}\nabla w - p \cdot \mathbf{a}\nabla w + q \cdot \nabla w \right) \\ &= - \int_U \left(-\frac{1}{2}\nabla h \cdot \mathbf{s}\nabla h - \nabla v \cdot \mathbf{s}\nabla h - p \cdot \mathbf{a}\nabla h + q \cdot \nabla h \right). \end{aligned}$$

Applying the optimality identity (2.4) with $h \in \mathcal{A}(U; \mathbf{a})$, we get

$$\oint_U (\nabla v \cdot \mathbf{s} \nabla h + p \cdot \mathbf{a} \nabla h - q \cdot \nabla h) = 0.$$

Substituting this into the previous computation yields

$$J(U, p, q; \mathbf{a}) - \oint_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{s} \nabla w - p \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) = \oint_U \frac{1}{2} \nabla h \cdot \mathbf{s} \nabla h,$$

which is exactly (2.13).

To prove (2.14), we evaluate (2.4) at $w = v$. This gives

$$q \cdot \oint_U \nabla v - p \cdot \oint_U \mathbf{a} \nabla v = \oint_U \nabla v \cdot \mathbf{s} \nabla v.$$

Substituting this identity into the definition of $J(U, p, q; \mathbf{a})$ at the maximizer v gives

$$J(U, p, q; \mathbf{a}) = \oint_U \frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v,$$

which is (2.14).

We now prove (2.15). Let $p', q' \in \mathbb{R}^d$ be arbitrary, and set

$$v' := v(\cdot, U, p', q'; \mathbf{a}).$$

Applying (2.4) to the maximizer v' with test function $w = v$, and using the symmetry of $\mathbf{s}$, we obtain

$$q' \cdot \oint_U \nabla v - p' \cdot \oint_U \mathbf{a} \nabla v = \oint_U \nabla v' \cdot \mathbf{s} \nabla v. \quad (2.16)$$

By the linearity of the maximizing gradient established in Lemma 2.1, we have

$$\nabla v(\cdot, U, p - p', q - q'; \mathbf{a}) = \nabla v - \nabla v'.$$

Using this and (2.14), polarization gives

$$\begin{aligned} \oint_U \nabla v' \cdot \mathbf{s} \nabla v &= \oint_U \left(-\frac{1}{2} (\nabla v - \nabla v') \cdot \mathbf{s} (\nabla v - \nabla v') + \frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v + \frac{1}{2} \nabla v' \cdot \mathbf{s} \nabla v' \right) \\ &= J(U, p, q; \mathbf{a}) + J(U, p', q'; \mathbf{a}) - J(U, p - p', q - q'; \mathbf{a}). \end{aligned}$$

Using (2.9) to expand the right-hand side, we find

$$\begin{aligned} \oint_U \nabla v' \cdot \mathbf{s} \nabla v &= p' \cdot \mathbf{s}(U; \mathbf{a})p + (q' + \mathbf{k}(U; \mathbf{a})p') \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p) - p' \cdot q - q' \cdot p \\ &= q' \cdot (\mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p) - p) \\ &\quad - p' \cdot (-\mathbf{s}(U; \mathbf{a})p - \mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p) + q). \end{aligned}$$

Comparing this with (2.16) and using the arbitrariness of p' and q' , we obtain

$$\oint_U \nabla v = -p + \mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p)$$

and

$$\int_U \mathbf{a} \nabla v = q - \mathbf{s}(U; \mathbf{a})p - \mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p).$$

The second formula may be rewritten as

$$\int_U \mathbf{a} \nabla v = (\mathbf{I}_d - \mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a}))q - (\mathbf{s}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a})\mathbf{k}(U; \mathbf{a}))p,$$

which is exactly the second line of (2.15). This completes the proof. $\square$

2.3.2 Symmetric coefficients: the Dirichlet–Neumann split

The definition of the response functional is designed for nonsymmetric coefficient fields, where one cannot separate the response into two scalar energy principles for the original operator. When the coefficient field is symmetric, however, the familiar Dirichlet and Neumann variational problems reappear. In this case the variable p belongs to the affine Dirichlet problem, while the variable q belongs to the Neumann problem with prescribed constant flux.

Throughout this subsection, assume that $\mathbf{a} = \mathbf{a}^t$. Thus $\mathbf{s} = \mathbf{a}$ and the microscopic antisymmetric part vanishes. For $p, q \in \mathbb{R}^d$, define

$$\begin{cases} \nu_D(U, p; \mathbf{a}) := \inf_{u \in p \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u, \\ \nu_N(U, q; \mathbf{a}) := \sup_{u \in H^1(U)} \int_U \left(q \cdot \nabla u - \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u \right). \end{cases} \quad (2.17)$$

The minimizer in the definition of ν_D will be denoted by u_p^D , so that

$$u_p^D - p \cdot x \in H_0^1(U) \quad \text{and} \quad \int_U \nabla \phi \cdot \mathbf{a} \nabla u_p^D = 0 \quad \forall \phi \in H_0^1(U).$$

The supremum in the definition of ν_N is insensitive to additive constants; we denote by u_q^N the unique mean-zero maximizer. It satisfies

$$(u_q^N)_U = 0 \quad \text{and} \quad \int_U \nabla \phi \cdot \mathbf{a} \nabla u_q^N = \int_U q \cdot \nabla \phi \quad \forall \phi \in H^1(U).$$

Thus u_q^N is the weak solution of the homogeneous equation with conormal boundary condition

$$(\mathbf{a} \nabla u_q^N - q) \cdot n = 0 \quad \text{on } \partial U.$$

Lemma 2.3 (Dirichlet–Neumann splitting for symmetric coefficients). *Assume that $\mathbf{a} = \mathbf{a}^t$. Then, for every $p, q \in \mathbb{R}^d$, the maximizer in the definition of $J(U, p, q; \mathbf{a})$ is, up to additive constants,*

$$v(\cdot, U, p, q; \mathbf{a}) = u_q^N - u_p^D. \quad (2.18)$$

Consequently,

$$J(U, p, q; \mathbf{a}) = \nu_D(U, p; \mathbf{a}) + \nu_N(U, q; \mathbf{a}) - p \cdot q. \quad (2.19)$$

Moreover,

$$\nu_D(U, p; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a})p, \quad \nu_N(U, q; \mathbf{a}) = \frac{1}{2} q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})q, \quad \mathbf{k}(U; \mathbf{a}) = 0. \quad (2.20)$$

Proof. We first record two elementary consequences of the boundary conditions. Since $u_p^D - p \cdot x \in H_0^1(U)$,

$$\oint_U \nabla u_p^D = p. \quad (2.21)$$

Testing the Neumann problem with the coordinate functions gives

$$\oint_U \mathbf{a} \nabla u_q^N = q. \quad (2.22)$$

Also, testing the Neumann problem with u_q^N yields

$$\oint_U q \cdot \nabla u_q^N = \oint_U \nabla u_q^N \cdot \mathbf{a} \nabla u_q^N,$$

and therefore

$$\nu_N(U, q; \mathbf{a}) = \oint_U \frac{1}{2} \nabla u_q^N \cdot \mathbf{a} \nabla u_q^N. \quad (2.23)$$

Let $w \in \mathcal{A}(U; \mathbf{a})$. By the variational equation for u_q^N ,

$$\oint_U \nabla w \cdot \mathbf{a} \nabla u_q^N = q \cdot \oint_U \nabla w. \quad (2.24)$$

On the other hand, since $w \in \mathcal{A}(U; \mathbf{a})$ and $\mathbf{a}$ is symmetric, we may test the equation for w with $u_p^D - p \cdot x \in H_0^1(U)$ to get

$$\oint_U \nabla w \cdot \mathbf{a} \nabla (u_p^D - p \cdot x) = 0.$$

Thus

$$\oint_U \nabla w \cdot \mathbf{a} \nabla u_p^D = p \cdot \oint_U \mathbf{a} \nabla w. \quad (2.25)$$

Combining (2.24) and (2.25), we obtain

$$\oint_U \nabla w \cdot \mathbf{a} \nabla (u_q^N - u_p^D) = q \cdot \oint_U \nabla w - p \cdot \oint_U \mathbf{a} \nabla w.$$

Since $\mathbf{s} = \mathbf{a}$, this is exactly the first-variation identity (2.4) for the maximizer of $J(U, p, q; \mathbf{a})$. The maximizer is unique up to additive constants, and this proves (2.18).

Using (2.14) and the preceding identification of the maximizer, and writing $\delta_{p,q} := u_q^N - u_p^D$, we get

$$J(U, p, q; \mathbf{a}) = \oint_U \frac{1}{2} \nabla \delta_{p,q} \cdot \mathbf{a} \nabla \delta_{p,q}. \quad (2.26)$$

The cross term in this energy is

$$\oint_U \nabla u_p^D \cdot \mathbf{a} \nabla u_q^N = \oint_U q \cdot \nabla u_p^D = p \cdot q,$$

where we used the Neumann variational equation with test function u_p^D and then (2.21). Expanding (2.26), using the definitions of ν_D and (2.23), gives (2.19).

It remains to compare this splitting with the general quadratic representation (2.9). The pure q terms give

$$\nu_N(U, q; \mathbf{a}) = \frac{1}{2} q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) q.$$

The mixed term in (2.9) is $q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p - p \cdot q$, whereas the mixed term in (2.19) is just $-p \cdot q$. Hence

$$q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p = 0 \quad \forall p, q \in \mathbb{R}^d.$$

Since $\mathbf{s}_*(U; \mathbf{a})$ is positive definite, this implies $\mathbf{k}(U; \mathbf{a}) = 0$. Comparing the pure p terms then gives

$$\nu_D(U, p; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p.$$

This proves (2.20). $\square$

The lemma gives a concrete interpretation of the two symmetric coarse-grained matrices. Applying (2.15) to $v(\cdot, U, p, 0; \mathbf{a}) = -u_p^D$ and to $v(\cdot, U, 0, q; \mathbf{a}) = u_q^N$, and using $\mathbf{k}(U; \mathbf{a}) = 0$, we obtain

$$\begin{cases} \oint_U \nabla u_p^D = p, & \oint_U \mathbf{a} \nabla u_p^D = \mathbf{s}(U; \mathbf{a}) p, \\ \oint_U \mathbf{a} \nabla u_q^N = q, & \oint_U \nabla u_q^N = \mathbf{s}_*^{-1}(U; \mathbf{a}) q. \end{cases} \quad (2.27)$$

Thus $\mathbf{s}(U; \mathbf{a})$ is the finite-volume Dirichlet matrix, while $\mathbf{s}_*(U; \mathbf{a})$ is the finite-volume Neumann matrix. The derived matrices also simplify:

$$\mathbf{a}(U; \mathbf{a}) = \mathbf{s}(U; \mathbf{a}), \quad \mathbf{a}_*(U; \mathbf{a}) = \mathbf{s}_*(U; \mathbf{a}), \quad \mathbf{b}(U; \mathbf{a}) = \mathbf{s}(U; \mathbf{a}). \quad (2.28)$$

In particular, the ellipticity bounds (2.12) reduce to the classical Dirichlet–Neumann bracketing

$$\left(\oint_U \mathbf{a}^{-1}(x) dx \right)^{-1} \leq \mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a}) \leq \oint_U \mathbf{a}(x) dx. \quad (2.29)$$

Completing the square in (2.19) gives the equivalent form

$$J(U, p, q; \mathbf{a}) = \frac{1}{2} p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})) p + \frac{1}{2} (q - \mathbf{s}_*(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q - \mathbf{s}_*(U; \mathbf{a}) p). \quad (2.30)$$

The choice $q = \mathbf{s}_*(U; \mathbf{a}) p$ is the Neumann flux for which the Neumann solution has average gradient p . If

$$\delta_p := u_{\mathbf{s}_*(U; \mathbf{a}) p}^N - u_p^D,$$

then

$$J(U, p, \mathbf{s}_*(U; \mathbf{a}) p; \mathbf{a}) = \oint_U \frac{1}{2} \nabla \delta_p \cdot \mathbf{a} \nabla \delta_p = \frac{1}{2} p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})) p. \quad (2.31)$$

Thus, in the symmetric case, the response functional measures the energy distance between the Dirichlet and Neumann problems. The finite-volume Dirichlet–Neumann gap is small precisely when these two boundary conditions produce nearly the same coarse response. The energy estimates for Dirichlet and Neumann problems in Chapter 3 can be read as quantitative versions of these two sides of the split, with right-hand sides and rough boundary data included. In the nonsymmetric case, this simple splitting is lost; the matrix $\mathbf{k}(U; \mathbf{a})$ records the operator-dependent mixing of the gradient and flux variables, and the doubled block formalism below restores a variational structure for that coupled response.

2.3.3 Subadditivity and scaling

We next record the structural properties which describe how the response functional behaves under partition of the domain and under the simplest operations on the coefficient field.

Lemma 2.4 (Subadditivity and scaling). *The following assertions hold for every bounded Lipschitz domain $U \subseteq \mathbb{R}^d$ and, in the subadditivity statement, for every pair of triadic scales $n < m$.*

1. For every $p, q \in \mathbb{R}^d$, the response functional is subadditive across triadic cube partitions:

$$J(\square_m, p, q; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}). \quad (2.32)$$

2. Coarse-graining is homogeneous with respect to rescaling the coefficient field: for every $\lambda > 0$,

$$\mathbf{s}(U; \lambda \mathbf{a}) = \lambda \mathbf{s}(U; \mathbf{a}), \quad \mathbf{s}_*(U; \lambda \mathbf{a}) = \lambda \mathbf{s}_*(U; \mathbf{a}), \quad \mathbf{k}(U; \lambda \mathbf{a}) = \lambda \mathbf{k}(U; \mathbf{a}), \quad (2.33)$$

and, equivalently,

$$J(U, p, q; \lambda \mathbf{a}) = J(U, \lambda^{1/2} p, \lambda^{-1/2} q; \mathbf{a}). \quad (2.34)$$

Proof. We begin with (2.32). Fix $v \in \mathcal{A}(\square_m; \mathbf{a})$. For each cube $z + \square_n$ in the triadic partition of $\square_m$, the restriction of v to $z + \square_n$ belongs to $\mathcal{A}(z + \square_n; \mathbf{a})$. Therefore,

$$\begin{aligned} \int_{\square_m} \left(-\frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right) &= \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \int_{z + \square_n} \left(-\frac{1}{2} \nabla v \cdot \mathbf{s} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right) \\ &\leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}). \end{aligned}$$

Taking the supremum over $v \in \mathcal{A}(\square_m; \mathbf{a})$ yields the desired subadditivity.

We next prove the equivalent scaling relation (2.34). Since

$$-\nabla \cdot (\lambda \mathbf{a} \nabla u) = \lambda (-\nabla \cdot \mathbf{a} \nabla u),$$

the solution spaces for $\lambda \mathbf{a}$ and $\mathbf{a}$ coincide:

$$\mathcal{A}(U; \lambda \mathbf{a}) = \mathcal{A}(U; \mathbf{a}).$$

Since this space is linear, the change of variables $w = \lambda^{1/2} v$ gives

$$\begin{aligned} J(U, \lambda^{1/2} p, \lambda^{-1/2} q; \mathbf{a}) &= \sup_{w \in \mathcal{A}(U; \mathbf{a})} \int_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{s} \nabla w - \lambda^{1/2} p \cdot \mathbf{a} \nabla w + \lambda^{-1/2} q \cdot \nabla w \right) \\ &= \sup_{v \in \mathcal{A}(U; \mathbf{a})} \int_U \left(-\frac{\lambda}{2} \nabla v \cdot \mathbf{s} \nabla v - \lambda p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right) \\ &= J(U, p, q; \lambda \mathbf{a}), \end{aligned}$$

which is (2.34).

We now derive (2.33) from the definitions of the coarse-grained matrices. By (2.6), in particular,

$$\frac{1}{2} q \cdot \mathbf{s}_*^{-1}(U; \lambda \mathbf{a}) q = J(U, 0, q; \lambda \mathbf{a}) = J(U, 0, \lambda^{-1/2} q; \mathbf{a}) = \frac{1}{2} q \cdot \lambda^{-1} \mathbf{s}_*^{-1}(U; \mathbf{a}) q.$$

This implies $\mathbf{s}_*(U; \lambda \mathbf{a}) = \lambda \mathbf{s}_*(U; \mathbf{a})$. Likewise, by the definition (2.7) of $\mathbf{k}(U; \cdot)$ and (2.34), if we set

$$p' := \lambda^{1/2} p, \quad q' := \lambda^{-1/2} q,$$

then

$$\begin{aligned} q \cdot \mathbf{s}_*^{-1}(U; \lambda \mathbf{a}) \mathbf{k}(U; \lambda \mathbf{a}) p &= J(U, p, q; \lambda \mathbf{a}) - J(U, p, 0; \lambda \mathbf{a}) - J(U, 0, q; \lambda \mathbf{a}) + p \cdot q \\ &= J(U, p', q'; \mathbf{a}) - J(U, p', 0; \mathbf{a}) - J(U, 0, q'; \mathbf{a}) + p' \cdot q' \\ &= q' \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p' \\ &= q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p. \end{aligned}$$

Since $\mathbf{s}_*^{-1}(U; \lambda \mathbf{a}) = \lambda^{-1} \mathbf{s}_*^{-1}(U; \mathbf{a})$, we deduce that $\mathbf{k}(U; \lambda \mathbf{a}) = \lambda \mathbf{k}(U; \mathbf{a})$. Finally, using (2.8),

$$\begin{aligned} \frac{1}{2} p \cdot \mathbf{s}(U; \lambda \mathbf{a}) p &= J(U, p, 0; \lambda \mathbf{a}) - \frac{1}{2} p \cdot \mathbf{k}^t(U; \lambda \mathbf{a}) \mathbf{s}_*^{-1}(U; \lambda \mathbf{a}) \mathbf{k}(U; \lambda \mathbf{a}) p \\ &= J(U, \lambda^{1/2} p, 0; \mathbf{a}) - \frac{1}{2} p \cdot (\lambda \mathbf{k}(U; \mathbf{a}))^t (\lambda^{-1} \mathbf{s}_*^{-1}(U; \mathbf{a})) (\lambda \mathbf{k}(U; \mathbf{a})) p \\ &= \frac{1}{2} \lambda p \cdot \mathbf{s}(U; \mathbf{a}) p \\ &\quad + \frac{1}{2} \lambda p \cdot \mathbf{k}^t(U; \mathbf{a}) \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p - \frac{1}{2} \lambda p \cdot \mathbf{k}^t(U; \mathbf{a}) \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p \\ &= \frac{1}{2} p \cdot \lambda \mathbf{s}(U; \mathbf{a}) p. \end{aligned}$$

This proves (2.33). □

2.4 Block-matrix and double-variable formalism

2.4.1 Why the block formalism is needed

The response functional $J(U, p, q; \mathbf{a})$ already packages the coarse-scale interaction between gradients and fluxes, but some of its most important structural properties are not transparent from the definition alone. In particular, the compatibility with the adjoint field $\mathbf{a}^t$, the ordering $\mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a})$, and the control of the symmetric part of $\mathbf{k}(U; \mathbf{a})$ all become conceptually clearer only after one passes to a doubled $2d \times 2d$ formalism.

The reason is that a nonsymmetric divergence-form equation is still variational, but the right convex functional depends on two fields at once: a potential field and a solenoidal field. If we define, for $p, q \in \mathbb{R}^d$,

$$A(p, q, x) := \frac{1}{2} p \cdot \mathbf{s}(x) p + \frac{1}{2} (q - \mathbf{k}(x) p) \cdot \mathbf{s}^{-1}(x) (q - \mathbf{k}(x) p),$$

then one may view solutions of $-\nabla \cdot \mathbf{a} \nabla u = 0$ as local minimizers, among functions with fixed boundary trace, of the map

$$w \mapsto \inf_{\mathbf{g} \in L_{\text{sol}}^2(U)} \int_U A(\nabla w, \mathbf{g}, x) dx$$

This is the nonsymmetric analogue of the Dirichlet principle. The doubled block matrix introduced below is simply the matrix representation of this quadratic density, and the doubled response functional is the corresponding coarse-grained variational quantity.

2.4.2 The block matrix field

The quadratic density $A(p, q, x)$ may be written in block form by introducing

$$\mathbf{A}(x) := \begin{pmatrix} (\mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k})(x) & -(\mathbf{k}^t \mathbf{s}^{-1})(x) \\ -(\mathbf{s}^{-1} \mathbf{k})(x) & \mathbf{s}^{-1}(x) \end{pmatrix}. \quad (2.35)$$

We also write, for $\mathbf{h} \in \mathbb{R}^{d \times d}$,

$$\mathbf{G}_{\mathbf{h}} := \begin{pmatrix} \mathbf{I}_d & 0 \\ \mathbf{h} & \mathbf{I}_d \end{pmatrix}, \quad (2.36)$$

as well as

$$\mathbf{R} := \begin{pmatrix} 0 & \mathbf{I}_d \\ \mathbf{I}_d & 0 \end{pmatrix}. \quad (2.37)$$

With this notation,

$$A(p, q, x) = \frac{1}{2} \begin{pmatrix} p \\ q \end{pmatrix} \cdot \mathbf{A}(x) \begin{pmatrix} p \\ q \end{pmatrix}. \quad (2.38)$$

Lemma 2.5 (Basic algebra of the block matrix field). *The field $\mathbf{A}(x)$ is symmetric and positive definite for every $x \in \mathbb{R}^d$. Moreover,*

$$\mathbf{A} = \mathbf{G}_{-\mathbf{k}}^t \begin{pmatrix} \mathbf{s} & 0 \\ 0 & \mathbf{s}^{-1} \end{pmatrix} \mathbf{G}_{-\mathbf{k}}, \quad (2.39)$$

and therefore

$$\mathbf{A}^{-1} = \mathbf{G}_{\mathbf{k}} \begin{pmatrix} \mathbf{s}^{-1} & 0 \\ 0 & \mathbf{s} \end{pmatrix} \mathbf{G}_{\mathbf{k}}^t = \begin{pmatrix} \mathbf{s}^{-1} & -\mathbf{s}^{-1} \mathbf{k} \\ -\mathbf{k}^t \mathbf{s}^{-1} & \mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k} \end{pmatrix} = \mathbf{R} \mathbf{A} \mathbf{R}. \quad (2.40)$$

Proof. The symmetry is evident from the definition (2.35). The factorization (2.39) is obtained by direct matrix multiplication:

$$\mathbf{G}_{-\mathbf{k}}^t \begin{pmatrix} \mathbf{s} & 0 \\ 0 & \mathbf{s}^{-1} \end{pmatrix} \mathbf{G}_{-\mathbf{k}} = \begin{pmatrix} \mathbf{s} + \mathbf{k}^t \mathbf{s}^{-1} \mathbf{k} & -\mathbf{k}^t \mathbf{s}^{-1} \\ -\mathbf{s}^{-1} \mathbf{k} & \mathbf{s}^{-1} \end{pmatrix} = \mathbf{A}.$$

Since $\mathbf{s}(x)$ is uniformly positive definite, the middle diagonal matrix is positive definite, and so is $\mathbf{A}(x)$. The formula (2.40) follows by inverting the factorization, and the identity $\mathbf{A}^{-1} = \mathbf{R} \mathbf{A} \mathbf{R}$ is then immediate from the explicit formulas. $\square$

2.4.3 The variational quantity μ and the coarse-grained block matrix

We now recall the spaces $L_{\text{pot}}^2(U)$, $L_{\text{pot},0}^2(U)$, $L_{\text{sol}}^2(U)$ and $L_{\text{sol},0}^2(U)$ from Chapter 1. They provide the natural ambient spaces for the potential and solenoidal components of the doubled variational problem. Since U is bounded Lipschitz, Poincaré's inequality implies that the gradient maps

$$\nabla : H^1(U)/\mathbb{R} \rightarrow L^2(U; \mathbb{R}^d) \quad \text{and} \quad \nabla : H_0^1(U) \rightarrow L^2(U; \mathbb{R}^d)$$

have closed ranges. Consequently $L_{\text{pot}}^2(U)$ and $L_{\text{pot},0}^2(U)$ are closed subspaces of $L^2(U; \mathbb{R}^d)$, and therefore $L_{\text{sol}}^2(U) = L_{\text{pot},0}^2(U)^\perp$ and $L_{\text{sol},0}^2(U) = L_{\text{pot}}^2(U)^\perp$ are closed as well.

For each $P \in \mathbb{R}^{2d}$, we define

$$\mu(U, P; \mathbf{a}) := \inf \left\{ \int_U \frac{1}{2} X \cdot \mathbf{A} X : X \in P + L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U) \right\}. \quad (2.41)$$

Because the admissible set is a closed affine subspace of a Hilbert space and the integrand is uniformly convex, the infimum is attained by a unique minimizer. Moreover, the map $P \mapsto \mu(U, P; \mathbf{a})$ is quadratic. We therefore define the *coarse-grained block matrix* $\mathbf{A}(U; \mathbf{a})$ by the identity

$$\mu(U, P; \mathbf{a}) = \frac{1}{2} P \cdot \mathbf{A}(U; \mathbf{a}) P \quad \forall P \in \mathbb{R}^{2d}. \quad (2.42)$$

The main point is that $\mathbf{A}(U; \mathbf{a})$ is not an additional coarse-grained object. Rather, as we will show below, it is the natural $2d \times 2d$ package for the same information carried by $\mathbf{s}(U; \mathbf{a})$, $\mathbf{s}_*(U; \mathbf{a})$, and $\mathbf{k}(U; \mathbf{a})$.

2.4.4 The doubled response functional

We next introduce the doubled response space

$$\mathcal{S}(U; \mathbf{a}) := \left\{ X \in L^2_{\text{pot}}(U) \times L^2_{\text{sol}}(U) : \int_U Y \cdot \mathbf{A}X = 0 \quad \forall Y \in L^2_{\text{pot},0}(U) \times L^2_{\text{sol},0}(U) \right\}. \quad (2.43)$$

This is the natural double-variable analogue of the solution space $\mathcal{A}(U; \mathbf{a})$.

For $P, Q \in \mathbb{R}^{2d}$, we define the *doubled response functional*

$$\mathbf{J}(U, P, Q; \mathbf{a}) := \sup_{X \in \mathcal{S}(U; \mathbf{a})} \int_U \left(-\frac{1}{2} X \cdot \mathbf{A}X - P \cdot \mathbf{A}X + Q \cdot X \right). \quad (2.44)$$

Since $\mathcal{S}(U; \mathbf{a})$ is a closed subspace of $L^2(U; \mathbb{R}^{2d})$ and the integrand is strictly concave and coercive in X , the maximizer exists and is unique; we denote it by

$$S(\cdot, U, P, Q; \mathbf{a}) \in \mathcal{S}(U; \mathbf{a}).$$

The quantity $\mathbf{J}$ is the doubled analogue of J . It is the right object for revealing the hidden duality between gradients and fluxes, and it is also the form that survives naturally when forcing terms are introduced later.

2.4.5 Structural properties in block form

The doubled formalism packages the underlying deterministic structure far more cleanly than the original response functional J . The first lemma identifies the doubled response space and relates $\mathbf{J}$ to J and its adjoint. We therefore first record the adjoint response functional

$$J^*(U, p, q; \mathbf{a}) := J(U, p, q; \mathbf{a}^t) = \sup_{v^* \in \mathcal{A}^*(U; \mathbf{a})} \int_U \left(-\frac{1}{2} \nabla v^* \cdot \mathbf{s} \nabla v^* - p \cdot \mathbf{a}^t \nabla v^* + q \cdot \nabla v^* \right). \quad (2.45)$$

Lemma 2.6 (The doubled response space and functional). *The following assertions hold for every bounded Lipschitz domain $U \subseteq \mathbb{R}^d$.*

1. *The doubled response space satisfies*

$$\mathcal{S}(U; \mathbf{a}) = \left\{ (\nabla v + \nabla v^*, \mathbf{a} \nabla v - \mathbf{a}^t \nabla v^*) : v \in \mathcal{A}(U; \mathbf{a}), v^* \in \mathcal{A}^*(U; \mathbf{a}) \right\}. \quad (2.46)$$

2. *For every $p, p^*, q, q^* \in \mathbb{R}^d$, we have*

$$\mathbf{J}\left(U, \begin{pmatrix} p \\ q \end{pmatrix}, \begin{pmatrix} q^* \\ p^* \end{pmatrix}; \mathbf{a}\right) = \frac{1}{2} J(U, p - p^*, q^* - q; \mathbf{a}) + \frac{1}{2} J^*(U, p^* + p, q^* + q; \mathbf{a}). \quad (2.47)$$

3. The maximizer of (2.44) is given by

$$S\left(\cdot, U, \begin{pmatrix} p \\ q \end{pmatrix}, \begin{pmatrix} q^* \\ p^* \end{pmatrix}; \mathbf{a}\right) = \frac{1}{2} \left(\nabla v(\cdot, U, p - p^*, q^* - q; \mathbf{a}) + \nabla v^*(\cdot, U, p^* + p, q^* + q; \mathbf{a}) \right. \\ \left. - \mathbf{a} \nabla v(\cdot, U, p - p^*, q^* - q; \mathbf{a}) - \mathbf{a}^t \nabla v^*(\cdot, U, p^* + p, q^* + q; \mathbf{a}) \right). \quad (2.48)$$

4. The map $(P, Q) \mapsto S(\cdot, U, P, Q; \mathbf{a})$ is linear, and for every $T \in \mathcal{S}(U; \mathbf{a})$ we have the first-variation identity

$$\int_U (Q \cdot T - P \cdot \mathbf{A}T) = \int_U T \cdot \mathbf{A}S(\cdot, U, P, Q; \mathbf{a}). \quad (2.49)$$

Proof. We first record the identities

$$\mathbf{A} \begin{pmatrix} \xi \\ \mathbf{a}\xi \end{pmatrix} = \begin{pmatrix} \mathbf{a}\xi \\ \xi \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} \eta \\ -\mathbf{a}^t \eta \end{pmatrix} = \begin{pmatrix} \mathbf{a}^t \eta \\ -\eta \end{pmatrix} \quad \forall \xi, \eta \in \mathbb{R}^d, \quad (2.50)$$

which follow directly from (2.35), using $\mathbf{a} = \mathbf{s} + \mathbf{k}$, $\mathbf{a}^t = \mathbf{s} - \mathbf{k}$ and $\mathbf{k}^t = -\mathbf{k}$.

We begin with (2.46). For the inclusion “ $\supseteq$ ”, let $v \in \mathcal{A}(U; \mathbf{a})$ and $v^* \in \mathcal{A}^*(U; \mathbf{a})$, and set

$$X := \begin{pmatrix} \nabla v + \nabla v^* \\ \mathbf{a} \nabla v - \mathbf{a}^t \nabla v^* \end{pmatrix}.$$

Then $X \in L_{\text{pot}}^2(U) \times L_{\text{sol}}^2(U)$, and by (2.50),

$$\mathbf{A}X = \begin{pmatrix} \mathbf{a} \nabla v + \mathbf{a}^t \nabla v^* \\ \nabla v - \nabla v^* \end{pmatrix}.$$

Therefore, for every $Y = (\nabla w, \mathbf{g}) \in L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U)$,

$$\int_U Y \cdot \mathbf{A}X = \int_U \nabla w \cdot \mathbf{a} \nabla v + \int_U \nabla w \cdot \mathbf{a}^t \nabla v^* + \int_U \mathbf{g} \cdot \nabla v - \int_U \mathbf{g} \cdot \nabla v^* = 0.$$

Indeed, the first two terms vanish because $v \in \mathcal{A}(U; \mathbf{a})$ and $v^* \in \mathcal{A}^*(U; \mathbf{a})$, while the last two vanish because $\mathbf{g} \in L_{\text{sol},0}^2(U) = L_{\text{pot}}^2(U)^\perp$. Hence $X \in \mathcal{S}(U; \mathbf{a})$.

For the reverse inclusion, let

$$X = \begin{pmatrix} \nabla u \\ \mathbf{h} \end{pmatrix} \in \mathcal{S}(U; \mathbf{a}).$$

Define

$$r := \mathbf{s}^{-1}(\mathbf{h} - \mathbf{k} \nabla u).$$

A direct multiplication shows that

$$\mathbf{A}X = \begin{pmatrix} \mathbf{s} \nabla u + \mathbf{k} r \\ r \end{pmatrix}.$$

Taking in the definition of $\mathcal{S}(U; \mathbf{a})$ test fields of the form $Y = (0, \mathbf{g})$ with $\mathbf{g} \in L_{\text{sol},0}^2(U)$, we obtain

$$\int_U \mathbf{g} \cdot r = 0 \quad \forall \mathbf{g} \in L_{\text{sol},0}^2(U).$$

Since $L_{\text{sol},0}^2(U) = L_{\text{pot}}^2(U)^\perp$, this implies $r \in L_{\text{pot}}^2(U)$, so there exists $\tilde{u} \in H^1(U)$ such that $r = \nabla \tilde{u}$. Taking instead $Y = (\nabla w, 0)$ with $w \in H_0^1(U)$, we find

$$\int_U \nabla w \cdot (\mathbf{s} \nabla u + \mathbf{k} \nabla \tilde{u}) = 0 \quad \forall w \in H_0^1(U),$$

that is,

$$\mathbf{s}\nabla u + \mathbf{k}\nabla\tilde{u} \in L_{\text{sol}}^2(U).$$

Now define

$$v := \frac{1}{2}(u + \tilde{u}), \quad v^* := \frac{1}{2}(u - \tilde{u}).$$

Then

$$\nabla v + \nabla v^* = \nabla u, \quad \mathbf{a}\nabla v - \mathbf{a}^t\nabla v^* = \mathbf{s}\nabla\tilde{u} + \mathbf{k}\nabla u = \mathbf{h}.$$

Moreover,

$$\mathbf{a}\nabla v = \frac{1}{2}(\mathbf{s}\nabla u + \mathbf{k}\nabla\tilde{u}) + \frac{1}{2}(\mathbf{s}\nabla\tilde{u} + \mathbf{k}\nabla u),$$

and both terms on the right belong to $L_{\text{sol}}^2(U)$, the first by the previous display and the second because it is exactly $\mathbf{h} \in L_{\text{sol}}^2(U)$. Hence $\mathbf{a}\nabla v \in L_{\text{sol}}^2(U)$, so $v \in \mathcal{A}(U; \mathbf{a})$. Likewise,

$$\mathbf{a}^t\nabla v^* = \frac{1}{2}(\mathbf{s}\nabla u + \mathbf{k}\nabla\tilde{u}) - \frac{1}{2}(\mathbf{s}\nabla\tilde{u} + \mathbf{k}\nabla u) \in L_{\text{sol}}^2(U),$$

and therefore $v^* \in \mathcal{A}^*(U; \mathbf{a})$. This proves (2.46).

We next prove (2.47). Since $\mathcal{A}(U; \mathbf{a})$ and $\mathcal{A}^*(U; \mathbf{a})$ are linear spaces, the representation (2.46) is equivalent to

$$\mathcal{S}(U; \mathbf{a}) = \left\{ \frac{1}{2} \begin{pmatrix} \nabla v + \nabla v^* \\ \mathbf{a}\nabla v - \mathbf{a}^t\nabla v^* \end{pmatrix} : v \in \mathcal{A}(U; \mathbf{a}), v^* \in \mathcal{A}^*(U; \mathbf{a}) \right\}.$$

Fix $p, p^*, q, q^* \in \mathbb{R}^d$, and abbreviate

$$P := \begin{pmatrix} p \\ q \end{pmatrix}, \quad Q := \begin{pmatrix} q^* \\ p^* \end{pmatrix}.$$

For

$$X = \frac{1}{2} \begin{pmatrix} \nabla v + \nabla v^* \\ \mathbf{a}\nabla v - \mathbf{a}^t\nabla v^* \end{pmatrix} \in \mathcal{S}(U; \mathbf{a}),$$

the identities (2.50) give

$$\mathbf{A}X = \frac{1}{2} \begin{pmatrix} \mathbf{a}\nabla v + \mathbf{a}^t\nabla v^* \\ \nabla v - \nabla v^* \end{pmatrix}.$$

Hence

$$\begin{aligned} & -\frac{1}{2}X \cdot \mathbf{A}X - P \cdot \mathbf{A}X + Q \cdot X \\ &= \frac{1}{2} \left(-\frac{1}{2}\nabla v \cdot \mathbf{s}\nabla v - (p - p^*) \cdot \mathbf{a}\nabla v + (q^* - q) \cdot \nabla v \right) \\ & \quad + \frac{1}{2} \left(-\frac{1}{2}\nabla v^* \cdot \mathbf{s}\nabla v^* - (p + p^*) \cdot \mathbf{a}^t\nabla v^* + (q^* + q) \cdot \nabla v^* \right). \end{aligned}$$

Taking $\oint_U$ and then the supremum over $(v, v^*) \in \mathcal{A}(U; \mathbf{a}) \times \mathcal{A}^*(U; \mathbf{a})$, we obtain (2.47).

The maximizer formula (2.48) is now immediate. The right side of (2.47) is the sum of two independent variational problems, and each has a unique maximizer, namely $v(\cdot, U, p - p^*, q^* - q; \mathbf{a})$ and $v^*(\cdot, U, p^* + p, q^* + q; \mathbf{a})$. Substituting these into the preceding representation of X yields (2.48).

Finally, the linearity of $(P, Q) \mapsto S(\cdot, U, P, Q; \mathbf{a})$ follows from (2.48), together with the linearity of $(p, q) \mapsto \nabla v(\cdot, U, p, q; \mathbf{a})$ from Lemma 2.1 and the same statement applied to the adjoint field $\mathbf{a}^t$. To prove (2.49), fix $T \in \mathcal{S}(U; \mathbf{a})$. Since $\mathcal{S}(U; \mathbf{a})$ is a linear space, the function

$$t \mapsto \int_U \left(-\frac{1}{2}(S + tT) \cdot \mathbf{A}(S + tT) - P \cdot \mathbf{A}(S + tT) + Q \cdot (S + tT) \right)$$

has a maximum at $t = 0$. Differentiating at $t = 0$ gives (2.49). $\square$

The second lemma identifies the previously defined coarse-grained block matrix $\mathbf{A}(U; \mathbf{a})$, introduces its dual companion $\mathbf{A}_*(U; \mathbf{a})$, and records the main formulas that connect them back to the matrices $\mathbf{s}(U; \mathbf{a})$, $\mathbf{s}_*(U; \mathbf{a})$ and $\mathbf{k}(U; \mathbf{a})$.

Lemma 2.7 (The coarse-grained block matrices). *The following assertions hold for every bounded Lipschitz domain $U \subseteq \mathbb{R}^d$.*

1. *There exists a symmetric positive definite matrix $\mathbf{A}_*(U; \mathbf{a})$ such that the matrix $\mathbf{A}(U; \mathbf{a})$ defined in (2.42) satisfies*

$$\mathbf{J}(U, P, Q; \mathbf{a}) = \frac{1}{2}P \cdot \mathbf{A}(U; \mathbf{a})P + \frac{1}{2}Q \cdot \mathbf{A}_*^{-1}(U; \mathbf{a})Q - P \cdot Q \quad \forall P, Q \in \mathbb{R}^{2d}. \quad (2.51)$$

2. *The matrices $\mathbf{A}(U; \mathbf{a})$ and $\mathbf{A}_*^{-1}(U; \mathbf{a})$ are given by*

$$\begin{cases} \mathbf{A}(U; \mathbf{a}) = \begin{pmatrix} \mathbf{b}(U; \mathbf{a}) & -\mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a}) \\ -\mathbf{s}_*^{-1}(U; \mathbf{a})\mathbf{k}(U; \mathbf{a}) & \mathbf{s}_*^{-1}(U; \mathbf{a}) \end{pmatrix}, \\ \mathbf{A}_*^{-1}(U; \mathbf{a}) = \begin{pmatrix} \mathbf{s}_*^{-1}(U; \mathbf{a}) & -\mathbf{s}_*^{-1}(U; \mathbf{a})\mathbf{k}(U; \mathbf{a}) \\ -\mathbf{k}^t(U; \mathbf{a})\mathbf{s}_*^{-1}(U; \mathbf{a}) & \mathbf{b}(U; \mathbf{a}) \end{pmatrix} = \mathbf{R}\mathbf{A}(U; \mathbf{a})\mathbf{R}. \end{cases} \quad (2.52)$$

3. *For every pair of triadic scales $n < m$, the block matrices are subadditive:*

$$\mathbf{A}(\square_m; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{A}(z + \square_n; \mathbf{a}), \quad (2.53)$$

and

$$\mathbf{A}_*^{-1}(\square_m; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{A}_*^{-1}(z + \square_n; \mathbf{a}). \quad (2.54)$$

4. *Coarse-graining is compatible with transpose:*

$$\mathbf{s}(U; \mathbf{a}^t) = \mathbf{s}(U; \mathbf{a}), \quad \mathbf{s}_*(U; \mathbf{a}^t) = \mathbf{s}_*(U; \mathbf{a}), \quad \mathbf{k}(U; \mathbf{a}^t) = -\mathbf{k}(U; \mathbf{a}). \quad (2.55)$$

Proof. We begin with the part of the doubled theory corresponding to the Dirichlet boundary-value problem. Fix $P \in \mathbb{R}^{2d}$, and let

$$X_P \in P + L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U)$$

denote the unique minimizer in the definition (2.41) of $\mu(U, P; \mathbf{a})$. By first variation of that minimization problem, for every $Y \in L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U)$ we have

$$\int_U Y \cdot \mathbf{A}X_P = 0,$$

and therefore $X_P \in \mathcal{S}(U; \mathbf{a})$.

Now let $S \in \mathcal{S}(U; \mathbf{a})$ be arbitrary. Since $X_P - P \in L^2_{\text{pot},0}(U) \times L^2_{\text{sol},0}(U)$ and $S \in \mathcal{S}(U; \mathbf{a})$, we have

$$\int_U (X_P - P) \cdot \mathbf{A}S = 0,$$

that is,

$$\int_U P \cdot \mathbf{A}S = \int_U X_P \cdot \mathbf{A}S.$$

Substituting this into the definition of $\mathbf{J}(U, P, 0; \mathbf{a})$ gives

$$\begin{aligned} \oint_U \left(-\frac{1}{2} S \cdot \mathbf{A}S - P \cdot \mathbf{A}S \right) &= \oint_U \left(-\frac{1}{2} S \cdot \mathbf{A}S - X_P \cdot \mathbf{A}S \right) \\ &= \oint_U \left(\frac{1}{2} X_P \cdot \mathbf{A}X_P - \frac{1}{2} (S + X_P) \cdot \mathbf{A}(S + X_P) \right) \\ &\leq \oint_U \frac{1}{2} X_P \cdot \mathbf{A}X_P = \mu(U, P; \mathbf{a}). \end{aligned}$$

Taking the supremum over $S \in \mathcal{S}(U; \mathbf{a})$, and then testing with the admissible choice $S = -X_P \in \mathcal{S}(U; \mathbf{a})$, we obtain

$$\mathbf{J}(U, P, 0; \mathbf{a}) = \mu(U, P; \mathbf{a}) = \frac{1}{2} P \cdot \mathbf{A}(U; \mathbf{a}) P.$$

This shows that the quadratic coefficient of P in the splitting formula is exactly the matrix $\mathbf{A}(U; \mathbf{a})$ already defined by (2.42). Moreover, by uniqueness of the maximizer in (2.44),

$$S(\cdot, U, P, 0; \mathbf{a}) = -X_P.$$

Since every field in $L^2_{\text{pot},0}(U) \times L^2_{\text{sol},0}(U)$ has zero average (for $L^2_{\text{pot},0}(U)$ this follows from the zero trace, and for $L^2_{\text{sol},0}(U)$ by testing against the coordinate functions), we deduce that

$$\oint_U S(\cdot, U, P, 0; \mathbf{a}) = -P. \quad (2.56)$$

Next, Lemma 2.6 implies that the map $Q \mapsto \mathbf{J}(U, 0, Q; \mathbf{a})$ is quadratic. We therefore define the symmetric matrix $\mathbf{A}_*^{-1}(U; \mathbf{a})$ by

$$\mathbf{J}(U, 0, Q; \mathbf{a}) = \frac{1}{2} Q \cdot \mathbf{A}_*^{-1}(U; \mathbf{a}) Q \quad \forall Q \in \mathbb{R}^{2d}.$$

We now prove (2.51). Let $S(\cdot, U, P, Q; \mathbf{a})$ be abbreviated by $S_{P,Q}$. By the first-variation identity (2.49) with $T = S_{P,Q}$, we have

$$\mathbf{J}(U, P, Q; \mathbf{a}) = \oint_U \frac{1}{2} S_{P,Q} \cdot \mathbf{A}S_{P,Q}.$$

Using the linearity of $(P, Q) \mapsto S(\cdot, U, P, Q; \mathbf{a})$ from Lemma 2.6, we get $S_{P,Q} = S_{P,0} + S_{0,Q}$ and therefore

$$\begin{aligned} \mathbf{J}(U, P, Q; \mathbf{a}) &= \oint_U \frac{1}{2} S_{P,0} \cdot \mathbf{A}S_{P,0} + \oint_U \frac{1}{2} S_{0,Q} \cdot \mathbf{A}S_{0,Q} + \oint_U S_{P,0} \cdot \mathbf{A}S_{0,Q} \\ &= \mathbf{J}(U, P, 0; \mathbf{a}) + \mathbf{J}(U, 0, Q; \mathbf{a}) + \oint_U S_{P,0} \cdot \mathbf{A}S_{0,Q}. \end{aligned}$$

Applying (2.49) to $S_{0,Q}$ with $T = S_{P,0}$ gives

$$\oint_U S_{P,0} \cdot \mathbf{A} S_{0,Q} = Q \cdot \oint_U S_{P,0} = -P \cdot Q,$$

where we used (2.56) in the last step. This shows that

$$\mathbf{J}(U, P, Q; \mathbf{a}) = \mathbf{J}(U, P, 0; \mathbf{a}) + \mathbf{J}(U, 0, Q; \mathbf{a}) - P \cdot Q.$$

Combining this with

$$\mathbf{J}(U, P, 0; \mathbf{a}) = \frac{1}{2} P \cdot \mathbf{A}(U; \mathbf{a}) P \quad \text{and} \quad \mathbf{J}(U, 0, Q; \mathbf{a}) = \frac{1}{2} Q \cdot \mathbf{A}_*^{-1}(U; \mathbf{a}) Q$$

gives

$$\mathbf{J}(U, P, Q; \mathbf{a}) = \frac{1}{2} P \cdot \mathbf{A}(U; \mathbf{a}) P + \frac{1}{2} Q \cdot \mathbf{A}_*^{-1}(U; \mathbf{a}) Q - P \cdot Q,$$

which is exactly (2.51). In particular, the quadratic coefficient of P in the splitting formula is the same matrix $\mathbf{A}(U; \mathbf{a})$ that was defined earlier by (2.42).

We next derive the explicit formulas in (2.52). We first identify the adjoint response functional. By Lemma 2.6 and (2.51),

$$2J(U, p, 0; \mathbf{a}) = \mathbf{J}\left(U, \begin{pmatrix} p \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -p \end{pmatrix}; \mathbf{a}\right) = \mathbf{J}\left(U, \begin{pmatrix} p \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ p \end{pmatrix}; \mathbf{a}\right) = 2J^*(U, p, 0; \mathbf{a}),$$

since the splitting formula depends on Q only through the quadratic term $\frac{1}{2} Q \cdot \mathbf{A}_*^{-1}(U; \mathbf{a}) Q$ and the inner product $P \cdot Q$, which both are unchanged when $Q = \begin{pmatrix} 0 \\ -p \end{pmatrix}$ is replaced by $Q = \begin{pmatrix} 0 \\ p \end{pmatrix}$. Likewise,

$$2J(U, 0, q; \mathbf{a}) = \mathbf{J}\left(U, \begin{pmatrix} 0 \\ -q \end{pmatrix}, \begin{pmatrix} q \\ 0 \end{pmatrix}; \mathbf{a}\right) = \mathbf{J}\left(U, \begin{pmatrix} 0 \\ q \end{pmatrix}, \begin{pmatrix} q \\ 0 \end{pmatrix}; \mathbf{a}\right) = 2J^*(U, 0, q; \mathbf{a}).$$

$$\begin{aligned} J(U, p, q; \mathbf{a}) + J^*(U, p, q; \mathbf{a}) &= 2\mathbf{J}\left(U, \begin{pmatrix} p \\ 0 \end{pmatrix}, \begin{pmatrix} q \\ 0 \end{pmatrix}; \mathbf{a}\right) \\ &= \begin{pmatrix} p \\ 0 \end{pmatrix} \cdot \mathbf{A}(U; \mathbf{a}) \begin{pmatrix} p \\ 0 \end{pmatrix} + \begin{pmatrix} q \\ 0 \end{pmatrix} \cdot \mathbf{A}_*^{-1}(U; \mathbf{a}) \begin{pmatrix} q \\ 0 \end{pmatrix} - 2p \cdot q \\ &= 2J(U, p, 0; \mathbf{a}) + 2J(U, 0, q; \mathbf{a}) - 2p \cdot q. \end{aligned}$$

Rearranging, and using the definitions (2.6), (2.7), and (2.10), we obtain

$$J^*(U, p, q; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{b}(U; \mathbf{a}) p + \frac{1}{2} q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) q - q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}) p - p \cdot q,$$

that is,

$$J^*(U, p, q; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p + \frac{1}{2} (q - \mathbf{k}(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q - \mathbf{k}(U; \mathbf{a}) p) - p \cdot q.$$

Now let $p, q \in \mathbb{R}^d$. By Lemma 2.6, (2.51), and the preceding display,

$$\begin{aligned} \frac{1}{2} \begin{pmatrix} p \\ q \end{pmatrix} \cdot \mathbf{A}(U; \mathbf{a}) \begin{pmatrix} p \\ q \end{pmatrix} &= \mathbf{J}\left(U, \begin{pmatrix} p \\ q \end{pmatrix}, 0; \mathbf{a}\right) \\ &= \frac{1}{2} J(U, p, -q; \mathbf{a}) + \frac{1}{2} J^*(U, p, q; \mathbf{a}) \\ &= \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p + \frac{1}{2} (q - \mathbf{k}(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q - \mathbf{k}(U; \mathbf{a}) p). \end{aligned}$$

Expanding the right side yields the formula for $\mathbf{A}(U; \mathbf{a})$ in (2.52). The identity for $\mathbf{A}_*^{-1}(U; \mathbf{a})$ then follows immediately from the definition of $\mathbf{R}$ in (2.37). In particular, since $\mathbf{A}(U; \mathbf{a})$ is positive definite, so is $\mathbf{A}_*^{-1}(U; \mathbf{a})$, and this completes the proof of the first two items.

We turn to subadditivity. Let $n < m$ and $P \in \mathbb{R}^{2d}$. For each $z \in 3^n \mathbb{Z}^d \cap \square_m$, let X_z be the minimizer in $\mu(z + \square_n, P; \mathbf{a})$. Define a field X on $\square_m$ by setting $X|_{z + \square_n} = X_z$. Since $X_z - P \in L_{\text{pot},0}^2(z + \square_n) \times L_{\text{sol},0}^2(z + \square_n)$ for every z , the piecewise-defined field $X - P$ belongs to $L_{\text{pot},0}^2(\square_m) \times L_{\text{sol},0}^2(\square_m)$ after extending the local potentials by zero and observing that elements of $L_{\text{sol},0}^2(z + \square_n)$ remain orthogonal to all gradients after extension by zero. Hence X is an admissible competitor for $\mu(\square_m, P; \mathbf{a})$, and therefore

$$\begin{aligned} \frac{1}{2} P \cdot \mathbf{A}(\square_m; \mathbf{a}) P &= \mu(\square_m, P; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mu(z + \square_n, P; \mathbf{a}) \\ &= \frac{1}{2} P \cdot \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{A}(z + \square_n; \mathbf{a}) \right) P. \end{aligned}$$

Since this holds for every $P \in \mathbb{R}^{2d}$, we obtain (2.53). Conjugating by $\mathbf{R}$ and using (2.52), we get (2.54).

Finally, we prove (2.55). By definition, $J(U, p, q; \mathbf{a}^t) = J^*(U, p, q; \mathbf{a})$, and the previous part gives

$$J(U, p, q; \mathbf{a}^t) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}) p + \frac{1}{2} (q - \mathbf{k}(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q - \mathbf{k}(U; \mathbf{a}) p) - p \cdot q.$$

On the other hand, applying (2.9) to the coefficient field $\mathbf{a}^t$ yields

$$J(U, p, q; \mathbf{a}^t) = \frac{1}{2} p \cdot \mathbf{s}(U; \mathbf{a}^t) p + \frac{1}{2} (q + \mathbf{k}(U; \mathbf{a}^t) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}^t) (q + \mathbf{k}(U; \mathbf{a}^t) p) - p \cdot q.$$

Comparing the pure q terms, then the mixed terms, and finally the pure p terms in these two quadratic expressions gives

$$\mathbf{s}_*(U; \mathbf{a}^t) = \mathbf{s}_*(U; \mathbf{a}), \quad \mathbf{k}(U; \mathbf{a}^t) = -\mathbf{k}(U; \mathbf{a}), \quad \mathbf{s}(U; \mathbf{a}^t) = \mathbf{s}(U; \mathbf{a}).$$

This completes the proof. $\square$

2.4.6 Useful algebraic normal forms

With the block-matrix formalism in hand, the response functional admits several useful algebraic normal forms. The identities below should be deduced from the formulas relating $\mathbf{J}$ to J and J^* and from the block-matrix formulas for $\mathbf{A}(U)$ and $\mathbf{A}_*^{-1}(U)$.

Lemma 2.8 (Adjoint formulas and magic identities). *For every $p, q, h \in \mathbb{R}^d$, the following assertions hold.*

1. *The response functional may be rewritten in the form*

$$\begin{aligned} J(U, p, q; \mathbf{a}) &= \frac{1}{2} p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})) p + \frac{1}{2} p \cdot (\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a})) p \\ &\quad + \frac{1}{2} (q - (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a}) p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (q - (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a}) p). \end{aligned} \quad (2.57)$$

2. The adjoint response functional is given by

$$J^*(U, p, q; \mathbf{a}) = \frac{1}{2}p \cdot \mathbf{s}(U; \mathbf{a})p + \frac{1}{2}(q - \mathbf{k}(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q - \mathbf{k}(U; \mathbf{a})p) - p \cdot q. \quad (2.58)$$

3. The sum of J and J^* satisfies the identity

$$\begin{aligned} J(U, p, q - h; \mathbf{a}) + J^*(U, p, q + h; \mathbf{a}) &= p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))p \\ &\quad + (q - \mathbf{s}_*(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q - \mathbf{s}_*(U; \mathbf{a})p) \\ &\quad + (h - \mathbf{k}(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(h - \mathbf{k}(U; \mathbf{a})p). \end{aligned} \quad (2.59)$$

4. In particular, for every $e \in \mathbb{R}^d$,

$$J(U, e, (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})e; \mathbf{a}) + J^*(U, e, (\mathbf{s}_* + \mathbf{k})(U; \mathbf{a})e; \mathbf{a}) = e \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))e. \quad (2.60)$$

Hence

$$\mathbf{s}_*(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a}). \quad (2.61)$$

5. The symmetric part of $\mathbf{k}(U; \mathbf{a})$ is controlled by the defect $\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})$:

$$\pm(\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a})) \leq \mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}). \quad (2.62)$$

Proof. The formula (2.58) was already obtained in the proof of Lemma 2.7. We therefore begin by proving (2.57). Starting from the defining formula (2.9), we have

$$\begin{aligned} J(U, p, q; \mathbf{a}) &= \frac{1}{2}p \cdot \mathbf{s}(U; \mathbf{a})p + \frac{1}{2}(q + \mathbf{k}(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q + \mathbf{k}(U; \mathbf{a})p) - p \cdot q \\ &= \frac{1}{2}p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))p + \frac{1}{2}p \cdot (\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}))p \\ &\quad + \frac{1}{2}(q - (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q - (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})p). \end{aligned}$$

Indeed, expanding the last quadratic term gives

$$\frac{1}{2}q \cdot \mathbf{s}_*^{-1}q - p \cdot q + q \cdot \mathbf{s}_*^{-1}\mathbf{k}p + \frac{1}{2}p \cdot \mathbf{s}_*p - p \cdot \mathbf{k}p + \frac{1}{2}p \cdot \mathbf{k}^t\mathbf{s}_*^{-1}\mathbf{k}p,$$

and since $p \cdot \mathbf{k}p = \frac{1}{2}p \cdot (\mathbf{k} + \mathbf{k}^t)p$, this is exactly (2.9). This proves (2.57).

To prepare for the sum identity, we rewrite (2.58) in the analogous completed-square form

$$J^*(U, p, q; \mathbf{a}) = \frac{1}{2}p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))p - \frac{1}{2}p \cdot (\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}))p \quad (2.63)$$

$$+ \frac{1}{2}(q - (\mathbf{s}_* + \mathbf{k})(U; \mathbf{a})p) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(q - (\mathbf{s}_* + \mathbf{k})(U; \mathbf{a})p). \quad (2.64)$$

This is proved by the same expansion as above.

We now prove (2.59). Applying (2.57) with $q - h$ in place of q and (2.63) with $q + h$ in place of q , we obtain

$$\begin{aligned} J(U, p, q - h; \mathbf{a}) + J^*(U, p, q + h; \mathbf{a}) &= p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))p + \frac{1}{2}(r - s) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(r - s) + \frac{1}{2}(r + s) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})(r + s), \end{aligned}$$

where

$$r := q - \mathbf{s}_*(U; \mathbf{a})p, \quad s := h - \mathbf{k}(U; \mathbf{a})p.$$

The polarization identity

$$\frac{1}{2}(r - s) \cdot M(r - s) + \frac{1}{2}(r + s) \cdot M(r + s) = r \cdot Mr + s \cdot Ms$$

for symmetric M yields exactly (2.59).

The diagonal identity (2.60) is the special case of (2.59) obtained by setting

$$p = e, \quad q = \mathbf{s}_*(U; \mathbf{a})e, \quad h = \mathbf{k}(U; \mathbf{a})e.$$

The ordering (2.61) then follows immediately from the nonnegativity of J and J^* .

Finally, we prove (2.62). Evaluating (2.57) at $q = (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})e$ gives

$$J(U, e, (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))e + \frac{1}{2}e \cdot (\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}))e.$$

Combining this with (2.60), we get

$$\begin{aligned} \frac{1}{2}e \cdot (\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}))e &= J(U, e, (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})e; \mathbf{a}) - \frac{1}{2}e \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))e \\ &= \frac{1}{2}e \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))e - J^*(U, e, (\mathbf{s}_* + \mathbf{k})(U; \mathbf{a})e; \mathbf{a}) \\ &\leq \frac{1}{2}e \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))e. \end{aligned}$$

Since this holds for every $e \in \mathbb{R}^d$, we obtain

$$\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a}) \leq \mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}).$$

Applying instead (2.63) at $q = (\mathbf{s}_* + \mathbf{k})(U; \mathbf{a})e$ and arguing symmetrically gives

$$-(\mathbf{k}(U; \mathbf{a}) + \mathbf{k}^t(U; \mathbf{a})) \leq \mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}).$$

This proves (2.62). □

The identity (2.57) is the main algebraic normal form to keep in mind. It shows that $J(U, p, q; \mathbf{a})$ is small precisely when two things happen simultaneously:

1. the defect $\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})$ is small, and
2. the dual variable q lies close to $(\mathbf{s}_* - \mathbf{k})(U; \mathbf{a})p$.

This is the nonsymmetric analogue of the convex-duality defect from the symmetric theory, and it is one of the main reasons to regard J as the master coarse-grained quantity.

2.4.7 Response and coarse-graining estimates

With the block-matrix identities in hand, we can now derive the first quantitative consequences of the response formalism. They compare the true averaged gradients and fluxes of solutions with the coarse-grained objects defined above.

Lemma 2.9 (Linear-response and coarse-graining estimates). *The following assertions hold for every bounded Lipschitz domain $U \subseteq \mathbb{R}^d$.*

1. *For every $p, q \in \mathbb{R}^d$ and every $w \in \mathcal{A}(U; \mathbf{a})$, we have the linear-response inequality*

$$\left| \int_U (p \cdot \mathbf{a} \nabla w - q \cdot \nabla w) \right| \leq \left(\int_U \nabla w \cdot \mathbf{s} \nabla w \right)^{1/2} (2J(U, p, q; \mathbf{a}))^{1/2}. \quad (2.65)$$

2. *For every $p \in \mathbb{R}^d$ and every $w \in \mathcal{A}(U; \mathbf{a})$, we have the coarse-graining inequality*

$$\left| p \cdot \int_U (\mathbf{a}_*(U; \mathbf{a}) - \mathbf{a}) \nabla w \right| \leq 2^{1/2} (p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a})) p)^{1/2} \left(\int_U \nabla w \cdot \mathbf{s} \nabla w \right)^{1/2}. \quad (2.66)$$

3. *For every $w \in \mathcal{A}(U; \mathbf{a})$, the average gradient and average flux satisfy the energy-averaging bounds*

$$\frac{1}{2} \left(\int_U \nabla w \right) \cdot \mathbf{s}_*(U; \mathbf{a}) \left(\int_U \nabla w \right) \leq \int_U \frac{1}{2} \nabla w \cdot \mathbf{s} \nabla w \quad (2.67)$$

and

$$\frac{1}{2} \left(\int_U \mathbf{a} \nabla w \right) \cdot \mathbf{b}^{-1}(U; \mathbf{a}) \left(\int_U \mathbf{a} \nabla w \right) \leq \int_U \frac{1}{2} \nabla w \cdot \mathbf{s} \nabla w. \quad (2.68)$$

Proof. We first prove (2.65). Denote $v := v(\cdot, U, p, q; \mathbf{a})$. By the optimality identity (2.4),

$$\int_U (p \cdot \mathbf{a} \nabla w - q \cdot \nabla w) = - \int_U \nabla w \cdot \mathbf{s} \nabla v.$$

Applying the Cauchy–Schwarz inequality in the scalar product induced by $\mathbf{s}$ and then using (2.14), we obtain

$$\begin{aligned} \left| \int_U (p \cdot \mathbf{a} \nabla w - q \cdot \nabla w) \right| &\leq \left(\int_U \nabla w \cdot \mathbf{s} \nabla w \right)^{1/2} \left(\int_U \nabla v \cdot \mathbf{s} \nabla v \right)^{1/2} \\ &= \left(\int_U \nabla w \cdot \mathbf{s} \nabla w \right)^{1/2} (2J(U, p, q; \mathbf{a}))^{1/2}, \end{aligned}$$

which is the desired inequality.

To prove (2.66), we choose

$$q := \mathbf{a}_*^t(U; \mathbf{a}) p = (\mathbf{s}_* - \mathbf{k})(U; \mathbf{a}) p.$$

Since $\mathbf{a}_*(U; \mathbf{a}) = \mathbf{s}_*(U; \mathbf{a}) - \mathbf{k}^t(U; \mathbf{a})$, we have

$$q \cdot \int_U \nabla w = p \cdot \mathbf{a}_*(U; \mathbf{a}) \int_U \nabla w = p \cdot \int_U \mathbf{a}_*(U; \mathbf{a}) \nabla w.$$

Therefore

$$p \cdot \int_U (\mathbf{a}_*(U; \mathbf{a}) - \mathbf{a}) \nabla w = q \cdot \int_U \nabla w - p \cdot \int_U \mathbf{a} \nabla w.$$

Applying (2.65) with this choice of q , we get

$$\left| p \cdot \int_U (\mathbf{a}_*(U; \mathbf{a}) - \mathbf{a}) \nabla w \right| \leq \left(\int_U \nabla w \cdot \mathbf{s} \nabla w \right)^{1/2} (2J(U, p, \mathbf{a}_*^t(U; \mathbf{a}) p; \mathbf{a}))^{1/2}.$$

Now (2.60) and the nonnegativity of J^* imply that

$$J(U, p, \mathbf{a}_*^t(U; \mathbf{a})p; \mathbf{a}) \leq p \cdot (\mathbf{s}(U; \mathbf{a}) - \mathbf{s}_*(U; \mathbf{a}))p.$$

Substituting this bound into the previous display yields (2.66).

We next prove (2.67). Set

$$\bar{p} := \int_U \nabla w.$$

Since $J(U, 0, q; \mathbf{a}) = \frac{1}{2}q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})q$, convex duality gives

$$\begin{aligned} \frac{1}{2}\bar{p} \cdot \mathbf{s}_*(U; \mathbf{a})\bar{p} &= \sup_{q \in \mathbb{R}^d} \left(\bar{p} \cdot q - \frac{1}{2}q \cdot \mathbf{s}_*^{-1}(U; \mathbf{a})q \right) \\ &= \sup_{q \in \mathbb{R}^d} (\bar{p} \cdot q - J(U, 0, q; \mathbf{a})) \\ &= \sup_{q \in \mathbb{R}^d} \inf_{v \in \mathcal{A}(U; \mathbf{a})} \int_U \left(q \cdot (\nabla w - \nabla v) + \frac{1}{2}\nabla v \cdot \mathbf{s}\nabla v \right) \\ &\leq \int_U \frac{1}{2}\nabla w \cdot \mathbf{s}\nabla w, \end{aligned}$$

where in the last step we simply choose $v = w$.

Finally, to prove (2.68), we set

$$\bar{q} := -\int_U \mathbf{a}\nabla w.$$

By (2.10) and (2.9) with $q = 0$, we have

$$J(U, p, 0; \mathbf{a}) = \frac{1}{2}p \cdot \mathbf{b}(U; \mathbf{a})p.$$

Therefore another convex-duality computation yields

$$\begin{aligned} \frac{1}{2}\bar{q} \cdot \mathbf{b}^{-1}(U; \mathbf{a})\bar{q} &= \sup_{p \in \mathbb{R}^d} \left(p \cdot \bar{q} - \frac{1}{2}p \cdot \mathbf{b}(U; \mathbf{a})p \right) \\ &= \sup_{p \in \mathbb{R}^d} (p \cdot \bar{q} - J(U, p, 0; \mathbf{a})) \\ &= \sup_{p \in \mathbb{R}^d} \inf_{v \in \mathcal{A}(U; \mathbf{a})} \int_U \left(p \cdot (\mathbf{a}\nabla v - \mathbf{a}\nabla w) + \frac{1}{2}\nabla v \cdot \mathbf{s}\nabla v \right) \\ &\leq \int_U \frac{1}{2}\nabla w \cdot \mathbf{s}\nabla w, \end{aligned}$$

again by choosing $v = w$. Since the left-hand side is unchanged if $\bar{q}$ is replaced by $-\bar{q}$, this is exactly (2.68). $\square$

2.5 Multiscale quantities on cubes

2.5.1 Coarse-grained ellipticity constants

The one-cube matrices $\mathbf{s}_*(U; \mathbf{a})$ and $\mathbf{b}(U; \mathbf{a})$ measure the effective ellipticity of the operator on a single domain. To use them in a multiscale argument, we need scale-local quantities which aggregate the same information over all subcubes of a given triadic cube. Following [ABK], we do this by taking geometrically discounted averages over the scales.

Definition 2.10 (Coarse-grained ellipticity). Let $s \in (0, \infty)$, $q \in [1, \infty]$, $m \in \mathbb{Z}$ and $\mathbf{a} \in \Omega(\square_m)$. For $q \in [1, \infty)$, we set

$$\mathbf{c}_{sq} := 1 - 3^{-sq},$$

so that $n \mapsto \mathbf{c}_{sq} 3^{-sqn}$ is a probability density on $\mathbb{N}_0$. We then define

$$\begin{cases} \Lambda_{s,q}(\square_m; \mathbf{a}) := \left(\mathbf{c}_{sq} \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{b}(z + \square_k; \mathbf{a})|^{q/2} \right)^{2/q}, \\ \lambda_{s,q}(\square_m; \mathbf{a}) := \left(\mathbf{c}_{sq} \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})|^{q/2} \right)^{-2/q}. \end{cases} \quad (2.69)$$

For $q = \infty$, we replace the geometric average by the supremum:

$$\begin{cases} \Lambda_{s,\infty}(\square_m; \mathbf{a}) := \sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{b}(z + \square_k; \mathbf{a})|, \\ \lambda_{s,\infty}(\square_m; \mathbf{a}) := \left(\sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \right)^{-1}. \end{cases} \quad (2.70)$$

If the exponent q is omitted, then it is understood to be equal to 1. We extend these definitions to translated cubes $y + \square_m$ in the obvious way. For $s, t \in (0, \infty)$, we also define the coarse-grained ellipticity ratio

$$\Theta_{s,t}(\square_m; \mathbf{a}) := \frac{\Lambda_{s,1}(\square_m; \mathbf{a})}{\lambda_{t,1}(\square_m; \mathbf{a})}. \quad (2.71)$$

2.5.2 Basic properties of the multiscale ellipticity constants

We first show that the pair $(\lambda_{s,q}, \Lambda_{s,q})$ is a scale-local interpolation between the microscopic ellipticity of $\mathbf{a}$ and the coarse-grained matrices on the cube $\square_m$ itself.

Lemma 2.11 (Basic properties of the multiscale ellipticity constants). *The following assertions hold for every $m \in \mathbb{Z}$, $q \in [1, \infty]$, and $t, s \in (0, \infty)$ with $t < s$.*

1. *The multiscale ellipticity constants satisfy*

$$\lambda_{t,q}(\square_m; \mathbf{a}) \leq \lambda_{s,q}(\square_m; \mathbf{a}) \leq |\mathbf{s}_*^{-1}(\square_m; \mathbf{a})|^{-1} \leq |\mathbf{b}(\square_m; \mathbf{a})| \leq \Lambda_{s,q}(\square_m; \mathbf{a}) \leq \Lambda_{t,q}(\square_m; \mathbf{a}). \quad (2.72)$$

2. *For every $k \in (-\infty, m] \cap \mathbb{Z}$,*

$$\begin{cases} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{b}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \Lambda_{s,q}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \Lambda_{s,q}(\square_m; \mathbf{a}), \\ \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \lambda_{s,q}^{-1}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \lambda_{s,q}^{-1}(\square_m; \mathbf{a}). \end{cases} \quad (2.73)$$

3. *If moreover $s \in (0, 1]$, then there exists $C < \infty$ such that, for every $1 \leq p \leq q < \infty$,*

$$\begin{cases} \Lambda_{s,q}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \Lambda_{s,p}(\square_m; \mathbf{a}), \\ \lambda_{s,q}^{-1}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \lambda_{s,p}^{-1}(\square_m; \mathbf{a}). \end{cases} \quad (2.74)$$

Proof. For each $n \in \mathbb{N}_0$, define

$$M_n := \max_{z \in 3^{m-n}\mathbb{Z}^d \cap \square_m} |\mathbf{b}(z + \square_{m-n}; \mathbf{a})|, \quad N_n := \max_{z \in 3^{m-n}\mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_{m-n}; \mathbf{a})|.$$

By the subadditivity of $J(\cdot, p, 0; \mathbf{a})$ and $J(\cdot, 0, q; \mathbf{a})$ on triadic cubes, we have

$$\mathbf{b}(z + \square_{m-n+1}; \mathbf{a}) \leq \sum_{z' \in z + 3^{m-n}\mathbb{Z}^d \cap (z + \square_{m-n+1})} \mathbf{b}(z' + \square_{m-n}; \mathbf{a})$$

and likewise

$$\mathbf{s}_*^{-1}(z + \square_{m-n+1}; \mathbf{a}) \leq \sum_{z' \in z + 3^{m-n}\mathbb{Z}^d \cap (z + \square_{m-n+1})} \mathbf{s}_*^{-1}(z' + \square_{m-n}; \mathbf{a}).$$

Taking operator norms and then maxima, we obtain that the sequences $(M_n)_{n \in \mathbb{N}_0}$ and $(N_n)_{n \in \mathbb{N}_0}$ are nondecreasing.

For $q < \infty$, the definitions may therefore be rewritten as

$$\Lambda_{s,q}(\square_m; \mathbf{a}) = \left(\mathbf{c}_{sq} \sum_{n=0}^{\infty} 3^{-sqn} M_n^{q/2} \right)^{2/q}, \quad \lambda_{s,q}^{-1}(\square_m; \mathbf{a}) = \left(\mathbf{c}_{sq} \sum_{n=0}^{\infty} 3^{-sqn} N_n^{q/2} \right)^{2/q}.$$

Using the identity

$$\mathbf{c}_{sq} \sum_{n=0}^{\infty} 3^{-sqn} \alpha_n = \alpha_0 + \sum_{n=1}^{\infty} 3^{-sqn} (\alpha_n - \alpha_{n-1})$$

valid for every sequence $(\alpha_n)_{n \in \mathbb{N}_0}$, we see that for nondecreasing α_n the geometric average on the left is nonincreasing in s . Applying this to $\alpha_n = M_n^{q/2}$ and $\alpha_n = N_n^{q/2}$ proves the monotonicity in s for finite q , and the case $q = \infty$ is immediate from (2.70). The second and fourth inequalities in (2.72) are simply the contributions of the term $n = 0$, while the middle inequality follows from (2.12).

We next prove (2.73). The first inequality in each line is again the one-cube bound from the previous item, now applied to the subcube $z + \square_k$. For the second inequality, fix such a cube and suppose first that $q < \infty$. Then

$$\begin{aligned} \Lambda_{s,q}(z + \square_k; \mathbf{a})^{q/2} &= \mathbf{c}_{sq} \sum_{l=-\infty}^k 3^{-sq(k-l)} \max_{z' \in 3^l \mathbb{Z}^d \cap (z + \square_k)} |\mathbf{b}(z' + \square_l; \mathbf{a})|^{q/2} \\ &\leq 3^{sq(m-k)} \mathbf{c}_{sq} \sum_{l=-\infty}^k 3^{-sq(m-l)} \max_{z' \in 3^l \mathbb{Z}^d \cap \square_m} |\mathbf{b}(z' + \square_l; \mathbf{a})|^{q/2} \\ &\leq 3^{sq(m-k)} \Lambda_{s,q}(\square_m; \mathbf{a})^{q/2}, \end{aligned}$$

which yields the claimed bound for $\Lambda_{s,q}$. The proof for $\lambda_{s,q}^{-1}$ is identical, and the case $q = \infty$ follows by the same argument with sums replaced by suprema.

Finally, we prove (2.74), assuming throughout that $s \in (0, 1]$. For $q < \infty$,

$$\Lambda_{s,q}(\square_m; \mathbf{a}) = \mathbf{c}_{sq}^{2/q} \left\| (3^{-sn} M_n^{1/2})_{n \in \mathbb{N}_0} \right\|_{\ell^q}^2 \leq \mathbf{c}_{sq}^{2/q} \mathbf{c}_{sp}^{-2/p} \Lambda_{s,p}(\square_m; \mathbf{a}),$$

since $\ell^p(\mathbb{N}_0) \subseteq \ell^q(\mathbb{N}_0)$ whenever $p \leq q$. The elementary bounds on the geometric discount give, for $s \in (0, 1]$ and $r \geq 1$,

$$cs \leq \mathbf{c}_{sr} \leq Csr, \quad r^{1/r} \leq C.$$

Therefore, for $1 \leq p \leq q$,

$$\mathbf{c}_{sq}^{2/q} \mathbf{c}_{sp}^{-2/p} \leq Cs^{2/q-2/p}.$$

This implies

$$\Lambda_{s,q}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \Lambda_{s,p}(\square_m; \mathbf{a}).$$

The proof for $\lambda_{s,q}^{-1}$ is the same. $\square$

The pair $(\lambda_{s,q}(\square_m; \mathbf{a}), \Lambda_{s,q}(\square_m; \mathbf{a}))$ therefore interpolates between the microscopic ellipticity of the coefficient field $\mathbf{a}$ and the one-cube coarse-grained matrices $\mathbf{s}_*(\square_m; \mathbf{a})$ and $\mathbf{b}(\square_m; \mathbf{a})$.

2.5.3 The homogenization error

In parallel with the multiscale ellipticity constants, we need a quantity which measures not merely the size of the coarse-grained response, but its discrepancy from a chosen constant-coefficient model. As in [ABK], the natural normalization is expressed in the block formalism.

Definition 2.12 (Multiscale homogenization error). Let $s \in (0, 1]$, $p, q \in [1, \infty]$, $m, n \in \mathbb{Z}$ with $n \leq m$, let $\mathbf{a} \in \Omega(\square_m)$, and let $\mathbf{a}_0 \in \mathbb{R}^{d \times d}$ be a constant uniformly elliptic matrix. Writing

$$\mathbf{s}_0 := \frac{1}{2}(\mathbf{a}_0 + \mathbf{a}_0^t), \quad \mathbf{k}_0 := \frac{1}{2}(\mathbf{a}_0 - \mathbf{a}_0^t),$$

we define the associated constant block matrix by

$$\mathbf{A}_0 := \begin{pmatrix} \mathbf{s}_0 + \mathbf{k}_0^t \mathbf{s}_0^{-1} \mathbf{k}_0 & -\mathbf{k}_0^t \mathbf{s}_0^{-1} \\ -\mathbf{s}_0^{-1} \mathbf{k}_0 & \mathbf{s}_0^{-1} \end{pmatrix}. \quad (2.75)$$

By Lemma 2.5, the matrix $\mathbf{A}_0$ is symmetric positive definite, so $\mathbf{A}_0^{\pm 1/2}$ are well-defined. In the maxima below, the vector e ranges over $\mathbb{R}^{2d}$ with $|e| = 1$. We then define the truncated multiscale homogenization error by

$$\mathcal{E}_{s,p,q}(\square_m, n; \mathbf{a}, \mathbf{a}_0) := \begin{cases} \left(\mathbf{c}_{sq} \sum_{l=-\infty}^n 3^{-sq(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{p/2} \right)^{q/p} \right)^{1/q}, & p, q \in [1, \infty), \\ \left(\mathbf{c}_{sq} \sum_{l=-\infty}^n 3^{-sq(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{q/2} \right)^{1/q}, & p = \infty, q \in [1, \infty), \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{p/2} \right)^{1/p}, & p \in [1, \infty), q = \infty, \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{1/2}, & p = q = \infty. \end{cases} \quad (2.76)$$

If the argument n is omitted, then it is understood that $n = m$:

$$\mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0) := \mathcal{E}_{s,p,q}(\square_m, m; \mathbf{a}, \mathbf{a}_0).$$

We extend the definition to translated cubes $y + \square_m$ in the obvious way.

The quantity $\mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0)$ is therefore the scale-local defect in the block response when the heterogeneous operator $-\nabla \cdot \mathbf{a} \nabla$ is compared to the constant-coefficient operator $-\nabla \cdot \mathbf{a}_0 \nabla$.

2.5.4 Basic properties of the homogenization error

The case $p = \infty$ is the most directly scale-local one, and it is the version which enters first in the later deterministic and renormalization arguments. We therefore record its first properties now.

Lemma 2.13 (Basic properties of the homogenization error). *Let $q \in [1, \infty]$, $m \in \mathbb{Z}$, $s, t \in (0, 1]$ with $t < s$, let $\mathbf{a} \in \Omega(\square_m)$, and let $\mathbf{a}_0 \in \mathbb{R}^{d \times d}$ be constant and uniformly elliptic. Then:*

1. *The one-cube defect is bounded by the multiscale one, and the latter is monotone in s :*

$$\max_{|e|=1} \mathbf{J}(\square_m, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{1/2} \leq \mathcal{E}_{s, \infty, q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \mathcal{E}_{t, \infty, q}(\square_m; \mathbf{a}, \mathbf{a}_0). \quad (2.77)$$

2. *For every $k \in (-\infty, m] \cap \mathbb{Z}$,*

$$\max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s, \infty, q}(z + \square_k; \mathbf{a}, \mathbf{a}_0) \leq 3^{s(m-k)} \mathcal{E}_{s, \infty, q}(\square_m; \mathbf{a}, \mathbf{a}_0). \quad (2.78)$$

Proof. For $n \in \mathbb{N}_0$, define

$$H_n := \max_{z \in 3^{m-n} \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z + \square_{m-n}, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{1/2}.$$

By the subadditivity of $\mathbf{A}$ and $\mathbf{A}_*^{-1}$ on triadic cubes, the splitting formula (2.51) implies that for every fixed $P, Q \in \mathbb{R}^{2d}$ the map $U \mapsto \mathbf{J}(U, P, Q; \mathbf{a})$ is subadditive across triadic partitions. Applying this with

$$P = \mathbf{A}_0^{-1/2} e, \quad Q = \mathbf{A}_0^{1/2} e, \quad |e| = 1,$$

we obtain that $(H_n)_{n \in \mathbb{N}_0}$ is nondecreasing.

For $q < \infty$, the definition of $\mathcal{E}_{s, \infty, q}$ may therefore be rewritten as

$$\mathcal{E}_{s, \infty, q}(\square_m; \mathbf{a}, \mathbf{a}_0) = \left(\mathbf{c}_{sq} \sum_{n=0}^{\infty} 3^{-sqn} H_n^q \right)^{1/q}.$$

The first inequality in (2.77) is then just the contribution of the term $n = 0$. The monotonicity in s follows from the same elementary weighted-average argument used in the proof of Lemma 2.11, and the case $q = \infty$ is immediate from the supremum formula in (2.76).

We next prove (2.78). Fix a subcube $z + \square_k$. If $q < \infty$, then

$$\begin{aligned} \mathcal{E}_{s, \infty, q}(z + \square_k; \mathbf{a}, \mathbf{a}_0)^q &= \mathbf{c}_{sq} \sum_{l=-\infty}^k 3^{-sq(k-l)} \max_{z' \in 3^l \mathbb{Z}^d \cap (z + \square_k)} \max_{|e|=1} \mathbf{J}(z' + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{q/2} \\ &\leq 3^{sq(m-k)} \mathbf{c}_{sq} \sum_{l=-\infty}^k 3^{-sq(m-l)} \max_{z' \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} \mathbf{J}(z' + \square_l, \mathbf{A}_0^{-1/2} e, \mathbf{A}_0^{1/2} e; \mathbf{a})^{q/2} \\ &\leq 3^{sq(m-k)} \mathcal{E}_{s, \infty, q}(\square_m; \mathbf{a}, \mathbf{a}_0)^q, \end{aligned}$$

which is exactly the desired bound. The proof for $q = \infty$ is the same, with sums replaced by suprema. $\square$

The quantity $\mathcal{E}_{s, \infty, q}$ is thus the scale-local state variable measuring how far the operator is from the constant model $\mathbf{a}_0$ across all subcubes of $\square_m$. This is precisely the form needed for renormalization arguments: it records the operator defect at the same scales on which one iterates.

The same state variable also controls the coarse-grained ellipticity constants, provided the comparison matrix is scalar. This deterministic observation is used later to remove the ellipticity factors from the quenched homogenization estimate.

Lemma 2.14 ($\mathcal{E}$ controls coarse-grained ellipticity). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $s \in (0, \infty)$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, and every $\sigma_0 > 0$,*

$$\max\{\sigma_0^{-1}\Lambda_{s,2}(\square_m; \mathbf{a}), \sigma_0\lambda_{s,2}^{-1}(\square_m; \mathbf{a})\} \leq C(1 + \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}, \sigma_0\mathbf{I}_d)^2). \quad (2.79)$$

Proof. It is enough to prove the corresponding one-scale estimate and then average over the geometric weights in the definitions. On a triadic subcube U , the normalized block response

$$\max_{|e|=1} \mathbf{J}(U, \mathbf{A}_0^{-1/2}e, \mathbf{A}_0^{1/2}e; \mathbf{a}), \quad \mathbf{a}_0 = \sigma_0\mathbf{I}_d,$$

controls both normalized diagonal blocks of the coarse-grained block matrix. In particular,

$$\sigma_0^{-1}|\mathbf{b}(U; \mathbf{a})| + \sigma_0|\mathbf{s}_*^{-1}(U; \mathbf{a})| \leq C\left(1 + \max_{|e|=1} \mathbf{J}(U, \mathbf{A}_0^{-1/2}e, \mathbf{A}_0^{1/2}e; \mathbf{a})\right).$$

Multiplying by the geometric weights in the definition with $q = 2$, taking the maximum over the subcubes $U = z + \square_k \subseteq \square_m$, and summing in $k \leq m$ gives (2.79). The normalizing constant in the geometric average is harmless and is absorbed into $C(d)$. $\square$

Chapter 3

Coarse-graining: deterministic theory

Chapter 2 attached to each cube a family of response functionals, coarse-grained matrices, multiscale ellipticity constants, and homogenization errors. The purpose of the present chapter is to show that these quantities control actual solutions of the equation. Everything in this chapter is deterministic: no stationarity, no independence, and no probability enter here. The goal is to produce a package of deterministic estimates that the later chapters can quote as black boxes.

3.1 Coarse-grained elliptic and functional inequalities

The multiscale quantities $\lambda_{s,q}(\square_m; \mathbf{a})$ and $\Lambda_{s,q}(\square_m; \mathbf{a})$ defined in Definition 2.10 are not merely a convenient way of packaging the one-cube matrices from Chapter 2. They imply genuine elliptic estimates at the scale of the cube $\square_m$. The first of these is a coarse-grained Poincaré inequality, which controls weak norms of the averaged gradient and flux of a solution by its energy.

3.1.1 Coarse-grained Poincaré inequality

Proposition 3.1 (Coarse-grained Poincaré inequality). *For every $m \in \mathbb{Z}$, every $s \in (0, 1]$ and $q \in [1, \infty]$, with $s = 1$ allowed only when $q = 1$, and every solution $u \in \mathcal{A}(\square_m; \mathbf{a})$, we have*

$$3^{-sm} [\nabla u]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq \mathfrak{c}_{sq}^{-1/q} \lambda_{s,q}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \quad (3.1)$$

and

$$3^{-sm} [\mathbf{a} \nabla u]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq \mathfrak{c}_{sq}^{-1/q} \Lambda_{s,q}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}. \quad (3.2)$$

When $q = \infty$, the factor $\mathfrak{c}_{sq}^{-1/q}$ is interpreted as 1.

Proof. We first prove (3.1). Assume first that $q < \infty$. By the definition (1.8) of the negative Besov seminorm,

$$3^{-sqm} [\nabla u]_{\dot{B}_{2,q}^{-s}(\square_m)}^q = \sum_{k=-\infty}^m 3^{sq(k-m)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} |(\nabla u)_{z+\square_k}|^2 \right)^{q/2}. \quad (3.3)$$

For each $k \in (-\infty, m] \cap \mathbb{Z}$ and $z \in 3^k \mathbb{Z}^d \cap \square_m$, the restriction of u to the subcube $z + \square_k$ belongs to $\mathcal{A}(z + \square_k; \mathbf{a})$. Applying (2.67) in the cube $z + \square_k$, we obtain

$$|(\nabla u)_{z+\square_k}|^2 \leq |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \int_{z+\square_k} \nabla u \cdot \mathbf{s} \nabla u.$$

Therefore,

$$\begin{aligned}
\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} |(\nabla u)_{z+\square_k}|^2 &\leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \int_{z+\square_k} \nabla u \cdot \mathbf{s} \nabla u \\
&= \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \int_{\square_m} \nabla u \cdot \mathbf{s} \nabla u \\
&= \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})| \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}^2.
\end{aligned}$$

Substituting this into (3.3), we get

$$\begin{aligned}
3^{-sqm} [\nabla u]_{\underline{B}_{2,q}^{-s}(\square_m)}^q &\leq \sum_{k=-\infty}^m 3^{sq(k-m)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})|^{q/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}^q \\
&= \mathbf{c}_{sq}^{-1} \lambda_{s,q}^{-q/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}^q.
\end{aligned}$$

Taking the power $1/q$ yields (3.1) for $q < \infty$.

If $q = \infty$, then by the definition of $[\cdot]_{\underline{B}_{2,\infty}^{-s}(\square_m)}$,

$$3^{-sm} [\nabla u]_{\underline{B}_{2,\infty}^{-s}(\square_m)} = \sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-s(m-k)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} |(\nabla u)_{z+\square_k}|^2 \right)^{1/2}.$$

Using the previous bound on the averaged gradients, we find

$$\begin{aligned}
3^{-sm} [\nabla u]_{\underline{B}_{2,\infty}^{-s}(\square_m)} &\leq \sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \\
&= \lambda_{s,\infty}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}.
\end{aligned}$$

This completes the proof of (3.1).

The proof of (3.2) is identical, except that we apply (2.68) on each subcube $z + \square_k$ to get

$$|(\mathbf{a} \nabla u)_{z+\square_k}|^2 \leq |\mathbf{b}(z + \square_k; \mathbf{a})| \int_{z+\square_k} \nabla u \cdot \mathbf{s} \nabla u,$$

and then use the definition of $\Lambda_{s,q}(\square_m; \mathbf{a})$ in place of that of $\lambda_{s,q}^{-1}(\square_m; \mathbf{a})$. $\square$

3.1.2 Coarse-grained Caccioppoli inequality

The most useful form of the coarse-grained Caccioppoli estimate is a boundary version. The proof is based on testing the equation with a cutoff and then estimating the error term by the Besov duality and cutoff lemmas from Chapter 1.

Proposition 3.2 (Coarse-grained Caccioppoli inequality, boundary version). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $s, t \in (0, 1)$ with $s + t < 1$, every $x \in \square_m$, and every $u \in H^1(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = 0 & \text{in } \square_m, \\ u = 0 & \text{on } (\partial \square_m) \cap (x + \square_{m-1}), \end{cases}$$

we have

$$\begin{aligned}
&\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{m-2}) \cap \square_m)}^2 \\
&\leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} s^{-\frac{2s}{1-s-t}} \left(\frac{\Lambda_{s,1}(\square_m; \mathbf{a})}{\lambda_{t,1}(\square_m; \mathbf{a})} \right)^{\frac{s}{1-s-t}} \Lambda_{s,1}(\square_m; \mathbf{a}) 3^{-2m} \|u\|_{\underline{L}^2(\square_m)}^2. \tag{3.4}
\end{aligned}$$

Proof. By translation and dilation, we may reduce to the case $m = 0$. Set

$$\sigma := 1 - s - t.$$

Since $\sigma > 0$, we have $t < 1 - s$.

We first prove a recursive estimate on intermediate cubes. Fix parameters $\rho_1, \rho_2 \in [1/3, 1)$ with $\rho_1 < \rho_2$. Choose integers $k, h \in \mathbb{N}$ satisfying

$$3^{-4}(\rho_2 - \rho_1) \leq 3^{-k} \leq 3^{-3}(\rho_2 - \rho_1) \quad \text{and} \quad h \geq k + 4.$$

Let $\varphi \in C^\infty(\mathbb{R}^d)$ satisfy

$$\mathbb{1}_{x+\rho_1\Box_{-1}} \leq \varphi \leq \mathbb{1}_{x+\frac{1}{2}(\rho_1+\rho_2)\Box_{-1}}$$

and, for $j \in \{1, 2\}$,

$$\|\nabla^j \varphi\|_{L^\infty(\mathbb{R}^d)} \leq C 3^{kj} \leq C(\rho_2 - \rho_1)^{-j}.$$

Since $u = 0$ on $(\partial\Box_0) \cap (x + \Box_{-1})$ and $\text{supp } \varphi \subseteq x + \Box_{-1}$, we have $u\varphi \in H_0^1(\Box_0)$. Testing the equation with $u\varphi$ yields

$$\int_{\Box_0} \varphi \nabla u \cdot \mathbf{a} \nabla u = - \int_{\Box_0} u \nabla \varphi \cdot \mathbf{a} \nabla u. \quad (3.5)$$

For each $z \in 3^{-h}\mathbb{Z}^d \cap \Box_0$, let

$$S := \{z \in 3^{-h}\mathbb{Z}^d \cap \Box_0 : \text{supp } (\nabla \varphi) \cap (z + \Box_{-h}) \neq \emptyset\}.$$

Since the cubes $\{z + \Box_{-h}\}_{z \in 3^{-h}\mathbb{Z}^d \cap \Box_0}$ partition $\Box_0$ and $|\Box_0| = 1$, we have

$$\int_{\Box_0} u \nabla \varphi \cdot \mathbf{a} \nabla u = \sum_{z \in 3^{-h}\mathbb{Z}^d \cap \Box_0} \mathbb{1}_S(z) \int_{z+\Box_{-h}} u \nabla \varphi \cdot \mathbf{a} \nabla u.$$

Fix $z \in S$. Decompose

$$u = (u)_{z+\Box_{-h}} + (u - (u)_{z+\Box_{-h}}).$$

Applying the cube-wise Besov pairing (1.24) with $r = 1$ to the first term and with $r = s$ to the second, we obtain

$$\begin{aligned} \left| \int_{z+\Box_{-h}} u \nabla \varphi \cdot \mathbf{a} \nabla u \right| &\leq C |(u)_{z+\Box_{-h}}| \|\nabla \varphi\|_{\underline{B}_{2,\infty}^1(z+\Box_{-h})} [\mathbf{a} \nabla u]_{\dot{\underline{B}}_{2,1}^{-1}(z+\Box_{-h})} \\ &\quad + C \|(u - (u)_{z+\Box_{-h}}) \nabla \varphi\|_{\underline{B}_{2,\infty}^s(z+\Box_{-h})} [\mathbf{a} \nabla u]_{\dot{\underline{B}}_{2,1}^{-s}(z+\Box_{-h})}. \end{aligned} \quad (3.6)$$

We next estimate the positive-order Besov factors. By the definition of the $\underline{B}_{2,\infty}^1$ norm, the Poincaré inequality on subcubes, and the bounds on φ ,

$$\|\nabla \varphi\|_{\underline{B}_{2,\infty}^1(z+\Box_{-h})} \leq C (\|\nabla^2 \varphi\|_{L^\infty(z+\Box_{-h})} + 3^h \|\nabla \varphi\|_{L^\infty(z+\Box_{-h})}) \leq C 3^{k+h}.$$

Also, since

$$3^{-h} \|\nabla \varphi\|_{\underline{W}^{1,\infty}(z+\Box_{-h})} \leq C (\|\nabla \varphi\|_{L^\infty(z+\Box_{-h})} + 3^{-h} \|\nabla^2 \varphi\|_{L^\infty(z+\Box_{-h})}) \leq C 3^k,$$

the cutoff/product estimate (1.27) gives

$$\|(u - (u)_{z+\Box_{-h}}) \nabla \varphi\|_{\underline{B}_{2,\infty}^s(z+\Box_{-h})} \leq C 3^k [\nabla u]_{\dot{\underline{B}}_{2,1}^{-s}(z+\Box_{-h})}. \quad (3.7)$$

Now apply the coarse-grained Poincaré inequality on the cube $z + \square_{-h}$. For the gradient term, we use (3.1) with exponent $1 - s \in (0, 1)$, together with the elementary bound $\mathfrak{c}_{(1-s)1}^{-1} \leq C(1-s)^{-1}$, to obtain

$$[\nabla u]_{\dot{B}_{2,1}^{s-1}(z+\square_{-h})} \leq \frac{C}{1-s} 3^{-(1-s)h} \lambda_{1-s,1}^{-1/2}(z + \square_{-h}; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_{-h})}. \quad (3.8)$$

Similarly, for $r \in \{s, 1\}$, the flux estimate (3.2) yields

$$[\mathbf{a} \nabla u]_{\dot{B}_{2,1}^{-r}(z+\square_{-h})} \leq \frac{C}{r} 3^{-rh} \Lambda_{r,1}^{1/2}(z + \square_{-h}; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_{-h})}. \quad (3.9)$$

Substituting (3.7), (3.8), and (3.9) into (3.6), we get

$$\begin{aligned} \left| \int_{z+\square_{-h}} u \nabla \varphi \cdot \mathbf{a} \nabla u \right| &\leq C 3^k \Lambda_{1,1}^{1/2}(z + \square_{-h}; \mathbf{a}) |(u)_{z+\square_{-h}}| \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_{-h})} \\ &\quad + \frac{C 3^{k-h}}{s(1-s)} \left(\frac{\Lambda_{s,1}(z + \square_{-h}; \mathbf{a})}{\lambda_{1-s,1}(z + \square_{-h}; \mathbf{a})} \right)^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_{-h})}^2. \end{aligned} \quad (3.10)$$

Since $|(u)_{z+\square_{-h}}| \leq \|u\|_{\underline{L}^2(z+\square_{-h})}$, the first term may be estimated by Cauchy-Schwarz after summing over $z \in S$. For the coefficient factors, we use the monotonicity and localization bounds from (2.72) and (2.73). Since $s < 1$, we have

$$\Lambda_{1,1}(z + \square_{-h}; \mathbf{a}) \leq \Lambda_{s,1}(z + \square_{-h}; \mathbf{a}) \leq 3^{2sh} \Lambda_{s,1}(\square_0; \mathbf{a}),$$

and, because $t < 1 - s$,

$$\lambda_{1-s,1}^{-1}(z + \square_{-h}; \mathbf{a}) \leq \lambda_{t,1}^{-1}(z + \square_{-h}; \mathbf{a}) \leq 3^{2th} \lambda_{t,1}^{-1}(\square_0; \mathbf{a}).$$

Therefore,

$$\begin{cases} \Lambda_{1,1}^{1/2}(z + \square_{-h}; \mathbf{a}) \leq 3^{sh} \Lambda_{s,1}^{1/2}(\square_0; \mathbf{a}), \\ 3^{-h} \left(\frac{\Lambda_{s,1}(z + \square_{-h}; \mathbf{a})}{\lambda_{1-s,1}(z + \square_{-h}; \mathbf{a})} \right)^{1/2} \leq 3^{-\sigma h} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2}. \end{cases} \quad (3.11)$$

Because $h \geq k + 4$ and $3^{-k} \leq 3^{-3}(\rho_2 - \rho_1)$, every cube $z + \square_{-h}$ with $z \in S$ is contained in $(x + \rho_2 \square_{-1}) \cap \square_0$. Combining (3.5), (3.10), and (3.11), and then using Cauchy-Schwarz in the averaged sum over z , we obtain

$$\begin{aligned} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho_1 \square_{-1}) \cap \square_0)}^2 &\leq \frac{C 3^k}{s(1-s)} 3^{-\sigma h} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho_2 \square_{-1}) \cap \square_0)}^2 \\ &\quad + C 3^k 3^{sh} \Lambda_{s,1}^{1/2}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho_2 \square_{-1}) \cap \square_0)}. \end{aligned} \quad (3.12)$$

Here we absorbed the harmless volume factors coming from the intersections $(x + \rho_j \square_{-1}) \cap \square_0$ into the constant $C(d)$, since $\rho_j \in [1/3, 1]$.

We now choose h so that the first term on the right can be absorbed. Set

$$h := \max \left\{ k + 4, \left\lceil \sigma^{-1} \log_3 \left(\frac{4C(3.12) 3^k}{s(1-s)} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \right) \right\rceil \right\}.$$

Then

$$\frac{C 3^k}{s(1-s)} 3^{-\sigma h} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \leq \frac{1}{4},$$

and Young's inequality gives

$$\begin{aligned} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho_1 \square_{-1}) \cap \square_0)}^2 &\leq \frac{1}{2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho_2 \square_{-1}) \cap \square_0)}^2 \\ &\quad + C(\rho_2 - \rho_1)^{-\frac{2(s+\sigma)}{\sigma}} \left(\frac{C}{s(1-s)} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \right)^{\frac{2s}{\sigma}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2. \end{aligned} \quad (3.13)$$

Indeed, since h is the maximum of the two quantities above, we have

$$3^{sh} \leq C 3^{sk} + C \left(\frac{C 3^k}{s(1-s)} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \right)^{\frac{s}{\sigma}}.$$

Using $3^k \leq C(\rho_2 - \rho_1)^{-1}$, $\Theta_{s,t}(\square_0; \mathbf{a}) \geq 1$, and $s(1-s) \leq 1$, the first term on the right contributes only

$$C(\rho_2 - \rho_1)^{-2(1+s)} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2,$$

which is bounded by the right side of (3.13) after enlarging C , because $2(1+s) \leq \frac{2(s+\sigma)}{\sigma}$.

To apply the standard radius iteration lemma, define

$$F(\rho) := \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\rho \square_{-1}) \cap \square_0)}^2 \quad \text{for } \rho \in [1/3, 1].$$

Let

$$\beta := \frac{2(s+\sigma)}{\sigma} \quad \text{and} \quad A := C \left(\frac{C}{s(1-s)} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \right)^{\frac{2s}{\sigma}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2.$$

Then (3.13) says that

$$F(\rho_1) \leq \frac{1}{2} F(\rho_2) + A(\rho_2 - \rho_1)^{-\beta} \quad \forall \rho_1, \rho_2 \in [1/3, 1], \quad \rho_1 < \rho_2.$$

Applying Lemma 1.13 on the interval $[1/3, 1]$, we obtain

$$F(1/3) \leq (C(1+\beta))^\beta A.$$

Since $\beta = \frac{2(s+\sigma)}{\sigma} \leq \frac{2}{\sigma}$, this implies

$$F(1/3) \leq \left(\frac{C}{\sigma} \right)^\beta A.$$

Because $\frac{1}{3} \square_{-1} = \square_{-2}$, this yields

$$\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \leq \left(\frac{C}{\sigma} \right)^\beta \left(\frac{C}{s(1-s)} \Theta_{s,t}(\square_0; \mathbf{a})^{1/2} \right)^{\frac{2s}{\sigma}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2.$$

Finally, using only $(1-s)^{-1} \leq \sigma^{-1}$ and $\sigma = 1-s-t$, we obtain

$$\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} s^{-\frac{2s}{1-s-t}} \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{s}{1-s-t}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2,$$

which is exactly (3.4) in the case $m = 0$. Scaling back yields the general case. $\square$

Corollary 3.3. *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $s, t \in (0, 1)$ with $s + t < 1$, every $x \in \mathbb{R}^d$, and every $u \in \mathcal{A}(x + \square_m; \mathbf{a})$, we have*

$$\begin{aligned} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(x + \square_{m-2})}^2 &\leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} s^{-\frac{2s}{1-s-t}} \left(\frac{\Lambda_{s,1}(x + \square_m; \mathbf{a})}{\lambda_{t,1}(x + \square_m; \mathbf{a})} \right)^{\frac{s}{1-s-t}} \\ &\quad \times \Lambda_{s,1}(x + \square_m; \mathbf{a}) 3^{-2m} \|u - (u)_{x+\square_m}\|_{\underline{L}^2(x+\square_m)}^2. \end{aligned} \quad (3.14)$$

Proof. By translation and dilation, it is enough to treat the case $x = 0$ and $m = 0$. Set

$$v := u - (u)_{\square_0}.$$

Then $v \in \mathcal{A}(\square_0; \mathbf{a})$, $\nabla v = \nabla u$, and

$$\|v\|_{\underline{L}^2(\square_0)} = \|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)}.$$

We now repeat the proof of Proposition 3.2 with v in place of u . The only point at which the boundary assumption was used there was to justify that the test function belongs to $H_0^1(\square_0)$. In the present interior setting, the cutoff φ is supported in $\square_{-1}$, so we have $v\varphi \in H_0^1(\square_0)$ automatically. The rest of the argument is unchanged, and yields

$$\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_{-2})}^2 \leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} s^{-\frac{2s}{1-s-t}} \left(\frac{\Lambda_{s,1}(\square_0; \mathbf{a})}{\lambda_{t,1}(\square_0; \mathbf{a})} \right)^{\frac{s}{1-s-t}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)}^2.$$

Scaling back gives (3.14). $\square$

3.1.3 Coarse-grained flux-response estimate

The next estimate is the first direct comparison between the heterogeneous operator $-\nabla \cdot \mathbf{a} \nabla$ and a chosen constant-coefficient model. It controls the weak norm of the flux defect $(\mathbf{a} - \mathbf{a}_0) \nabla u$ by the energy of u and the multiscale homogenization error from Chapter 2.

Lemma 3.4 (Coarse-grained flux-response estimate). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $s \in (0, 1]$, every scalar constant-coefficient matrix $\mathbf{a}_0 = \sigma_0 \mathbf{I}_d$ with $\sigma_0 > 0$, and every solution $u \in \mathcal{A}(\square_m; \mathbf{a})$, we have*

$$3^{-sm} [(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\underline{B}_{2,1}^{-s}(\square_m)} \leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{a}_0). \quad (3.15)$$

Proof. Since $u \in \mathcal{A}(\square_m; \mathbf{a})$, its restriction to each subcube $z + \square_k$ belongs to $\mathcal{A}(z + \square_k; \mathbf{a})$.

Fix $z \in 3^k \mathbb{Z}^d \cap \square_m$ and $e \in \mathbb{R}^d$ with $|e| = 1$. Since $\mathbf{a}_0$ is symmetric,

$$(\mathbf{a}_0^{-1/2} e) \cdot ((\mathbf{a} - \mathbf{a}_0) \nabla u)_{z+\square_k} = \int_{z+\square_k} (\mathbf{a}_0^{-1/2} e) \cdot \mathbf{a} \nabla u - (\mathbf{a}_0^{1/2} e) \cdot \nabla u.$$

Applying the linear-response inequality (2.65) in the cube $z + \square_k$ with

$$p := \mathbf{a}_0^{-1/2} e \quad \text{and} \quad q := \mathbf{a}_0^{1/2} e,$$

we obtain

$$|\mathbf{a}_0^{-1/2} e \cdot ((\mathbf{a} - \mathbf{a}_0) \nabla u)_{z+\square_k}| \leq (2J(z + \square_k, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_k)}.$$

Since

$$\begin{aligned} |((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}| &= |\mathbf{a}_0^{1/2} \mathbf{a}_0^{-1/2} ((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}| \\ &\leq |\mathbf{a}_0|^{1/2} |\mathbf{a}_0^{-1/2} ((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}| \\ &= |\mathbf{a}_0|^{1/2} \max_{|e|=1} |\mathbf{a}_0^{-1/2} e \cdot ((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}|. \end{aligned}$$

we deduce that

$$|((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}| \leq 2^{1/2} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_k)} \max_{|e|=1} J(z + \square_k, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})^{1/2}. \quad (3.16)$$

We next connect the scalar response quantity on the right side to the block response entering $\mathcal{E}_{s,\infty,1}$. Since $\mathbf{a}_0$ is symmetric, the associated block matrix from (2.75) is

$$\mathbf{A}_0 = \begin{pmatrix} \mathbf{a}_0 & 0 \\ 0 & \mathbf{a}_0^{-1} \end{pmatrix}.$$

For each $e \in \mathbb{R}^d$ with $|e| = 1$, define

$$\mathbf{e} := \frac{1}{\sqrt{2}} \begin{pmatrix} e \\ -e \end{pmatrix} \in \mathbb{R}^{2d}.$$

Then $|\mathbf{e}| = 1$ and

$$\mathbf{A}_0^{-1/2} \mathbf{e} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{a}_0^{-1/2} e \\ -\mathbf{a}_0^{1/2} e \end{pmatrix}, \quad \mathbf{A}_0^{1/2} \mathbf{e} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{a}_0^{1/2} e \\ -\mathbf{a}_0^{-1/2} e \end{pmatrix}.$$

Applying (2.47) with

$$p := \frac{1}{\sqrt{2}} \mathbf{a}_0^{-1/2} e, \quad q := -\frac{1}{\sqrt{2}} \mathbf{a}_0^{1/2} e, \quad q^* := \frac{1}{\sqrt{2}} \mathbf{a}_0^{1/2} e, \quad p^* := -\frac{1}{\sqrt{2}} \mathbf{a}_0^{-1/2} e,$$

we find

$$\begin{aligned} \mathbf{J}(z + \square_k, \mathbf{A}_0^{-1/2} \mathbf{e}, \mathbf{A}_0^{1/2} \mathbf{e}; \mathbf{a}) &= \frac{1}{2} J(z + \square_k, \sqrt{2} \mathbf{a}_0^{-1/2} e, \sqrt{2} \mathbf{a}_0^{1/2} e; \mathbf{a}) + \frac{1}{2} J^*(z + \square_k, 0, 0; \mathbf{a}) \\ &= J(z + \square_k, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}), \end{aligned}$$

since $J^*(z + \square_k, 0, 0; \mathbf{a}) = 0$ and $J(z + \square_k, \cdot, \cdot; \mathbf{a})$ is quadratic. Therefore,

$$\max_{|e|=1} J(z + \square_k, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})^{1/2} \leq \max_{|\mathbf{e}|=1} \mathbf{J}(z + \square_k, \mathbf{A}_0^{-1/2} \mathbf{e}, \mathbf{A}_0^{1/2} \mathbf{e}; \mathbf{a})^{1/2}. \quad (3.17)$$

By the definition (1.8) of the concrete negative Besov seminorm,

$$3^{-sm} [(\mathbf{a} - \mathbf{a}_0)\nabla u]_{\dot{B}_{2,1}^{-s}(\square_m)} = \sum_{k=-\infty}^m 3^{s(k-m)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} |((\mathbf{a} - \mathbf{a}_0)\nabla u)_{z+\square_k}|^2 \right)^{1/2}.$$

Using (3.16) and then (3.17), we infer

$$\begin{aligned} &3^{-sm} [(\mathbf{a} - \mathbf{a}_0)\nabla u]_{\dot{B}_{2,1}^{-s}(\square_m)} \\ &\leq 2^{1/2} |\mathbf{a}_0|^{1/2} \sum_{k=-\infty}^m 3^{s(k-m)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_k)}^2 \max_{|\mathbf{e}|=1} \mathbf{J}(z + \square_k, \mathbf{A}_0^{-1/2} \mathbf{e}, \mathbf{A}_0^{1/2} \mathbf{e}; \mathbf{a}) \right)^{1/2} \\ &\leq 2^{1/2} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \sum_{k=-\infty}^m 3^{s(k-m)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \max_{|\mathbf{e}|=1} \mathbf{J}(z + \square_k, \mathbf{A}_0^{-1/2} \mathbf{e}, \mathbf{A}_0^{1/2} \mathbf{e}; \mathbf{a})^{1/2}, \end{aligned}$$

because the cubes $\{z + \square_k\}_{z \in 3^k \mathbb{Z}^d \cap \square_m}$ partition $\square_m$ and hence

$$\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z + \square_k)}^2 = \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}^2.$$

Using the definition (2.76) with $p = \infty$ and $q = 1$, we get

$$\begin{aligned} 3^{-sm}[(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\dot{B}_{2,1}^{-s}(\square_m)} &\leq 2^{1/2} \mathbf{c}_s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{a}_0) \\ &\leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{a}_0), \end{aligned}$$

since $\mathbf{c}_s = 1 - 3^{-s} \geq cs$ for every $s \in (0, 1]$. This is exactly (3.15). $\square$

3.2 Inhomogeneous equations

We now upgrade the coarse-graining estimates from harmonic functions to solutions of

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g}.$$

The key new idea is local correction of the forcing. On a subcube Q , we solve a zero-Dirichlet problem with right-hand side $\nabla \cdot \mathbf{g}$ in Q and subtract that solution from u . The remainder then solves the homogeneous equation in Q , so the results of Section 3.1 become available again. This is the right place to separate the genuinely multiscale estimate from its energy consequences: we first prove a coarse-grained Poincaré estimate for ∇u , and only afterward deduce the forcing-dependent energy bounds.

3.2.1 Coarse-grained Poincaré inequality with right-hand side

The proof of the next proposition is organized around local Dirichlet correctors. On each subcube Q , we solve the zero-boundary problem with forcing $\nabla \cdot (\mathbf{g} - (\mathbf{g})_Q)$ in Q . Subtracting the average of $\mathbf{g}$ does not change the equation and makes the local corrector compatible with the Besov seminorms. Then the remainder is $\mathbf{a}$ -harmonic in Q , so the homogeneous coarse-grained Poincaré inequality applies. This yields a one-step recurrence across scales for the weak norm of ∇u , which is the multiscale heart of the argument.

Proposition 3.5 (Coarse-grained Poincaré inequality with right-hand side). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $s \in (0, 1)$, every vector field $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and every solution $u \in H^1(\square_m)$ of the equation*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

we have

$$3^{-sm}[\nabla u]_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_m; \mathbf{a}) 3^{sm}[\mathbf{g}]_{\dot{B}_{2,2}^s(\square_m)}. \quad (3.18)$$

Proof. By the scale invariance of all the quantities in the statement, it is enough to prove the estimate in the case $m = 0$.

We divide the proof into three steps.

Step 1. Local correction of the forcing. Fix $n \in (-\infty, 0] \cap \mathbb{Z}$ and $z \in 3^n \mathbb{Z}^d \cap \square_0$. Since the restriction of $\mathbf{a}$ to $z + \square_n$ belongs to $\Omega(z + \square_n)$, there exists a unique function $\rho_z \in H_0^1(z + \square_n)$ such that

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla \rho_z = \nabla \cdot (\mathbf{g} - (\mathbf{g})_{z+\square_n}) & \text{in } z + \square_n, \\ \rho_z = 0 & \text{on } \partial(z + \square_n). \end{cases} \quad (3.19)$$

Set

$$w_z := u - \rho_z.$$

Since $\nabla \cdot (\mathbf{g} - (\mathbf{g})_{z+\square_n}) = \nabla \cdot \mathbf{g}$ and

$$[\mathbf{g} - (\mathbf{g})_{z+\square_n}]_{\underline{B}_{2,2}^s(z+\square_n)} = [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)},$$

the function w_z satisfies

$$-\nabla \cdot \mathbf{a} \nabla w_z = 0 \quad \text{in } z + \square_n,$$

and therefore $w_z \in \mathcal{A}(z + \square_n; \mathbf{a})$.

Applying the homogeneous coarse-grained Poincaré inequality (3.1) with $q = 2$ on $z + \square_n$, and using

$$\mathbf{c}_{2s}^{-1/2} \leq C s^{-1/2},$$

we find

$$3^{-sn} [\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \leq C s^{-1/2} \lambda_{s,2}^{-1/2}(z + \square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla w_z\|_{\underline{L}^2(z+\square_n)}. \quad (3.20)$$

On the other hand, testing (3.19) with ρ_z gives

$$\|\mathbf{s}^{1/2} \nabla \rho_z\|_{\underline{L}^2(z+\square_n)}^2 = - \int_{z+\square_n} (\mathbf{g} - (\mathbf{g})_{z+\square_n}) \cdot \nabla \rho_z.$$

Since $(\mathbf{g} - (\mathbf{g})_{z+\square_n})_{z+\square_n} = 0$, the normalized Besov norm on the right is just the seminorm, and the cube-wise Besov pairing (1.24) with the choices $p = q = 2$ yields

$$\|\mathbf{s}^{1/2} \nabla \rho_z\|_{\underline{L}^2(z+\square_n)}^2 \leq C [\nabla \rho_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}. \quad (3.21)$$

Since $\rho_z \in H_0^1(z + \square_n)$, the average of each weak derivative of ρ_z vanishes. Indeed, this is clear for smooth compactly supported functions, and the general case follows by density. Thus

$$(\nabla \rho_z)_{z+\square_n} = 0.$$

Consequently,

$$(\nabla u)_{z+\square_n} = (\nabla w_z)_{z+\square_n}.$$

Moreover,

$$\nabla \rho_z = \nabla u - \nabla w_z,$$

and so

$$[\nabla \rho_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \leq [\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} + [\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)}. \quad (3.22)$$

Step 2. A one-step recurrence across scales. Combining (3.20) with the triangle inequality

$$\|\mathbf{s}^{1/2} \nabla w_z\|_{\underline{L}^2(z+\square_n)} \leq \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)} + \|\mathbf{s}^{1/2} \nabla \rho_z\|_{\underline{L}^2(z+\square_n)},$$

we obtain

$$3^{-sn} [\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \leq C s^{-1/2} \lambda_{s,2}^{-1/2}(z + \square_n; \mathbf{a}) \left(\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)} + \|\mathbf{s}^{1/2} \nabla \rho_z\|_{\underline{L}^2(z+\square_n)} \right). \quad (3.23)$$

Likewise, (3.21) and (3.22) imply

$$\|\mathbf{s}^{1/2}\nabla\rho_z\|_{\underline{L}^2(z+\square_n)}^2 \leq C3^{sn}[\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)} \left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} + 3^{-sn}[\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right). \quad (3.24)$$

Substituting this into (3.23), squaring, and using Young's inequality twice, we get that, for every $\eta \in (0, 1)$,

$$\begin{aligned} & \left(3^{-sn}[\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 \\ & \leq Cs^{-1}\lambda_{s,2}^{-1}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)}^2 + \eta \left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 \\ & \quad + C\eta^{-1}s^{-2}\lambda_{s,2}^{-2}(z+\square_n; \mathbf{a}) 3^{2sn}[\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.25)$$

We next compare $[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)}$ and $[\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)}$. By the definition (1.8) of the concrete negative Besov seminorm,

$$\left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 = |(\nabla u)_{z+\square_n}|^2 + 3^{-2s} \sum_{z' \in z+3^{n-1}\mathbb{Z}^d \cap (z+\square_n)} 3^{-2s(n-1)} [\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z'+\square_{n-1})}^2. \quad (3.26)$$

This is simply the definition of the concrete negative Besov seminorm on $z+\square_n$, split into the top-scale average and the contribution of its triadic children. Since $(\nabla u)_{z+\square_n} = (\nabla w_z)_{z+\square_n}$ and the first term in the definition of the negative Besov seminorm is the square of the average on the cube itself, we have

$$|(\nabla u)_{z+\square_n}|^2 = |(\nabla w_z)_{z+\square_n}|^2 \leq \left(3^{-sn}[\nabla w_z]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2.$$

Therefore, (3.25) and (3.26) yield

$$\begin{aligned} \left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 & \leq 3^{-2s} \sum_{z' \in z+3^{n-1}\mathbb{Z}^d \cap (z+\square_n)} 3^{-2s(n-1)} [\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z'+\square_{n-1})}^2 \\ & \quad + Cs^{-1}\lambda_{s,2}^{-1}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)}^2 + \eta \left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 \\ & \quad + C\eta^{-1}s^{-2}\lambda_{s,2}^{-2}(z+\square_n; \mathbf{a}) 3^{2sn}[\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.27)$$

Choose

$$\eta := \mathbf{c}_{s/2} = 1 - 3^{-s/2}.$$

Then $\eta \geq cs$, $\eta^{-1} \leq Cs^{-1}$, and

$$\frac{3^{-2s}}{1-\eta} = \frac{3^{-2s}}{3^{-s/2}} = 3^{-3s/2}.$$

Dividing (3.27) by $1-\eta$, we arrive at

$$\begin{aligned} \left(3^{-sn}[\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z+\square_n)} \right)^2 & \leq 3^{-3s/2} \sum_{z' \in z+3^{n-1}\mathbb{Z}^d \cap (z+\square_n)} 3^{-2s(n-1)} [\nabla u]_{\dot{\underline{B}}_{2,2}^{-s}(z'+\square_{n-1})}^2 \\ & \quad + Cs^{-2}\lambda_{s,2}^{-1}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ & \quad + Cs^{-4}\lambda_{s,2}^{-2}(z+\square_n; \mathbf{a}) 3^{2sn}[\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.28)$$

Step 3. Summation over scales and iteration. For each $n \in (-\infty, 0] \cap \mathbb{Z}$, define

$$R_n := \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} 3^{-2sn} [\nabla u]_{\underline{B}_{2,2}^{-s}(z+\square_n)}^2$$

Summing (3.28) over $z \in 3^n \mathbb{Z}^d \cap \square_0$, and using that each scale- $(n-1)$ cube appears exactly once as a child of a scale- n cube, we obtain

$$\begin{aligned} R_n &\leq 3^{-3s/2} R_{n-1} + C s^{-2} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \lambda_{s,2}^{-1}(z + \square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &\quad + C s^{-4} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \lambda_{s,2}^{-2}(z + \square_n; \mathbf{a}) 3^{2sn} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.29)$$

By (2.72) and (2.73),

$$\lambda_{s,2}^{-1}(z + \square_n; \mathbf{a}) \leq \lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a}) \leq 3^{-sn} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}), \quad (3.30)$$

and therefore

$$\begin{aligned} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \lambda_{s,2}^{-1}(z + \square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 &\leq 3^{-sn} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &= 3^{-sn} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2. \end{aligned} \quad (3.31)$$

Similarly,

$$\lambda_{s,2}^{-2}(z + \square_n; \mathbf{a}) \leq 3^{-2sn} \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}). \quad (3.32)$$

Expanding the definition (1.6) of the positive Besov seminorm, and using that for each $l \leq n$ the families of scale- l subcubes of the cubes $\{z + \square_n\}_{z \in 3^n \mathbb{Z}^d \cap \square_0}$ partition the family of all scale- l subcubes of $\square_0$, we also get

$$\sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2 \leq [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2. \quad (3.33)$$

(3.32) and (3.33) therefore yield

$$\begin{aligned} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \lambda_{s,2}^{-2}(z + \square_n; \mathbf{a}) 3^{2sn} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2 &\leq \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2 \\ &\leq \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2. \end{aligned} \quad (3.34)$$

Substituting (3.31) and (3.34) into (3.29), we find

$$R_n \leq 3^{-3s/2} R_{n-1} + C s^{-2} 3^{-sn} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 + C s^{-4} \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2. \quad (3.35)$$

Now set

$$S_n := 3^{sn} R_n.$$

Then (3.35) becomes

$$S_n \leq 3^{-s/2} S_{n-1} + C s^{-2} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 + C s^{-4} \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2, \quad (3.36)$$

because $3^{sn} \leq 1$ for $n \leq 0$.

We next note that $(S_n)_{n \leq 0}$ is uniformly bounded. Indeed, by Jensen's inequality, the definition (1.8), and the same reindexing over the partition $\{z + \square_n\}_{z \in 3^n \mathbb{Z}^d \cap \square_0}$,

$$\begin{aligned} S_n &= 3^{-sn} \sum_{l=-\infty}^n 3^{2sl} \left(\sum_{y \in 3^l \mathbb{Z}^d \cap \square_0} |(\nabla u)_{y+\square_l}|^2 \right) \\ &\leq \sum_{l=-\infty}^n 3^{-sn} 3^{2sl} \|\nabla u\|_{\underline{L}^2(\square_0)}^2 \\ &\leq \sum_{l=-\infty}^n 3^{s(l-n)} \|\nabla u\|_{\underline{L}^2(\square_0)}^2 \leq \mathfrak{c}_s^{-1} \|\nabla u\|_{\underline{L}^2(\square_0)}^2. \end{aligned} \quad (3.37)$$

Fix $N \in \mathbb{N}$. Iterating (3.36) from $n = -N$ up to $n = 0$, we obtain

$$S_0 \leq 3^{-sN/2} S_{-N} + \sum_{j=0}^{N-1} 3^{-sj/2} \left(C s^{-2} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 + C s^{-4} \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2 \right).$$

By (3.37), the first term on the right tends to 0 as $N \rightarrow \infty$. Since

$$\sum_{j=0}^{\infty} 3^{-sj/2} = \mathfrak{c}_{s/2}^{-1} \leq C s^{-1},$$

we conclude that

$$[\nabla u]_{\underline{B}_{2,2}^{-s}(\square_0)}^2 = S_0 \leq C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 + C s^{-5} \lambda_{s/2,2}^{-2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2. \quad (3.38)$$

Taking square roots and using $s \in (0, 1)$, we obtain

$$[\nabla u]_{\underline{B}_{2,2}^{-s}(\square_0)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}.$$

Rescaling back to the cube $\square_m$ gives exactly (3.18). $\square$

The previous proposition immediately yields the first energy consequence for forced equations. We record it separately, since it will be used repeatedly in the next subsections.

Lemma 3.6 (Auxiliary zero-Dirichlet energy estimate). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $t \in (0, 1/2)$, every vector field $\mathbf{g} \in H^{2t}(\square_m; \mathbb{R}^d)$, and every solution $v \in H_0^1(\square_m)$ of the equation*

$$-\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

we have

$$\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_m)} \leq C t^{-3/2} \lambda_{t,2}^{-1/2}(\square_m; \mathbf{a}) 3^{2tm} [\mathbf{g}]_{\underline{B}_{2,2}^{2t}(\square_m)}. \quad (3.39)$$

Proof. By the scale invariance of the quantities in the statement, it is enough to consider the case $m = 0$. Since $v \in H_0^1(\square_0)$, the average of each weak derivative of v vanishes. Indeed, this is clear for smooth compactly supported functions, and the general case follows by density. In particular,

$$(\nabla v)_{\square_0} = 0.$$

Testing the equation with v therefore gives

$$\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}^2 = -\int_{\square_0} (\mathbf{g} - (\mathbf{g})_{\square_0}) \cdot \nabla v.$$

Since $\mathbf{g} - (\mathbf{g})_{\square_0}$ has zero average on $\square_0$, the normalized Besov norm on the right is just the seminorm. Applying the cube-wise Besov pairing (1.24) with exponent $2t$, we obtain

$$\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}^2 \leq C[\nabla v]_{\dot{B}_{2,2}^{-2t}(\square_0)}[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)}. \quad (3.40)$$

We now apply Proposition 3.5 to v with $s = 2t$. Since $2t \in (0, 1)$, this gives

$$[\nabla v]_{\dot{B}_{2,2}^{-2t}(\square_0)} \leq Ct^{-3/2}\lambda_{t,2}^{-1/2}(\square_0; \mathbf{a})\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)} + Ct^{-3}\lambda_{t,2}^{-1}(\square_0; \mathbf{a})[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)}. \quad (3.41)$$

Substituting (3.41) into (3.40), we find

$$\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}^2 \leq Ct^{-3/2}\lambda_{t,2}^{-1/2}(\square_0; \mathbf{a})\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)} + Ct^{-3}\lambda_{t,2}^{-1}(\square_0; \mathbf{a})[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)}^2.$$

Using Young's inequality on the first term on the right, we get

$$\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}^2 \leq \frac{1}{2}\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)}^2 + Ct^{-3}\lambda_{t,2}^{-1}(\square_0; \mathbf{a})[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)}^2.$$

Hence

$$\|\mathbf{s}^{1/2}\nabla v\|_{\underline{L}^2(\square_0)} \leq Ct^{-3/2}\lambda_{t,2}^{-1/2}(\square_0; \mathbf{a})[\mathbf{g}]_{B_{2,2}^{2t}(\square_0)},$$

which is exactly (3.39) in the case of $\square_0$. Scaling back yields the general result. $\square$

3.2.2 Coarse-grained Caccioppoli inequality with right-hand side

The homogeneous coarse-grained Caccioppoli inequality extends to equations with forcing by subtracting the auxiliary zero-Dirichlet corrector from Lemma 3.6. This reduces the problem to the homogeneous estimate, and all of the forcing dependence is carried by the auxiliary corrector.

Proposition 3.7 (Coarse-grained Caccioppoli inequality with right-hand side, boundary version). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $s \in (0, 1)$, every $t \in (0, 1/2)$ satisfying $s + t < 1$, every $x \in \square_m$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every vector field $\mathbf{g} \in H^{2t}(\square_m; \mathbb{R}^d)$, and every solution $u \in H^1(\square_m)$ of the equation*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = 0 & \text{on } (\partial \square_m) \cap (x + \square_{m-1}), \end{cases}$$

we have

$$\begin{aligned} & \|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2((x+\square_{m-2})\cap\square_m)}^2 \\ & \leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} s^{-\frac{2s}{1-s-t}} \Theta_{s,t}(\square_m; \mathbf{a})^{\frac{1-t}{1-s-t}} \\ & \quad \times \left(\lambda_t(\square_m; \mathbf{a}) 3^{-2m} \|u\|_{\underline{L}^2(\square_m)}^2 + \frac{t^{-8}}{1-2t} \lambda_t^{-1}(\square_m; \mathbf{a}) 3^{4tm} [\mathbf{g}]_{B_{2,2}^{2t}(\square_m)}^2 \right). \end{aligned} \quad (3.42)$$

Proof. By translation and dilation, we may reduce to the case $m = 0$. Set

$$\sigma := 1 - s - t.$$

Let $v \in H_0^1(\square_0)$ be the solution of the Dirichlet problem

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_0, \\ v = 0 & \text{on } \partial \square_0. \end{cases}$$

Define

$$w := u - v.$$

Then

$$-\nabla \cdot \mathbf{a} \nabla w = 0 \quad \text{in } \square_0,$$

and, since $v = 0$ on $\partial \square_0$ and $u = 0$ on $(\partial \square_0) \cap (x + \square_{-1})$, we have

$$w = 0 \quad \text{on } (\partial \square_0) \cap (x + \square_{-1}).$$

Applying Proposition 3.2 to w , we obtain

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \leq \left(\frac{C}{\sigma}\right)^{2+\frac{4s}{\sigma}} s^{-\frac{2s}{\sigma}} \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{s}{\sigma}} \Lambda_s(\square_0; \mathbf{a}) \|w\|_{\underline{L}^2(\square_0)}^2. \quad (3.43)$$

We next estimate the auxiliary corrector v . Lemma 3.6 and (2.74) imply

$$\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)}^2 \leq Ct^{-4} \lambda_t^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^{2t}(\square_0)}^2. \quad (3.44)$$

Since $v \in H_0^1(\square_0)$, the usual Poincaré inequality on $\square_0$, together with the uniform ellipticity of $\mathbf{a}$, gives

$$\|v\|_{\underline{L}^2(\square_0)}^2 \leq C \|\nabla v\|_{\underline{L}^2(\square_0)}^2 \leq C \|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)}^2 \leq Ct^{-4} \lambda_t^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^{2t}(\square_0)}^2. \quad (3.45)$$

Therefore,

$$\|w\|_{\underline{L}^2(\square_0)}^2 \leq 2\|u\|_{\underline{L}^2(\square_0)}^2 + 2\|v\|_{\underline{L}^2(\square_0)}^2,$$

and so (3.43) gives

$$\begin{aligned} \|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 &\leq \left(\frac{C}{\sigma}\right)^{2+\frac{4s}{\sigma}} s^{-\frac{2s}{\sigma}} \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{s}{\sigma}} \Lambda_s(\square_0; \mathbf{a}) \left(\|u\|_{\underline{L}^2(\square_0)}^2 \right. \\ &\quad \left. + Ct^{-4} \lambda_t^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^{2t}(\square_0)}^2 \right). \end{aligned} \quad (3.46)$$

Since

$$\Theta_{s,t}(\square_0; \mathbf{a})^{\frac{s}{\sigma}} \Lambda_s(\square_0; \mathbf{a}) = \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{s+\sigma}{\sigma}} \lambda_t(\square_0; \mathbf{a}) = \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{1-t}{\sigma}} \lambda_t(\square_0; \mathbf{a}),$$

we may rewrite (3.46) as

$$\begin{aligned} \|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 &\leq \left(\frac{C}{\sigma}\right)^{2+\frac{4s}{\sigma}} s^{-\frac{2s}{\sigma}} \Theta_{s,t}(\square_0; \mathbf{a})^{\frac{1-t}{\sigma}} \\ &\quad \times \left(\lambda_t(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2 + Ct^{-4} [\mathbf{g}]_{\underline{B}_{2,2}^{2t}(\square_0)}^2 \right). \end{aligned} \quad (3.47)$$

Since $t \in (0, 1/2)$ and $\lambda_t(\square_0; \mathbf{a})$ is bounded above by a constant depending only on the ambient ellipticity, we have

$$\frac{t^{-8}}{1-2t} \lambda_t^{-1}(\square_0; \mathbf{a}) \geq Ct^{-4}.$$

Since also $\sigma \leq 1$, $s \in (0, 1)$, and $\Theta_{s,t}(\square_0; \mathbf{a}) \geq 1$, the right-hand side of (3.42) in the case $m = 0$ dominates both (3.47) and (3.44). Therefore, by the triangle inequality,

$$\|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \leq 2\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 + 2\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)}^2,$$

and (3.42) follows in the case of $\square_0$. Scaling back yields the general result. $\square$

3.2.3 Weak flux estimate with right-hand side

The flux counterpart of Proposition 3.5 is most naturally written at the level $q = 2$. The local correction is now of Neumann type: on each subcube we solve away the forcing by a mean-zero variational problem whose corrected flux belongs to $L_{\text{sol},0}^2$. This is exactly what is needed to make the concrete negative Besov seminorm split cleanly across scales.

Proposition 3.8 (Weak flux estimate with right-hand side). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $s \in (0, 1)$, every vector field $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and every solution $u \in H^1(\square_m)$ of the equation*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

we have

$$3^{-sm} [\mathbf{a} \nabla u]_{\dot{B}_{2,2}^{-s}(\square_m)} \leq Cs^{-1} \Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + Cs^{-5/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})} 3^{sm} [\mathbf{g}]_{\dot{B}_{2,2}^s(\square_m)}. \quad (3.48)$$

Proof. By the scale invariance of the quantities in the statement, it is enough to consider the case $m = 0$.

Step 1. Local mean-zero Neumann correction. Fix $n \in (-\infty, 0] \cap \mathbb{Z}$ and $z \in 3^n \mathbb{Z}^d \cap \square_0$. By Lax-Milgram on the mean-zero subspace of $H^1(z + \square_n)$, there exists a unique function $\omega_z \in H^1(z + \square_n)$ satisfying $(\omega_z)_{z+\square_n} = 0$ and

$$\int_{z+\square_n} \nabla \phi \cdot \mathbf{a} \nabla \omega_z = - \int_{z+\square_n} (\mathbf{g} - (\mathbf{g})_{z+\square_n}) \cdot \nabla \phi \quad \forall \phi \in H^1(z + \square_n). \quad (3.49)$$

Set

$$v_z := u - \omega_z.$$

Since (3.49) holds in particular for every $\phi \in H_0^1(z + \square_n)$, the function v_z satisfies

$$-\nabla \cdot \mathbf{a} \nabla v_z = 0 \quad \text{in } z + \square_n,$$

and therefore $v_z \in \mathcal{A}(z + \square_n; \mathbf{a})$.

Moreover, by the definition (1.5) of the space $L_{\text{sol},0}^2(z + \square_n)$, the variational identity (3.49) is equivalent to

$$\mathbf{a} \nabla \omega_z + \mathbf{g} - (\mathbf{g})_{z+\square_n} \in L_{\text{sol},0}^2(z + \square_n).$$

Every element of $L^2_{\text{sol},0}(z + \square_n)$ has zero average, so

$$(\mathbf{a}\nabla\omega_z + \mathbf{g} - (\mathbf{g})_{z+\square_n})_{z+\square_n} = 0.$$

Since $(\mathbf{g} - (\mathbf{g})_{z+\square_n})_{z+\square_n} = 0$, we conclude that

$$(\mathbf{a}\nabla\omega_z)_{z+\square_n} = 0. \quad (3.50)$$

Step 2. Local bounds for the corrector and the corrected flux. Testing (3.49) with $\phi = \omega_z$ gives

$$\|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}^2 = -\int_{z+\square_n} (\mathbf{g} - (\mathbf{g})_{z+\square_n}) \cdot \nabla\omega_z.$$

Since $(\mathbf{g} - (\mathbf{g})_{z+\square_n})_{z+\square_n} = 0$, the normalized Besov norm on the right is just the seminorm, and therefore the cube-wise Besov pairing (1.24) yields

$$\|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}^2 \leq C[\nabla\omega_z]_{\dot{B}_{2,2}^{-s}(z+\square_n)}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)}. \quad (3.51)$$

Applying Proposition 3.5 to ω_z on the cube $z + \square_n$, we obtain

$$\begin{aligned} 3^{-sn}[\nabla\omega_z]_{\dot{B}_{2,2}^{-s}(z+\square_n)} &\leq Cs^{-3/2}\lambda_{s/2,2}^{-1/2}(z + \square_n; \mathbf{a})\|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)} \\ &\quad + Cs^{-3}\lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a})3^{sn}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)}. \end{aligned}$$

Substituting this into (3.51), we find

$$\begin{aligned} \|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}^2 &\leq Cs^{-3/2}\lambda_{s/2,2}^{-1/2}(z + \square_n; \mathbf{a})\|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}3^{sn}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)} \\ &\quad + Cs^{-3}\lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a})3^{2sn}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)}^2. \end{aligned}$$

Using Young's inequality on the first term on the right yields

$$\|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}^2 \leq Cs^{-3}\lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a})3^{2sn}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)}^2. \quad (3.52)$$

Applying the homogeneous coarse-grained flux estimate (3.2) with $q = 2$ to v_z , and then using

$$\|\mathbf{s}^{1/2}\nabla v_z\|_{\underline{L}^2(z+\square_n)} \leq \|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)} + \|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)},$$

we get

$$3^{-sn}[\mathbf{a}\nabla v_z]_{\dot{B}_{2,2}^{-s}(z+\square_n)} \leq Cs^{-1/2}\Lambda_{s,2}^{1/2}(z + \square_n; \mathbf{a})\left(\|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)} + \|\mathbf{s}^{1/2}\nabla\omega_z\|_{\underline{L}^2(z+\square_n)}\right).$$

Squaring this estimate and then invoking (3.52), we arrive at

$$\begin{aligned} \left(3^{-sn}[\mathbf{a}\nabla v_z]_{\dot{B}_{2,2}^{-s}(z+\square_n)}\right)^2 &\leq Cs^{-1}\Lambda_{s,2}(z + \square_n; \mathbf{a})\|\mathbf{s}^{1/2}\nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &\quad + Cs^{-4}\Lambda_{s,2}(z + \square_n; \mathbf{a})\lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a})3^{2sn}[\mathbf{g}]_{B_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.53)$$

Step 3. A one-step recurrence across scales. By the definition (1.8) of the concrete negative Besov seminorm,

$$\begin{aligned} \left(3^{-sn}[\mathbf{a}\nabla u]_{\dot{B}_{2,2}^{-s}(z+\square_n)}\right)^2 &= |(\mathbf{a}\nabla u)_{z+\square_n}|^2 \\ &\quad + 3^{-2s} \sum_{z' \in z+3^{n-1}\mathbb{Z}^d \cap (z+\square_n)} 3^{-2s(n-1)}[\mathbf{a}\nabla u]_{\dot{B}_{2,2}^{-s}(z'+\square_{n-1})}^2. \end{aligned} \quad (3.54)$$

On the other hand, (3.50) implies

$$(\mathbf{a}\nabla u)_{z+\square_n} = (\mathbf{a}\nabla v_z)_{z+\square_n},$$

and the first term in the concrete negative Besov seminorm is exactly the square of this average. Hence

$$|(\mathbf{a}\nabla u)_{z+\square_n}|^2 \leq \left(3^{-sn} [\mathbf{a}\nabla v_z]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right)^2.$$

Combining this with (3.53), we obtain

$$\begin{aligned} \left(3^{-sn} [\mathbf{a}\nabla u]_{\underline{B}_{2,2}^{-s}(z+\square_n)} \right)^2 &\leq 3^{-2s} \sum_{z' \in z+3^{n-1}\mathbb{Z}^d \cap (z+\square_n)} 3^{-2s(n-1)} [\mathbf{a}\nabla u]_{\underline{B}_{2,2}^{-s}(z'+\square_{n-1})}^2 \\ &\quad + Cs^{-1} \Lambda_{s,2}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &\quad + Cs^{-4} \Lambda_{s,2}(z+\square_n; \mathbf{a}) \lambda_{s/2,2}^{-1}(z+\square_n; \mathbf{a}) 3^{2sn} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.55)$$

Step 4. Summation over scales and iteration. For each $n \in (-\infty, 0] \cap \mathbb{Z}$, define

$$D_n := \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} 3^{-2sn} [\mathbf{a}\nabla u]_{\underline{B}_{2,2}^{-s}(z+\square_n)}^2$$

Summing (3.55) over $z \in 3^n \mathbb{Z}^d \cap \square_0$, and using that every scale- $(n-1)$ cube appears exactly once as a child of a scale- n cube, yields

$$\begin{aligned} D_n &\leq 3^{-2s} D_{n-1} \\ &\quad + Cs^{-1} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \Lambda_{s,2}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &\quad + Cs^{-4} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \Lambda_{s,2}(z+\square_n; \mathbf{a}) \lambda_{s/2,2}^{-1}(z+\square_n; \mathbf{a}) 3^{2sn} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2. \end{aligned} \quad (3.56)$$

By (2.72) and (2.73),

$$\Lambda_{s,2}(z+\square_n; \mathbf{a}) \leq \Lambda_{s/2,2}(z+\square_n; \mathbf{a}) \leq 3^{-sn} \Lambda_{s/2,2}(\square_0; \mathbf{a}),$$

and likewise

$$\lambda_{s/2,2}^{-1}(z+\square_n; \mathbf{a}) \leq 3^{-sn} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}).$$

Therefore,

$$\begin{aligned} \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \Lambda_{s,2}(z+\square_n; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 &\leq 3^{-sn} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)}^2 \\ &= 3^{-sn} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2. \end{aligned} \quad (3.57)$$

Also, exactly as in the proof of Proposition 3.5,

$$\sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} [\mathbf{g}]_{\underline{B}_{2,2}^s(z+\square_n)}^2 \leq [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2. \quad (3.58)$$

(3.58) therefore gives

$$\begin{aligned}
& \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \Lambda_{s,2}(z + \square_n; \mathbf{a}) \lambda_{s/2,2}^{-1}(z + \square_n; \mathbf{a}) 3^{2sn} [\mathbf{g}]_{\underline{B}_{2,2}^s(z + \square_n)}^2 \\
& \leq \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} [\mathbf{g}]_{\underline{B}_{2,2}^s(z + \square_n)}^2 \\
& \leq \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) \\
& \quad \times [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2.
\end{aligned} \tag{3.59}$$

Substituting (3.57) and (3.59) into (3.56), we find

$$\begin{aligned}
D_n & \leq 3^{-2s} D_{n-1} + C s^{-1} 3^{-sn} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 \\
& \quad + C s^{-4} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2.
\end{aligned} \tag{3.60}$$

Now set

$$T_n := 3^{sn} D_n.$$

Then (3.60) becomes

$$\begin{aligned}
T_n & \leq 3^{-s} T_{n-1} + C s^{-1} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 \\
& \quad + C s^{-4} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2,
\end{aligned} \tag{3.61}$$

because $3^{sn} \leq 1$ for $n \leq 0$.

We next note that $(T_n)_{n \leq 0}$ is uniformly bounded. Indeed, by Jensen's inequality and the definition (1.8),

$$\begin{aligned}
T_n & = 3^{-sn} \sum_{l=-\infty}^n 3^{2sl} \sum_{y \in 3^l \mathbb{Z}^d \cap \square_0} |(\mathbf{a} \nabla u)_{y + \square_l}|^2 \\
& \leq \sum_{l=-\infty}^n 3^{-sn} 3^{2sl} \|\mathbf{a} \nabla u\|_{\underline{L}^2(\square_0)}^2 \\
& \leq \sum_{l=-\infty}^n 3^{s(l-n)} \|\mathbf{a} \nabla u\|_{\underline{L}^2(\square_0)}^2 \leq \mathbf{c}_s^{-1} \|\mathbf{a} \nabla u\|_{\underline{L}^2(\square_0)}^2.
\end{aligned} \tag{3.62}$$

Fix $N \in \mathbb{N}$. Iterating (3.61) from $n = -N$ up to $n = 0$, we obtain

$$\begin{aligned}
T_0 & \leq 3^{-sN} T_{-N} + \sum_{j=0}^{N-1} 3^{-sj} \left(C s^{-1} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 \right. \\
& \quad \left. + C s^{-4} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2 \right).
\end{aligned}$$

By (3.62), the first term on the right tends to 0 as $N \rightarrow \infty$. Since

$$\sum_{j=0}^{\infty} 3^{-sj} = \mathbf{c}_s^{-1} \leq C s^{-1},$$

we conclude that

$$\begin{aligned}
[\mathbf{a} \nabla u]_{\underline{B}_{2,2}^{-s}(\square_0)}^2 & = T_0 \leq C s^{-2} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}^2 \\
& \quad + C s^{-5} \Lambda_{s/2,2}(\square_0; \mathbf{a}) \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}^2.
\end{aligned} \tag{3.63}$$

Taking square roots and using $s \in (0, 1)$, we obtain

$$[\mathbf{a} \nabla u]_{\dot{B}_{2,2}^{-s}(\square_0)} \leq C s^{-1} \Lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)} + C s^{-5/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})} [\mathbf{g}]_{B_{2,2}^s(\square_0)}.$$

This is exactly (3.48) in the case of $\square_0$. Scaling back yields the general result. $\square$

3.2.4 Coarse-grained flux-response estimate with right-hand side

For the inhomogeneous operator estimate, the target is the concrete negative Besov seminorm from Section 3.1.3. This is the norm which localizes across triadic subcubes and therefore feeds directly into the duality lemma and the general coarse-graining theorem.

Lemma 3.9 (Coarse-grained flux-response estimate with right-hand side). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $s \in (0, 1)$, every scalar constant-coefficient matrix $\mathbf{a}_0 = \sigma_0 \mathbf{I}_d$ with $\sigma_0 > 0$, every vector field $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and every solution $u \in H^1(\square_m)$ of the equation*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

we have

$$\begin{aligned} 3^{-sm} [(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\dot{B}_{2,2}^{-s}(\square_m)} &\leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{a}_0) \\ &\quad + C \left(s^{-5/2} |\mathbf{a}_0|^{1/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{a}_0) \right. \\ &\quad \left. + s^{-5/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})} + s^{-3} |\mathbf{a}_0| \lambda_{s/2,2}^{-1}(\square_m; \mathbf{a}) \right) \\ &\quad \times 3^{sm} [\mathbf{g}]_{B_{2,2}^s(\square_m)}. \end{aligned} \quad (3.64)$$

Proof. By the scale invariance of all the quantities in the statement, it is enough to consider the case $m = 0$. Let $v \in H_0^1(\square_0)$ be the solution of the Dirichlet problem

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_0, \\ v = 0 & \text{on } \partial \square_0. \end{cases}$$

Define

$$w := u - v.$$

Then

$$-\nabla \cdot \mathbf{a} \nabla w = 0 \quad \text{in } \square_0,$$

so $w \in \mathcal{A}(\square_0; \mathbf{a})$.

By the triangle inequality,

$$[(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\dot{B}_{2,2}^{-s}(\square_0)} \leq [(\mathbf{a} - \mathbf{a}_0) \nabla w]_{\dot{B}_{2,2}^{-s}(\square_0)} + [\mathbf{a} \nabla v]_{\dot{B}_{2,2}^{-s}(\square_0)} + |\mathbf{a}_0| [\nabla v]_{\dot{B}_{2,2}^{-s}(\square_0)}. \quad (3.65)$$

By the elementary inequality $[\cdot]_{\dot{B}_{2,2}^{-s}(\square_0)} \leq [\cdot]_{\dot{B}_{2,1}^{-s}(\square_0)}$ and Lemma 3.4 applied to w , we get

$$\begin{aligned} [(\mathbf{a} - \mathbf{a}_0) \nabla w]_{\dot{B}_{2,2}^{-s}(\square_0)} &\leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)} \mathcal{E}_{s,\infty,1}(\square_0; \mathbf{a}, \mathbf{a}_0) \\ &\leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)} \mathcal{E}_{s,\infty,1}(\square_0; \mathbf{a}, \mathbf{a}_0) \\ &\quad + C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)} \mathcal{E}_{s,\infty,1}(\square_0; \mathbf{a}, \mathbf{a}_0). \end{aligned} \quad (3.66)$$

Using Lemma 3.6 with $t = s/2$, we find

$$\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}. \quad (3.67)$$

Substituting this into (3.66), we obtain

$$\begin{aligned} [(\mathbf{a} - \mathbf{a}_0) \nabla w]_{\dot{B}_{2,2}^{-s}(\square_0)} &\leq C s^{-1} |\mathbf{a}_0|^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)} \mathcal{E}_{s,\infty,1}(\square_0; \mathbf{a}, \mathbf{a}_0) \\ &\quad + C s^{-5/2} |\mathbf{a}_0|^{1/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) \mathcal{E}_{s,\infty,1}(\square_0; \mathbf{a}, \mathbf{a}_0) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}. \end{aligned} \quad (3.68)$$

Applying Proposition 3.8 to v and then using (3.67), we get

$$\begin{aligned} [\mathbf{a} \nabla v]_{\dot{B}_{2,2}^{-s}(\square_0)} &\leq C s^{-1} \Lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)} \\ &\quad + C s^{-5/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})} [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)} \\ &\leq C s^{-5/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_0; \mathbf{a})} [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}. \end{aligned} \quad (3.69)$$

Likewise, Proposition 3.5 applied to v , together with (3.67), gives

$$\begin{aligned} [\nabla v]_{\dot{B}_{2,2}^{-s}(\square_0)} &\leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)} \\ &\leq C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_0)}. \end{aligned} \quad (3.70)$$

Substituting (3.68), (3.69), and (3.70) into (3.65), we obtain (3.64) in the case of $\square_0$. Scaling back yields the general result. $\square$

3.2.5 Energy consequences for Dirichlet and Neumann problems

The multiscale work in Section 3.2 is now complete. What remains is to package the previous estimates into a few energy black boxes that can be cited later without reopening the multiscale recurrence arguments. The first of these is the general Dirichlet estimate, which combines the zero-boundary forcing estimate from Section 3.2.1 with the homogeneous boundary-data estimate coming from the flux side of Section 3.1.

Proposition 3.10 (Dirichlet energy estimate with right-hand side and boundary data). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $s \in (0, 1)$, every vector field $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, every function $h \in H^{1+s}(\square_m)$, and every solution $v \in H^1(\square_m)$ of the Dirichlet problem*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ v - h \in H_0^1(\square_m), \end{cases}$$

we have

$$\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_m)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_m)} + C s^{-1/2} \Lambda_{s,2}^{1/2}(\square_m; \mathbf{a}) 3^{sm} \|\nabla h\|_{\underline{B}_{2,2}^s(\square_m)}. \quad (3.71)$$

Proof. By the scale invariance of all the quantities in the statement, it is enough to consider the case $m = 0$.

Let $v_0 \in H_0^1(\square_0)$ be the solution of

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla v_0 = \nabla \cdot \mathbf{g} & \text{in } \square_0, \\ v_0 = 0 & \text{on } \partial \square_0, \end{cases}$$

and define

$$\tilde{h} := v - v_0.$$

Then $\tilde{h} - h \in H_0^1(\square_0)$ and

$$-\nabla \cdot \mathbf{a} \nabla \tilde{h} = 0 \quad \text{in } \square_0,$$

so $\tilde{h} \in \mathcal{A}(\square_0; \mathbf{a})$. By the triangle inequality,

$$\|\mathbf{s}^{1/2} \nabla v\|_{\underline{L}^2(\square_0)} \leq \|\mathbf{s}^{1/2} \nabla v_0\|_{\underline{L}^2(\square_0)} + \|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)}. \quad (3.72)$$

The first term is controlled by Lemma 3.6 with $t = s/2$:

$$\|\mathbf{s}^{1/2} \nabla v_0\|_{\underline{L}^2(\square_0)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) [\mathbf{g}]_{B_{2,2}^s(\square_0)}. \quad (3.73)$$

We next estimate $\tilde{h}$. Testing the homogeneous equation for $\tilde{h}$ with $\tilde{h} - h \in H_0^1(\square_0)$, we get

$$\int_{\square_0} \nabla(\tilde{h} - h) \cdot \mathbf{a} \nabla \tilde{h} = 0.$$

Since $\mathbf{a} = \mathbf{s} + \mathbf{k}$ with $\mathbf{k}$ skew-symmetric, this gives

$$\|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)}^2 = \int_{\square_0} \nabla h \cdot \mathbf{a} \nabla \tilde{h}.$$

Applying the cube-wise Besov pairing (1.24) with $f = \mathbf{a} \nabla \tilde{h}$, $g = \nabla h$, $p = q = 2$, we obtain

$$\|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)}^2 \leq C [\mathbf{a} \nabla \tilde{h}]_{\dot{B}_{2,2}^{-s}(\square_0)} \|\nabla h\|_{B_{2,2}^s(\square_0)}. \quad (3.74)$$

Since $\tilde{h} \in \mathcal{A}(\square_0; \mathbf{a})$, the homogeneous coarse-grained flux estimate (3.2) with $q = 2$ yields

$$[\mathbf{a} \nabla \tilde{h}]_{\dot{B}_{2,2}^{-s}(\square_0)} \leq c_{2s}^{-1/2} \Lambda_{s,2}^{1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)} \leq C s^{-1/2} \Lambda_{s,2}^{1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)}.$$

Substituting this into (3.74) we find that either

$$\|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)} = 0,$$

in which case there is nothing to prove, or else, after dividing by $\|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)}$,

$$\|\mathbf{s}^{1/2} \nabla \tilde{h}\|_{\underline{L}^2(\square_0)} \leq C s^{-1/2} \Lambda_{s,2}^{1/2}(\square_0; \mathbf{a}) \|\nabla h\|_{B_{2,2}^s(\square_0)}. \quad (3.75)$$

Combining (3.72), (3.73), and (3.75), we obtain (3.71) in the case of $\square_0$. Scaling back yields the general result. $\square$

The Neumann analogue is even shorter: once the boundary condition allows us to test the equation with the solution itself, the weak gradient estimate with right-hand side immediately upgrades to an energy bound.

Proposition 3.11 (Mean-zero Neumann energy estimate with right-hand side). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every $s \in (0, 1)$, every vector field $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and every mean-zero function $w \in H^1(\square_m)$ satisfying the variational Neumann problem*

$$\begin{cases} \int_{\square_m} \nabla \phi \cdot \mathbf{a} \nabla w = - \int_{\square_m} (\mathbf{g} - (\mathbf{g})_{\square_m}) \cdot \nabla \phi & \forall \phi \in H^1(\square_m), \\ (w)_{\square_m} = 0, \end{cases}$$

we have

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_m)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{B}_{2,2}^s(\square_m)}. \quad (3.76)$$

Proof. By the scale invariance of the quantities in the statement, it is enough to consider the case $m = 0$. The weak formulation of the Neumann problem reads

$$\int_{\square_0} \nabla \phi \cdot \mathbf{a} \nabla w = - \int_{\square_0} (\mathbf{g} - (\mathbf{g})_{\square_0}) \cdot \nabla \phi \quad \forall \phi \in H^1(\square_0).$$

Taking $\phi = w$, we obtain

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)}^2 = - \int_{\square_0} (\mathbf{g} - (\mathbf{g})_{\square_0}) \cdot \nabla w.$$

Since $(\mathbf{g} - (\mathbf{g})_{\square_0})_{\square_0} = 0$, the normalized Besov norm on the right is just the seminorm. Applying the cube-wise Besov pairing (1.24), we find

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)}^2 \leq C [\nabla w]_{\dot{B}_{2,2}^{-s}(\square_0)} [\mathbf{g}]_{B_{2,2}^s(\square_0)}. \quad (3.77)$$

Proposition 3.5 gives

$$[\nabla w]_{\dot{B}_{2,2}^{-s}(\square_0)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{B_{2,2}^s(\square_0)}.$$

Substituting this into (3.77), we obtain

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)}^2 \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)} [\mathbf{g}]_{B_{2,2}^s(\square_0)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{B_{2,2}^s(\square_0)}^2.$$

Using Young's inequality to absorb the mixed term yields

$$\|\mathbf{s}^{1/2} \nabla w\|_{\underline{L}^2(\square_0)}^2 \leq C s^{-3} \lambda_{s/2,2}^{-1}(\square_0; \mathbf{a}) [\mathbf{g}]_{B_{2,2}^s(\square_0)}^2.$$

Taking square roots gives (3.76) in the case of $\square_0$, and scaling back yields the general result. $\square$

3.3 Deterministic homogenization black boxes

Sections 3.1 and 3.2 gave us local control of the coarse-grained flux defect $(\mathbf{a} - \mathbf{a}_0) \nabla u$. The task in this section is to turn that local information into direct comparison estimates between the heterogeneous solution and the corresponding constant-coefficient one. The bridge is a deterministic duality argument: once we know how to control the flux defect on mesoscopic subcubes, the difference $u - v$ and the flux error $\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v$ follow by testing against solutions of a constant-coefficient adjoint problem.

3.3.1 A duality lemma

We begin with the purely deterministic step which converts local control of the flux defect into global control of the solution and flux differences. Since the local coarse-grained operator estimate from Section 3.2.3 is presently written in the genuine dual negative Besov norm, we state the lemma in that same language. Throughout this subsection, when a Besov norm is applied to an $\mathbb{R}^d$ -valued field, we mean the natural componentwise extension with the Euclidean structure on the target.

Lemma 3.12 (Duality estimate from flux defect to solution comparison). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $n \in \mathbb{Z}$ with $n \leq m$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every scalar constant-coefficient matrix $\mathbf{a}_0 = \sigma_0 \mathbf{I}_d$ with $\sigma_0 > 0$, every pair of exponents s, t satisfying $0 < t < 1/2s < 1/2$, and every pair $u, v \in H^1(\square_m)$ satisfying*

$$\begin{cases} \nabla \cdot (\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v) = 0 & \text{in } \square_m, \\ u - v \in H_0^1(\square_m), \end{cases}$$

we have

$$\begin{aligned} & 3^{-sm} [\mathbf{a}_0(\nabla u - \nabla v)]_{\underline{B}_{2,2}^{-s}(\square_m)} + 3^{-sm} [\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v]_{\underline{B}_{2,2}^{-s}(\square_m)} \\ & \leq C s^{-1} t^{-2} (1/2 - t)^{-1} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(3^{-tn} [(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\underline{B}_{2,2}^{-t}(z + \square_n)} \right)^2 \right)^{1/2}. \end{aligned} \quad (3.78)$$

Proof. By the scale invariance of all the quantities in the statement, it is enough to consider the case $m = 0$. Then

$$\nabla \cdot (\mathbf{a}_0(\nabla u - \nabla v) + (\mathbf{a} - \mathbf{a}_0) \nabla u) = 0 \quad \text{in } \square_0, \quad u - v \in H_0^1(\square_0).$$

The proof is the usual duality argument, but carried out at the lower exponent t . Solving the scalar constant-coefficient Dirichlet adjoint problem and applying Lemma 1.6, the standard positive-to-overlapping localization step gives

$$\|\nabla w\|_{\underline{B}_{2,2,\text{overlap}}^t(\square_0)} \leq C t^{-1} (1/2 - t)^{-1} \|h\|_{\underline{B}_{2,2}^t(\square_0)}.$$

Here the factor $(1/2 - t)^{-1}$ is the boundary-overlap summation loss, and it is finite because $t < 1/2$. Testing the adjoint equation against $u - v$, splitting over the cubes $z + \square_n$, and using Cauchy's inequality yields

$$[\mathbf{a}_0(\nabla u - \nabla v)]_{\underline{B}_{2,2,\text{dual}}^{-t}(\square_0)} \leq C t^{-1} (1/2 - t)^{-1} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \left(3^{-tn} [(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\underline{B}_{2,2}^{-t}(z + \square_n)} \right)^2 \right)^{1/2}.$$

The same argument applied to $\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v = (\mathbf{a} - \mathbf{a}_0) \nabla u + \mathbf{a}_0(\nabla u - \nabla v)$ gives the identical bound for the flux-comparison term. The embedding from the genuine dual negative norm at exponent t to the concrete negative Besov seminorm at exponent s costs $C s^{-1} t^{-1}$ when $0 < t < 1/2s$. Combining the last two estimates gives the claim on $\square_0$, and scaling back to $\square_m$ yields (3.78). $\square$

3.3.2 The general coarse-graining theorem in L^2

We can now combine the local operator estimate from Section 3.2.3 with the duality lemma to obtain the first genuine homogenization black box of the chapter. The result compares the heterogeneous solution with the corresponding constant-coefficient one in a global negative Besov norm, with an error measured entirely by the state variables from Chapter 2 and the energy of u .

Proposition 3.13 (General coarse-graining estimate in L^2). *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{Z}$, every $n \in \mathbb{Z}$ with $n \leq m$, every coefficient field $\mathbf{a} \in \Omega(\square_m)$, every pair of exponents s, t, t_2 satisfying $0 < t < 1/2s < 1/2$ and $t < t_2$, every scalar constant-coefficient matrix $\mathbf{a}_0 = \sigma_0 \mathbf{I}_d$ with $\sigma_0 > 0$, every vector field $\mathbf{g} \in H^{t_2}(\square_m; \mathbb{R}^d)$, and every pair $u, v \in H^1(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ -\nabla \cdot \mathbf{a}_0 \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u - v \in H_0^1(\square_m), \end{cases}$$

we have

$$\begin{aligned} & 3^{-sm} [\mathbf{a}_0(\nabla u - \nabla v)]_{\dot{B}_{2,2}^{-s}(\square_m)} + 3^{-sm} [\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v]_{\dot{B}_{2,2}^{-s}(\square_m)} \\ & \leq C s^{-1} t^{-2} (1/2 - t)^{-1} \left[t^{-1} |\mathbf{a}_0|^{1/2} \mathcal{E}_{t,\infty,1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} \right. \\ & \quad + \left(t^{-5/2} |\mathbf{a}_0|^{1/2} 3^{t/2(m-n)} \lambda_{t/2,2}^{-1/2}(\square_m; \mathbf{a}) \mathcal{E}_{t,\infty,1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \right. \\ & \quad \left. \left. + t^{-5/2} 3^{t(m-n)} \frac{\Lambda_{t/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{t/2,2}^{1/2}(\square_m; \mathbf{a})} + t^{-3} 3^{t(m-n)} |\mathbf{a}_0| \lambda_{t/2,2}^{-1}(\square_m; \mathbf{a}) \right) 3^{-t_2(m-n)} 3^{t_2 m} [\mathbf{g}]_{\underline{B}_{2,2}^{t_2}(\square_m)} \right]. \end{aligned} \quad (3.79)$$

Proof. By the scale invariance of all the quantities in the statement, it is enough to consider the case $m = 0$. We partition $\square_0$ into the triadic cubes $z + \square_n$, where $z \in 3^n \mathbb{Z}^d \cap \square_0$.

Since u solves $-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g}$ in $\square_0$, it also solves the same equation in each subcube $z + \square_n$. Applying Lemma 3.9 on each such cube at exponent t , then using the definitions of the parent-scale quantities, (2.73), Minkowski's inequality, the energy partition, and the positive Besov localization of $\mathbf{g}$ at exponent t_2 , gives

$$\begin{aligned} & \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_0} \left(3^{-tn} [(\mathbf{a} - \mathbf{a}_0) \nabla u]_{\dot{B}_{2,2}^{-t}(z + \square_n)} \right)^2 \right)^{1/2} \\ & \leq C \left[t^{-1} |\mathbf{a}_0|^{1/2} \mathcal{E}_{t,\infty,1}(\square_0, n; \mathbf{a}, \mathbf{a}_0) \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)} \right. \\ & \quad + \left(t^{-5/2} |\mathbf{a}_0|^{1/2} 3^{-t/n} \lambda_{t/2,2}^{-1/2}(\square_0; \mathbf{a}) \mathcal{E}_{t,\infty,1}(\square_0, n; \mathbf{a}, \mathbf{a}_0) \right. \\ & \quad \left. \left. + t^{-5/2} 3^{-tn} \frac{\Lambda_{t/2,2}^{1/2}(\square_0; \mathbf{a})}{\lambda_{t/2,2}^{1/2}(\square_0; \mathbf{a})} + t^{-3} 3^{-tn} |\mathbf{a}_0| \lambda_{t/2,2}^{-1}(\square_0; \mathbf{a}) \right) 3^{t_2 n} [\mathbf{g}]_{\underline{B}_{2,2}^{t_2}(\square_0)} \right]. \end{aligned}$$

Applying Lemma 3.12 to this bound gives (3.79) in the case of $\square_0$. Scaling back yields the general estimate. $\square$

The stronger L^2 harmonic-approximation statements used in the excess-decay argument require two additional ingredients that are not yet present here: first, quantitative smallness of $\mathcal{E}$ on a large family of scales, and second, an efficient way of converting the negative-order comparison estimate above into an excess bound. In the homogeneous case, Proposition 3.13 applies with $\mathbf{g} = 0$ and $n = m$, and the same negative-order estimate may of course be applied on any mesoscopic subcube to the corresponding scalar constant-coefficient harmonic replacement. We defer that upgrade to the large-scale regularity chapter, where these estimates will be paired with the renormalization input.

3.4 Stability and sensitivity under perturbations

This section will be written later. The deterministic homogenization black boxes proved above depend only on the coarse-grained estimates already established in Sections [3.1](#) and [3.2](#), so we may cleanly defer the perturbative theory for now.

Chapter 4

Stationary random fields and concentration

This chapter collects the probabilistic formalism used in the next chapter and the concentration inequalities that make the stochastic part of the theory manageable. The emphasis is on stationary random coefficient fields, local observables attached to cubes, and concentration estimates for sums of independent random variables.

The chapter begins by fixing the stochastic setting and showing that the main coarse-grained quantities from Chapters 2 and 3 are local random variables. Under unit-range dependence, these observables become independent after a deterministic coloring of the cube partition. The middle section then gathers a small concentration toolkit, formulated in terms of the weak Orlicz notation $\mathcal{O}_\Psi$ and Rosenthal's inequality, and the final section applies it to partition averages and annealed quantities.

The next chapter will combine these probabilistic ingredients with the deterministic coarse-graining theory of Chapters 2 and 3 in order to prove the quantitative homogenization-scale theorem under unit-range dependence and isotropy.

4.1 Probabilistic setup, local observables, and independence

4.1.1 Stationary random coefficient fields

Throughout this chapter, we work on the measurable space $(\Omega, \mathcal{F})$ of locally uniformly elliptic coefficient fields introduced in Chapter 2. For each Borel set $U \subseteq \mathbb{R}^d$, we denote by $\mathcal{F}(U)$ the sigma-algebra generated by the family of random variables

$$\mathbf{a} \mapsto \int_{\mathbb{R}^d} e' \cdot \mathbf{a}(x) e \varphi(x) dx, \quad e, e' \in \mathbb{R}^d, \quad \varphi \in C_c^\infty(U).$$

We also set $\mathcal{F} := \mathcal{F}(\mathbb{R}^d)$.

For the coefficient-field observables, we assume throughout the baseline law-facing hypotheses needed to treat the deterministic quantities below as random variables: $\mathbb{P}$ is a probability measure carried by locally uniformly elliptic fields, local-test events are $\mathbb{P}$ -null measurable, and the cube observables have the corresponding a.e. measurable local representatives when expectations are taken. These background support and measurability hypotheses are separate from the structural assumptions below; the latter are invoked explicitly in the statements that use them.

For each $z \in \mathbb{Z}^d$, we let $\tau_z : \Omega \rightarrow \Omega$ denote the translation action

$$(\tau_z \mathbf{a})(x) := \mathbf{a}(x + z), \quad x \in \mathbb{R}^d.$$

The first structural assumption is that the law is stationary with respect to this action. We invoke it explicitly in the statements which use it.

(P1) *Stationarity with respect to $\mathbb{Z}^d$ -translations.* The probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ satisfies

$$\mathbb{P} \circ \tau_z = \mathbb{P}, \quad \forall z \in \mathbb{Z}^d.$$

The main random quantities introduced in Chapters 2 and 3, such as $J(z + \square_m, p, q)$, $\lambda_{s,q}(z + \square_m)$, $\Lambda_{s,q}(z + \square_m)$, and $\mathcal{E}_{s,p,q}(z + \square_m; \mathbf{a}, \mathbf{a}_0)$, depend only on the restriction of the field to the corresponding cube. In particular, these become local random variables on $(\Omega, \mathcal{F})$, and stationarity implies that their laws do not depend on the location of the cube under integer translations. When a scale- n descendant cube is compared with the origin cube using this stationarity, we assume $0 \leq n$ so that the relevant triadic shifts are integer vectors.

4.1.2 Unit-range dependence, isotropy, and scalarization

The second structural assumption is a quantitative ergodicity hypothesis in its simplest form.

(P2) *Unit-range dependence.* Whenever $U, V \subseteq \mathbb{R}^d$ are Borel sets satisfying $\text{dist}(U, V) \geq 1$, the sigma-algebras $\mathcal{F}(U)$ and $\mathcal{F}(V)$ are independent.

The third assumption is an isotropy condition in law, together with invariance under passage to the adjoint field. For each signed permutation matrix $R \in \mathbb{R}^{d \times d}$, we let $\rho_R : \Omega \rightarrow \Omega$ denote the action

$$(\rho_R \mathbf{a})(x) := R^t \mathbf{a}(Rx), \quad x \in \mathbb{R}^d.$$

We also let $\iota : \Omega \rightarrow \Omega$ denote the adjoint involution

$$(\iota \mathbf{a})(x) := \mathbf{a}(x)^t, \quad x \in \mathbb{R}^d.$$

(P3) *Isotropy and adjoint symmetry in law.* The probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ satisfies

$$\mathbb{P} \circ \rho_R = \mathbb{P}, \quad \forall R \in \mathbb{R}^{d \times d} \text{ signed permutation matrix},$$

and

$$\mathbb{P} \circ \iota = \mathbb{P}.$$

The point of this assumption is that, although the coarse-grained matrices are matrix-valued random variables, the corresponding annealed objects reduce to scalar quantities on cubes.

Lemma 4.1 (Scalarization of the annealed coarse-grained matrices). *Assume (P3). At each scale considered below, assume that the displayed expectations are finite. For such $n \in \mathbb{Z}$, define the deterministic matrices*

$$\begin{cases} \bar{\mathbf{s}}_{*,n} := \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a})]^{-1}, \\ \bar{\mathbf{k}}_n := \bar{\mathbf{s}}_{*,n} \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})], \\ \bar{\mathbf{b}}_n := \mathbb{E}[\mathbf{b}(\square_n; \mathbf{a})], \\ \bar{\mathbf{s}}_n := \bar{\mathbf{b}}_n - \bar{\mathbf{k}}_n^t \bar{\mathbf{s}}_{*,n}^{-1} \bar{\mathbf{k}}_n. \end{cases} \quad (4.1)$$

These quantities are defined so that the corresponding annealed block matrix is simply the expectation of the doubled coarse-grained matrix, namely

$$\overline{\mathbf{A}}_n := \mathbb{E}[\mathbf{A}(\square_n; \mathbf{a})] = \begin{pmatrix} \overline{\mathbf{s}}_n + \overline{\mathbf{k}}_n^t \overline{\mathbf{s}}_{*,n}^{-1} \overline{\mathbf{k}}_n & -\overline{\mathbf{k}}_n^t \overline{\mathbf{s}}_{*,n}^{-1} \\ -\overline{\mathbf{s}}_{*,n}^{-1} \overline{\mathbf{k}}_n & \overline{\mathbf{s}}_{*,n}^{-1} \end{pmatrix}. \quad (4.2)$$

We also define the corresponding annealed starred block matrix by

$$\overline{\mathbf{A}}_{*,n}^{-1} := \mathbb{E}[\mathbf{A}_*^{-1}(\square_n; \mathbf{a})] = \mathbf{R} \overline{\mathbf{A}}_n \mathbf{R}. \quad (4.3)$$

Then there exist scalars $\overline{\sigma}_n, \overline{\sigma}_{*,n} \in (0, \infty)$ such that

$$\overline{\mathbf{s}}_n = \overline{\sigma}_n \mathbf{I}_d \quad \text{and} \quad \overline{\mathbf{s}}_{*,n} = \overline{\sigma}_{*,n} \mathbf{I}_d. \quad (4.4)$$

Moreover,

$$\overline{\mathbf{k}}_n = 0. \quad (4.5)$$

Proof. Fix $n \in \mathbb{Z}$ and let R be a signed permutation matrix. Since $R\square_n = \square_n$, a change of variables in the definition of the response functional gives, for every $p, q \in \mathbb{R}^d$,

$$J(\square_n, p, q; R^t \mathbf{a}(R \cdot) R) = J(\square_n, Rp, Rq; \mathbf{a}).$$

By the defining formulas of the coarse-grained matrices, it follows that

$$\mathbf{s}_*(\square_n; R^t \mathbf{a}(R \cdot) R) = R^t \mathbf{s}_*(\square_n; \mathbf{a}) R,$$

$$\mathbf{k}(\square_n; R^t \mathbf{a}(R \cdot) R) = R^t \mathbf{k}(\square_n; \mathbf{a}) R,$$

and

$$\mathbf{b}(\square_n; R^t \mathbf{a}(R \cdot) R) = R^t \mathbf{b}(\square_n; \mathbf{a}) R.$$

Hence

$$\mathbf{s}_*^{-1}(\square_n; R^t \mathbf{a}(R \cdot) R) = R^t \mathbf{s}_*^{-1}(\square_n; \mathbf{a}) R,$$

and therefore, using the invariance in law,

$$\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a})] = R^t \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a})] R,$$

$$\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})] = R^t \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})] R,$$

and

$$\mathbb{E}[\mathbf{b}(\square_n; \mathbf{a})] = R^t \mathbb{E}[\mathbf{b}(\square_n; \mathbf{a})] R$$

for every signed permutation matrix R .

Any matrix commuting with all diagonal sign matrices must be diagonal, since the off-diagonal entries change sign under conjugation by a sign flip. If a diagonal matrix also commutes with all coordinate permutations, then all of its diagonal entries are equal, and it is therefore a scalar multiple of the identity. We deduce that $\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a})]$ and $\mathbb{E}[\mathbf{b}(\square_n; \mathbf{a})]$ are scalar matrices, and therefore so are $\overline{\mathbf{s}}_{*,n}$ and $\overline{\mathbf{b}}_n$.

To prove (4.5), we use the adjoint identity (2.55). Since $\mathbf{a}$ and $\mathbf{a}^t$ have the same law, and since

$$\mathbf{s}_*(\square_n; \mathbf{a}^t) = \mathbf{s}_*(\square_n; \mathbf{a}) \quad \text{and} \quad \mathbf{k}(\square_n; \mathbf{a}^t) = -\mathbf{k}(\square_n; \mathbf{a}),$$

we obtain

$$\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})] = -\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})].$$

Hence $\mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})] = 0$, and therefore $\overline{\mathbf{k}}_n = 0$. Since $\overline{\mathbf{s}}_n = \overline{\mathbf{b}}_n - \overline{\mathbf{k}}_n^t \overline{\mathbf{s}}_{*,n}^{-1} \overline{\mathbf{k}}_n$, it follows that $\overline{\mathbf{s}}_n = \overline{\mathbf{b}}_n$, so $\overline{\mathbf{s}}_n$ is also a scalar matrix. This proves (4.4). $\square$

4.1.3 Cube observables as local random variables

We next record that the deterministic quantities introduced in Chapters 2 and 3 are honest local observables on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The cleanest starting point is the block variational quantity $\mu(U, P; \mathbf{a})$, because its admissible class does not depend on $\mathbf{a}$.

Lemma 4.2 (Locality and translation covariance of the block observables). *Let $U \subseteq \mathbb{R}^d$ be a bounded Lipschitz domain and $P, Q \in \mathbb{R}^{2d}$.*

(i) *The observables*

$$\mathbf{a} \mapsto \mu(U, P; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{A}(U; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{A}_*^{-1}(U; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{J}(U, P, Q; \mathbf{a})$$

are $\mathcal{F}(U)$ -measurable.

(ii) *These observables are translation covariant: for every $z \in \mathbb{Z}^d$,*

$$\begin{cases} \mu(z + U, P; \mathbf{a}) = \mu(U, P; \tau_z \mathbf{a}), \\ \mathbf{A}(z + U; \mathbf{a}) = \mathbf{A}(U; \tau_z \mathbf{a}), \\ \mathbf{A}_*^{-1}(z + U; \mathbf{a}) = \mathbf{A}_*^{-1}(U; \tau_z \mathbf{a}), \\ \mathbf{J}(z + U, P, Q; \mathbf{a}) = \mathbf{J}(U, P, Q; \tau_z \mathbf{a}). \end{cases} \quad (4.6)$$

Proof. Let

$$\mathcal{H}(U) := L_{\text{pot},0}^2(U) \times L_{\text{sol},0}^2(U) \subseteq L^2(U; \mathbb{R}^{2d}).$$

Since $\mathcal{H}(U)$ is a separable Hilbert space, we may fix a countable dense subset $\{Y_n\}_{n \in \mathbb{N}} \subseteq \mathcal{H}(U)$.

For each $n \in \mathbb{N}$, define

$$F_n(\mathbf{a}) := \int_U \frac{1}{2} (P + Y_n) \cdot \mathbf{A}(P + Y_n).$$

Because $\mathbf{A}(x)$ is obtained from $\mathbf{a}(x)$ by algebraic operations and inversion of the symmetric part, the map $\mathbf{a} \mapsto F_n(\mathbf{a})$ depends only on the restriction of $\mathbf{a}$ to U and is $\mathcal{F}(U)$ -measurable. By the definition of μ in (2.41),

$$\mu(U, P; \mathbf{a}) = \inf_{Y \in \mathcal{H}(U)} \int_U \frac{1}{2} (P + Y) \cdot \mathbf{A}(P + Y) = \inf_{n \in \mathbb{N}} F_n(\mathbf{a}),$$

where the second equality follows from the density of $\{Y_n\}_{n \in \mathbb{N}}$ in $\mathcal{H}(U)$. Hence $\mu(U, P; \mathbf{a})$ is $\mathcal{F}(U)$ -measurable.

We next prove the translation identities. Define the translation operator

$$(\mathcal{T}_z Y)(x) := Y(x - z), \quad x \in z + U.$$

Then $\mathcal{T}_z$ maps $\mathcal{H}(U)$ bijectively onto $\mathcal{H}(z + U)$, and for every $Y \in \mathcal{H}(U)$ we have

$$\int_{z+U} \frac{1}{2} (P + \mathcal{T}_z Y) \cdot \mathbf{A}(P + \mathcal{T}_z Y) = \int_U \frac{1}{2} (P + Y) \cdot \tau_z \mathbf{A}(P + Y).$$

Taking the infimum over $Y \in \mathcal{H}(U)$ yields the first identity in (4.6).

To prove the measurability of $\mathbf{A}(U; \mathbf{a})$, let $\{E_1, \dots, E_{2d}\}$ be the standard basis of $\mathbb{R}^{2d}$. Since

$$\mu(U, P; \mathbf{a}) = \frac{1}{2} P \cdot \mathbf{A}(U; \mathbf{a}) P,$$

the entries of $\mathbf{A}(U; \mathbf{a})$ are recovered by polarization:

$$E_i \cdot \mathbf{A}(U; \mathbf{a}) E_i = 2\mu(U, E_i; \mathbf{a}),$$

and, for $i \neq j$,

$$E_i \cdot \mathbf{A}(U; \mathbf{a}) E_j = \mu(U, E_i + E_j; \mathbf{a}) - \mu(U, E_i; \mathbf{a}) - \mu(U, E_j; \mathbf{a}).$$

Hence every entry of $\mathbf{A}(U; \mathbf{a})$ is $\mathcal{F}(U)$ -measurable. The translation identity for $\mathbf{A}$ follows immediately from the corresponding identity for μ .

By (2.52),

$$\mathbf{A}_*^{-1}(U; \mathbf{a}) = \mathbf{R} \mathbf{A}(U; \mathbf{a}) \mathbf{R},$$

and therefore $\mathbf{A}_*^{-1}(U; \mathbf{a})$ is $\mathcal{F}(U)$ -measurable and satisfies the same translation identity.

Finally, by the splitting formula (2.51), the quantity $\mathbf{J}(U, P, Q; \mathbf{a})$ is a quadratic polynomial in the entries of $\mathbf{A}(U; \mathbf{a})$ and $\mathbf{A}_*^{-1}(U; \mathbf{a})$. It is therefore $\mathcal{F}(U)$ -measurable, and its translation identity follows from those of $\mathbf{A}(U; \mathbf{a})$ and $\mathbf{A}_*^{-1}(U; \mathbf{a})$. $\square$

Lemma 4.3 (Locality of the Chapter 2 and 3 cube observables). *Let $m, n \in \mathbb{Z}$ with $n \leq m$, let $y \in \mathbb{R}^d$, let $s \in (0, \infty)$, let $p, q \in [1, \infty]$, let $\mathbf{p}, \mathbf{q} \in \mathbb{R}^d$, let $P, Q \in \mathbb{R}^{2d}$, and let $\mathbf{a}_0 \in \mathbb{R}^{d \times d}$ be a constant uniformly elliptic matrix.*

(i) *The one-cube quantities*

$$\mathbf{a} \mapsto J(y + \square_m, \mathbf{p}, \mathbf{q}; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{s}(y + \square_m; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{s}_*(y + \square_m; \mathbf{a}),$$

$$\mathbf{a} \mapsto \mathbf{k}(y + \square_m; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{b}(y + \square_m; \mathbf{a}), \quad \mathbf{a} \mapsto \mathbf{J}(y + \square_m, P, Q; \mathbf{a})$$

are $\mathcal{F}(y + \square_m)$ -measurable.

(ii) *The multiscale quantities*

$$\mathbf{a} \mapsto \Lambda_{s,q}(y + \square_m; \mathbf{a}), \quad \mathbf{a} \mapsto \lambda_{s,q}(y + \square_m; \mathbf{a}), \quad \mathbf{a} \mapsto \mathcal{E}_{s,p,q}(y + \square_m, n; \mathbf{a}, \mathbf{a}_0)$$

are $\mathcal{F}(y + \square_m)$ -measurable.

(iii) *If $z \in \mathbb{Z}^d$, then all of the observables above are translation covariant. In particular,*

$$\Lambda_{s,q}(z + \square_m; \mathbf{a}) = \Lambda_{s,q}(\square_m; \tau_z \mathbf{a}), \quad \lambda_{s,q}(z + \square_m; \mathbf{a}) = \lambda_{s,q}(\square_m; \tau_z \mathbf{a}), \quad (4.7)$$

and

$$\mathcal{E}_{s,p,q}(z + \square_m, n; \mathbf{a}, \mathbf{a}_0) = \mathcal{E}_{s,p,q}(\square_m, n; \tau_z \mathbf{a}, \mathbf{a}_0). \quad (4.8)$$

Proof. Fix the cube $U := y + \square_m$. By (2.52), the matrices

$$\mathbf{b}(U; \mathbf{a}), \quad \mathbf{s}_*^{-1}(U; \mathbf{a}), \quad \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a})$$

are precisely the blocks of $\mathbf{A}(U; \mathbf{a})$. Hence $\mathbf{b}(U; \mathbf{a})$, $\mathbf{s}_*(U; \mathbf{a})$, and $\mathbf{k}(U; \mathbf{a})$ are $\mathcal{F}(U)$ -measurable. Since

$$\mathbf{s}(U; \mathbf{a}) = \mathbf{b}(U; \mathbf{a}) - \mathbf{k}(U; \mathbf{a})^t \mathbf{s}_*^{-1}(U; \mathbf{a}) \mathbf{k}(U; \mathbf{a}),$$

the same is true for $\mathbf{s}(U; \mathbf{a})$. The formula

$$J(U, \mathbf{p}, \mathbf{q}; \mathbf{a}) = \frac{1}{2} \mathbf{p} \cdot \mathbf{s}(U; \mathbf{a}) \mathbf{p} + \frac{1}{2} (\mathbf{q} + \mathbf{k}(U; \mathbf{a}) \mathbf{p}) \cdot \mathbf{s}_*^{-1}(U; \mathbf{a}) (\mathbf{q} + \mathbf{k}(U; \mathbf{a}) \mathbf{p}) - \mathbf{p} \cdot \mathbf{q}$$

then shows that $J(U, \mathbf{p}, \mathbf{q}; \mathbf{a})$ is $\mathcal{F}(U)$ -measurable. The measurability of $\mathbf{J}(U, P, Q; \mathbf{a})$ was proved in Lemma 4.2. This proves part (i).

Part (ii) follows directly from the definitions in (2.69), (2.70), and (2.76): each quantity is obtained from countably many one-cube observables on subcubes of U by taking maxima, geometric sums, and suprema. Since every subcube appearing in these definitions is contained in U , the resulting random variables are $\mathcal{F}(U)$ -measurable.

Finally, part (iii) follows from (4.6) and the fact that the families of subcubes entering the definitions of $\lambda_{s,q}$, $\Lambda_{s,q}$, and $\mathcal{E}_{s,p,q}$ are translated into one another by z . $\square$

4.1.4 Independence for separated cubes

Once locality has been established, the unit-range dependence assumption immediately turns geometric separation into probabilistic independence.

Lemma 4.4 (Independence of separated local observables). *Assume (P2). Let $N \in \mathbb{N}$, and let $U_1, \dots, U_N \subseteq \mathbb{R}^d$ be Borel sets satisfying*

$$\text{dist}(U_i, U_j) \geq 1, \quad \forall i \neq j,$$

and let X_i be an $\mathcal{F}(U_i)$ -measurable random variable for every $i \in \{1, \dots, N\}$. Then the family $\{X_i\}_{i=1}^N$ is independent.

Proof. We argue by induction on N . The case $N = 1$ is trivial, and the case $N = 2$ is exactly the unit-range dependence assumption (P2).

Assume the statement has been proved for some $N \in \mathbb{N}$, and consider a family $U_1, \dots, U_{N+1}$ as above. Since

$$\text{dist}\left(U_1, \bigcup_{i=2}^{N+1} U_i\right) \geq 1,$$

the sigma-algebra $\mathcal{F}(U_1)$ is independent of

$$\mathcal{G} := \mathcal{F}\left(\bigcup_{i=2}^{N+1} U_i\right).$$

Since each X_i with $i \geq 2$ is $\mathcal{G}$ -measurable, the random variable X_1 is independent of the vector $(X_2, \dots, X_{N+1})$. The induction hypothesis gives independence of the family $\{X_i\}_{i=2}^{N+1}$, and therefore the full family $\{X_i\}_{i=1}^{N+1}$ is independent. $\square$

For cube partitions at a fixed scale, the unit-range dependence assumption is not enough to give independence of the full partition, because neighboring cubes touch. The remedy is a deterministic coloring of the lattice into a bounded number of classes whose cubes are pairwise separated by distance at least 1.

Lemma 4.5 (Deterministic coloring of a triadic cube partition). *For each $n \in \mathbb{Z}$, define*

$$r_n := 1 + \lceil 3^{-n} \rceil \quad \text{and} \quad \Xi_n := \{0, \dots, r_n - 1\}^d.$$

For every $\xi = (\xi_1, \dots, \xi_d) \in \Xi_n$, set

$$\mathcal{Z}_{n,\xi} := \{z \in 3^n \mathbb{Z}^d : 3^{-n} z_i \equiv \xi_i \pmod{r_n}, \forall i \in \{1, \dots, d\}\}.$$

Then:

(i) The family $\{\mathcal{Z}_{n,\xi}\}_{\xi \in \Xi_n}$ is a partition of $3^n \mathbb{Z}^d$.

(ii) If $\xi \in \Xi_n$ and $z, z' \in \mathcal{Z}_{n,\xi}$ are distinct, then

$$\text{dist}(z + \square_n, z' + \square_n) \geq 1. \quad (4.9)$$

(iii) There exists a constant $C(d) < \infty$ such that

$$\#\Xi_n = r_n^d \leq C(d)(1 + 3^{-n})^d. \quad (4.10)$$

Proof. Part (i) is immediate from the definition.

Fix $\xi \in \Xi_n$, and let $z, z' \in \mathcal{Z}_{n,\xi}$ be distinct. Then there exists $i \in \{1, \dots, d\}$ such that

$$|z_i - z'_i| \geq r_n 3^n.$$

Since the side length of $\square_n$ is 3^n , we obtain

$$\text{dist}(z + \square_n, z' + \square_n) \geq (r_n - 1)3^n.$$

By definition of r_n ,

$$(r_n - 1)3^n = \lceil 3^{-n} \rceil 3^n \geq 1,$$

which proves (4.9).

Finally,

$$r_n = 1 + \lceil 3^{-n} \rceil \leq 2 + 3^{-n} \leq 3(1 + 3^{-n}),$$

and therefore

$$\#\Xi_n = r_n^d \leq 3^d(1 + 3^{-n})^d.$$

This proves (4.10). □

4.2 Concentration inequalities for independent sums

In this section we collect a small concentration toolkit for sums of independent random variables. The organizing notation is the weak Orlicz shorthand $X = \mathcal{O}_\Psi(A)$, which allows us to treat polynomial moments, stretched-exponential tails, and log-normal-type tails within a single framework.

The section is organized from general notation to increasingly structured concentration statements. We begin with the weak Orlicz classes and their basic calculus, then turn to the finite-moment branch through Rosenthal's inequality, and finally return to tail classes through direct Chernoff bounds, the general truncation argument, and their stretched-exponential and log-normal consequences. These estimates will be applied in the final section to partition averages of local cube observables.

4.2.1 Weak Orlicz quasi-norms and basic examples

We begin with the tail notation that will be used throughout the probabilistic chapters.

Definition 4.6 (The $\mathcal{O}_\Psi$ notation). Let $\Psi : [0, \infty) \rightarrow [1, \infty)$ be an increasing function satisfying

$$\lim_{t \rightarrow \infty} \frac{\Psi(t)}{t} = +\infty.$$

If X is a random variable and $A \in [0, \infty)$, then we write

$$X \leq \mathcal{O}_\Psi(A)$$

to mean that

$$\mathbb{P}[X > At] \leq \frac{1}{\Psi(t)}, \quad \forall t \in [1, \infty). \quad (4.11)$$

We write

$$X = \mathcal{O}_\Psi(A)$$

to mean that $|X| \leq \mathcal{O}_\Psi(A)$.

If $\Psi_1, \dots, \Psi_N$ are increasing functions and $A_1, \dots, A_N \in [0, \infty)$, then

$$X \leq \mathcal{O}_{\Psi_1}(A_1) + \dots + \mathcal{O}_{\Psi_N}(A_N)$$

means that there exist random variables $X_1, \dots, X_N$ such that

$$X \leq X_1 + \dots + X_N \quad \text{and} \quad X_i \leq \mathcal{O}_{\Psi_i}(A_i), \quad \forall i \in \{1, \dots, N\}.$$

This notation is simply a convenient way of encoding weak tail bounds. The point is that later algebraic rules, concentration inequalities, and finite-coloring arguments can all be stated in a uniform language.

The two model families of tail functions that we will use most often are the stretched-exponential class and the log-normal class.

$$\Gamma_\sigma(t) := \exp(t^\sigma), \quad t \in [0, \infty), \quad \sigma \in (0, \infty), \quad (4.12)$$

and

$$\Psi_\sigma(t) := \exp\left(\frac{1}{\sigma^2} \log^2(1 + \sigma t)\right), \quad t \in [0, \infty), \quad \sigma \in (0, \infty). \quad (4.13)$$

Remark 4.7 (Interpretation of the model classes). Let $A \in (0, \infty)$ and let X be a random variable.

- The relation $X = \mathcal{O}_{\Gamma_\sigma}(A)$ is equivalent to the stretched-exponential tail bound

$$\mathbb{P}[|X| > At] \leq \exp(-t^\sigma), \quad \forall t \in [1, \infty).$$

The case $\sigma = 2$ corresponds to subgaussian tails, while $\sigma = 1$ corresponds to exponential tails.

- The relation $X = \mathcal{O}_{\Psi_\sigma}(A)$ describes a log-normal-type tail. Indeed, since

$$\Psi_\sigma(t)^{-1} = \exp\left(-\frac{1}{\sigma^2} \log^2(1 + \sigma t)\right),$$

the decay is weaker than stretched exponential but still much stronger than any power law.

4.2.2 Calculus of $\mathcal{O}_\Psi$ and $\mathcal{O}_{\Gamma_\sigma}$

For the general weak Orlicz notation, the most useful hypothesis is the following mild growth condition.

$$t\Psi(t) \leq \Psi(K_\Psi t), \quad \forall t \in [1, \infty), \quad (4.14)$$

for some constant $K_\Psi \in [2, \infty)$. Under (4.14), powers of t can be absorbed by dilating the argument of Ψ , and this yields both moment bounds and a generalized triangle inequality.

Lemma 4.8 (Basic calculus for admissible Ψ). *Let $\Psi : [0, \infty) \rightarrow [1, \infty)$ be increasing and suppose that (4.14) holds for some $K_\Psi \in [2, \infty)$. Then the following statements hold.*

(i) For every $p \in [1, \infty)$,

$$\frac{t^p}{\Psi(t)} \leq \frac{1}{\Psi(K_\Psi^{-4[p]}t)}, \quad \forall t \in [K_\Psi^{4[p]}, \infty). \quad (4.15)$$

(ii) The function Ψ satisfies the minimal growth estimate

$$\Psi(t) \geq \exp\left(\frac{\log^2 t}{9 \log K_\Psi}\right), \quad \forall t \in [K_\Psi^2, \infty). \quad (4.16)$$

(iii) For every random variable X , every $a \in (0, \infty)$, and every $p \in [1, \infty)$,

$$X = \mathcal{O}_\Psi(a) \implies \mathbb{E}[|X|^p] \leq a^p \left(1 + 2pK_\Psi^{\lceil \frac{1}{2}p(p+1) \rceil} (1 + \log K_\Psi)\right). \quad (4.17)$$

(iv) For every finite family $X_1, \dots, X_N$ of random variables and every family $a_1, \dots, a_N \in (0, \infty)$,

$$X_k \leq \mathcal{O}_\Psi(a_k) \quad \forall k \in \{1, \dots, N\} \implies \sum_{k=1}^N X_k \leq \mathcal{O}_\Psi\left(4K_\Psi^{12} \sum_{k=1}^N a_k\right). \quad (4.18)$$

Proof. Step 1. Absorbing polynomial factors. We first show by induction that, for every $k \in \mathbb{N}$,

$$t^k \Psi(t) \leq K_\Psi^{-\frac{1}{2}k(k-1)} \Psi(K_\Psi^k t), \quad \forall t \in [1, \infty). \quad (4.19)$$

The case $k = 1$ is exactly (4.14). Assuming (4.19) holds for some $k \in \mathbb{N}$, we obtain

$$\begin{aligned} t^{k+1} \Psi(t) &\leq K_\Psi^{-\frac{1}{2}k(k-1)} t \Psi(K_\Psi^k t) \\ &\leq K_\Psi^{-\frac{1}{2}k(k-1)} K_\Psi^{-k} \Psi(K_\Psi^{k+1} t) = K_\Psi^{-\frac{1}{2}(k+1)k} \Psi(K_\Psi^{k+1} t), \end{aligned}$$

which closes the induction.

Taking $k = [p]$ in (4.19), we find

$$t^p \Psi(t) \leq \Psi(K_\Psi^{[p]} t), \quad \forall t \in [1, \infty). \quad (4.20)$$

Applying (4.20) with $K_\Psi^{-[2p]}t$ in place of t yields

$$\frac{t^p}{\Psi(t)} \leq (K_\Psi^{2[2p]} t^{-1})^p \frac{1}{\Psi(K_\Psi^{-[2p]+[p]} t)}, \quad \forall t \in [K_\Psi^{[2p]}, \infty).$$

If $t \geq K_\Psi^{4[p]}$, then the prefactor is at most 1 and $-[2p] + [p] \geq -4[p]$, so (4.15) follows.

Step 2. Minimal growth of Ψ . Let $m \in \mathbb{N}$ and $t \in [K_\Psi^m, K_\Psi^{m+1}]$. Since Ψ is increasing and (4.19) holds, we have

$$\Psi(t) \geq \Psi(K_\Psi^m) \geq K_\Psi^{\frac{1}{2}m(m-1)}. \quad (4.21)$$

Using $t \leq K_\Psi^{m+1}$, we deduce

$$K_\Psi^{\frac{1}{2}m(m-1)} \geq \exp\left(\frac{m(m-1)}{2(m+1)^2} \frac{\log^2 t}{\log K_\Psi}\right).$$

If $m \geq 2$, then

$$\frac{m(m-1)}{2(m+1)^2} \geq \frac{1}{9}.$$

Since $t \geq K_\Psi^2$ implies $m \geq 2$, combining this with (4.21) yields (4.16).

Step 3. Moment bounds. Let $X = \mathcal{O}_\Psi(a)$ and let $p \in [1, \infty)$. By the layer-cake formula,

$$a^{-p} \mathbb{E}[|X|^p] = p \int_0^\infty t^{p-1} \mathbb{P}[|X| > at] dt \leq 1 + p \int_1^\infty \frac{t^{p-1}}{\Psi(t)} dt.$$

To estimate the integral, we decompose it dyadically in the multiplicative scale generated by K_Ψ :

$$\int_1^\infty \frac{t^{p-1}}{\Psi(t)} dt \leq \sum_{j=0}^{[p]} \int_{K_\Psi^j}^{K_\Psi^{j+1}} \frac{t^{p-1}}{\Psi(t)} dt + \int_{K_\Psi^{[p]+1}}^\infty \frac{t^{p-1}}{\Psi(t)} dt.$$

By (4.19),

$$\Psi(t) \geq K_\Psi^{-\frac{1}{2}j(j+1)} t^j, \quad \forall t \in [K_\Psi^j, \infty),$$

and therefore

$$\begin{aligned} \int_1^\infty \frac{t^{p-1}}{\Psi(t)} dt &\leq \sum_{j=0}^{[p]} K_\Psi^{\frac{1}{2}j(j+1)} \int_{K_\Psi^j}^{K_\Psi^{j+1}} t^{p-j-1} dt \\ &\quad + K_\Psi^{\frac{1}{2}([p]+1)([p]+2)} \int_{K_\Psi^{[p]+1}}^\infty t^{p-[p]-2} dt. \end{aligned}$$

Since $p - [p] \leq 0$, the final integral is bounded by $(1 + \log K_\Psi) K_\Psi^{\frac{1}{2}[p]([p]+1)}$, and the finite sum is bounded in the same way up to a factor 2. This gives (4.17).

Step 4. A doubling consequence. We claim that for every $q \in [2, \infty)$,

$$s^q \leq K_\Psi^{3q^2} \frac{\Psi(ts)}{\Psi(t)}, \quad \forall t, s \in [1, \infty). \quad (4.22)$$

Indeed, arguing as in Step 2, for every integer $m \geq 1$ and every $s \in [K_\Psi^m, K_\Psi^{m+1}]$,

$$\frac{\Psi(ts)}{\Psi(t)} \geq K_\Psi^{\frac{1}{2}m(m-1)}.$$

Since $s \leq K_\Psi^{m+1}$, this implies

$$\frac{\Psi(ts)}{\Psi(t)} \geq K_\Psi^{-\frac{m^2}{3}} s^{m/3}.$$

Choosing an integer $m \in [5q, 5q+1]$ yields (4.22).

Step 5. Generalized triangle inequality. We first prove a slightly more abstract statement. Assume that for some $q \in (1, \infty)$ and $C_0 \in (1, \infty)$,

$$s^q \leq C_0 \frac{\Psi(ts)}{\Psi(t)}, \quad \forall t, s \in [1, \infty). \quad (4.23)$$

Let $X_1, \dots, X_N$ and $a_1, \dots, a_N$ be as in the statement and set $X := \sum_{k=1}^N X_k$ and $a := \sum_{k=1}^N a_k$. Fix $t \geq 1$. By Markov's inequality,

$$\begin{aligned} \mathbb{P}[X > ta] &\leq \mathbb{P}\left[\sum_{k=1}^N X_k \mathbb{1}_{\{X_k > \frac{1}{2}ta_k\}} > \frac{1}{2}ta\right] \\ &\leq \frac{2}{ta} \sum_{k=1}^N \mathbb{E}[X_k \mathbb{1}_{\{X_k > \frac{1}{2}ta_k\}}]. \end{aligned}$$

For each k , the layer-cake formula gives

$$\begin{aligned} \mathbb{E}[X_k \mathbb{1}_{\{X_k > \frac{1}{2}ta_k\}}] &\leq \frac{1}{2}ta_k \mathbb{P}[X_k > \frac{1}{2}ta_k] + \int_{\frac{1}{2}ta_k}^{\infty} \mathbb{P}[X_k > s] ds \\ &\leq \frac{1}{2}ta_k \Psi(t/2)^{-1} + \int_{\frac{1}{2}ta_k}^{\infty} \Psi(s/a_k)^{-1} ds. \end{aligned}$$

Using (4.23) with $s/(a_k t/2)$ in place of s , we obtain

$$\frac{1}{\Psi(s/a_k)} \leq \frac{C_0}{\Psi(t/2)} \left(\frac{2s}{a_k t}\right)^{-q}, \quad \forall s \geq \frac{1}{2}ta_k.$$

Hence

$$\begin{aligned} \mathbb{E}[X_k \mathbb{1}_{\{X_k > \frac{1}{2}ta_k\}}] &\leq \frac{1}{\Psi(t/2)} \left(\frac{ta_k}{2} + C_0 \int_{\frac{1}{2}ta_k}^{\infty} \left(\frac{2s}{a_k t}\right)^{-q} ds\right) \\ &\leq \frac{ta_k}{2\Psi(t/2)} \left(1 + \frac{C_0}{q-1}\right). \end{aligned}$$

Summing over k yields

$$\mathbb{P}[X > ta] \leq \frac{1 + C_0(q-1)^{-1}}{\Psi(t/2)}.$$

Applying (4.23) once more, we can absorb the prefactor by dilating t by a constant $C = C(q, C_0) < \infty$. We therefore obtain

$$\sum_{k=1}^N X_k \leq \mathcal{O}_{\Psi}(Ca).$$

Finally, applying this with (4.22), $q = 2$, and $C_0 = K_{\Psi}^{12}$ proves (4.18). $\square$

The previous lemma is the general engine. For the stretched-exponential classes Γ_{σ} , it yields a convenient concrete calculus.

Lemma 4.9 (From moment growth to stretched-exponential tails). *Let $\sigma \in (0, \infty)$, let $M \in (0, \infty)$, and let Y be a nonnegative random variable satisfying*

$$\mathbb{E}[Y^p]^{1/p} \leq Mp^{1/\sigma}, \quad \forall p \in [1, \infty).$$

Then

$$Y \leq \mathcal{O}_{\Gamma_{\sigma}}(eM).$$

Proof. Fix $t \geq 1$ and choose $p = t^\sigma$. By Markov's inequality,

$$\mathbb{P}[Y > eMt] \leq \frac{\mathbb{E}[Y^p]}{(eMt)^p} \leq \left(\frac{Mp^{1/\sigma}}{eMt} \right)^p = e^{-p} = e^{-t^\sigma}.$$

This is exactly the definition of $Y \leq \mathcal{O}_{\Gamma_\sigma}(eM)$. $\square$

Proposition 4.10 (Calculus of $\mathcal{O}_{\Gamma_\sigma}$). *The following statements hold.*

(i) *For every $\sigma \in (0, \infty)$, there exists a constant $C(\sigma) < \infty$ such that, whenever $X_1, \dots, X_N$ is a finite family of random variables and $a_1, \dots, a_N \in (0, \infty)$, we have*

$$X_k \leq \mathcal{O}_{\Gamma_\sigma}(a_k) \quad \forall k \in \{1, \dots, N\} \implies \sum_{k=1}^N X_k \leq \mathcal{O}_{\Gamma_\sigma} \left(C(\sigma) \sum_{k=1}^N a_k \right). \quad (4.24)$$

(ii) *For every $\sigma_1, \sigma_2 \in (0, \infty)$, there exists a constant $C(\sigma_1, \sigma_2) < \infty$ such that, for every pair of nonnegative random variables X_1, X_2 ,*

$$X_1 \leq \mathcal{O}_{\Gamma_{\sigma_1}}(A_1) \quad \text{and} \quad X_2 \leq \mathcal{O}_{\Gamma_{\sigma_2}}(A_2) \implies X_1 X_2 \leq \mathcal{O}_{\Gamma_{\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}}}(C(\sigma_1, \sigma_2) A_1 A_2). \quad (4.25)$$

In particular, for every $\sigma, p, K \in (0, \infty)$ and nonnegative random variable X ,

$$X \leq \mathcal{O}_{\Gamma_\sigma}(K) \iff X^p \leq \mathcal{O}_{\Gamma_{\sigma/p}}(K^p). \quad (4.26)$$

(iii) *For every $\sigma, A \in (0, \infty)$ and every finite family $X_1, \dots, X_N$ with $N \geq 2$,*

$$X_i \leq \mathcal{O}_{\Gamma_\sigma}(A) \quad \forall i \in \{1, \dots, N\} \implies \max_{1 \leq i \leq N} X_i \leq \mathcal{O}_{\Gamma_\sigma}((3 \log N)^{1/\sigma} A). \quad (4.27)$$

(iv) *For every event E with $0 < \mathbb{P}[E] < 1$ and every $\sigma \in (0, \infty)$,*

$$\mathbb{1}_E \leq \mathcal{O}_{\Gamma_\sigma}(|\log \mathbb{P}[E]|^{-1/\sigma}). \quad (4.28)$$

(v) *For every $\sigma \in (0, \infty)$, there exists a constant $C(\sigma) < \infty$ such that, for every $K \in (0, \infty)$, every random variable X satisfying $X = \mathcal{O}_{\Gamma_\sigma}(K)$, and every $p \in [1, \infty)$,*

$$\mathbb{E}[|X|^p]^{1/p} \leq C(\sigma) p^{1/\sigma} K. \quad (4.29)$$

Proof. Step 1. Moment bounds. Let $X = \mathcal{O}_{\Gamma_\sigma}(K)$. Then

$$\mathbb{P}[|X| > Kt] \leq e^{-t^\sigma}, \quad \forall t \in [1, \infty).$$

By the layer-cake formula, for every $p \in [1, \infty)$,

$$\begin{aligned} K^{-p} \mathbb{E}[|X|^p] &= p \int_0^\infty t^{p-1} \mathbb{P}[|X| > Kt] dt \\ &\leq 1 + p \int_1^\infty t^{p-1} e^{-t^\sigma} dt \\ &= 1 + \frac{p}{\sigma} \int_1^\infty u^{\frac{p}{\sigma}-1} e^{-u} du \leq 1 + \Gamma\left(\frac{p}{\sigma} + 1\right), \end{aligned}$$

after the change of variables $u = t^\sigma$. Stirling's inequality implies that

$$\Gamma\left(\frac{p}{\sigma} + 1\right)^{1/p} \leq C(\sigma)p^{1/\sigma},$$

and this proves (4.29).

Step 2. The generalized triangle inequality. Let

$$Y := \left(\sum_{k=1}^N X_k\right)_+.$$

Since $Y \leq \sum_{k=1}^N (X_k)_+$, Minkowski's inequality and Step 1 give, for every $p \in [1, \infty)$,

$$\mathbb{E}[Y^p]^{1/p} \leq \sum_{k=1}^N \mathbb{E}[(X_k)_+^p]^{1/p} \leq C(\sigma)p^{1/\sigma} \sum_{k=1}^N a_k.$$

Applying Lemma 4.9 proves (4.24).

Step 3. Products and powers. Let

$$\tau := \frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}.$$

If X_1 and X_2 are nonnegative, then by Hölder's inequality and Step 1,

$$\begin{aligned} \mathbb{E}[(X_1 X_2)^p]^{1/p} &\leq \mathbb{E}[X_1^{2p}]^{1/2p} \mathbb{E}[X_2^{2p}]^{1/2p} \\ &\leq C(\sigma_1, \sigma_2) p^{\frac{1}{\sigma_1} + \frac{1}{\sigma_2}} A_1 A_2 = C(\sigma_1, \sigma_2) p^{1/\tau} A_1 A_2. \end{aligned}$$

Lemma 4.9 therefore gives (4.25).

The power rule (4.26) is immediate from the definition:

$$\mathbb{P}[X^p > K^p t] = \mathbb{P}[X > K t^{1/p}] \leq \exp(-t^{\sigma/p}), \quad \forall t \geq 1.$$

The converse implication is obtained by replacing t with t^p .

To prove (4.27), let $t \geq 1$. By a union bound,

$$\begin{aligned} \mathbb{P}\left[\max_{1 \leq i \leq N} X_i > (3 \log N)^{1/\sigma} A t\right] &\leq \sum_{i=1}^N \mathbb{P}[X_i > (3 \log N)^{1/\sigma} A t] \\ &\leq N \exp(-3t^\sigma \log N) \leq \exp(-t^\sigma), \end{aligned}$$

which is exactly (4.27).

To prove (4.28), fix $t \geq 1$. If $t |\log \mathbb{P}[E]|^{-1/\sigma} \geq 1$, then

$$\mathbb{P}[\mathbb{1}_E > t |\log \mathbb{P}[E]|^{-1/\sigma}] = 0.$$

Otherwise, the event on the left-hand side is just E , and therefore

$$\mathbb{P}[\mathbb{1}_E > t |\log \mathbb{P}[E]|^{-1/\sigma}] = \mathbb{P}[E] = \exp(-|\log \mathbb{P}[E]|) \leq \exp(-t^\sigma).$$

This proves (4.28). □

Remark 4.11 (The log-normal scale). The family Ψ_σ is the natural Orlicz scale associated with log-normal random variables. For every $\sigma \in [1, \infty)$,

$$t\Psi_\sigma(t) \leq \Psi_\sigma(K_{\Psi_\sigma}t), \quad \forall t \in [1, \infty),$$

with

$$K_{\Psi_\sigma} := 2\exp(2\sigma^2). \quad (4.30)$$

Moreover,

$$X \leq \mathcal{O}_{\Gamma_2}(\sigma) \iff \exp(X) - 1 \leq \mathcal{O}_{\Psi_\sigma}(\sigma). \quad (4.31)$$

Proof. To prove (4.30), let $t \geq 1$. Since

$$\log^2(1 + K_{\Psi_\sigma}\sigma t) - \log^2(1 + \sigma t) = (\log(1 + K_{\Psi_\sigma}\sigma t) - \log(1 + \sigma t))(\log(1 + K_{\Psi_\sigma}\sigma t) + \log(1 + \sigma t))$$

and the second factor is at least $\log(1 + K_{\Psi_\sigma}\sigma t)$, it is enough to show that

$$\sigma^2 \log t \leq \log(1 + K_{\Psi_\sigma}\sigma t)(\log(1 + K_{\Psi_\sigma}\sigma t) - \log(1 + \sigma t)).$$

Now

$$\log(1 + K_{\Psi_\sigma}\sigma t) \geq \log t,$$

because $K_{\Psi_\sigma}\sigma \geq 1$, and moreover

$$\log(1 + K_{\Psi_\sigma}\sigma t) - \log(1 + \sigma t) = \log\left(\frac{1 + K_{\Psi_\sigma}\sigma t}{1 + \sigma t}\right) \geq \log\left(\frac{K_{\Psi_\sigma}\sigma t}{2\sigma t}\right) = 2\sigma^2,$$

using $\sigma \geq 1$ and $K_{\Psi_\sigma} = 2\exp(2\sigma^2)$. Hence

$$\log(1 + K_{\Psi_\sigma}\sigma t)(\log(1 + K_{\Psi_\sigma}\sigma t) - \log(1 + \sigma t)) \geq 2\sigma^2 \log t,$$

which proves (4.30).

For (4.31), observe that

$$\exp(X) - 1 > \sigma t \iff X > \log(1 + \sigma t).$$

If $X \leq \mathcal{O}_{\Gamma_2}(\sigma)$, then

$$\mathbb{P}[\exp(X) - 1 > \sigma t] = \mathbb{P}[X > \log(1 + \sigma t)] \leq \exp\left(-\frac{1}{\sigma^2} \log^2(1 + \sigma t)\right) = \frac{1}{\Psi_\sigma(t)}$$

for every $t \geq 1$. This proves the forward implication. Conversely, if $\exp(X) - 1 \leq \mathcal{O}_{\Psi_\sigma}(\sigma)$, then for every $t \geq 1$,

$$\begin{aligned} \mathbb{P}[X > \sigma t] &= \mathbb{P}[\exp(X) - 1 > e^{\sigma t} - 1] \\ &\leq \frac{1}{\Psi_\sigma((e^{\sigma t} - 1)/\sigma)} = \exp(-t^2). \end{aligned}$$

Thus $X \leq \mathcal{O}_{\Gamma_2}(\sigma)$. □

The next subsection turns from the algebra of tail classes to sums of centered independent random variables, beginning with Rosenthal's inequality and its polynomial-moment corollaries.

4.2.3 Rosenthal inequality and polynomial-moment concentration

We begin with the classical Rosenthal bound in a form convenient for later applications to partition averages.

Proposition 4.12 (Rosenthal's inequality). *There exists a universal constant $C < \infty$ such that, for every $p \in [2, \infty)$ and every finite family of independent random variables $X_1, \dots, X_N \in L^p(\mathbb{P})$ satisfying*

$$\mathbb{E}[X_i] = 0, \quad \forall i \in \{1, \dots, N\},$$

we have

$$\mathbb{E} \left[\left| \sum_{i=1}^N X_i \right|^p \right]^{1/p} \leq C \left(\sqrt{p} \left(\sum_{i=1}^N \mathbb{E}[X_i^2] \right)^{1/2} + p \mathbb{E} \left[\max_{1 \leq i \leq N} |X_i|^p \right]^{1/p} \right). \quad (4.32)$$

The proof can still be organized around the same truncation philosophy as the other concentration arguments in this chapter. The only genuinely new input is a sharper Laplace-transform estimate for bounded centered sums, namely Bennett's inequality.

Lemma 4.13 (Symmetrization reduction). *Let $p \in [1, \infty)$, and let $X_1, \dots, X_N \in L^p(\mathbb{P})$ be independent random variables with $\mathbb{E}[X_i] = 0$ for every $i \in \{1, \dots, N\}$. Let $X'_1, \dots, X'_N$ be an independent copy, and set*

$$S := \sum_{i=1}^N X_i \quad \text{and} \quad S' := \sum_{i=1}^N X'_i.$$

Then

$$\mathbb{E}[|S|^p]^{1/p} \leq \mathbb{E}[|S - S'|^p]^{1/p}. \quad (4.33)$$

Proof. Since $\mathbb{E}[X'_i] = 0$ for every i , we have

$$S = \mathbb{E}[S - S' \mid X_1, \dots, X_N].$$

By conditional Jensen's inequality,

$$|S|^p \leq \mathbb{E}[|S - S'|^p \mid X_1, \dots, X_N].$$

Taking expectation proves (4.33). $\square$

Lemma 4.14 (Bennett inequality for bounded centered sums). *Let $y \in (0, \infty)$, and let $Y_1, \dots, Y_N$ be independent centered random variables satisfying*

$$|Y_i| \leq y \quad \text{almost surely for every } i \in \{1, \dots, N\},$$

and define

$$T := \sum_{i=1}^N Y_i \quad \text{and} \quad v := \sum_{i=1}^N \mathbb{E}[Y_i^2].$$

If $v = 0$, then $T = 0$ almost surely. If $v > 0$, then, for every $t > 0$,

$$\mathbb{P}[T > t] \leq \exp \left(-\frac{v}{y^2} h \left(\frac{ty}{v} \right) \right), \quad (4.34)$$

where

$$h(r) := (1 + r) \log(1 + r) - r, \quad \forall r \in [0, \infty). \quad (4.35)$$

Consequently, if $v > 0$, then

$$\mathbb{P}[|T| > t] \leq 2 \exp \left(-\frac{v}{y^2} h \left(\frac{ty}{v} \right) \right). \quad (4.36)$$

Proof. If $v = 0$, then $\mathbb{E}[Y_i^2] = 0$ for every $i \in \{1, \dots, N\}$, and therefore $Y_i = 0$ almost surely for every i . Hence $T = 0$ almost surely, so the first assertion holds. We may therefore assume from now on that $v > 0$ in order to prove (4.34) and (4.36).

Fix $\lambda > 0$. Since $\mathbb{E}[Y_i] = 0$, we have

$$\mathbb{E}[e^{\lambda Y_i}] = 1 + \sum_{n=2}^{\infty} \frac{\lambda^n}{n!} \mathbb{E}[Y_i^n].$$

Because $|Y_i| \leq y$, for every integer $n \geq 2$,

$$\mathbb{E}[Y_i^n] \leq \mathbb{E}[|Y_i|^n] \leq y^{n-2} \mathbb{E}[Y_i^2].$$

Therefore

$$\begin{aligned} \mathbb{E}[e^{\lambda Y_i}] &\leq 1 + \frac{\mathbb{E}[Y_i^2]}{y^2} \sum_{n=2}^{\infty} \frac{(\lambda y)^n}{n!} = 1 + \frac{\mathbb{E}[Y_i^2]}{y^2} (e^{\lambda y} - 1 - \lambda y) \\ &\leq \exp\left(\frac{\mathbb{E}[Y_i^2]}{y^2} (e^{\lambda y} - 1 - \lambda y)\right). \end{aligned}$$

Multiplying over i and using independence yields

$$\mathbb{E}[e^{\lambda T}] \leq \exp\left(\frac{v}{y^2} (e^{\lambda y} - 1 - \lambda y)\right).$$

By Chernoff's bound,

$$\mathbb{P}[T > t] \leq \exp\left(-\lambda t + \frac{v}{y^2} (e^{\lambda y} - 1 - \lambda y)\right).$$

The derivative of the exponent with respect to λ is

$$-t + \frac{v}{y} (e^{\lambda y} - 1),$$

so the minimizer is

$$\lambda_* = \frac{1}{y} \log\left(1 + \frac{ty}{v}\right).$$

Substituting λ_* back into the exponent gives exactly (4.34). Applying the same bound to $-T$ and using a union bound yields (4.36). $\square$

Lemma 4.15 (Elementary properties of the Bennett function). *Define*

$$\beta(r) := \frac{h(r)}{r}, \quad \forall r \in (0, \infty), \quad (4.37)$$

with h as in (4.35). Then the following statements hold.

(i) The function β is increasing on $(0, \infty)$.

(ii) For every $r \in [0, 1]$,

$$h(r) \geq \frac{1}{4} r^2. \quad (4.38)$$

(iii) There exists a universal constant $r_0 \in [2, \infty)$ such that, for every $r \in [r_0, \infty)$,

$$\beta(r) \geq \frac{3}{4} \log r. \quad (4.39)$$

Proof. For $r > 0$,

$$\beta'(r) = \frac{r - \log(1+r)}{r^2}.$$

Since $\log(1+r) < r$ for every $r > 0$, this proves part (i).

Next, since

$$h''(r) = \frac{1}{1+r},$$

we have $h''(r) \geq 1/2$ on $[0, 1]$. Using also $h(0) = h'(0) = 0$, we obtain

$$h(r) = \int_0^r (r-s)h''(s) ds \geq \frac{1}{2} \int_0^r (r-s) ds = \frac{1}{4}r^2,$$

which proves part (ii).

Finally,

$$\lim_{r \rightarrow \infty} \frac{\beta(r)}{\log r} = 1.$$

Therefore there exists a universal $r_0 \geq 2$ such that (4.39) holds for every $r \in [r_0, \infty)$. $\square$

Proof of Proposition 4.12. Set

$$S := \sum_{i=1}^N X_i \quad \text{and} \quad \sigma^2 := \sum_{i=1}^N \mathbb{E}[X_i^2] \quad \text{and} \quad M := \max_{1 \leq i \leq N} |X_i|.$$

If $\sigma = 0$, then $\mathbb{E}[X_i^2] = 0$ for every $i \in \{1, \dots, N\}$, and hence $X_i = 0$ almost surely for every i . Therefore $S = 0$ almost surely, and the conclusion is trivial. We may therefore assume from now on that $\sigma > 0$.

Step 1. Reduction to the symmetric case. Let $X'_1, \dots, X'_N$ be an independent copy of $X_1, \dots, X_N$, and define

$$Y_i := X_i - X'_i \quad \text{and} \quad M' := \max_{1 \leq i \leq N} |X'_i|.$$

By Lemma 4.13,

$$\mathbb{E}[|S|^p]^{1/p} \leq \mathbb{E}\left[\left|\sum_{i=1}^N Y_i\right|^p\right]^{1/p}.$$

The random variables $Y_1, \dots, Y_N$ are independent and symmetric. Moreover,

$$\sum_{i=1}^N \mathbb{E}[Y_i^2] = 2 \sum_{i=1}^N \mathbb{E}[X_i^2] = 2\sigma^2,$$

and

$$\max_{1 \leq i \leq N} |Y_i| \leq M + M'.$$

Therefore it is enough to prove the estimate under the additional assumption that $X_1, \dots, X_N$ are symmetric, since at the end we can use

$$\mathbb{E}[(M + M')^p]^{1/p} \leq 2 \mathbb{E}[M^p]^{1/p}.$$

The factors coming from $\sum_{i=1}^N \mathbb{E}[Y_i^2] = 2\sigma^2$ and $\max_{1 \leq i \leq N} |Y_i| \leq M + M'$ are absorbed into the universal constant when we pass back from the symmetric case to the general case. From now on, we therefore assume that $X_1, \dots, X_N$ are symmetric.

Step 2. Truncation at the moment-adapted scale. Fix $t > 0$ and define

$$y := \frac{t}{p}.$$

Set

$$Y_i^{(y)} := X_i \mathbb{1}_{\{|X_i| \leq y\}} \quad \text{and} \quad T_y := \sum_{i=1}^N Y_i^{(y)}.$$

Since each X_i is symmetric, each $Y_i^{(y)}$ is still symmetric and hence centered. Also,

$$|Y_i^{(y)}| \leq y \quad \text{and} \quad \sum_{i=1}^N \mathbb{E}[(Y_i^{(y)})^2] \leq \sigma^2.$$

On the event $\{M \leq y\}$, we have $T_y = S$. Therefore

$$\{|S| > t\} \subseteq \{M > y\} \cup \{|T_y| > t\},$$

and thus

$$\mathbb{P}[|S| > t] \leq \mathbb{P}\left[M > \frac{t}{p}\right] + \mathbb{P}[|T_y| > t]. \quad (4.40)$$

Step 3. Bennett estimate for the truncated sum. Set

$$v := \sum_{i=1}^N \mathbb{E}[(Y_i^{(y)})^2].$$

If $v = 0$, then $Y_i^{(y)} = 0$ almost surely for every i , so $T_y = 0$ almost surely. In particular,

$$\mathbb{P}[|T_y| > t] = 0,$$

and (4.42) is trivial. We may therefore assume from now on that $v > 0$. Since $v \leq \sigma^2$, Lemma 4.14 and then Lemma 4.15(i) yield

$$\begin{aligned} \mathbb{P}[|T_y| > t] &\leq 2 \exp\left(-\frac{v}{y^2} h\left(\frac{ty}{v}\right)\right) \\ &= 2 \exp\left(-\frac{t}{y} \beta\left(\frac{ty}{v}\right)\right) \\ &\leq 2 \exp\left(-\frac{t}{y} \beta\left(\frac{ty}{\sigma^2}\right)\right). \end{aligned} \quad (4.41)$$

Since $y = t/p$, this becomes

$$\mathbb{P}[|T_y| > t] \leq 2 \exp\left(-p \beta\left(\frac{t^2}{p\sigma^2}\right)\right). \quad (4.42)$$

Combining this with (4.40), we conclude that

$$\mathbb{P}[|S| > t] \leq \mathbb{P}\left[M > \frac{t}{p}\right] + 2 \exp\left(-p \beta\left(\frac{t^2}{p\sigma^2}\right)\right). \quad (4.43)$$

Step 4. Tail integration. Using the layer-cake formula,

$$\mathbb{E}[|S|^p] = p \int_0^\infty t^{p-1} \mathbb{P}[|S| > t] dt.$$

Inserting (4.43), we obtain

$$\mathbb{E}[|S|^p] \leq I_1 + I_2,$$

where

$$I_1 := p \int_0^\infty t^{p-1} \mathbb{P}\left[M > \frac{t}{p}\right] dt$$

and

$$I_2 := 2p \int_0^\infty t^{p-1} \exp\left(-p\beta\left(\frac{t^2}{p\sigma^2}\right)\right) dt.$$

For I_1 , the change of variables $t = ps$ gives

$$I_1 = p^p \mathbb{E}[M^p].$$

Hence

$$I_1^{1/p} = p \mathbb{E}[M^p]^{1/p}. \quad (4.44)$$

For I_2 , we change variables $t = \sqrt{p}\sigma r$ and obtain

$$I_2 = 2p(\sqrt{p}\sigma)^p \int_0^\infty r^{p-1} e^{-p\beta(r^2)} dr.$$

We claim that there exists a universal constant $C < \infty$ such that

$$p \int_0^\infty r^{p-1} e^{-p\beta(r^2)} dr \leq C^p, \quad \forall p \in [2, \infty). \quad (4.45)$$

To prove this, we split the integral into three regions.

On $[0, 1]$, we simply use

$$p \int_0^1 r^{p-1} dr = 1.$$

On $[1, \sqrt{r_0}]$, where r_0 is the constant from Lemma 4.15(iii), we have

$$p \int_1^{\sqrt{r_0}} r^{p-1} dr \leq r_0^{p/2}.$$

On $[\sqrt{r_0}, \infty)$, Lemma 4.15 (iii) gives

$$\beta(r^2) \geq \frac{3}{2} \log r.$$

Therefore

$$p \int_{\sqrt{r_0}}^\infty r^{p-1} e^{-p\beta(r^2)} dr \leq p \int_{\sqrt{r_0}}^\infty r^{-p/2-1} dr \leq 2.$$

This proves (4.45). Consequently,

$$I_2 \leq (C\sqrt{p}\sigma)^p,$$

and hence

$$I_2^{1/p} \leq C\sqrt{p}\sigma. \quad (4.46)$$

Step 5. Conclusion. Combining (4.44) and (4.46), we obtain

$$\mathbb{E}[|S|^p]^{1/p} \leq C(\sqrt{p}\sigma + p \mathbb{E}[M^p]^{1/p})$$

in the symmetric case. By the symmetrization reduction from Step 1, this implies the general case after increasing the universal constant C . $\square$

Corollary 4.16 (Polynomial-moment corollaries of Rosenthal). *There exists a universal constant $C < \infty$ such that the following statements hold.*

(i) *Let $p \in [2, \infty)$, and let $X_1, \dots, X_N \in L^p(\mathbb{P})$ be a finite family of independent centered random variables. Then*

$$\mathbb{E} \left[\left| \sum_{i=1}^N X_i \right|^p \right]^{1/p} \leq C \left(\sqrt{p} \left(\sum_{i=1}^N \mathbb{E}[X_i^2] \right)^{1/2} + p \left(\sum_{i=1}^N \mathbb{E}[|X_i|^p] \right)^{1/p} \right). \quad (4.47)$$

(ii) *If moreover $K \in [0, \infty)$ and $\mathbb{E}[|X_i|^p]^{1/p} \leq K$ for every $i \in \{1, \dots, N\}$, then*

$$\mathbb{E} \left[\left| \sum_{i=1}^N X_i \right|^p \right]^{1/p} \leq C (\sqrt{p} N^{1/2} K + p N^{1/p} K). \quad (4.48)$$

Proof. Part (i) follows from Proposition 4.12 and the elementary bound

$$\mathbb{E} \left[\max_{1 \leq i \leq N} |X_i|^p \right] \leq \sum_{i=1}^N \mathbb{E}[|X_i|^p].$$

Part (ii) is immediate from part (i). □

4.2.4 Direct Chernoff bounds in the exponential regime

The first such argument applies to stretched-exponential tails with $\sigma \in [1, 2]$. In this range, the exponential moment is strong enough that one can estimate the Laplace transform directly, without any truncation.

Proposition 4.17 (Concentration for centered independent $\mathcal{O}_{\Gamma_\sigma}$ random variables in the exponential regime). *For every $\sigma \in [1, 2]$, there exists a constant $C(\sigma) < \infty$ such that, for every $m \in \mathbb{N}$ and every finite family of independent measurable integrable random variables $X_1, \dots, X_m$ satisfying*

$$\mathbb{E}[X_k] = 0 \quad \text{and} \quad X_k = \mathcal{O}_{\Gamma_\sigma}(1), \quad \forall k \in \{1, \dots, m\},$$

we have

$$\sum_{k=1}^m X_k = \mathcal{O}_{\Gamma_\sigma}(C(\sigma)m^{1/2}). \quad (4.49)$$

Proof. Set

$$S_m := \sum_{k=1}^m X_k.$$

By Chernoff's bound, for every $a > 0$ and every $\lambda > 0$,

$$\mathbb{P}[S_m > a] \leq \exp(-\lambda a) \prod_{k=1}^m \mathbb{E}[e^{\lambda X_k}]. \quad (4.50)$$

We therefore begin by estimating the Laplace transforms $\mathbb{E}[e^{\lambda X_k}]$.

Step 1. Small values of λ . Fix $k \in \{1, \dots, m\}$. Since $\mathbb{E}[X_k] = 0$, the elementary Taylor remainder estimate

$$|e^{\lambda x} - (1 + \lambda x)| \leq \frac{1}{2} \lambda^2 |x|^2 e^{\lambda x}, \quad \forall x \in \mathbb{R},$$

implies

$$\mathbb{E}[e^{\lambda X_k}] \leq 1 + \frac{1}{2}\lambda^2 \mathbb{E}[|X_k|^2 e^{\lambda(X_k)_+}]. \quad (4.51)$$

If $\lambda \in (0, 1/2]$, then $\mathbb{P}[X_k > t] \leq e^{-t^\sigma} \leq e^{-t}$ for every $t \geq 1$, and therefore

$$\begin{aligned} \mathbb{E}[|X_k|^2 e^{\lambda(X_k)_+}] &\leq \mathbb{E}[|X_k|^2] + \int_1^\infty (2t + \lambda t^2) e^{\lambda t} \mathbb{P}[X_k > t] dt \\ &\leq \mathbb{E}[|X_k|^2] + \int_1^\infty (2t + \lambda t^2) e^{-t/2} dt. \end{aligned}$$

By (4.29) with $p = 2$, we have $\mathbb{E}[|X_k|^2] \leq C$. The integral is also bounded by a universal constant. Hence, after increasing C , we obtain from (4.51) that

$$\mathbb{E}[e^{\lambda X_k}] \leq \exp(C\lambda^2), \quad \forall \lambda \in (0, 1/2]. \quad (4.52)$$

Step 2. Large values of λ when $\sigma > 1$. Assume now that $\sigma \in (1, 2]$ and that $\lambda \geq 1/2$. Using (4.51) again, it remains to estimate

$$\mathbb{E}[|X_k|^2 e^{\lambda(X_k)_+}].$$

For every $t, \lambda \geq 0$, Young's inequality gives

$$\lambda t \leq \frac{1}{\sigma} t^\sigma + \frac{\sigma - 1}{\sigma} \lambda^{\frac{\sigma}{\sigma-1}}.$$

Using also the tail bound $\mathbb{P}[X_k > t] \leq e^{-t^\sigma}$, we deduce

$$\begin{aligned} \int_1^\infty (2t + \lambda t^2) e^{\lambda t} \mathbb{P}[X_k > t] dt &\leq \exp(C(\sigma) \lambda^{\frac{\sigma}{\sigma-1}}) \int_1^\infty (2t + \lambda t^2) e^{-t^\sigma/\sigma} dt \\ &\leq C(\sigma)(1 + \lambda) \exp(C(\sigma) \lambda^{\frac{\sigma}{\sigma-1}}). \end{aligned}$$

Returning to (4.51), and using that $\lambda \geq 1/2$, we may absorb the polynomial factor $\lambda^2(1 + \lambda)$ into the exponential term. We obtain

$$\mathbb{E}[e^{\lambda X_k}] \leq \exp(C(\sigma) \lambda^{\frac{\sigma}{\sigma-1}}), \quad \forall \lambda \in [1/2, \infty). \quad (4.53)$$

Step 3. Tail bounds for S_m . If $\sigma = 1$, then combining (4.50) with (4.52) yields

$$\mathbb{P}[S_m > a] \leq \exp(-\lambda a + C\lambda^2 m), \quad \forall \lambda \in (0, 1/2].$$

Choosing $\lambda = \min\{1/2, \frac{a}{2Cm}\}$, we obtain

$$\mathbb{P}[S_m > a] \leq \exp(-c \min\{a^2 m^{-1}, a\}).$$

Taking $a = C\sqrt{m}t$ with $t \geq 1$ and increasing C , we get

$$\mathbb{P}[S_m > C\sqrt{m}t] \leq e^{-t},$$

which proves (4.49) in the case $\sigma = 1$.

Assume next that $\sigma \in (1, 2]$. From (4.50), (4.52), and (4.53), we obtain the two bounds

$$\mathbb{P}[S_m > a] \leq \exp(-\lambda a + C\lambda^2 m), \quad \forall \lambda \in (0, 1/2],$$

and

$$\mathbb{P}[S_m > a] \leq \exp\left(-\lambda a + C(\sigma)m\lambda^{\frac{\sigma}{\sigma-1}}\right), \quad \forall \lambda \in [1/2, \infty).$$

The first inequality gives

$$\mathbb{P}[S_m > a] \leq \exp(-ca^2m^{-1}) \quad \text{whenever } a \leq cm,$$

by taking $\lambda = \frac{a}{2Cm} \leq 1/2$. The second inequality gives

$$\mathbb{P}[S_m > a] \leq \exp(-c(\sigma)a^\sigma m^{1-\sigma}) \quad \text{whenever } a \geq C(\sigma)m,$$

by taking

$$\lambda := \left(\frac{a}{2C(\sigma)m}\right)^{\sigma-1},$$

which indeed satisfies $\lambda \geq 1/2$ in that regime.

We now set $a = B(\sigma)\sqrt{m}t$ with $t \geq 1$. If $t \leq C(\sigma)\sqrt{m}$, then the Gaussian estimate gives

$$\mathbb{P}[S_m > B(\sigma)\sqrt{m}t] \leq \exp(-cB(\sigma)^2t^2) \leq e^{-t^\sigma},$$

after increasing $B(\sigma)$. If instead $t \geq C(\sigma)\sqrt{m}$, then the large-deviation bound gives

$$\mathbb{P}[S_m > B(\sigma)\sqrt{m}t] \leq \exp(-c(\sigma)B(\sigma)^\sigma t^\sigma m^{1-\sigma/2}) \leq e^{-t^\sigma},$$

again after increasing $B(\sigma)$. This proves (4.49). $\square$

4.2.5 General concentration via truncation and Chernoff

The next result is the heavy-tail counterpart of the previous proposition. It uses the same Chernoff idea, but now the large values of the summands are first removed by truncation. The theorem is written in a flexible form so that stretched-exponential and log-normal concentration can both be recovered by a later choice of Ψ .

Proposition 4.18 (General concentration for centered independent $\mathcal{O}_\Psi$ variables). *Let $\Psi : [0, \infty) \rightarrow [1, \infty)$ be increasing and suppose that*

$$C_\Psi := \int_1^\infty \frac{t}{\Psi(t)} dt < \infty. \quad (4.54)$$

Let $m \in \mathbb{N}$ and let $X_1, \dots, X_m$ be independent measurable integrable random variables satisfying

$$\mathbb{E}[X_k] = 0 \quad \text{and} \quad X_k = \mathcal{O}_\Psi(1), \quad \forall k \in \{1, \dots, m\}.$$

Suppose that $M \in [1, \infty)$, $\lambda \in (0, 1]$, and $L \in [1, \infty)$ satisfy

$$\lambda t \leq \log \Psi(t) - 4 \log t + \log M, \quad \forall t \in [1, L]. \quad (4.55)$$

Then, for every $a > 0$,

$$\mathbb{P}\left[\sum_{k=1}^m X_k > a\right] \leq \frac{m}{\Psi(L)} + \exp(-\lambda a + \lambda^2 m(3 + M + C_\Psi)). \quad (4.56)$$

Proof. Set

$$S_m := \sum_{k=1}^m X_k \quad \text{and} \quad Y_k := \min\{X_k, L\}.$$

Then either $S_m = \sum_{k=1}^m Y_k$, or else at least one of the events $\{X_k > L\}$ occurs. Therefore

$$\mathbb{P}[S_m > a] \leq \mathbb{P}\left[\sum_{k=1}^m Y_k > a\right] + \mathbb{P}\left[\max_{1 \leq k \leq m} X_k > L\right]. \quad (4.57)$$

The second term is bounded by the union bound:

$$\mathbb{P}\left[\max_{1 \leq k \leq m} X_k > L\right] \leq \sum_{k=1}^m \mathbb{P}[X_k > L] \leq \frac{m}{\Psi(L)}. \quad (4.58)$$

It remains to estimate the truncated sum. By Markov's inequality and independence,

$$\mathbb{P}\left[\sum_{k=1}^m Y_k > a\right] \leq \exp(-\lambda a) \prod_{k=1}^m \mathbb{E}[e^{\lambda Y_k}]. \quad (4.59)$$

We now estimate each factor on the right-hand side.

Since $\mathbb{E}[X_k] = 0$ and $Y_k \leq X_k$, we have $\mathbb{E}[Y_k] \leq 0$. Therefore, by the same Taylor remainder bound used above,

$$\mathbb{E}[e^{\lambda Y_k}] \leq 1 + \frac{1}{2} \lambda^2 \mathbb{E}[Y_k^2 e^{\lambda(Y_k)_+}].$$

Using the layer-cake formula and the bound $Y_k \leq L$, we obtain

$$\begin{aligned} \mathbb{E}[Y_k^2 e^{\lambda(Y_k)_+}] &\leq \mathbb{E}[X_k^2] + \int_1^L (2t + \lambda t^2) e^{\lambda t} \mathbb{P}[X_k > t] dt \\ &\leq \mathbb{E}[X_k^2] + \int_1^L \frac{(2t + \lambda t^2) e^{\lambda t}}{\Psi(t)} dt. \end{aligned}$$

The assumption (4.54) implies

$$\mathbb{E}[X_k^2] \leq 1 + 2 \int_1^\infty \frac{t}{\Psi(t)} dt = 1 + 2C_\Psi.$$

Moreover, (4.55) gives

$$\frac{e^{\lambda t}}{\Psi(t)} \leq M t^{-4}, \quad \forall t \in [1, L].$$

Since $\lambda \leq 1$, we may therefore bound

$$\int_1^L \frac{(2t + \lambda t^2) e^{\lambda t}}{\Psi(t)} dt \leq M \int_1^\infty (2t + t^2) t^{-4} dt \leq M.$$

The omitted contribution from $(0, 1]$ in the layer-cake decomposition is bounded by a universal constant. Putting these estimates together, and absorbing that bounded term into the numerical constant, we get

$$\mathbb{E}[e^{\lambda Y_k}] \leq 1 + \lambda^2(3 + M + C_\Psi) \leq \exp(\lambda^2(3 + M + C_\Psi)).$$

Inserting this into (4.59), we find

$$\mathbb{P}\left[\sum_{k=1}^m Y_k > a\right] \leq \exp(-\lambda a + \lambda^2 m(3 + M + C_\Psi)).$$

Combining this with (4.57) and (4.58) completes the proof. $\square$

4.2.6 Stretched-exponential concentration

We now combine the two preceding propositions into a single concentration bound for centered independent stretched-exponential random variables in the full range $\sigma \in (0, 2]$.

Corollary 4.19 (Concentration for centered independent $\mathcal{O}_{\Gamma_\sigma}$ random variables). *For every $\sigma \in (0, 2]$, there exists a constant $C(\sigma) < \infty$ such that, for every $m \in \mathbb{N}$ and every finite family of independent measurable integrable random variables $X_1, \dots, X_m$ satisfying*

$$\mathbb{E}[X_k] = 0 \quad \text{and} \quad X_k = \mathcal{O}_{\Gamma_\sigma}(1), \quad \forall k \in \{1, \dots, m\},$$

we have

$$\sum_{k=1}^m X_k = \mathcal{O}_{\Gamma_\sigma}(C(\sigma)m^{1/2}). \quad (4.60)$$

Proof. If $\sigma \in [1, 2]$, then this is exactly Proposition 4.17. We may therefore assume that $\sigma \in (0, 1)$.

Set

$$S_m := \sum_{k=1}^m X_k.$$

Since $\Psi = \Gamma_\sigma$ satisfies

$$C_{\Gamma_\sigma} = \int_1^\infty t e^{-t^\sigma} dt < \infty,$$

Proposition 4.18 is applicable once we verify (4.55).

Choose $M_\sigma \in [1, \infty)$ so large that

$$4 \log t \leq \frac{1}{2} t^\sigma + \log M_\sigma, \quad \forall t \in [1, \infty). \quad (4.61)$$

Then every $\lambda \in (0, 1]$ and every $L \in [1, (2\lambda)^{-1/1-\sigma}]$ satisfy

$$\lambda t \leq \frac{1}{2} t^\sigma \leq t^\sigma - 4 \log t + \log M_\sigma, \quad \forall t \in [1, L].$$

In other words, (4.55) holds with $\Psi = \Gamma_\sigma$ and $M = M_\sigma$.

The case $m = 0$ is trivial, so we assume $m \geq 1$. Set

$$A_\sigma := 3 + M_\sigma + C_{\Gamma_\sigma} \quad \text{and} \quad B_\sigma := 16A_\sigma^{1/2} \quad \text{and} \quad r_0(m) := m^{\frac{\sigma}{2(2-\sigma)}}.$$

Fix $r \in [1, \infty)$ and set

$$a := B_\sigma m^{1/2} r.$$

Case 1: $1 \leq r \leq r_0(m)$. Set

$$\lambda := \frac{r}{4A_\sigma^{1/2} m^{1/2}}.$$

Since $A_\sigma \geq 4$ and $r_0(m) \leq m^{1/2}$, we have $\lambda \leq 1$. We then choose

$$L := (2\lambda)^{-1/1-\sigma}.$$

Proposition 4.18 yields

$$\begin{aligned} \mathbb{P}[S_m > a] &\leq m \exp(-L^\sigma) + \exp(-\lambda a + A_\sigma \lambda^2 m) \\ &\leq m \exp\left(-c_\sigma \left(\frac{m^{1/2}}{r}\right)^{\frac{\sigma}{1-\sigma}}\right) + \exp(-2r^2). \end{aligned}$$

Since $r \leq r_0(m)$, the first term is bounded by $\exp(-2r^\sigma)$ on the whole regime. The second term is also bounded by $\exp(-2r^\sigma)$, because $r \geq 1$ and $\sigma < 1$. Thus

$$\mathbb{P}[S_m > a] \leq e^{-r^\sigma}.$$

Case 2: $r \geq r_0(m)$. Set

$$\lambda := \frac{m^{\frac{\sigma-1}{2}} r^{\sigma-1}}{4A_\sigma^{1/2}}.$$

Since $\sigma - 1 < 0$, $m \geq 1$, $r \geq 1$, and $A_\sigma \geq 4$, we have $\lambda \leq 1$. We again set

$$L := (2\lambda)^{-1/(1-\sigma)}.$$

Applying Proposition 4.18, we obtain

$$\begin{aligned} \mathbb{P}[S_m > a] &\leq m \exp(-L^\sigma) + \exp(-\lambda a + A_\sigma \lambda^2 m) \\ &\leq m \exp(-c_\sigma A_\sigma^{\frac{\sigma}{2(1-\sigma)}} m^{\sigma/2} r^\sigma) + \exp(-\lambda a + A_\sigma \lambda^2 m). \end{aligned}$$

The prefactor m can be absorbed into the exponential after decreasing c_σ if necessary, since $m^{\sigma/2} r^\sigma \geq m^\sigma \geq \log(1+m)$. Moreover,

$$\lambda a = 4m^{\sigma/2} r^\sigma$$

while

$$A_\sigma \lambda^2 m = \frac{1}{16} m^\sigma r^{2\sigma-2}.$$

Because $r \geq r_0(m)$ and $\sigma < 1$, we have

$$m^\sigma r^{2\sigma-2} \leq m^{\sigma/2} r^\sigma.$$

We therefore obtain

$$-\lambda a + A_\sigma \lambda^2 m \leq -3m^{\sigma/2} r^\sigma \leq -r^\sigma.$$

This proves

$$\mathbb{P}[S_m > B_\sigma m^{1/2} r] \leq e^{-r^\sigma}, \quad \forall r \in [1, \infty),$$

which is exactly (4.60). □

4.2.7 Log-normal concentration

The log-normal scale is slightly subtler: one again uses the truncation argument, but the admissible truncation level grows like $\lambda^{-1} \log^2(\lambda^{-1})$ rather than a power of λ^{-1} .

Lemma 4.20 (Log-normal admissibility). *Let $\sigma \in [1, \infty)$ and set*

$$q_\sigma(t) := \log \Psi_\sigma(t) = \frac{1}{\sigma^2} \log^2(1 + \sigma t).$$

If $L \geq 1$, $0 \leq \lambda \leq 1$, and

$$\lambda \leq \frac{1}{2} \frac{q_\sigma(L)}{L},$$

then

$$\lambda t \leq \frac{1}{2} q_\sigma(t) + e^2, \quad \forall t \in [1, L].$$

Consequently the logarithmic constraint (4.55) holds with

$$M_\sigma^{\log} := \exp(e^2 + 8\sigma^2).$$

Proof. The point is that, on the large range, the needed comparison between the values at L and at t follows from the monotonicity of $\log x/\sqrt{x}$ on $[e^2, \infty)$. In the formal proof this is applied to $x = 1 + \sigma t$. Thus, if $1 + \sigma t \geq e^2$ and $t \leq L$, the hypothesis on λ gives

$$\lambda t \leq \frac{1}{2} q_\sigma(t).$$

On the remaining range $1 + \sigma t \leq e^2$, we use only $\lambda \leq 1$ and $\sigma \geq 1$, which imply $t \leq e^2$, and hence

$$\lambda t \leq e^2 \leq \frac{1}{2} q_\sigma(t) + e^2.$$

This proves the displayed local control.

Finally,

$$4 \log t \leq \frac{1}{2} q_\sigma(t) + 8\sigma^2, \quad t \geq 1,$$

so the local control implies

$$\lambda t \leq q_\sigma(t) - 4 \log t + e^2 + 8\sigma^2 = \log \Psi_\sigma(t) - 4 \log t + \log M_\sigma^{\log}.$$

□

Corollary 4.21 (Concentration for centered independent $\mathcal{O}_{\Psi_\sigma}$ random variables). *For every $\sigma \in [1, \infty)$, there exists a constant $C(\sigma) < \infty$ such that, for every $m \in \mathbb{N}$ and every finite family of independent measurable integrable random variables $X_1, \dots, X_m$ satisfying*

$$\mathbb{E}[X_k] = 0 \quad \text{and} \quad X_k = \mathcal{O}_{\Psi_\sigma}(1), \quad \forall k \in \{1, \dots, m\},$$

we have

$$\sum_{k=1}^m X_k = \mathcal{O}_{\Psi_\sigma}(C(\sigma)m^{1/2}). \quad (4.62)$$

Proof. Set

$$S_m := \sum_{k=1}^m X_k.$$

Since Ψ_σ satisfies (4.30), the quantity C_{Ψ_σ} in (4.54) is finite. Put

$$M_\sigma^{\log} := \exp(e^2 + 8\sigma^2), \quad \text{and} \quad D_\sigma := 3 + M_\sigma^{\log} + C_{\Psi_\sigma}, \quad \text{and} \quad B_\sigma := 128 \exp(\sigma^2/2) D_\sigma^{1/2}.$$

The case $m = 0$ is trivial, so assume $m \geq 1$. Fix $r \in [1, \infty)$ and set

$$a := B_\sigma m^{1/2} r, \quad q_\sigma(r) := \log \Psi_\sigma(r).$$

Choose

$$L := \frac{(1 + \sigma r) \exp(\sigma \sqrt{\log(4m)}) - 1}{\sigma} \quad \text{and} \quad \lambda := \frac{4(q_\sigma(r) + \log 2)}{B_\sigma m^{1/2} r}.$$

Then $L \geq 1$ and $\lambda \leq 1$. Indeed $q_\sigma(r) + \log 2 \leq 9r$ for $r \geq 1$, while $B_\sigma m^{1/2} \geq 36$.

We next check that Lemma 4.20 applies on $[1, L]$. Since

$$1 + \sigma L = (1 + \sigma r) \exp(\sigma \sqrt{\log(4m)}),$$

we have

$$q_\sigma(L) \geq q_\sigma(r) + \log(4m) = q_\sigma(r) + \log(2m) + \log 2. \quad (4.63)$$

Moreover the definition of B_σ gives

$$8L \leq B_\sigma m^{1/2} r.$$

Consequently

$$\lambda \leq \frac{1}{2} \frac{q_\sigma(L)}{L},$$

and the logarithmic constraint (4.55) holds with $M = M_\sigma^{\log}$.

Applying the symmetric form of Proposition 4.18 to X_k and to $-X_k$, we get

$$\mathbb{P}[|S_m| > a] \leq 2 \frac{m}{\Psi_\sigma(L)} + 2 \exp(-\lambda a + D_\sigma \lambda^2 m). \quad (4.64)$$

The two terms are absorbed by the following logarithmic inequalities:

$$\log 2 + q_\sigma(r) + \log 2 + D_\sigma \lambda^2 m \leq \lambda B_\sigma m^{1/2} r, \quad (4.65)$$

$$\log(2m) + q_\sigma(r) + \log 2 \leq q_\sigma(L). \quad (4.66)$$

The second is exactly (4.63). For the first, note that

$$\lambda B_\sigma m^{1/2} r = 4(q_\sigma(r) + \log 2)$$

and the chosen value of B_σ implies $D_\sigma \lambda^2 m \leq q_\sigma(r) + \log 2$. Combining (4.64) with (4.65) and (4.66) yields

$$\mathbb{P}[|S_m| > B_\sigma m^{1/2} r] \leq \exp(-q_\sigma(r)) = \frac{1}{\Psi_\sigma(r)}, \quad \forall r \in [1, \infty).$$

This is precisely (4.62). $\square$

4.3 Partition averages and annealed quantities

We now return to the local cube observables from Section 4.1. After a deterministic coloring of the partition, the concentration toolkit from Section 4.2 yields fluctuation bounds for partition averages. Taking expectations in the subadditivity inequalities then isolates the annealed quantities that drive the next chapter.

4.3.1 Fluctuation bounds for partition averages

We now combine the locality and coloring lemmas with the concentration estimates from Section 4.2. The result is that partition averages of local cube observables fluctuate on the scale of the square root of the number of effectively independent cubes.

Proposition 4.22 (Fluctuation bounds for partition averages of local observables). *Assume (P1) and (P2). Let $n, m \in \mathbb{Z}$ with $0 \leq n \leq m$, and let X be an integrable $\mathcal{F}(\square_n)$ -measurable random variable. For every $z \in 3^n \mathbb{Z}^d$, define*

$$X_z := X \circ \tau_z.$$

Then $\mathbb{E}[X_z] = \mathbb{E}[X]$ for every $z \in 3^n \mathbb{Z}^d$, and the following statements hold.

- (i) *For every $p \in [2, \infty)$ such that $\mathbb{E}[|X - \mathbb{E}[X]|^p] < \infty$, there exists a constant $C(d) < \infty$ such that*

$$\begin{aligned} \mathbb{E} \left[\left| \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (X_z - \mathbb{E}[X]) \right|^p \right]^{1/p} &\leq C(d) \left(\sqrt{p} (1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \right. \\ &\quad \left. + p (1 + 3^{-n})^{d(1-1/p)} 3^{-d(1-1/p)(m-n)} \right) \mathbb{E}[|X - \mathbb{E}[X]|^p]^{1/p}. \end{aligned} \quad (4.67)$$

(ii) For every $\sigma \in (0, 2]$ and $\theta \in (0, \infty)$, there exists a constant $C(d, \sigma) < \infty$ such that

$$X - \mathbb{E}[X] = \mathcal{O}_{\Gamma_\sigma}(\theta) \implies \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (X_z - \mathbb{E}[X]) = \mathcal{O}_{\Gamma_\sigma} \left(C(d, \sigma) (1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \theta \right). \quad (4.68)$$

(iii) For every $\sigma \in [1, \infty)$ and $\theta \in (0, \infty)$, there exists a constant $C(d, \sigma) < \infty$ such that

$$X - \mathbb{E}[X] = \mathcal{O}_{\Psi_\sigma}(\theta) \implies \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (X_z - \mathbb{E}[X]) = \mathcal{O}_{\Psi_\sigma} \left(C(d, \sigma) (1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \theta \right). \quad (4.69)$$

The integrability assumption makes the centering explicit; in the formal version of the log-normal clause, the corresponding L^1 hypotheses on the origin observable and on the translated descendants are also kept explicit.

Proof. Set

$$I := 3^n \mathbb{Z}^d \cap \square_m \quad \text{and} \quad N := \#I = 3^{d(m-n)}.$$

Also set

$$Y_z := X_z - \mathbb{E}[X], \quad \forall z \in I.$$

Since $0 \leq n$, every $z \in 3^n \mathbb{Z}^d$ is an integer translation vector. Since X is $\mathcal{F}(\square_n)$ -measurable, stationarity implies that each X_z is $\mathcal{F}(z + \square_n)$ -measurable and has the same law as X . In particular, each Y_z is centered, $\mathcal{F}(z + \square_n)$ -measurable, and has the same distribution as $X - \mathbb{E}[X]$.

Let

$$M := \#\Xi_n.$$

For every $\xi \in \Xi_n$, set

$$I_\xi := I \cap \mathcal{Z}_{n, \xi}.$$

By Lemma 4.5, the family $\{I_\xi\}_{\xi \in \Xi_n}$ is a partition of I , and (4.10) gives

$$M \leq C(d)(1 + 3^{-n})^d. \quad (4.70)$$

Moreover, by the same-color separation (4.9) and Lemma 4.4, the family $\{Y_z\}_{z \in I_\xi}$ is independent for every $\xi \in \Xi_n$.

To prove part (i), let

$$K_p := \mathbb{E}[|X - \mathbb{E}[X]|^p]^{1/p}.$$

By Corollary 4.16, for every $\xi \in \Xi_n$ we have

$$\mathbb{E} \left[\left| \sum_{z \in I_\xi} Y_z \right|^p \right]^{1/p} \leq C \left(\sqrt{p} |I_\xi|^{1/2} + p |I_\xi|^{1/p} \right) K_p.$$

Using the triangle inequality in $L^p(\Omega)$, we deduce that

$$\begin{aligned} \mathbb{E} \left[\left| \sum_{z \in I} Y_z \right|^p \right]^{1/p} &\leq N^{-1} \sum_{\xi \in \Xi_n} \mathbb{E} \left[\left| \sum_{z \in I_\xi} Y_z \right|^p \right]^{1/p} \\ &\leq CN^{-1} \left(\sqrt{p} \sum_{\xi \in \Xi_n} |I_\xi|^{1/2} + p \sum_{\xi \in \Xi_n} |I_\xi|^{1/p} \right) K_p. \end{aligned}$$

By Hölder's inequality,

$$\sum_{\xi \in \Xi_n} |I_\xi|^{1/2} \leq M^{1/2} N^{1/2} \quad \text{and} \quad \sum_{\xi \in \Xi_n} |I_\xi|^{1/p} \leq M^{1-1/p} N^{1/p}.$$

Inserting these bounds together with (4.70) yields (4.67).

For part (ii), assume that

$$X - \mathbb{E}[X] = \mathcal{O}_{\Gamma_\sigma}(\theta).$$

Then $Y_z = \mathcal{O}_{\Gamma_\sigma}(\theta)$ for every $z \in I$. Applying Corollary 4.19 on each color class, we obtain

$$\sum_{z \in I_\xi} Y_z = \mathcal{O}_{\Gamma_\sigma}(C(\sigma)|I_\xi|^{1/2}\theta), \quad \forall \xi \in \Xi_n.$$

Using the generalized triangle inequality in Proposition 4.10, we get

$$\sum_{z \in I} Y_z = \mathcal{O}_{\Gamma_\sigma}\left(C(\sigma) \sum_{\xi \in \Xi_n} |I_\xi|^{1/2}\theta\right).$$

Dividing by N and using the bounds above on $\sum_{\xi \in \Xi_n} |I_\xi|^{1/2}$ and M yields (4.68).

For part (iii), assume that

$$X - \mathbb{E}[X] = \mathcal{O}_{\Psi_\sigma}(\theta).$$

Applying Corollary 4.21 on each color class, we get

$$\sum_{z \in I_\xi} Y_z = \mathcal{O}_{\Psi_\sigma}(C(\sigma)|I_\xi|^{1/2}\theta), \quad \forall \xi \in \Xi_n.$$

Using the generalized triangle inequality (4.18), we infer that

$$\sum_{z \in I} Y_z = \mathcal{O}_{\Psi_\sigma}\left(C(\sigma) \sum_{\xi \in \Xi_n} |I_\xi|^{1/2}\theta\right).$$

After dividing by N and invoking the same counting bounds as above, we obtain (4.69). $\square$

We will also need a version of the partition-average bound that uses only a finite ξ th moment.

Corollary 4.23 (Finite-moment fluctuation of partition averages). *Assume (P1) and (P2). Let $\xi \in [2, \infty)$, $n, m \in \mathbb{Z}$ with $0 \leq n \leq m$, and let X be an $\mathcal{F}(\square_n)$ -measurable random variable satisfying $\mathbb{E}[|X|^\xi] < \infty$. For every $z \in 3^n \mathbb{Z}^d$, set*

$$X_z := X \circ \tau_z.$$

Then there exists a constant $C(d, \xi) < \infty$ such that

$$\mathbb{E}\left[\left|\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (X_z - \mathbb{E}[X])\right|\right] \leq C(d, \xi)(1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \mathbb{E}[|X - \mathbb{E}[X]|^\xi]^{1/\xi}. \quad (4.71)$$

Proof. Set

$$I := 3^n \mathbb{Z}^d \cap \square_m, \quad Y_z := X_z - \mathbb{E}[X].$$

By Jensen's inequality and (4.67) with $p = 2$,

$$\begin{aligned} \mathbb{E}\left[\left|\sum_{z \in I} Y_z\right|\right] &\leq \mathbb{E}\left[\left|\sum_{z \in I} Y_z\right|^2\right]^{1/2} \leq C(d)(1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \mathbb{E}[|X - \mathbb{E}[X]|^2]^{1/2} \\ &\leq C(d, \xi)(1 + 3^{-n})^{d/2} 3^{-d/2(m-n)} \mathbb{E}[|X - \mathbb{E}[X]|^\xi]^{1/\xi}, \end{aligned}$$

where the last step uses the monotonicity of L^p norms since $\xi \geq 2$. This proves (4.71). $\square$

The most important instance for us is the response functional J .

Corollary 4.24 (Fluctuations of partition averages of J). *Assume (P1) and (P2). Let $n, m \in \mathbb{Z}$ with $0 \leq n \leq m$, and let $p, q \in \mathbb{R}^d$. Assume that the response observables below are integrable on the origin cube and on all scale- n descendants of $\square_m$. Define*

$$\mathcal{J}_{n,m}(p, q) := \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}).$$

Then

$$\mathbb{E}[\mathcal{J}_{n,m}(p, q)] = \mathbb{E}[J(\square_n, p, q; \mathbf{a})].$$

Moreover, whenever the corresponding moment or tail hypotheses in Proposition 4.22 are satisfied by the observable

$$X(\omega) := J(\square_n, p, q; \mathbf{a}(\omega, \cdot)),$$

then all three conclusions of that proposition apply. For the log-normal conclusion, these L^1 hypotheses are exactly the explicit integrability assumptions kept in the formal statement.

Proof. The identity for the expectation follows from the translation covariance in (4.7), stationarity, and integrability. The rest is an immediate consequence of Lemma 4.3 and Proposition 4.22. $\square$

4.3.2 Annealed subadditivity

The deterministic subadditivity statements from Chapter 2 become, after taking expectations, monotonicity statements for the annealed quantities. These are the deterministic backbone of the stochastic argument in Chapter 5.

In the formal statement, the deterministic hypotheses are kept visible: at the two scales under comparison, and at all scale- n descendants of the coarse cube, we assume the open-cube elliptic recovery identities from Chapters 2–3 together with the measurability hypotheses and the L^1 integrability needed to take expectations of the response observables and of the entries of the doubled coarse block matrices. In the notation below we refer to this as the recovery/measurability/integrability package at the pair of scales (n, m) .

Proposition 4.25 (Annealed subadditivity of the response functional and the block matrices). *Assume (P1). Let $0 \leq n \leq m$ be triadic scales for which the recovery/measurability/integrability package holds at (n, m) . Then, for every $p, q \in \mathbb{R}^d$ and every $P \in \mathbb{R}^{2d}$, we have*

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a})] \leq \mathbb{E}[J(\square_n, p, q; \mathbf{a})], \quad (4.72)$$

$$P \cdot \overline{\mathbf{A}}_m P \leq P \cdot \overline{\mathbf{A}}_n P, \quad (4.73)$$

and

$$P \cdot \overline{\mathbf{A}}_{*,m}^{-1} P \leq P \cdot \overline{\mathbf{A}}_{*,n}^{-1} P. \quad (4.74)$$

Proof. The subadditivity of the response functional, (2.32), gives

$$J(\square_m, p, q; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}).$$

Taking expectations and using the translation covariance from (4.7), we obtain

$$\begin{aligned}\mathbb{E}[J(\square_m, p, q; \mathbf{a})] &\leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbb{E}[J(z + \square_n, p, q; \mathbf{a})] \\ &= \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbb{E}[J(\square_n, p, q; \tau_z \mathbf{a})] \\ &= \mathbb{E}[J(\square_n, p, q; \mathbf{a})].\end{aligned}$$

This proves (4.72).

Next, the block-matrix subadditivity (2.53) yields

$$\mathbf{A}(\square_m; \mathbf{a}) \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{A}(z + \square_n; \mathbf{a}).$$

Taking expectations and using (4.6), we get

$$\overline{\mathbf{A}}_m \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbb{E}[\mathbf{A}(z + \square_n; \mathbf{a})] = \overline{\mathbf{A}}_n.$$

Testing this matrix inequality against $P \in \mathbb{R}^{2d}$ gives (4.73). The proof of (4.74) is identical, using (2.54) instead. $\square$

Corollary 4.26 (Monotonicity of the annealed coarse-grained matrices). *Assume (P1). Let $0 \leq n \leq m$ be triadic scales for which the recovery/measurability/integrability package holds at (n, m) . Then*

$$\overline{\mathbf{s}}_{*,m}^{-1} \leq \overline{\mathbf{s}}_{*,n}^{-1} \quad \text{and} \quad \overline{\mathbf{b}}_m \leq \overline{\mathbf{b}}_n. \quad (4.75)$$

If, in addition, (P3) holds, then

$$\overline{\mathbf{s}}_{*,n} \leq \overline{\mathbf{s}}_{*,m} \leq \overline{\mathbf{s}}_m \leq \overline{\mathbf{s}}_n. \quad (4.76)$$

Equivalently, if

$$\overline{\mathbf{s}}_n = \overline{\sigma}_n \mathbf{I}_d \quad \text{and} \quad \overline{\mathbf{s}}_{*,n} = \overline{\sigma}_{*,n} \mathbf{I}_d,$$

as in (4.4), then

$$\overline{\sigma}_{*,n} \leq \overline{\sigma}_{*,m} \leq \overline{\sigma}_m \leq \overline{\sigma}_n. \quad (4.77)$$

In particular, the scalar contrast ratio

$$\overline{\sigma}_n \overline{\sigma}_{*,n}^{-1} \quad (4.78)$$

is nonincreasing in n and always at least 1.

Proof. Testing the block-matrix inequality (4.73) on vectors supported in the lower-right and upper-left components gives exactly (4.75), since the lower-right block of $\overline{\mathbf{A}}_j$ is $\overline{\mathbf{s}}_{*,j}^{-1}$ and the upper-left block is $\overline{\mathbf{b}}_j$.

Assume now also (P3). By Lemma 4.1, we have

$$\overline{\mathbf{k}}_n = \overline{\mathbf{k}}_m = 0.$$

Therefore $\overline{\mathbf{s}}_j = \overline{\mathbf{b}}_j$ for $j \in \{n, m\}$. The first inequality in (4.75) gives

$$\overline{\mathbf{s}}_{*,n} \leq \overline{\mathbf{s}}_{*,m}.$$

The second gives $\overline{\mathbf{s}}_m \leq \overline{\mathbf{s}}_n$.

It remains to prove that $\bar{\mathbf{s}}_{*,m} \leq \bar{\mathbf{s}}_m$. By the operator convexity of matrix inversion,

$$\bar{\mathbf{s}}_{*,m} = \mathbb{E}[\mathbf{s}_*^{-1}(\square_m; \mathbf{a})]^{-1} \leq \mathbb{E}[\mathbf{s}_*(\square_m; \mathbf{a})].$$

Since

$$\mathbf{s}_*(\square_m; \mathbf{a}) \leq \mathbf{s}(\square_m; \mathbf{a}) \leq \mathbf{b}(\square_m; \mathbf{a})$$

by (2.61) and the definition of $\mathbf{b}$, we obtain

$$\bar{\mathbf{s}}_{*,m} \leq \mathbb{E}[\mathbf{b}(\square_m; \mathbf{a})].$$

Finally, because $\bar{\mathbf{k}}_m = 0$, the definition (4.1) gives

$$\bar{\mathbf{s}}_m = \bar{\mathbf{b}}_m = \mathbb{E}[\mathbf{b}(\square_m; \mathbf{a})].$$

This proves (4.76). The scalar formulation (4.77) is just (4.76) written in the basis of (4.4). The final claim about the ratio in (4.78) is then immediate. $\square$

We next show that the fluctuations of the normalized block response can be strongly controlled once the corresponding one-cell quantity has a stretched-exponential tail.

Lemma 4.27 (Concentration of normalized block responses). *Assume (P1) and (P2). Let $\sigma \in (0, 2]$, $\theta \in (0, \infty)$, and $\mathbf{B} \in \mathbb{R}_{\text{sym},+}^{2d \times 2d}$. Suppose that*

$$\mathbf{J}(\square_0, \mathbf{B}^{-1/2}e, \mathbf{B}^{1/2}e; \mathbf{a}) = \mathcal{O}_{\Gamma_\sigma}(\theta), \quad \forall e \in \mathbb{R}^{2d}, \quad |e| = 1. \quad (4.79)$$

Then there exists $C(d, \sigma) < \infty$ such that, for every $n, m \in \mathbb{Z}$ with $0 \leq n < m$ and every $e \in \mathbb{R}^{2d}$ with $|e| = 1$,

$$\mathbf{J}(\square_m, \mathbf{B}^{-1/2}e, \mathbf{B}^{1/2}e; \mathbf{a}) \leq \mathbb{E}[\mathbf{J}(\square_n, \mathbf{B}^{-1/2}e, \mathbf{B}^{1/2}e; \mathbf{a})] + \mathcal{O}_{\Gamma_\sigma}(C3^{-\frac{d}{2}(m-n)}\theta). \quad (4.80)$$

Proof. Fix $e \in \mathbb{R}^{2d}$ with $|e| = 1$, and set $P_e := \mathbf{B}^{-1/2}e$ and $Q_e := \mathbf{B}^{1/2}e$. Subadditivity of $U \mapsto \mathbf{J}(U, P_e, Q_e; \mathbf{a})$ gives

$$\mathbf{J}(\square_n, P_e, Q_e; \mathbf{a}) \leq \sum_{y \in \mathbb{Z}^d \cap \square_n} \mathbf{J}(y + \square_0, P_e, Q_e; \mathbf{a}).$$

Each summand on the right has the same law as $\mathbf{J}(\square_0, P_e, Q_e; \mathbf{a})$ by stationarity and the translation covariance in (4.6). Applying the triangle inequality for $\mathcal{O}_{\Gamma_\sigma}$ to the scaled summands, and using the nonnegativity of the block response, we obtain, for a constant $C(d, \sigma) < \infty$,

$$\mathbf{J}(\square_n, P_e, Q_e; \mathbf{a}) = \mathcal{O}_{\Gamma_\sigma}(C\theta). \quad (4.81)$$

Using subadditivity again, we have $\square_m$, we have

$$\begin{aligned} \mathbf{J}(\square_m, P_e, Q_e; \mathbf{a}) &\leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{J}(z + \square_n, P_e, Q_e; \mathbf{a}) \\ &= \mathbb{E}[\mathbf{J}(\square_n, P_e, Q_e; \mathbf{a})] + \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (\mathbf{J}(z + \square_n, P_e, Q_e; \mathbf{a}) - \mathbb{E}[\mathbf{J}(\square_n, P_e, Q_e; \mathbf{a})]). \end{aligned}$$

By Lemma 4.3, the observable $X(\mathbf{a}) := \mathbf{J}(\square_n, P_e, Q_e; \mathbf{a})$ is $\mathcal{F}(\square_n)$ -measurable, and it is integrable by (4.81) and (4.29). Its translated copies are the summands above by the translation covariance in (4.6). Applying (4.68) with this X and using (4.81), the nonnegativity of $\mathbf{J}$ and absorbing the factor $(1 + 3^{-n})^{d/2} \leq 2^{d/2}$ into $C(d, \sigma)$, we obtain

$$\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (\mathbf{J}(z + \square_n, P_e, Q_e; \mathbf{a}) - \mathbb{E}[\mathbf{J}(\square_n, P_e, Q_e; \mathbf{a})]) = \mathcal{O}_{\Gamma_\sigma}(C3^{-\frac{d}{2}(m-n)}\theta).$$

This completes the proof. $\square$

Chapter 5

The homogenization scale under unit-range dependence and isotropy

5.1 The scalar quantities to be tracked and the main theorem

The probabilistic assumptions of stationarity, unit-range dependence, and isotropy were fixed in Chapter 4. We now add the quantitative input used in the annealed argument. Since the one-dimensional case is trivial, we assume throughout this chapter that $d \geq 2$.

(P4) *Quantitative coarse-grained ellipticity on the unit cube.* There exist exponents $s_1, s_2 \in [0, 1)$ and $\xi \in (2d, \infty)$ ¹ satisfying

$$s_1 + s_2 < 1 \quad \text{and} \quad s_1 \wedge s_2 > \frac{d}{\xi}$$

such that

$$\mathbb{E}[\Lambda_{s_1,1}^\xi(\square_0; \mathbf{a})]^\frac{1}{\xi} \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_0; \mathbf{a})]^\frac{1}{\xi} < \infty. \quad (5.1)$$

We fix one such triple (s_1, s_2, ξ) for the rest of the chapter.

For every triadic scale n , recall the annealed coarse-grained matrices

$$\begin{cases} \bar{\mathbf{s}}_{*,n} := \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a})]^{-1}, \\ \bar{\mathbf{k}}_n := \bar{\mathbf{s}}_{*,n} \mathbb{E}[\mathbf{s}_*^{-1}(\square_n; \mathbf{a}) \mathbf{k}(\square_n; \mathbf{a})], \\ \bar{\mathbf{b}}_n := \mathbb{E}[\mathbf{b}(\square_n; \mathbf{a})], \\ \bar{\mathbf{s}}_n := \bar{\mathbf{b}}_n - \bar{\mathbf{k}}_n^t \bar{\mathbf{s}}_{*,n}^{-1} \bar{\mathbf{k}}_n. \end{cases}$$

By Lemma 4.1,

$$\bar{\mathbf{s}}_n = \bar{\sigma}_n \mathbf{I}_d, \quad \bar{\mathbf{s}}_{*,n} = \bar{\sigma}_{*,n} \mathbf{I}_d, \quad \bar{\mathbf{k}}_n = 0. \quad (5.2)$$

The annealed contrast is

$$\Theta_n := \bar{\sigma}_n \bar{\sigma}_{*,n}^{-1}. \quad (5.3)$$

The high-moment quantity we track in parallel is

$$\tilde{\Theta}_n := \mathbb{E}[\Lambda_{s_1,1}^\xi(\square_n; \mathbf{a})]^\frac{1}{\xi} \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_n; \mathbf{a})]^\frac{1}{\xi}. \quad (5.4)$$

¹In the Lean formalization, the moment exponent ξ is assumed to be an integer satisfying $\xi > 2d$. The manuscript keeps the customary real-exponent notation.

We also define, for $k \leq n$, the annealed additivity defect

$$\tau_{n,k}(p, q) := \mathbb{E}[J(\square_k, p, q; \mathbf{a})] - \mathbb{E}[J(\square_n, p, q; \mathbf{a})]. \quad (5.5)$$

By the annealed subadditivity of J , we have

$$\tau_{n,k}(p, q) \geq 0.$$

Theorem 5.1 (Convergence of the annealed contrast). *Assume (P1), (P2), (P3), and (P4). Then there exists a constant $C(d, s_1, s_2, \xi) < \infty$ and an exponent $\alpha(d, s_1, s_2, \xi) > 0$ such that, if we define*

$$N_0 := \lceil C \log^2(2 + \tilde{\Theta}_0) \rceil + \lceil C \xi \log(2 + C \xi \tilde{\Theta}_0) \rceil, \quad (5.6)$$

then, for every $n \in \mathbb{N}$,

$$\Theta_{N_0+n} \leq 1 + 3^{-\alpha n}. \quad (5.7)$$

5.2 Annealed scalar identities and moment bounds

We begin by recording the soft scalar consequences of the annealed block subadditivity from Chapter 4. These are the deterministic quantities that the rest of the chapter will track.

Lemma 5.2 (Basic scalar consequences of annealed subadditivity). *Assume (P1), (P2), (P3), and (P4). Then, for every pair of triadic scales $n \leq m$,*

$$1 \leq \Theta_m \leq \Theta_n \leq \tilde{\Theta}_n. \quad (5.8)$$

Moreover, for every $n \in \mathbb{N}$ and every $p, q \in \mathbb{R}^d$,

$$\mathbb{E}[J(\square_n, p, q; \mathbf{a})] = \frac{1}{2} p \cdot \bar{\mathbf{s}}_n p - p \cdot q + \frac{1}{2} q \cdot \bar{\mathbf{s}}_{*,n}^{-1} q, \quad (5.9)$$

and therefore, for every $k \leq n$,

$$\tau_{n,k}(p, q) = \frac{1}{2} p \cdot (\bar{\mathbf{s}}_k - \bar{\mathbf{s}}_n) p + \frac{1}{2} q \cdot (\bar{\mathbf{s}}_{*,k}^{-1} - \bar{\mathbf{s}}_{*,n}^{-1}) q. \quad (5.10)$$

In particular,

$$\tau_{n,k}(p, q) \geq 0. \quad (5.11)$$

Proof. The monotonicity

$$1 \leq \Theta_m \leq \Theta_n$$

is exactly the scalar form (4.78) of Corollary 4.26.

To prove the bound $\Theta_n \leq \tilde{\Theta}_n$, we use (2.72), which gives

$$|\mathbf{b}(\square_n; \mathbf{a})| \leq \Lambda_{s_1,1}(\square_n; \mathbf{a}) \quad \text{and} \quad |\mathbf{s}_*^{-1}(\square_n; \mathbf{a})| \leq \lambda_{s_2,1}^{-1}(\square_n; \mathbf{a}).$$

Taking expectations, using the definitions of $\bar{\mathbf{s}}_n$ and $\bar{\mathbf{s}}_{*,n}$, and then applying Hölder's inequality, we obtain

$$\bar{\sigma}_n \leq \mathbb{E}[\Lambda_{s_1,1}(\square_n; \mathbf{a})] \leq \mathbb{E}[\Lambda_{s_1,1}^\xi(\square_n; \mathbf{a})]^{1/\xi},$$

$$\bar{\sigma}_{*,n}^{-1} \leq \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_n; \mathbf{a})]^{1/\xi}.$$

Multiplying the two inequalities yields

$$\Theta_n = \bar{\sigma}_n \bar{\sigma}_{*,n}^{-1} \leq \tilde{\Theta}_n.$$

This proves (5.8).

Next, by (2.52),

$$J(\square_n, p, q; \mathbf{a}) = \frac{1}{2} \begin{pmatrix} -q \\ p \end{pmatrix} \cdot \mathbf{A}_*^{-1}(\square_n; \mathbf{a}) \begin{pmatrix} -q \\ p \end{pmatrix} - p \cdot q.$$

Taking expectations and using (4.2), (4.4), (4.5), and the identity

$$\mathbf{A}_*^{-1}(\square_n; \mathbf{a}) = \mathbf{R} \mathbf{A}(\square_n; \mathbf{a}) \mathbf{R},$$

we get

$$\overline{\mathbf{A}}_{*,n}^{-1} = \begin{pmatrix} \bar{\mathbf{s}}_{*,n}^{-1} & 0 \\ 0 & \bar{\mathbf{s}}_n \end{pmatrix}.$$

Substituting this into the previous formula proves (5.9). Finally, (5.10) is simply the definition (5.5), and the nonnegativity (5.11) follows from (4.72). $\square$

We next isolate the algebraic quantity that measures the defect of the annealed flow at a fixed scale. In the symmetric theory one can encode this with a single centered response. In the present nonsymmetric setting it is cleaner to write the centered primal and adjoint responses side by side.

For every $m \in \mathbb{N}$, we set

$$\hat{\sigma}_m := (\bar{\sigma}_m \bar{\sigma}_{*,m})^{1/2}. \quad (5.12)$$

For every unit vector $e \in \mathbb{R}^d$, we then define

$$p_e := \hat{\sigma}_m^{-1/2} e, \quad q_e := \hat{\sigma}_m^{1/2} e. \quad (5.13)$$

Finally, for every bounded Lipschitz domain $U \subseteq \mathbb{R}^d$ and every $p, q \in \mathbb{R}^d$, we define the centered responses

$$\tilde{J}_m(U, p, q; \mathbf{a}) := J(U, p, q; \mathbf{a}) - \frac{1}{2} (\bar{\mathbf{s}}_{*,m}^{-1} q - p) \cdot (q - \bar{\mathbf{s}}_m p) \quad (5.14)$$

and

$$\tilde{J}_m^*(U, p, q; \mathbf{a}) := J^*(U, p, q; \mathbf{a}) - \frac{1}{2} (\bar{\mathbf{s}}_{*,m}^{-1} q - p) \cdot (q - \bar{\mathbf{s}}_m p). \quad (5.15)$$

Lemma 5.3 (Centered primal and adjoint responses). *Assume (P1), (P2), (P3), and (P4). Then, for every $m \in \mathbb{N}$ and every $p, q \in \mathbb{R}^d$,*

$$\mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] = \mathbb{E}[\tilde{J}_m^*(\square_m, p, q; \mathbf{a})] = \frac{1}{2} p \cdot (\bar{\mathbf{s}}_m \bar{\mathbf{s}}_{*,m}^{-1} - \mathbf{I}_d) q. \quad (5.16)$$

In particular, for every $e \in \mathbb{R}^d$ with $|e| = 1$,

$$\Theta_m - 1 = \mathbb{E}[\tilde{J}_m(\square_m, p_e, q_e; \mathbf{a}) + \tilde{J}_m^*(\square_m, p_e, q_e; \mathbf{a})]. \quad (5.17)$$

Proof. By (2.9),

$$J(\square_m, p, q; \mathbf{a}) = \frac{1}{2} p \cdot \mathbf{b}(\square_m; \mathbf{a}) p + \frac{1}{2} q \cdot \mathbf{s}_*^{-1}(\square_m; \mathbf{a}) q + q \cdot \mathbf{s}_*^{-1}(\square_m; \mathbf{a}) \mathbf{k}(\square_m; \mathbf{a}) p - p \cdot q.$$

Taking expectations and using (4.1) and (4.5), we obtain

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a})] = \frac{1}{2}p \cdot \bar{\mathbf{s}}_m p - p \cdot q + \frac{1}{2}q \cdot \bar{\mathbf{s}}_{*,m}^{-1} q. \quad (5.18)$$

Likewise, by the adjoint formula for J^* ,

$$J^*(\square_m, p, q; \mathbf{a}) = \frac{1}{2}p \cdot \mathbf{b}(\square_m; \mathbf{a})p + \frac{1}{2}q \cdot \mathbf{s}_*^{-1}(\square_m; \mathbf{a})q - q \cdot \mathbf{s}_*^{-1}(\square_m; \mathbf{a})\mathbf{k}(\square_m; \mathbf{a})p - p \cdot q,$$

and therefore

$$\mathbb{E}[J^*(\square_m, p, q; \mathbf{a})] = \frac{1}{2}p \cdot \bar{\mathbf{s}}_m p - p \cdot q + \frac{1}{2}q \cdot \bar{\mathbf{s}}_{*,m}^{-1} q. \quad (5.19)$$

Expanding the centering term in (5.14) gives

$$\frac{1}{2}(\bar{\mathbf{s}}_{*,m}^{-1} q - p) \cdot (q - \bar{\mathbf{s}}_m p) = \frac{1}{2}q \cdot \bar{\mathbf{s}}_{*,m}^{-1} q - \frac{1}{2}p \cdot q - \frac{1}{2}q \cdot \bar{\mathbf{s}}_{*,m}^{-1} \bar{\mathbf{s}}_m p + \frac{1}{2}p \cdot \bar{\mathbf{s}}_m p.$$

Subtracting this identity from (5.18) and using that $\bar{\mathbf{s}}_m$ and $\bar{\mathbf{s}}_{*,m}$ are scalar matrices yields

$$\mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] = \frac{1}{2}p \cdot (\bar{\mathbf{s}}_m \bar{\mathbf{s}}_{*,m}^{-1} - \mathbf{I}_d)q.$$

The same computation with (5.19) gives the identical formula for $\tilde{J}_m^*$, and this proves (5.16).

Finally, for the special choice (5.13), we have

$$p_e \cdot q_e = 1 \quad \text{and} \quad p_e \cdot \bar{\mathbf{s}}_m \bar{\mathbf{s}}_{*,m}^{-1} q_e = \Theta_m.$$

Therefore each expectation on the right side of (5.16) is equal to $\frac{1}{2}(\Theta_m - 1)$, and (5.17) follows. $\square$

Before turning to the analytic reduction, we record a soft consequence of the concentration machinery of Chapter 4. It shows that the multiscale ellipticity moments improve across scales at a rate compatible with the baseline moment assumption.

Lemma 5.4 (Moment bounds for the coarse-grained ellipticity constants). *Assume (P1), (P2), (P3), and (P4). Then there exists $C(d) < \infty$ such that, for every r, s satisfying*

$$\frac{d}{\xi} < s < r < 1,$$

and set

$$\Gamma_{s,r,\xi} := s^{-2} \left(\frac{\xi}{\frac{d}{2} + \frac{d}{\xi} - r} + (r - s)^{-2} \right). \quad (5.20)$$

Then, for every $m \in \mathbb{N}$,

$$\mathbb{E} \left[\left(\lambda_{r,1}^{-1}(\square_m; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \right)_+^\xi \right]^{1/\xi} \leq C \Gamma_{s,r,\xi} 3^{-(r-\frac{d}{\xi})m} \mathbb{E} \left[\lambda_{s,1}^{-\xi}(\square_0; \mathbf{a}) \right]^{1/\xi} \quad (5.21)$$

and

$$\mathbb{E} \left[\left(\Lambda_{r,1}(\square_m; \mathbf{a}) - \bar{\sigma}_0 \right)_+^\xi \right]^{1/\xi} \leq C \Gamma_{s,r,\xi} 3^{-(r-\frac{d}{\xi})m} \mathbb{E} \left[\Lambda_{s,1}^\xi(\square_0; \mathbf{a}) \right]^{1/\xi}. \quad (5.22)$$

Proof. We prove (5.21); the proof of (5.22) is identical, with $\mathbf{b}$ and the unstarred block subadditivity inequality in place of $\mathbf{s}_*^{-1}$ and the starred one.

Write the square-root representation of $\lambda_{r,1}^{-1}$ and split it into the scales $k \in \{1, \dots, m\}$ and $k \leq 0$. Since the weights in (2.69) form a probability density, Jensen's inequality gives the large-scale bound

$$\lambda_{r,1}^{-1}(\square_m; \mathbf{a}) \leq \mathbf{c}_r \sum_{k=-\infty}^m 3^{-r(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_k; \mathbf{a})|. \quad (5.23)$$

The small scales are estimated before squaring. Grouping them inside unit cubes and using the definition of $\lambda_{s,\infty}^{-1}$ gives

$$\begin{aligned} \mathbf{c}_r \sum_{k=-\infty}^0 3^{-r(m-k)} \max_{y \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(y + \square_k; \mathbf{a})|^{1/2} \\ \leq 3^{-rm} \max_{z \in \mathbb{Z}^d \cap \square_m} \sum_{j=0}^{\infty} 3^{-rj} \max_{y \in 3^{-j} \mathbb{Z}^d \cap (z + \square_0)} |\mathbf{s}_*^{-1}(y + \square_{-j}; \mathbf{a})|^{1/2} \\ \leq C(d)(r-s)^{-1} 3^{-rm} \max_{z \in \mathbb{Z}^d \cap \square_m} \lambda_{s,\infty}^{-1/2}(z + \square_0; \mathbf{a}) \\ \leq C(d)s^{-1}(r-s)^{-1} 3^{-rm} \max_{z \in \mathbb{Z}^d \cap \square_m} \lambda_{s,1}^{-1/2}(z + \square_0; \mathbf{a}). \end{aligned} \quad (5.24)$$

Consequently the small scales contribute, in the estimate of $(\lambda_{r,1}^{-1}(\square_m; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1})_+$, at most

$$C(d)s^{-2}(r-s)^{-2} 3^{-2rm} \max_{z \in \mathbb{Z}^d \cap \square_m} \lambda_{s,1}^{-1}(z + \square_0; \mathbf{a}).$$

Now let $n \in \{1, \dots, m\}$. By (2.54), taking the first diagonal block yields

$$\mathbf{s}_*^{-1}(z + \square_n; \mathbf{a}) \leq \sum_{z' \in \mathbb{Z}^d \cap (z + \square_n)} \mathbf{s}_*^{-1}(z' + \square_0; \mathbf{a}) \quad \forall z \in 3^n \mathbb{Z}^d \cap \square_m.$$

Hence

$$\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_n; \mathbf{a})| \leq \bar{\sigma}_{*,0}^{-1} + \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \sum_{z' \in \mathbb{Z}^d \cap (z + \square_n)} \left(\mathbf{s}_*^{-1}(z' + \square_0; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \mathbf{I}_d \right) \right|. \quad (5.25)$$

We apply Proposition 4.22 entrywise to the local observable

$$X(\omega) := \mathbf{s}_*^{-1}(\square_0; \mathbf{a}(\omega, \cdot)) - \bar{\sigma}_{*,0}^{-1} \mathbf{I}_d.$$

By finite-dimensional norm equivalence and stationarity,

$$\begin{aligned} \mathbb{E} \left[\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \sum_{z' \in \mathbb{Z}^d \cap (z + \square_n)} \left(\mathbf{s}_*^{-1}(z' + \square_0; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \mathbf{I}_d \right) \right|^\xi \right]^{1/\xi} \\ \leq C(d) 3^{\frac{d}{\xi}(m-n)} \xi 3^{-\frac{d}{2}n} \mathbb{E} \left[|\mathbf{s}_*^{-1}(\square_0; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \mathbf{I}_d|^\xi \right]^{1/\xi} \\ \leq C(d) \xi s^{-2} 3^{-\frac{d}{2}n + \frac{d}{\xi}(m-n)} \mathbb{E} \left[\lambda_{s,1}^{-\xi}(\square_0; \mathbf{a}) \right]^{1/\xi}, \end{aligned} \quad (5.26)$$

where the final inequality follows from (2.73) on the unit cube and the elementary bound $1 - 3^{-s} \geq cs$ for $s \in (0, 1)$.

Combining (5.23), (5.24), (5.25), and (5.26), and then using the triangle inequality in $L^\xi(\Omega)$, we obtain

$$\begin{aligned}
& \mathbb{E} \left[\left(\lambda_{r,1}^{-1}(\square_m; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \right)_+^\xi \right]^{1/\xi} \\
& \leq C(d) \sum_{n=1}^m 3^{-r(m-n)} \mathbb{E} \left[\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \sum_{z' \in \mathbb{Z}^d \cap (z + \square_n)} \left(\mathbf{s}_*^{-1}(z' + \square_0; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1} \mathbf{I}_d \right) \right|^\xi \right]^{1/\xi} \\
& \quad + C(d) s^{-2} (r-s)^{-2} 3^{-2rm} \mathbb{E} \left[\max_{z \in \mathbb{Z}^d \cap \square_m} \lambda_{s,1}^{-\xi}(z + \square_0; \mathbf{a}) \right]^{1/\xi} \\
& \leq C(d) s^{-2} \xi \mathbb{E} \left[\lambda_{s,1}^{-\xi}(\square_0; \mathbf{a}) \right]^{1/\xi} \sum_{n=1}^m 3^{-r(m-n)} 3^{-\frac{d}{2}n + \frac{d}{\xi}(m-n)} \\
& \quad + C(d) s^{-2} (r-s)^{-2} 3^{-2rm + \frac{d}{\xi}m} \mathbb{E} \left[\lambda_{s,1}^{-\xi}(\square_0; \mathbf{a}) \right]^{1/\xi}.
\end{aligned}$$

Since

$$3^{-r(m-n)} 3^{-\frac{d}{2}n + \frac{d}{\xi}(m-n)} = 3^{-(r-\frac{d}{\xi})m} 3^{-(\frac{d}{2} + \frac{d}{\xi} - r)n},$$

and since $r < \frac{d}{2} + \frac{d}{\xi}$, we have

$$\frac{d}{2} + \frac{d}{\xi} - r > 0,$$

so the geometric series is bounded by

$$C(d) \left(\frac{d}{2} + \frac{d}{\xi} - r \right)^{-1}.$$

The small-scale term is bounded by the same right side as (5.21), since

$$3^{-2rm + \frac{d}{\xi}m} \leq 3^{-(r-\frac{d}{\xi})m}.$$

This proves (5.21). □

5.3 Analytic reduction to coarse-grained fluctuations

The next lemma is the soft analytic input of the contraction argument. It controls the annealed response by the additivity defect and by weak norms of the gradient and flux of the maximizer. In the nonsymmetric setting it is best to state the lemma only for J ; the identical adjoint estimate for J^* is then obtained by applying it to the transposed field $\mathbf{a}^t$.

Lemma 5.5 (Upper bound for J by weak norms). *Assume (P1). There exists a constant $C(d) < \infty$ such that, for every $m, k \in \mathbb{N}$ with $k < m$, every $p, q, p_0, q_0 \in \mathbb{R}^d$, and every $s, t \in (0, 1)$ satisfying $s + t \leq 1$, the following statement holds. If*

$$v_m := v(\cdot, \square_m, p, q; \mathbf{a}),$$

then

$$\begin{aligned}
& \mathbb{E}[J(\square_m, p, q; \mathbf{a})] - \frac{1}{2}p_0 \cdot q_0 \\
& \leq C\tau_{m,k}(p, q)^{1/2} \mathbb{E}[J(\square_k, p, q; \mathbf{a})]^{1/2} + C3^{-(m-k)} \mathbb{E}[J(\square_m, p, q; \mathbf{a})] \\
& \quad + C|q_0| \mathbb{E} \left[3^{-sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)} \right] \\
& \quad + C|p_0| \mathbb{E} \left[3^{-tm} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-t}(\square_m)} \right] \\
& \quad + C \mathbb{E} \left[3^{-2sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)}^2 \right]^{1/2} \mathbb{E} \left[3^{-2tm} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-t}(\square_m)}^2 \right]^{1/2}. \tag{5.27}
\end{aligned}$$

Proof. For every $z \in 3^k \mathbb{Z}^d \cap \square_m$, set

$$v_{k,z} := v(\cdot, z + \square_k, p, q; \mathbf{a}).$$

Choose a nonnegative function $\varphi \in C_c^\infty(\square_m)$ such that

$$(\varphi)_{\square_m} = 1 \quad \text{and} \quad \|\varphi\|_{L^\infty(\square_m)} + 3^m \|\nabla \varphi\|_{L^\infty(\square_m)} + 3^{2m} \|\nabla^2 \varphi\|_{L^\infty(\square_m)} \leq C.$$

Let $\ell_{p_0}(x) := p_0 \cdot x + c_0$ be the affine function with slope p_0 chosen so that

$$(v_m - \ell_{p_0})_{\square_m} = 0.$$

By (2.14) and the identity

$$\frac{1}{2} \nabla v_m \cdot \mathbf{s} \nabla v_m - \frac{1}{2} p_0 \cdot q_0 = \frac{1}{2} (\nabla v_m - p_0) \cdot (\mathbf{a} \nabla v_m - q_0) + \frac{1}{2} q_0 \cdot (\nabla v_m - p_0) + \frac{1}{2} p_0 \cdot (\mathbf{a} \nabla v_m - q_0),$$

we obtain

$$\begin{aligned}
& J(\square_m, p, q; \mathbf{a}) - \frac{1}{2} p_0 \cdot q_0 \\
& = \int_{\square_m} \frac{1}{2} \varphi (\nabla v_m - p_0) \cdot (\mathbf{a} \nabla v_m - q_0) \\
& \quad + \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \int_{z + \square_k} \frac{1}{2} ((\varphi)_{z + \square_k} - \varphi) \nabla v_m \cdot \mathbf{s} \nabla v_m \\
& \quad + \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} (1 - (\varphi)_{z + \square_k}) \int_{z + \square_k} \frac{1}{2} \left((\nabla v_m - \nabla v_{k,z}) \cdot \mathbf{s} (\nabla v_m + \nabla v_{k,z}) + \nabla v_{k,z} \cdot \mathbf{s} \nabla v_{k,z} \right) \\
& \quad + \frac{1}{2} q_0 \cdot \int_{\square_m} \varphi (\nabla v_m - p_0) + \frac{1}{2} p_0 \cdot \int_{\square_m} \varphi (\mathbf{a} \nabla v_m - q_0). \tag{5.28}
\end{aligned}$$

We estimate the first term on the right side of (5.28) by integrating by parts. Since $\nabla(v_m - \ell_{p_0}) = \nabla v_m - p_0$,

$$(v_m - \ell_{p_0})_{\square_m} = 0 \quad \text{and} \quad \mathbf{a} \nabla v_m - q_0 \in L_{\text{sol}}^2(\square_m),$$

$$\int_{\square_m} \frac{1}{2} \varphi (\nabla v_m - p_0) \cdot (\mathbf{a} \nabla v_m - q_0) = - \int_{\square_m} \frac{1}{2} (v_m - \ell_{p_0}) \nabla \varphi \cdot (\mathbf{a} \nabla v_m - q_0).$$

Applying the Besov duality estimate (1.24) with exponent $1 - s$, then (1.27) with $u = v_m - \ell_{p_0}$ and parameter $1 - s$, we find

$$\begin{aligned} & \left| \int_{\square_m} \frac{1}{2} \varphi (\nabla v_m - p_0) \cdot (\mathbf{a} \nabla v_m - q_0) \right| \\ & \leq C \| (v_m - \ell_{p_0}) \nabla \varphi \|_{\dot{B}_{2,\infty}^{1-s}(\square_m)} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{s-1}(\square_m)} \\ & \leq C 3^m \| \nabla \varphi \|_{W^{1,\infty}(\square_m)} \| \nabla v_m - p_0 \|_{\dot{B}_{2,1}^{-s}(\square_m)} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{s-1}(\square_m)} \\ & \leq C 3^{-m} \| \nabla v_m - p_0 \|_{\dot{B}_{2,1}^{-s}(\square_m)} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{s-1}(\square_m)}. \end{aligned}$$

Since $s + t \leq 1$, we have $1 - s \geq t$, and therefore the definition of the negative Besov seminorm gives

$$\| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{s-1}(\square_m)} \leq C 3^{(1-s-t)m} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{-t}(\square_m)}.$$

Consequently,

$$\left| \int_{\square_m} \frac{1}{2} \varphi (\nabla v_m - p_0) \cdot (\mathbf{a} \nabla v_m - q_0) \right| \leq C 3^{-sm} \| \nabla v_m - p_0 \|_{\dot{B}_{2,1}^{-s}(\square_m)} 3^{-tm} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{-t}(\square_m)}. \quad (5.29)$$

For the second term on the right side of (5.28), we use the smoothness of the cutoff:

$$\begin{aligned} & \left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \int_{z + \square_k} \frac{1}{2} ((\varphi)_{z + \square_k} - \varphi) \nabla v_m \cdot \mathbf{s} \nabla v_m \right| \\ & \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \| \varphi - (\varphi)_{z + \square_k} \|_{L^\infty(z + \square_k)} \int_{\square_m} \frac{1}{2} \nabla v_m \cdot \mathbf{s} \nabla v_m \leq C 3^{-(m-k)} J(\square_m, p, q; \mathbf{a}). \quad (5.30) \end{aligned}$$

To estimate the third term on the right side of (5.28), we use Cauchy's inequality, the second-variation identity (2.13) on each cube $z + \square_k$, and the energy identity (2.14). This gives

$$\begin{aligned} & \left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} (1 - (\varphi)_{z + \square_k}) \int_{z + \square_k} \frac{1}{2} (\nabla v_m - \nabla v_{k,z}) \cdot \mathbf{s} (\nabla v_m + \nabla v_{k,z}) \right| \\ & \leq C \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \| \mathbf{s}^{1/2} (\nabla v_m - \nabla v_{k,z}) \|_{\underline{L}^2(z + \square_k)}^2 \right)^{1/2} \\ & \quad \times \left(\| \mathbf{s}^{1/2} \nabla v_m \|_{\underline{L}^2(\square_m)}^2 + \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \| \mathbf{s}^{1/2} \nabla v_{k,z} \|_{\underline{L}^2(z + \square_k)}^2 \right)^{1/2} \\ & \leq C \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) \right)^{1/2}. \end{aligned}$$

Finally, the last two terms in (5.28) are estimated by Besov duality. Since the chosen cutoff satisfies

$$\| \varphi \|_{\dot{B}_{2,\infty}^r(\square_m)} \leq C 3^{-rm} \quad \forall r \in (0, 1),$$

we obtain

$$\begin{aligned} & \left| \frac{1}{2} q_0 \cdot \int_{\square_m} \varphi (\nabla v_m - p_0) \right| \leq C |q_0| 3^{-sm} \| \nabla v_m - p_0 \|_{\dot{B}_{2,1}^{-s}(\square_m)}, \\ & \left| \frac{1}{2} p_0 \cdot \int_{\square_m} \varphi (\mathbf{a} \nabla v_m - q_0) \right| \leq C |p_0| 3^{-tm} \| \mathbf{a} \nabla v_m - q_0 \|_{\dot{B}_{2,1}^{-t}(\square_m)}. \quad (5.31) \end{aligned}$$

Combining the previous estimates with (5.28), and recalling that

$$\oint_{z+\square_k} \frac{1}{2} \nabla v_{k,z} \cdot \mathbf{s} \nabla v_{k,z} = J(z + \square_k, p, q; \mathbf{a}),$$

we arrive at

$$\begin{aligned} & \left| J(\square_m, p, q; \mathbf{a}) - \frac{1}{2} p_0 \cdot q_0 - \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} (1 - (\varphi)_{z+\square_k}) J(z + \square_k, p, q; \mathbf{a}) \right| \\ & \leq C \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) \right)^{1/2} \\ & \quad + C 3^{-(m-k)} J(\square_m, p, q; \mathbf{a}) \\ & \quad + C |q_0| 3^{-sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)} + C |p_0| 3^{-tm} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-t}(\square_m)} \\ & \quad + C 3^{-sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)} 3^{-tm} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-t}(\square_m)}. \end{aligned}$$

We now take expectations. Since $(\varphi)_{\square_m} = 1$, stationarity gives

$$\begin{aligned} \mathbb{E} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} (1 - (\varphi)_{z+\square_k}) J(z + \square_k, p, q; \mathbf{a}) \right] &= \mathbb{E}[J(\square_k, p, q; \mathbf{a})] \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} (1 - (\varphi)_{z+\square_k}) \\ &= \mathbb{E}[J(\square_k, p, q; \mathbf{a})] (1 - (\varphi)_{\square_m}) \\ &= 0. \end{aligned}$$

Using the definition of $\tau_{m,k}(p, q)$, the bound (2.32), and Cauchy's inequality for the last product term, we obtain (5.27). $\square$

Applying Lemma 5.5 to the transposed field $\mathbf{a}^t$, and using the compatibility statements (2.55) and (2.45), gives the identical estimate with J^* in place of J . We will use this adjoint version implicitly below.

We next estimate the weak Besov norms of the gradient and flux of the maximizer itself.

Lemma 5.6 (Deterministic weak-norm bounds for the maximizer). *There exists a constant $C(d) < \infty$ such that the following statement holds. Let $m, k \in \mathbb{N}$ satisfy $k < m$, let $p, q, p_0, q_0 \in \mathbb{R}^d$, let $s, t \in (0, 1]$, and let $s' \in [1/2s, s)$ and $t' \in [1/2t, t)$. If*

$$v_m := v(\cdot, \square_m, p, q; \mathbf{a}),$$

and, for every $j \leq m$ and $z \in 3^j \mathbb{Z}^d \cap \square_m$,

$$v_{j,z} := v(\cdot, z + \square_j, p, q; \mathbf{a}),$$

then

$$\begin{aligned} & 3^{-sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)} \\ & \leq \sum_{j=k+1}^m 3^{-s(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \left| \oint_{z+\square_j} \nabla v_{j,z} - p_0 \right|^2 \right)^{1/2} \\ & \quad + C \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) \sum_{j=k+1}^m 3^{-(s-s')(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \\ & \quad + C (s - s')^{-1} 3^{-(s-s')(m-k)} \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})^{1/2} \\ & \quad + C s^{-1} 3^{-s(m-k)} |p_0| \end{aligned} \tag{5.32}$$

and

$$\begin{aligned}
& 3^{-tm} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-t}(\square_m)} \\
& \leq \sum_{j=k+1}^m 3^{-t(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \left| \int_{z+\square_j} \mathbf{a} \nabla v_{j,z} - q_0 \right|^2 \right)^{1/2} \\
& \quad + C \Lambda_{t',1}^{1/2}(\square_m; \mathbf{a}) \sum_{j=k+1}^m 3^{-(t-t')(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \\
& \quad + C(t-t')^{-1} 3^{-(t-t')(m-k)} \Lambda_{t',1}^{1/2}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})^{1/2} \\
& \quad + C t^{-1} 3^{-t(m-k)} |q_0|. \tag{5.33}
\end{aligned}$$

Proof. We first prove (5.32). By the definition of the negative Besov seminorm,

$$\begin{aligned}
3^{-sm} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-s}(\square_m)} &= \sum_{j=-\infty}^m 3^{-s(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_j} - p_0|^2 \right)^{1/2} \\
&\leq \sum_{j=k+1}^m 3^{-s(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_j} - p_0|^2 \right)^{1/2} \\
&\quad + \sum_{j=-\infty}^k 3^{-s(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_j}|^2 \right)^{1/2} + C s^{-1} 3^{-s(m-k)} |p_0|.
\end{aligned}$$

For every $j \in \{k+1, \dots, m\}$,

$$\begin{aligned}
& \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_j} - p_0|^2 \right)^{1/2} \\
& \leq \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \left| \int_{z+\square_j} \nabla v_{j,z} - p_0 \right|^2 \right)^{1/2} + \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m - \nabla v_{j,z})_{z+\square_j}|^2 \right)^{1/2}. \tag{5.34}
\end{aligned}$$

To bound the second term in (5.34), we apply (2.67) on each cube $z + \square_j$ to the harmonic function $v_m - v_{j,z} \in \mathcal{A}(z + \square_j; \mathbf{a})$:

$$\begin{aligned}
& \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m - \nabla v_{j,z})_{z+\square_j}|^2 \\
& \leq C \max_{z \in 3^j \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_j; \mathbf{a})| \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \int_{z+\square_j} \frac{1}{2} (\nabla v_m - \nabla v_{j,z}) \cdot \mathbf{s} (\nabla v_m - \nabla v_{j,z}).
\end{aligned}$$

Now apply the second-variation identity (2.13) on each cube $z + \square_j$, with maximizer $v_{j,z}$ and competitor given by the restriction of v_m , to get

$$\begin{aligned}
& \int_{z+\square_j} \frac{1}{2} (\nabla v_m - \nabla v_{j,z}) \cdot \mathbf{s} (\nabla v_m - \nabla v_{j,z}) \\
& = J(z + \square_j, p, q; \mathbf{a}) - \int_{z+\square_j} \left(-\frac{1}{2} \nabla v_m \cdot \mathbf{s} \nabla v_m - p \cdot \mathbf{a} \nabla v_m + q \cdot \nabla v_m \right).
\end{aligned}$$

Averaging over z and using the partition identity gives

$$\begin{aligned}
& \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \int_{z + \square_j} \frac{1}{2} (\nabla v_m - \nabla v_{j,z}) \cdot \mathbf{s} (\nabla v_m - \nabla v_{j,z}) \\
&= \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - \int_{\square_m} \left(-\frac{1}{2} \nabla v_m \cdot \mathbf{s} \nabla v_m - p \cdot \mathbf{a} \nabla v_m + q \cdot \nabla v_m \right) \\
&= \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}).
\end{aligned}$$

Using the single-scale bound (2.73), we deduce

$$\begin{aligned}
& \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m - \nabla v_{j,z})_{z + \square_j}|^2 \right)^{1/2} \\
& \leq C 3^{s'(m-j)} \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2}. \quad (5.35)
\end{aligned}$$

Combining (5.34) and (5.35), then summing over $j = k + 1, \dots, m$, yields the first two lines of (5.32).

It remains to treat the low scales $j \leq k$. Applying (2.67) directly to the harmonic function v_m on each cube $z + \square_j$, and then using (2.14), gives

$$\begin{aligned}
\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z + \square_j}|^2 & \leq C \max_{z \in 3^j \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_j; \mathbf{a})| \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} \int_{z + \square_j} \frac{1}{2} \nabla v_m \cdot \mathbf{s} \nabla v_m \\
& \leq C \max_{z \in 3^j \mathbb{Z}^d \cap \square_m} |\mathbf{s}_*^{-1}(z + \square_j; \mathbf{a})| J(\square_m, p, q; \mathbf{a}).
\end{aligned}$$

Using again (2.73), we obtain

$$\left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z + \square_j}|^2 \right)^{1/2} \leq C 3^{s'(m-j)} \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})^{1/2}.$$

Therefore

$$\begin{aligned}
& \sum_{j=-\infty}^k 3^{-s(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z + \square_j}|^2 \right)^{1/2} \\
& \leq C \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})^{1/2} \sum_{j=-\infty}^k 3^{-(s-s')(m-j)} \\
& \leq C (s - s')^{-1} 3^{-(s-s')(m-k)} \lambda_{s',1}^{-1/2}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})^{1/2}.
\end{aligned}$$

This completes the proof of (5.32).

The proof of (5.33) is identical. One replaces (2.67) by (2.68), and invokes the upper bound on $\mathbf{b}$ from (2.73) instead of the lower bound on $\mathbf{s}_*$. This gives (5.33). $\square$

Combining the previous two lemmas removes the weak norms from the upper bound for the centered response. We only state the version needed in the contraction argument, namely for the special vectors p_e and q_e .

Lemma 5.7 (Upper bound for the centered response). *Assume (P1), (P2), (P3), and (P4). Let $m, k \in \mathbb{N}$ with $k < m$, let $e \in \mathbb{R}^d$ satisfy $|e| = 1$, and define p_e, q_e by (5.13). Set*

$$p_{0,e} := \bar{\mathbf{s}}_{*,m}^{-1} q_e - p_e, \quad q_{0,e} := q_e - \bar{\mathbf{s}}_m p_e. \quad (5.36)$$

Put

$$\beta := \frac{1}{4} \min \left\{ 1 - s_1 - s_2, s_1, s_2, s_1 - \frac{d}{\xi}, s_2 - \frac{d}{\xi} \right\}, \quad \zeta := \frac{\xi}{\xi - 1}.$$

There exists a constant $C(d, \xi, \beta) < \infty$ such that, for every $\varepsilon \in (0, 1]$,

$$\begin{aligned} & \mathbb{E}[\tilde{J}_m(\square_m, p_e, q_e; \mathbf{a})] \\ & \leq C \tau_{m,k}(p_e, q_e)^{1/2} \mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})]^{1/2} \\ & \quad + C\varepsilon (\Theta_m^{1/2} - 1)^2 \\ & \quad + C\varepsilon^{-1} \Theta_m \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \bar{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \bar{\mathbf{A}}_m) \bar{\mathbf{A}}_m^{-1/2} \right|^2 \right] \\ & \quad + C\varepsilon^{-1} (\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0) \sum_{j=k+1}^m 3^{-\beta(m-j)} \tau_{m,j}(p_e, q_e) \\ & \quad + C\xi \varepsilon^{-1} 3^{-\beta m} \left(\hat{\sigma}_m \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_0; \mathbf{a})]^{1/\xi} + \hat{\sigma}_m^{-1} \mathbb{E}[\Lambda_{s_1,1}^\xi(\square_0; \mathbf{a})]^{1/\xi} \right) \mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})]^{1/\zeta} \\ & \quad + C\varepsilon^{-1} 3^{-2\beta(m-k)} (\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0) (\Theta_m - 1). \end{aligned} \quad (5.37)$$

Proof. Set $p := p_e$, $q := q_e$, $p_0 := p_{0,e}$, $q_0 := q_{0,e}$, and $v_m := v(\cdot, \square_m, p, q; \mathbf{a})$. By (5.14) and (5.36),

$$\tilde{J}_m(\square_m, p, q; \mathbf{a}) = J(\square_m, p, q; \mathbf{a}) - \frac{1}{2} p_0 \cdot q_0.$$

Moreover,

$$\hat{\sigma}_m |p_0|^2 = \hat{\sigma}_m^{-1} |q_0|^2 = (\Theta_m^{1/2} - 1)^2.$$

The definition of β ensures that

$$s_1 + s_2 + 4\beta \leq 1, \quad s_i + \beta \in [1/2(s_i + 2\beta), s_i + 2\beta) \quad \text{for } i \in \{1, 2\}.$$

Applying Lemma 5.5 with $s = s_2 + 2\beta$ and $t = s_1 + 2\beta$, and then Young's inequality with $\theta = \hat{\sigma}_m$, gives

$$\begin{aligned} \mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] & \leq C \tau_{m,k}(p, q)^{1/2} \mathbb{E}[J(\square_k, p, q; \mathbf{a})]^{1/2} + C 3^{-(m-k)} \mathbb{E}[J(\square_m, p, q; \mathbf{a})] \\ & \quad + C\varepsilon (\Theta_m^{1/2} - 1)^2 + C\varepsilon^{-1} (\hat{\sigma}_m \mathbb{E}[X^2] + \hat{\sigma}_m^{-1} \mathbb{E}[Y^2]), \end{aligned}$$

where

$$X := 3^{-(s_2+2\beta)m} \|\nabla v_m - p_0\|_{\dot{B}_{2,1}^{-(s_2+2\beta)}(\square_m)} \quad \text{and} \quad Y := 3^{-(s_1+2\beta)m} \|\mathbf{a} \nabla v_m - q_0\|_{\dot{B}_{2,1}^{-(s_1+2\beta)}(\square_m)}.$$

We estimate $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ by Lemma 5.6, with $s' = s_2 + \beta$ and $t' = s_1 + \beta$. Squaring the two deterministic bounds and using

$$(a_1 + a_2 + a_3 + a_4)^2 \leq 4(a_1^2 + a_2^2 + a_3^2 + a_4^2)$$

gives the low-scale centering tails and the ellipticity-weighted terms. The two coarse-average terms collapse to the block-matrix fluctuation. Indeed, set

$$P_e := \begin{pmatrix} -q_e \\ p_e \end{pmatrix}.$$

Using (2.15), (2.52), and (5.36), we have, for every triadic subcube $U \subseteq \square_m$,

$$\begin{pmatrix} -\left(\oint_U \nabla v(\cdot, U, p, q; \mathbf{a}) - p_0\right) \\ -\left(\oint_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) - q_0\right) \end{pmatrix} = (\mathbf{A}_*^{-1}(U; \mathbf{a}) - \overline{\mathbf{A}}_{*,m}^{-1}) P_e.$$

Since

$$\widehat{\sigma}_m |W_1|^2 + \widehat{\sigma}_m^{-1} |W_2|^2 = \Theta_m^{1/2} |(\overline{\mathbf{A}}_{*,m}^{-1})^{-1/2} W|^2 \quad \forall W = (W_1, W_2) \in \mathbb{R}^d \times \mathbb{R}^d$$

and

$$|(\overline{\mathbf{A}}_{*,m}^{-1})^{1/2} P_e|^2 = 2\Theta_m^{1/2},$$

it follows that

$$\begin{aligned} \widehat{\sigma}_m \left| \oint_U \nabla v(\cdot, U, p, q; \mathbf{a}) - p_0 \right|^2 + \widehat{\sigma}_m^{-1} \left| \oint_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) - q_0 \right|^2 \\ \leq C\Theta_m |\overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(U; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2}|^2. \end{aligned}$$

Here we used $\mathbf{A}_*^{-1}(U; \mathbf{a}) = \mathbf{R}\mathbf{A}(U; \mathbf{a})\mathbf{R}$ and $\overline{\mathbf{A}}_{*,m}^{-1} = \mathbf{R}\overline{\mathbf{A}}_m\mathbf{R}$. Weighted Cauchy's inequality gives

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{j=k+1}^m (3^{-(s_2+2\beta)(m-j)} + 3^{-(s_1+2\beta)(m-j)}) \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |\overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(z + \square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2}|^2 \right)^{1/2} \right)^2 \right] \\ \leq C\beta^{-1} \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |\overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(z + \square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2}|^2 \right] \\ = C\beta^{-1} \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[|\overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2}|^2 \right], \end{aligned}$$

where the last step is stationarity. Multiplying by the prefactor $C\varepsilon^{-1}\Theta_m$ gives the matrix-variance term in (5.37). For the ellipticity-weighted terms, we use the weighted Cauchy inequality

$$\begin{aligned} \left(\sum_{j=k+1}^m 3^{-\beta(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \right)^2 \\ \leq C\beta^{-1} \sum_{j=k+1}^m 3^{-\beta(m-j)} \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right), \end{aligned}$$

and the analogous estimate for the flux contribution. For the remaining ellipticity factors, we do not apply Hölder's inequality to the whole coarse-grained ellipticity. Instead, we split off the deterministic baseline:

$$\lambda_{s_2,1}^{-1}(\square_m; \mathbf{a}) \leq \bar{\sigma}_{*,0}^{-1} + (\lambda_{s_2,1}^{-1}(\square_m; \mathbf{a}) - \bar{\sigma}_{*,0}^{-1})_+$$

and

$$\Lambda_{s_1,1}(\square_m; \mathbf{a}) \leq \bar{\sigma}_0 + (\Lambda_{s_1,1}(\square_m; \mathbf{a}) - \bar{\sigma}_0)_+.$$

By subadditivity (2.32) and the energy identity (2.14), the additivity difference above is nonnegative and bounded by the local average of J on the scale j . Therefore stationarity and Jensen's inequality give

$$\mathbb{E} \left[\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right] = \mathbb{E}[J(\square_j, p, q; \mathbf{a})] - \mathbb{E}[J(\square_m, p, q; \mathbf{a})] = \tau_{m,j}(p, q),$$

and

$$\mathbb{E} \left[\left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^\zeta \right]^{1/\zeta} \leq \mathbb{E}[J(\square_j, p, q; \mathbf{a})^\zeta]^{1/\zeta}.$$

Applying Hölder's inequality only to these positive parts, and then using Lemma 5.4, the definition of β , and absorbing the harmless denominators into C , gives, for every $j \in \{k+1, \dots, m\}$,

$$\begin{aligned} \mathbb{E} \left[\lambda_{s_2,1}^{-1}(\square_m; \mathbf{a}) \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right) \right] \\ \leq \bar{\sigma}_{*,0}^{-1} \tau_{m,j}(p, q) + C \xi 3^{-\beta m} \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_0; \mathbf{a})]^{1/\xi} \times \mathbb{E}[J(\square_j, p, q; \mathbf{a})^\zeta]^{1/\zeta}. \end{aligned}$$

$$\begin{aligned} \mathbb{E} \left[\Lambda_{s_1,1}(\square_m; \mathbf{a}) \left(\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} J(z + \square_j, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right) \right] \\ \leq \bar{\sigma}_0 \tau_{m,j}(p, q) + C \xi 3^{-\beta m} \mathbb{E}[\Lambda_{s_1,1}^\xi(\square_0; \mathbf{a})]^{1/\xi} \times \mathbb{E}[J(\square_j, p, q; \mathbf{a})^\zeta]^{1/\zeta}. \end{aligned}$$

The same argument, with the additivity difference replaced by $J(\square_m, p, q; \mathbf{a})$, gives

$$\begin{aligned} \mathbb{E}[\lambda_{s_2,1}^{-1}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})] &\leq \bar{\sigma}_{*,0}^{-1} \mathbb{E}[J(\square_m, p, q; \mathbf{a})] \\ &\quad + C \xi 3^{-\beta m} \mathbb{E}[\lambda_{s_2,1}^{-\xi}(\square_0; \mathbf{a})]^{1/\xi} \mathbb{E}[J(\square_m, p, q; \mathbf{a})^\zeta]^{1/\zeta}, \\ \mathbb{E}[\Lambda_{s_1,1}(\square_m; \mathbf{a}) J(\square_m, p, q; \mathbf{a})] &\leq \bar{\sigma}_0 \mathbb{E}[J(\square_m, p, q; \mathbf{a})] \\ &\quad + C \xi 3^{-\beta m} \mathbb{E}[\Lambda_{s_1,1}^\xi(\square_0; \mathbf{a})]^{1/\xi} \mathbb{E}[J(\square_m, p, q; \mathbf{a})^\zeta]^{1/\zeta}. \end{aligned}$$

Iterating subadditivity, and then using Jensen's inequality and stationarity, gives, for every $j \in \{k, \dots, m\}$,

$$\mathbb{E}[J(\square_j, p, q; \mathbf{a})^\zeta]^{1/\zeta} \leq \mathbb{E}[J(\square_k, p, q; \mathbf{a})^\zeta]^{1/\zeta}.$$

Moreover, by (5.9) and the special choice of p and q ,

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a})] = \Theta_m^{1/2} - 1 \leq \Theta_m - 1.$$

The definition of β makes it no larger than the two exponent gaps, the two moment gains $s_i - d/\xi$, and the two negative-norm exponents. Therefore the geometric sums are bounded by the corresponding β -weighted sums. The coarse-average terms have already produced the variance term, with one factor of β^{-1} coming from weighted Cauchy's inequality. The baseline parts of the ellipticity-weighted additivity differences give the

$$\beta^{-2} (\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0) \sum_{j=k+1}^m 3^{-\beta(m-j)} \tau_{m,j}(p, q)$$

term after the same weighted Cauchy step and a harmless weakening of the geometric factor. The positive-part moment errors carry one further collapsed summation factor, and are therefore bounded by the single k -scale moment term with coefficient $C\xi\varepsilon^{-1}\beta^{-3}3^{-\beta m}$ in (5.37). Finally, the cutoff contribution and the low-scale centering tails are bounded by the final scale-separation term, using the previous display and

$$(\Theta_m^{1/2} - 1)^2 \leq \Theta_m - 1.$$

We also used the monotonicity $\bar{\sigma}_{*,0} \leq \bar{\sigma}_{*,m} \leq \bar{\sigma}_m \leq \bar{\sigma}_0$, which implies

$$\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} \geq 1 \quad \text{and} \quad \hat{\sigma}_m^{-1} \bar{\sigma}_0 \geq 1,$$

so the standalone constant is absorbed by these two factors. Inserting the previous estimates into the preceding bound proves (5.37). $\square$

5.4 Good scales and variance bounds

We now introduce the scale-selection input. The analytic estimates above did not use pigeonholing; it enters only here, where the annealed scalar flow is almost stationary across a prescribed scale gap unless the contrast has already contracted.

Lemma 5.8 (Pigeonhole lemma). *Assume (P1), (P2), and (P3), and (P4). Let $\delta, \sigma \in (0, 1/2]$ and let $N, h \in \mathbb{N}$ satisfy*

$$N \geq \left\lceil 2\delta^{-1} |\log \sigma| \right\rceil h. \quad (5.38)$$

Then at least one of the following statements holds.

- *There exists a scale parameter $n \in \{h, \dots, N\}$ such that*

$$\bar{\sigma}_{n-h} \leq (1 + \delta) \bar{\sigma}_n \quad \text{and} \quad \bar{\sigma}_{*,n-h}^{-1} \leq (1 + \delta) \bar{\sigma}_{*,n}^{-1}. \quad (5.39)$$

- *The annealed contrast has already decayed by the factor σ :*

$$\Theta_N \leq \sigma \Theta_0. \quad (5.40)$$

Proof. Let

$$k := \left\lceil 2\delta^{-1} |\log \sigma| \right\rceil.$$

If the first alternative fails, then at each scale jh at least one of the two backward ratios in (5.39) is larger than $1 + \delta$, while monotonicity keeps the other forward ratio at most 1. Therefore, by (4.77) and $N \geq kh$,

$$\Theta_N \Theta_0^{-1} = \bar{\sigma}_N \bar{\sigma}_0^{-1} \bar{\sigma}_{*,N}^{-1} \bar{\sigma}_{*,0} \leq \prod_{j=1}^k \bar{\sigma}_{jh} \bar{\sigma}_{(j-1)h}^{-1} \bar{\sigma}_{*,jh}^{-1} \bar{\sigma}_{*,(j-1)h} \leq (1 + \delta)^{-k} \leq e^{-\frac{1}{2}\delta k} \leq \sigma.$$

This is exactly (5.40). $\square$

The next lemma records the elementary bounds satisfied by the special choices (5.13) under the near-stationarity hypothesis furnished by the pigeonhole lemma.

Lemma 5.9 (Good-scale parameter bounds). *Assume (P1), (P2), (P3), and (P4). Let $\delta \in (0, 1/2]$ and $m \in \mathbb{N}$, and assume that*

$$\bar{\sigma}_0 \leq (1 + \delta)\bar{\sigma}_m \quad \text{and} \quad \bar{\sigma}_{*,0}^{-1} \leq (1 + \delta)\bar{\sigma}_{*,m}^{-1}. \quad (5.41)$$

For every $e \in \mathbb{R}^d$ with $|e| = 1$, define p_e and q_e by (5.13), and then set

$$p_{0,e} := \bar{\mathbf{s}}_{*,m}^{-1} q_e - p_e, \quad q_{0,e} := q_e - \bar{\mathbf{s}}_m p_e. \quad (5.42)$$

Then the following statements hold.

1.

$$|\hat{\sigma}_m^{1/2} p_{0,e}| = |\hat{\sigma}_m^{-1/2} q_{0,e}| = |\Theta_m^{1/2} - 1| \leq \Theta_0^{1/2}. \quad (5.43)$$

2.

$$\hat{\sigma}_m^{-1} \bar{\sigma}_0 \leq (1 + \delta)\Theta_0^{1/2} \quad \text{and} \quad \hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} \leq (1 + \delta)\Theta_0^{1/2}. \quad (5.44)$$

3. For every $k \in \{0, \dots, m\}$,

$$\mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})] \leq (1 + \delta)\Theta_0^{1/2} \quad (5.45)$$

and

$$\tau_{m,k}(p_e, q_e) \leq \delta\Theta_0^{1/2}. \quad (5.46)$$

Proof. Using (5.13) and (5.42), we compute

$$\hat{\sigma}_m^{1/2} p_{0,e} = \hat{\sigma}_m^{1/2} (\bar{\sigma}_{*,m}^{-1} \hat{\sigma}_m^{1/2} - \hat{\sigma}_m^{-1/2}) e = (\Theta_m^{1/2} - 1) e$$

and

$$\hat{\sigma}_m^{-1/2} q_{0,e} = \hat{\sigma}_m^{-1/2} (\hat{\sigma}_m^{1/2} - \bar{\sigma}_m \hat{\sigma}_m^{-1/2}) e = (1 - \Theta_m^{1/2}) e.$$

This proves the first two identities in (5.43). The final inequality in (5.43) follows from the monotonicity of Θ_n in (4.78).

Next, by (5.12),

$$\hat{\sigma}_m^{-1} \bar{\sigma}_0 = \bar{\sigma}_0 \bar{\sigma}_m^{-1/2} \bar{\sigma}_{*,m}^{-1/2} \leq (1 + \delta) \bar{\sigma}_m^{1/2} \bar{\sigma}_{*,m}^{-1/2} = (1 + \delta) \Theta_m^{1/2} \leq (1 + \delta) \Theta_0^{1/2},$$

and the second inequality in (5.44) is proved in the same way.

For the third part, combining (5.18) with the special choice of p_e and q_e yields

$$\mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})] = \frac{1}{2} \hat{\sigma}_m^{-1} \bar{\sigma}_k + \frac{1}{2} \hat{\sigma}_m \bar{\sigma}_{*,k}^{-1} - 1.$$

Since $k \leq m$, the monotonicity relations in (4.77) imply

$$\bar{\sigma}_k \leq \bar{\sigma}_0 \quad \text{and} \quad \bar{\sigma}_{*,k}^{-1} \leq \bar{\sigma}_{*,0}^{-1}.$$

Applying (5.44), we get

$$\mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})] \leq \frac{1}{2} (1 + \delta) \Theta_0^{1/2} + \frac{1}{2} (1 + \delta) \Theta_0^{1/2} \leq (1 + \delta) \Theta_0^{1/2},$$

which is (5.45).

Finally, by the definition of $\tau_{m,k}(p_e, q_e)$ and (5.18),

$$\tau_{m,k}(p_e, q_e) = \frac{1}{2}\hat{\sigma}_m^{-1}(\bar{\sigma}_k - \bar{\sigma}_m) + \frac{1}{2}\hat{\sigma}_m(\bar{\sigma}_{*,k}^{-1} - \bar{\sigma}_{*,m}^{-1}).$$

Using again $k \leq m$,

$$\bar{\sigma}_k - \bar{\sigma}_m \leq \bar{\sigma}_0 - \bar{\sigma}_m \leq \delta \bar{\sigma}_m$$

by (5.41), and similarly

$$\bar{\sigma}_{*,k}^{-1} - \bar{\sigma}_{*,m}^{-1} \leq \bar{\sigma}_{*,0}^{-1} - \bar{\sigma}_{*,m}^{-1} \leq \delta \bar{\sigma}_{*,m}^{-1}.$$

Therefore

$$\tau_{m,k}(p_e, q_e) \leq \frac{1}{2}\delta \hat{\sigma}_m^{-1} \bar{\sigma}_m + \frac{1}{2}\delta \hat{\sigma}_m \bar{\sigma}_{*,m}^{-1} = \delta \Theta_m^{1/2} \leq \delta \Theta_0^{1/2},$$

which is (5.46). $\square$

The next lemma is the probabilistic input that the analytic reduction needs: at a nearly stationary scale, the coarse-grained block matrices have small variance after normalization by the annealed matrix at that scale.

Lemma 5.10 (Variance bound at a nearly stationary scale). *Assume (P1), (P2), (P3), and (P4). Define*

$$\beta := \frac{1}{4} \min \left\{ 1 - s_1 - s_2, s_1, s_2, s_1 - \frac{d}{\xi}, s_2 - \frac{d}{\xi} \right\}.$$

There exists a constant $C(d, \xi, \beta) < \infty$ such that the following statement holds. Let $\delta \in (0, 1/2]$ and $m \in \mathbb{N}$ satisfy

$$m \geq C\xi \log(2 + \delta^{-1}\xi\tilde{\Theta}_0), \quad (5.47)$$

and assume that

$$\bar{\sigma}_0 \leq (1 + \delta)\bar{\sigma}_m \quad \text{and} \quad \bar{\sigma}_{*,0}^{-1} \leq (1 + \delta)\bar{\sigma}_{*,m}^{-1}. \quad (5.48)$$

Then

$$\sum_{j=1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \bar{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \bar{\mathbf{A}}_m) \bar{\mathbf{A}}_m^{-1/2} \right|^2 \right] \leq C\delta^{1/2}. \quad (5.49)$$

Proof. By scalarization and monotonicity,

$$\bar{\mathbf{A}}_m \leq \bar{\mathbf{A}}_j \leq \bar{\mathbf{A}}_0 \quad \forall j \in \{0, \dots, m\}.$$

The good-scale hypothesis gives the sharper upper bound

$$\bar{\mathbf{A}}_0 \leq (1 + \delta)\bar{\mathbf{A}}_m.$$

Thus, for every unit vector $Q \in \mathbb{R}^{2d}$ and every $j \leq m$,

$$1 \leq Q \cdot \bar{\mathbf{A}}_m^{-1/2} \bar{\mathbf{A}}_j \bar{\mathbf{A}}_m^{-1/2} Q \leq 1 + \delta. \quad (5.50)$$

Fix such a vector Q and a scale $j \in \{1, \dots, m\}$. Set

$$X_j := Q \cdot \bar{\mathbf{A}}_m^{-1/2} \mathbf{A}(\square_j; \mathbf{a}) \bar{\mathbf{A}}_m^{-1/2} Q.$$

By the block-matrix subadditivity (2.53),

$$X_j \leq Q \cdot \bar{\mathbf{A}}_m^{-1/2} \bar{\mathbf{A}}_0 \bar{\mathbf{A}}_m^{-1/2} Q + \sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y,$$

where

$$Y_y := Q \cdot \bar{\mathbf{A}}_m^{-1/2} (\mathbf{A}(y + \square_0; \mathbf{a}) - \bar{\mathbf{A}}_0) \bar{\mathbf{A}}_m^{-1/2} Q.$$

Therefore, by (5.50),

$$(X_j - 1)_+ \leq C\delta + \left| \sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y \right|.$$

Since $X_j \geq 0$ and $\mathbb{E}[X_j - 1] \geq 0$, we also have

$$\mathbb{E}[(X_j - 1)_-^2] \leq \mathbb{E}[(X_j - 1)_-] \leq \mathbb{E}[(X_j - 1)_+].$$

Consequently,

$$\mathbb{E}[|X_j - 1|^2] \leq C\delta + C\mathbb{E}\left[\left|\sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y\right|\right] + C\mathbb{E}\left[\left|\sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y\right|^2\right]. \quad (5.51)$$

We now estimate the partition average. By (2.52), (2.73), the positivity of $\mathbf{A}(\square_0; \mathbf{a})$, and the good-scale comparisons, we have

$$\mathbb{E}[|Y_0|^\xi]^{1/\xi} \leq C\tilde{\Theta}_0.$$

Indeed, scalarization reduces the normalization by $\bar{\mathbf{A}}_m^{-1/2}$ to ratios of $\bar{\sigma}_m, \bar{\sigma}_{*,m}$, and the good-scale hypothesis compares these ratios with the corresponding scale-zero quantities. The unit-cube block matrix is then controlled by $\Lambda_{s_1,1}(\square_0; \mathbf{a})$ and $\lambda_{s_2,1}^{-1}(\square_0; \mathbf{a})$. Applying the partition-average estimate (4.67) with $p = 2$, and then Jensen's inequality, gives

$$\mathbb{E}\left[\left|\sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y\right|^2\right]^{1/2} \leq C3^{-d/2j}\tilde{\Theta}_0$$

and

$$\mathbb{E}\left[\left|\sum_{y \in \mathbb{Z}^d \cap \square_j} Y_y\right|\right] \leq C3^{-d/2j}\tilde{\Theta}_0.$$

Inserting these estimates into (5.51) yields

$$\mathbb{E}[|X_j - 1|^2] \leq C\left(\delta + 3^{-d/2j}\tilde{\Theta}_0 + 3^{-dj}\tilde{\Theta}_0^2\right). \quad (5.52)$$

Choose a finite subset $\mathcal{N} \subseteq \partial B_1^{2d}$ such that

$$|\mathbf{B}|^2 \leq C \sum_{Q \in \mathcal{N}} |Q \cdot \mathbf{B}Q|^2 \quad \forall \mathbf{B} \in \mathbb{R}_{\text{sym}}^{2d \times 2d}.$$

Applying (5.52) to each $Q \in \mathcal{N}$ gives, for every $j \in \{1, \dots, m\}$,

$$\mathbb{E}\left[\left|\bar{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \bar{\mathbf{A}}_m) \bar{\mathbf{A}}_m^{-1/2}\right|^2\right] \leq C\left(\delta + 3^{-d/2j}\tilde{\Theta}_0 + 3^{-dj}\tilde{\Theta}_0^2\right). \quad (5.53)$$

Since $\beta < d/2$, the geometric sums satisfy

$$\sum_{j=1}^m 3^{-\beta(m-j)} 3^{-d/2j} + \sum_{j=1}^m 3^{-\beta(m-j)} 3^{-dj} \leq C3^{-\beta m}.$$

After increasing the constant in (5.47), we may assume

$$3^{-\beta m}(\tilde{\Theta}_0 + \tilde{\Theta}_0^2) \leq \delta.$$

Multiplying (5.53) by $3^{-\beta(m-j)}$, summing over j , and using $\delta \leq \delta^{1/2}$ proves (5.49). $\square$

We next combine the variance bound with the analytic estimate from Section 5.3. This is the point at which the good scale becomes a genuine contraction of the annealed contrast.

Proposition 5.11 (One-step contraction of the annealed flow). *Assume (P1), (P2), (P3), and (P4). There exists a constant $C(d, s_1, s_2, \xi) < \infty$ such that the following statement holds. Let $\delta \in (0, 1/2]$ and $m \in \mathbb{N}$ satisfy*

$$m \geq C\xi \log(2 + \delta^{-1}\xi\tilde{\Theta}_0), \quad (5.54)$$

and assume that

$$\bar{\sigma}_0 \leq (1 + \delta)\bar{\sigma}_m \quad \text{and} \quad \bar{\sigma}_{*,0}^{-1} \leq (1 + \delta)\bar{\sigma}_{*,m}^{-1}. \quad (5.55)$$

Then

$$\Theta_m \leq 1 + C\delta^{1/4}\Theta_0. \quad (5.56)$$

Proof. Fix a unit vector $e \in \mathbb{R}^d$, set $p := p_e$ and $q := q_e$, and let

$$\beta := \frac{1}{4} \min \left\{ 1 - s_1 - s_2, s_1, s_2, s_1 - \frac{d}{\xi}, s_2 - \frac{d}{\xi} \right\}.$$

By (5.17),

$$\Theta_m - 1 = \mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] + \mathbb{E}[\tilde{J}_m^*(\square_m, p, q; \mathbf{a})].$$

We first estimate the primal term. Apply Lemma 5.7 with $k = 0$. The good-scale bounds of Lemma 5.9 give

$$\tau_{m,j}(p, q) \leq \delta\Theta_0^{1/2}, \quad \mathbb{E}[J(\square_0, p, q; \mathbf{a})] \leq C\Theta_0^{1/2},$$

for every $j \leq m$, and also

$$\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0 \leq C\Theta_0^{1/2}.$$

The variance term in (5.37) is bounded by Lemma 5.10. The additivity-defect sum is also harmless: the displayed good-scale bounds give

$$(\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0) \sum_{j=1}^m 3^{-\beta(m-j)} \tau_{m,j}(p, q) \leq C\delta\Theta_0,$$

which is bounded by $C\delta^{1/2}\Theta_0$ since $\delta \leq 1/2$.

It remains only to record that the two scale-separation terms in (5.37) are harmless. The unit-scale bound (2.73), the special choice of p and q , and Hölder's inequality give

$$\mathbb{E}[J(\square_0, p, q; \mathbf{a})]^{1/\zeta} \leq C \left(\hat{\sigma}_m^{-1} \mathbb{E}[\Lambda_{s_{1,1}}^\xi(\square_0; \mathbf{a})]^{1/\xi} + \hat{\sigma}_m \mathbb{E}[\lambda_{s_{2,1}}^{-\xi}(\square_0; \mathbf{a})]^{1/\xi} \right),$$

where $\zeta = \xi/(\xi - 1)$. Combining this with the definition of $\tilde{\Theta}_0$, the good-scale comparisons, and $\Theta_0 \leq \tilde{\Theta}_0$, we get before reattaching the prefactor $C\varepsilon^{-1}\beta^{-3}$ from (5.37),

$$\begin{aligned} 3^{-\beta m} \left(\hat{\sigma}_m \mathbb{E}[\lambda_{s_{2,1}}^{-\xi}(\square_0; \mathbf{a})]^{1/\xi} + \hat{\sigma}_m^{-1} \mathbb{E}[\Lambda_{s_{1,1}}^\xi(\square_0; \mathbf{a})]^{1/\xi} \right) \mathbb{E}[J(\square_0, p, q; \mathbf{a})]^{1/\zeta} \\ + 3^{-2\beta m} (\hat{\sigma}_m \bar{\sigma}_{*,0}^{-1} + \hat{\sigma}_m^{-1} \bar{\sigma}_0) (\Theta_m - 1) \leq C3^{-\beta m} \tilde{\Theta}_0^2 \Theta_0. \end{aligned}$$

Increasing the constant in (5.54), we may therefore assume that these two scale-separation contributions are bounded by

$$C\varepsilon^{-1}\delta^{1/2}\Theta_0$$

for every $\varepsilon \in (0, 1]$. Combining the preceding estimates yields

$$\mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] \leq C\delta^{1/2}\Theta_0^{1/2} + C\varepsilon\Theta_0 + C\varepsilon^{-1}\delta^{1/2}\Theta_0.$$

Since $\Theta_0 \geq 1$ and $\delta \leq 1/2$, this becomes

$$\mathbb{E}[\tilde{J}_m(\square_m, p, q; \mathbf{a})] \leq C(\varepsilon + \varepsilon^{-1}\delta^{1/2})\Theta_0.$$

The same estimate applies to the adjoint term by applying the same argument to the transposed coefficient field $\mathbf{a}^t$. The hypotheses are invariant under transpose, the scalar annealed quantities are unchanged, and the block-matrix variance estimate is unchanged up to a fixed orthogonal conjugation. Choosing $\varepsilon = \delta^{1/4}$ and using the centered-response identity above proves (5.56). $\square$

5.5 Iteration and annealed convergence

This section iterates the one-step mechanism and proves Theorem 5.1. Throughout the section we set

$$\beta := \frac{1}{4} \min \left\{ 1 - s_1 - s_2, s_1, s_2, s_1 - \frac{d}{\xi}, s_2 - \frac{d}{\xi} \right\}.$$

The first step is to localize the soft control from Section 5.2. For $\eta > 0$, write

$$\tilde{\Theta}_n^{(\eta)} := \mathbb{E}[\Lambda_{s_1+\eta,1}^\xi(\square_n; \mathbf{a})]^{1/\xi} \mathbb{E}[\lambda_{s_2+\eta,1}^{-\xi}(\square_n; \mathbf{a})]^{1/\xi}.$$

Corollary 5.12 (Shifted localization of the annealed contrast). *Assume (P1), (P2), (P3), and (P4). Then there exists a constant $C(d, \xi, \beta) < \infty$ such that, for every $k \leq n$,*

$$\Theta_n \leq \tilde{\Theta}_n^{(2\beta)} \leq \Theta_k + C\xi^2 3^{-\beta(n-k)} \tilde{\Theta}_0. \quad (5.57)$$

Proof. Push forward the law by the dilation that sends $\square_k$ to $\square_0$. Applying Lemma 5.4 in the rescaled variables, with source exponents $s_1 + \beta, s_2 + \beta$ and target exponents $s_1 + 2\beta, s_2 + 2\beta$, we obtain

$$\mathbb{E} \left[\left(\lambda_{s_2+2\beta,1}^{-1}(\square_n; \mathbf{a}) - \bar{\sigma}_{*,k}^{-1} \right)_+^\xi \right]^{1/\xi} \leq C\xi 3^{-\beta(n-k)} \mathbb{E} \left[\lambda_{s_2+\beta,1}^{-\xi}(\square_k; \mathbf{a}) \right]^{1/\xi},$$

and similarly

$$\mathbb{E} \left[\left(\Lambda_{s_1+2\beta,1}(\square_n; \mathbf{a}) - \bar{\sigma}_k^\xi \right)_+^\xi \right]^{1/\xi} \leq C\xi 3^{-\beta(n-k)} \mathbb{E} \left[\Lambda_{s_1+\beta,1}^\xi(\square_k; \mathbf{a}) \right]^{1/\xi}.$$

Combining the four estimates with the scalar decomposition used in the proof of Section 5.2 gives

$$\tilde{\Theta}_n^{(2\beta)} \leq \Theta_k + C\xi^2 3^{-\beta(n-k)} \tilde{\Theta}_k^{(\beta)}.$$

Applying the same estimate on the window $[0, k]$ yields

$$\tilde{\Theta}_k^{(\beta)} \leq \Theta_0 + C\xi^2 3^{-\beta k} \tilde{\Theta}_0 \leq C\xi^2 \tilde{\Theta}_0,$$

where the second application uses source exponents s_1, s_2 and target exponents $s_1 + \beta, s_2 + \beta$, and we used $\Theta_0 \leq \tilde{\Theta}_0$. The first inequality in (5.57) is the bound $\Theta_n \leq \tilde{\Theta}_n^{(2\beta)}$, proved exactly as in Lemma 5.2. This gives (5.57). $\square$

The same localization applies to the one-step contraction.

Corollary 5.13 (Shifted one-step contraction). *Assume (P1), (P2), (P3), and (P4). Then there exists a constant $C(d, \xi, \beta) < \infty$ such that the following statement holds. Let $k < n$ and let $\delta \in (0, 1/2]$. If*

$$n - k \geq C\xi \log(2 + \delta^{-1}\xi\tilde{\Theta}_0) \quad (5.58)$$

and

$$\bar{\sigma}_k \leq (1 + \delta)\bar{\sigma}_n \quad \text{and} \quad \bar{\sigma}_{*,k}^{-1} \leq (1 + \delta)\bar{\sigma}_{*,n}^{-1}, \quad (5.59)$$

then

$$\Theta_n \leq 1 + C\delta^{1/4}\Theta_k. \quad (5.60)$$

Proof. We repeat the proof of Proposition 5.11 on the scale window $[k, n]$, with the scale-zero quantities replaced by their scale- k counterparts. More explicitly, the good-scale parameter bounds become

$$\tau_{n,j}(p, q) \leq \delta\Theta_k^{1/2}, \quad \mathbb{E}[J(\square_k, p, q; \mathbf{a})] \leq C\Theta_k^{1/2} \quad (k \leq j \leq n),$$

for the special vectors associated with scale n . The proof of Lemma 5.7 gives the same centered-response estimate with the sums over $j = k+1, \dots, n$, with $\bar{\sigma}_0, \bar{\sigma}_{*,0}^{-1}$, and $\tilde{\Theta}_0$ replaced by $\bar{\sigma}_k, \bar{\sigma}_{*,k}^{-1}$, and the corresponding shifted local moment budget. Corollary 5.12 controls this budget by $\tilde{\Theta}_0$, with the scale-separation factor $3^{-\beta(n-k)}$. The shifted form of Lemma 5.10 is obtained by the same proof on the window $[k, n]$, and the only global input needed for the scale-separation tails is Corollary 5.12. These shifted estimates reproduce the proof of Proposition 5.11 and give (5.60). $\square$

We now combine the pigeonhole lemma with the shifted contraction estimate to obtain one genuine improvement step for Θ_n , with $\tilde{\Theta}_0$ as the fixed global moment budget.

Lemma 5.14 (One improvement step for the annealed flow). *Assume (P1), (P2), (P3), and (P4). There exists a constant $C(d, \xi, \beta) < \infty$ such that, for every $\sigma \in (0, 1/2]$ and every pair of triadic scales $k < N$ satisfying*

$$N - k \geq C\xi\sigma^{-4}|\log \sigma| \log(2 + \sigma^{-4}\xi\tilde{\Theta}_0), \quad (5.61)$$

we have

$$\Theta_N \leq 1 + \sigma\Theta_k. \quad (5.62)$$

Proof. Let $C_0 = C_0(d, s_1, s_2, \xi)$ be large enough for the shifted estimates used below. Choose $c > 0$ small enough and set

$$\delta := c\sigma^4, \quad h := \lceil C_0\xi \log(2 + \delta^{-1}\xi\tilde{\Theta}_0) \rceil, \quad r := \lceil 2\delta^{-1}|\log \sigma| \rceil.$$

After increasing the constant C in the statement, the hypothesis (5.61) implies

$$N - k \geq (r + 1)h.$$

Apply Lemma 5.8 to the shifted scalar sequence

$$j \mapsto (\bar{\sigma}_{k+j}, \bar{\sigma}_{*,k+j})$$

on the interval $[k, N - h]$. If the contraction alternative holds, then

$$\Theta_{N-h} \leq \sigma\Theta_k.$$

Corollary 5.12 gives

$$\Theta_N \leq \Theta_{N-h} + C\xi^2 3^{-\beta h} \tilde{\Theta}_0.$$

Increasing C_0 in the definition of h yields

$$C\xi^2 3^{-\beta h} \tilde{\Theta}_0 \leq \frac{1}{2}\sigma.$$

Since $\sigma/2 \leq 1$, this proves $\Theta_N \leq 1 + \sigma\Theta_k$ in this case.

It remains to consider the good-scale alternative. Then there exists

$$n \in \{k + h, \dots, N - h\}$$

such that

$$\bar{\sigma}_{n-h} \leq (1 + \delta)\bar{\sigma}_n \quad \text{and} \quad \bar{\sigma}_{*,n-h}^{-1} \leq (1 + \delta)\bar{\sigma}_{*,n}^{-1}.$$

The choice of h allows us to apply Corollary 5.13 on the window $[n - h, n]$. We obtain

$$\Theta_n \leq 1 + C\delta^{1/4}\Theta_{n-h} \leq 1 + \frac{1}{2}\sigma\Theta_k,$$

after reducing c if necessary and using the monotonicity of Θ_j . Applying the shifted bound from n to N , and using $N - n \geq h$, gives

$$\Theta_N \leq \Theta_n + C\xi^2 3^{-\beta h} \tilde{\Theta}_0.$$

The definition of h implies

$$C\xi^2 3^{-\beta h} \tilde{\Theta}_0 \leq \frac{1}{2}\sigma.$$

Combining the last three displays, and using $\Theta_k \geq 1$, proves (5.62). $\square$

Proposition 5.15 (Annealed perturbative entry and convergence). *Assume (P1), (P2), (P3), and (P4). Then there exists a constant $C(d, s_1, s_2, \xi) < \infty$ such that, for every $\sigma \in (0, 1/2]$, if*

$$N_\sigma := \lceil C \log^2(2 + \tilde{\Theta}_0) \rceil + \lceil C\xi\sigma^{-4} |\log \sigma| \log(2 + \sigma^{-4}\xi\tilde{\Theta}_0) \rceil, \quad (5.63)$$

then

$$\Theta_{N_\sigma} \leq 1 + \sigma. \quad (5.64)$$

Proof. Let C be the constant in Lemma 5.14, increased as needed below. First apply the lemma repeatedly with the fixed value $\sigma_0 = 1/4$. Define $n_0 := 0$ and

$$n_{j+1} := n_j + \lceil C\xi \log(2 + \xi\tilde{\Theta}_0) \rceil.$$

An induction using (5.62) gives

$$\Theta_{n_j} \leq \frac{4}{3} + 4^{-j}\tilde{\Theta}_0 \quad \forall j \in \mathbb{N}.$$

The initial step uses $\Theta_0 \leq \tilde{\Theta}_0$. Choose

$$j_* := \left\lceil \log_4 \left(\frac{3}{2}(2 + \tilde{\Theta}_0) \right) \right\rceil.$$

Then $\Theta_{n_{j*}} \leq 2$, and

$$n_{j*} \leq C j_* \log(2 + \xi \tilde{\Theta}_0) \leq C \log^2(2 + \tilde{\Theta}_0),$$

where the dependence on ξ is absorbed into C .

Now fix $\sigma \in (0, 1/2]$ and let N_σ be defined by (5.63). Applying Lemma 5.14 with $k = n_{j*}$, $N = N_\sigma$, and σ replaced by $\sigma/2$ gives

$$\Theta_{N_\sigma} \leq 1 + \frac{\sigma}{2} \Theta_{n_{j*}} \leq 1 + \sigma,$$

after increasing C in (5.63). This is (5.64). $\square$

5.6 Algebraic convergence after the perturbative scale

Lemma 5.16 (Upper bound for J in small contrast). *Assume (P1), (P2), (P3), and (P4). Assume also that*

$$\tilde{\Theta}_0 \leq 2. \quad (5.65)$$

Let $e \in \mathbb{R}^d$ satisfy $|e| = 1$, and define p_e, q_e by (5.13). There exists $C(d, \xi, \beta) < \infty$ such that, for every $m, k \in \mathbb{N}$ with $m - k \geq C$,

$$\begin{aligned} \mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] &\leq C \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2 \right] \\ &\quad + C \tau_{m,k} + C 3^{-\beta m} + C (\Theta_m - 1)^2. \end{aligned} \quad (5.66)$$

Proof. We apply (5.37) with $\varepsilon = 1$ to obtain $C(d, \xi, \beta) < \infty$ such that

$$\begin{aligned} \mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] &\leq C \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2 \right] \\ &\quad + C \tau_{m,k} + C \sum_{j=k+1}^m 3^{-\beta(m-j)} \tau_{m,j}(p_e, q_e) \\ &\quad + C 3^{-\beta m} + C 3^{-2\beta(m-k)} (\Theta_m - 1) + C (\Theta_m - 1)^2. \end{aligned} \quad (5.67)$$

Here we used that (5.65) implies $\Theta_n \leq 2$ for all $n \in \mathbb{N}$, that

$$\mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] \leq \mathbb{E}[\tilde{J}_m(\square_m, p_e, q_e; \mathbf{a})] + C(\Theta_m - 1)^2,$$

that

$$\begin{aligned} \tau_{m,k}(p_e, q_e)^{1/2} \mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})]^{1/2} &\leq \frac{1}{2} \tau_{m,k}(p_e, q_e) + \frac{1}{2} \mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})] \\ &\leq \frac{3}{2} \tau_{m,k}(p_e, q_e) + \frac{1}{2} \mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] \end{aligned}$$

and that the moment bounds of Lemma 5.4 give

$$\mathbb{E}[J(\square_k, p_e, q_e; \mathbf{a})^\zeta]^{1/\zeta} \leq C(\Theta_0 - 1) + C 3^{-\beta m} \leq C(1 + 3^{-\beta m}).$$

We can simplify the right side of (5.67) more. Using that $\mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] \geq \frac{1}{4}(\Theta_m - 1)$, if $\beta(m - k) \geq C$ then we can reabsorb the middle term on the last line of (5.67). Since we allow constants to depend on β , we also have by monotonicity of $\tau_{m,k}$ in k that

$$C \sum_{j=k+1}^m 3^{-\beta(m-j)} \tau_{m,j}(p_e, q_e) \leq C \tau_{m,k}.$$

Substituting these into (5.67) yields (5.66). $\square$

Lemma 5.17 (Variance estimate with quadratic error). *For every $n, k \in \mathbb{Z}$ with $k \leq n$ and matrix $\mathbf{B} \in \mathbb{R}_{\text{sym},+}^{2d \times 2d}$,*

$$\begin{aligned} \text{var} [\mathbf{B}^{-1/2} \mathbf{A}(\square_n) \mathbf{B}^{-1/2}] &\leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{B}^{-1/2} \mathbf{A}(z + \square_k) \mathbf{B}^{-1/2} \right] \\ &\quad + 8 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^{2d} \mathbf{J}(z + \square_k, \mathbf{B}^{-1/2} e_i, \mathbf{B}^{1/2} e_i; \mathbf{a}) \right)^2 \right]. \end{aligned} \quad (5.68)$$

Proof. By the triangle inequality, for every $k \leq n$,

$$\begin{aligned} \text{var} [\mathbf{B}^{-1/2} \mathbf{A}(\square_n) \mathbf{B}^{-1/2}] &\leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{B}^{-1/2} \mathbf{A}(z + \square_k) \mathbf{B}^{-1/2} \right] \\ &\quad + 2 \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{B}^{-1/2} (\mathbf{A}(z + \square_k) - \mathbf{A}(\square_n)) \mathbf{B}^{-1/2} \right|^2 \right]. \end{aligned} \quad (5.69)$$

To estimate the second term on the right-side we use the fact that for every sequence $\mathbf{b}_1, \dots, \mathbf{b}_N$ of symmetric positive definite matrices, if we denote their mean by $\mathbf{m} := \sum_{i=1}^N \mathbf{b}_i$ and their harmonic mean by $\mathbf{h} := (\sum_{i=1}^N \mathbf{b}_i^{-1})^{-1}$, then we have, for any other symmetric matrix $\mathbf{g}$ of the same size,

$$\mathbf{m} = \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{g}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{g}) - \underbrace{(\mathbf{h} - \mathbf{g}) \mathbf{h}^{-1} (\mathbf{h} - \mathbf{g})}_{\geq 0} \leq \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{g}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{g}). \quad (5.70)$$

Applying the above display with $\{\mathbf{b}_i\}$ given by $\{\mathbf{A}_*(z + \square_k) : z \in 3^k \mathbb{Z}^d \cap \square_n\}$, we deduce that for any $\mathbf{B} \in \mathbb{R}_{\text{sym},+}^{2d \times 2d}$,

$$\begin{aligned} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{A}_*(z + \square_k) - \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{A}_*^{-1}(z + \square_k) \right)^{-1} \\ \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} (\mathbf{B} - \mathbf{A}_*(z + \square_k)) \mathbf{A}_*^{-1}(z + \square_k) (\mathbf{B} - \mathbf{A}_*(z + \square_k)). \end{aligned}$$

Using (2.51) we have that for any $\mathbf{B} \in \mathbb{R}_{\text{sym},+}^{2d \times 2d}$, $P \in \mathbb{R}^{2d}$ and Lipschitz domain $U \subseteq \mathbb{R}^d$,

$$\mathbf{J}(U, P, \mathbf{B}P; \mathbf{a}) = \frac{1}{2} P \cdot (\mathbf{A}(U) - \mathbf{A}_*(U)) P + \frac{1}{2} P \cdot (\mathbf{B} - \mathbf{A}_*(U)) \mathbf{A}_*^{-1}(U) (\mathbf{B} - \mathbf{A}_*(U)) P.$$

The previous two displays and Lemma 2.7 imply that, for any $\mathbf{B} \in \mathbb{R}_{\text{sym},+}^{2d \times 2d}$,

$$\begin{aligned}
& \mathbf{B}^{-1/2} \mathbf{A}(\square_n) \mathbf{B}^{-1/2} \\
& \leq \mathbf{B}^{-1/2} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{A}(z + \square_k) \mathbf{B}^{-1/2} \\
& \leq \mathbf{B}^{-1/2} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{A}_*^{-1}(z + \square_k) \right)^{-1} \mathbf{B}^{-1/2} + 2 \sum_{i=1}^{2d} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{J}(z + \square_k, \mathbf{B}^{-1/2} e_i, \mathbf{B}^{1/2} e_i; \mathbf{a}) \mathbf{I}_d \\
& \leq \mathbf{B}^{-1/2} \mathbf{A}(\square_n) \mathbf{B}^{-1/2} + 2 \sum_{i=1}^{2d} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{J}(z + \square_k, \mathbf{B}^{-1/2} e_i, \mathbf{B}^{1/2} e_i; \mathbf{a}) \mathbf{I}_d,
\end{aligned}$$

where to get the last line in the previous display we used

$$\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{A}_*^{-1}(z + \square_k) \right)^{-1} \leq \mathbf{A}_*(\square_n) \leq \mathbf{A}(\square_n).$$

We deduce that

$$0 \leq \left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{B}^{-1/2} (\mathbf{A}(z + \square_k) - \mathbf{A}(\square_n)) \mathbf{B}^{-1/2} \right| \leq 2 \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^{2d} \mathbf{J}(z + \square_k, \mathbf{B}^{-1/2} e_i, \mathbf{B}^{1/2} e_i; \mathbf{a})$$

and therefore

$$\begin{aligned}
& \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{B}^{-1/2} (\mathbf{A}(z + \square_k) - \mathbf{A}(\square_n)) \mathbf{B}^{-1/2} \right|^2 \right] \\
& \leq 4 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^{2d} \mathbf{J}(z + \square_k, \mathbf{B}^{-1/2} e_i, \mathbf{B}^{1/2} e_i; \mathbf{a}) \right)^2 \right].
\end{aligned}$$

Combining this estimate with (5.69) yields (5.68). $\square$

We next combine the previous two lemmas to arrive at the following.

Lemma 5.18. *Assume (P1), (P2), (P3), and (P4). Assume also that*

$$\tilde{\Theta}_0 \leq 2. \tag{5.71}$$

Then there exists $C(d, \xi, \beta) < \infty$ such that, for every $m, k, \ell \in \mathbb{N}$ with $\ell < k < m$ and $m - k \geq C$ and every $e \in \mathbb{R}^d$ with $|e| = 1$,

$$\mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] \leq C \tau_{m,k} + C 3^{-\beta m} + C 3^{-d(k-\ell)} + C(\Theta_\ell - 1)^2. \tag{5.72}$$

Proof. We recall (5.66), which was proved in Lemma 5.16:

$$\begin{aligned}
\mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] & \leq C \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2 \right] \\
& \quad + C \tau_{m,k} + C 3^{-\beta m} + C(\Theta_m - 1)^2.
\end{aligned} \tag{5.73}$$

We estimate the first term on the right using the triangle inequality as

$$\begin{aligned} & \sum_{j=k+1}^m 3^{-\beta(m-j)} \mathbb{E} \left[\left| \overline{\mathbf{A}}_m^{-1/2} (\mathbf{A}(\square_j; \mathbf{a}) - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2 \right] \\ & \leq \sum_{j=k+1}^m 3^{-\beta(m-j)} \text{var} [\overline{\mathbf{A}}_m^{-1/2} \mathbf{A}(\square_j; \mathbf{a}) \overline{\mathbf{A}}_m^{-1/2}] + \sum_{j=k+1}^m 3^{-\beta(m-j)} \left| \overline{\mathbf{A}}_m^{-1/2} (\overline{\mathbf{A}}_j - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2. \end{aligned}$$

For the second term on the left side, we use

$$\left| \overline{\mathbf{A}}_m^{-1/2} (\overline{\mathbf{A}}_j - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right|^2 \leq 2 \left| \overline{\mathbf{A}}_m^{-1/2} (\overline{\mathbf{A}}_j - \overline{\mathbf{A}}_m) \overline{\mathbf{A}}_m^{-1/2} \right| \leq C \tau_{m,j}.$$

and

$$\sum_{j=k+1}^m 3^{-\beta(m-j)} \tau_{m,j} \leq C \beta^{-1} \tau_{m,k} \leq C \tau_{m,k}.$$

For the first term, we use (5.65) and then apply Lemma 5.17, to get

$$\begin{aligned} \text{var} [\overline{\mathbf{A}}_m^{-1/2} \mathbf{A}(\square_j; \mathbf{a}) \overline{\mathbf{A}}_m^{-1/2}] & \leq 4 \text{var} [\overline{\mathbf{A}}_\ell^{-1/2} \mathbf{A}(\square_j; \mathbf{a}) \overline{\mathbf{A}}_\ell^{-1/2}] \\ & \leq C \text{var} \left[\sum_{z \in 3^\ell \mathbb{Z}^d \cap \square_j} \overline{\mathbf{A}}_\ell^{-1/2} \mathbf{A}(z + \square_\ell) \overline{\mathbf{A}}_\ell^{-1/2} \right] \\ & \quad + C \mathbb{E} \left[\left(\sum_{z \in 3^\ell \mathbb{Z}^d \cap \square_j} \sum_{i=1}^{2d} \mathbf{J}(z + \square_\ell, \overline{\mathbf{A}}_\ell^{-1/2} e_i, \overline{\mathbf{A}}_\ell^{1/2} e_i; \mathbf{a}) \right)^2 \right]. \end{aligned}$$

By independence, we have

$$\text{var} \left[\sum_{z \in 3^\ell \mathbb{Z}^d \cap \square_j} \overline{\mathbf{A}}_\ell^{-1/2} \mathbf{A}(z + \square_\ell) \overline{\mathbf{A}}_\ell^{-1/2} \right] \leq C 3^{-d(k-\ell)}$$

and

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{z \in 3^\ell \mathbb{Z}^d \cap \square_j} \sum_{i=1}^{2d} \mathbf{J}(z + \square_\ell, \overline{\mathbf{A}}_\ell^{-1/2} e_i, \overline{\mathbf{A}}_\ell^{1/2} e_i; \mathbf{a}) \right)^2 \right] \\ & = \text{var} \left[\sum_{z \in 3^\ell \mathbb{Z}^d \cap \square_j} \sum_{i=1}^{2d} \mathbf{J}(z + \square_\ell, \overline{\mathbf{A}}_\ell^{-1/2} e_i, \overline{\mathbf{A}}_\ell^{1/2} e_i; \mathbf{a}) \right] + \mathbb{E} \left[\sum_{i=1}^{2d} \mathbf{J}(\square_\ell, \overline{\mathbf{A}}_\ell^{-1/2} e_i, \overline{\mathbf{A}}_\ell^{1/2} e_i; \mathbf{a}) \right]^2 \\ & \leq C 3^{-d(k-\ell)} + C(\Theta_\ell - 1)^2. \end{aligned}$$

Combining these, we obtain (5.72). This completes the proof. $\square$

Proposition 5.19 (Algebraic decay in the small-contrast regime). *Assume (P1), (P2), (P3), and (P4). There exist constants $\alpha(d, \xi, \beta) > 0$, $\delta_0(d, \xi, \beta) > 0$, and $C(d, \xi, \beta) < \infty$ such that, if*

$$\tilde{\Theta}_0 - 1 \leq \delta_0,$$

then, for every $m \in \mathbb{N}$ and every $e \in \mathbb{R}^d$ with $|e| = 1$, with p_e, q_e defined by (5.13),

$$\mathbb{E} [J(\square_m, p_e, q_e; \mathbf{a})] \leq C 3^{-\alpha m}. \quad (5.74)$$

Proof. Set

$$F_m := \Theta_m - 1.$$

After reducing δ_0 if necessary, the assumption implies $\tilde{\Theta}_0 \leq 2$. By (5.8),

$$0 \leq F_m \leq F_0 \leq \tilde{\Theta}_0 - 1 \leq \delta_0, \quad \forall m \in \mathbb{N}.$$

Moreover, by (5.9) and (5.13),

$$\mathbb{E}[J(\square_m, p_e, q_e; \mathbf{a})] = \Theta_m^{1/2} - 1. \quad (5.75)$$

Since $\Theta_m \leq 2$, this quantity is comparable to F_m .

We first turn (5.72) into a scalar recursion. By (5.10), for $k < m$ and the special choice of p_e, q_e at scale m ,

$$2\tau_{m,k}(p_e, q_e) = \hat{\sigma}_m^{-1}(\bar{\sigma}_k - \bar{\sigma}_m) + \hat{\sigma}_m(\bar{\sigma}_{*,k}^{-1} - \bar{\sigma}_{*,m}^{-1}).$$

Let

$$r := \Theta_m^{1/2}, \quad x := \hat{\sigma}_m^{-1}\bar{\sigma}_k, \quad y := \hat{\sigma}_m\bar{\sigma}_{*,k}^{-1}.$$

By monotonicity, $x, y \geq r$, while $xy = \Theta_k$ and $r^2 = \Theta_m$. Since $r \geq 1$, we have

$$(x - r) + (y - r) \leq xy - r^2 = \Theta_k - \Theta_m = F_k - F_m.$$

Thus

$$\tau_{m,k}(p_e, q_e) \leq C(F_k - F_m). \quad (5.76)$$

Combining (5.72), (5.75), and (5.76), we get

$$F_m \leq C(F_k - F_m) + C3^{-\beta m} + C3^{-d(k-\ell)} + CF_\ell^2 \quad (5.77)$$

whenever $\ell < k < m$ and $m - k$ is sufficiently large. Absorbing the negative multiple of F_m into the left side gives, for some $\theta \in (0, 1)$,

$$F_m \leq \theta F_k + C3^{-\beta m} + C3^{-d(k-\ell)} + CF_\ell^2. \quad (5.78)$$

We now choose the scales. Let $L \in \mathbb{N}$ be large enough that (5.78) is valid with $k = m - L$, and fix $\kappa \in (0, 1/2)$. For all sufficiently large m we may take

$$k = m - L, \quad \ell = \lfloor (1 - \kappa)m \rfloor.$$

Then $\ell < k < m$ and

$$3^{-d(k-\ell)} \leq C3^{-d\kappa m/2}.$$

Thus, for some $\alpha_0 > 0$ and all sufficiently large m ,

$$F_m \leq \theta F_{m-L} + CF_{\lfloor (1-\kappa)m \rfloor}^2 + C3^{-\alpha_0 m}. \quad (5.79)$$

Choose $\gamma \in (0, \alpha_0)$ small enough that $\theta 3^{\gamma L} < 1$. We prove by induction that

$$F_m \leq K3^{-\gamma m} \quad \forall m \in \mathbb{N}$$

for a large constant K to be fixed. The finitely many initial scales are covered by reducing δ_0 and increasing K . Suppose the bound has been proved below scale m . The first term in (5.79) is bounded by

$$\theta K 3^{-\gamma(m-L)} = \theta 3^{\gamma L} K 3^{-\gamma m},$$

which is a strict fraction of $K3^{-\gamma m}$. The source term is absorbed by taking K large, since $\gamma < \alpha_0$. Finally, the quadratic term is absorbed by combining the induction hypothesis with the uniform smallness $F_j \leq \delta_0$. Indeed, choose $\rho > 0$ so small that $(2 - \rho)(1 - \kappa) > 1$. Then

$$F_{\lfloor (1-\kappa)m \rfloor}^2 \leq \delta_0^\rho (K3^{-\gamma \lfloor (1-\kappa)m \rfloor})^{2-\rho} \leq C\delta_0^\rho K^{2-\rho} 3^{-\gamma m}.$$

Reducing δ_0 once more makes this term smaller than the remaining fraction of $K3^{-\gamma m}$, and closes the induction. Therefore $F_m \leq K3^{-\gamma m}$ for every $m \in \mathbb{N}$. The desired estimate follows from (5.75), with $\alpha = \gamma$. $\square$

Proof of Theorem 5.1. We use the estimates of the previous sections with a slightly smaller value of β , still denoted by β , so that the shifted exponents

$$s'_1 = s_1 + 2\beta, \quad s'_2 = s_2 + 2\beta$$

still satisfy the deterministic exponent inequalities in (P4). Apply Proposition 5.19 with these shifted exponents, and let $\delta_0 > 0$, $\alpha_0 > 0$, and $K < \infty$ be the constants it gives. Set

$$\delta := \delta_0 \wedge \frac{1}{4}.$$

We first show that the scale N_0 in (5.6) can be chosen so that

$$\tilde{\Theta}_{N_0}^{(2\beta)} - 1 \leq \delta. \quad (5.80)$$

Indeed, apply Proposition 5.15 with a fixed $\sigma \in (0, 1/2]$, chosen depending only on δ , for instance $\sigma = \delta/2$. This gives a scale N_σ satisfying $\Theta_{N_\sigma} \leq 1 + \sigma$. Applying Corollary 5.12 between N_σ and a later scale N gives

$$\tilde{\Theta}_N^{(2\beta)} \leq 1 + \sigma + C3^{-\beta(N-N_\sigma)} \tilde{\Theta}_0.$$

Taking

$$N - N_\sigma \geq C \log(2 + \delta^{-1} \tilde{\Theta}_0)$$

makes the last term at most $\delta/2$. Since δ and β are fixed in terms of d, s_1, s_2, ξ , increasing the constant C in (5.6) ensures that the displayed definition of N_0 dominates both N_σ and this additional logarithmic separation. This proves (5.80).

Now rescale the environment so that $\square_{N_0}$ becomes the unit cube. In the rescaled variables, (5.80) is precisely

$$\tilde{\Theta}'_0 - 1 \leq \delta \leq \delta_0$$

for the shifted exponents s'_1, s'_2 . Therefore Proposition 5.19 gives, for every $m \in \mathbb{N}$ and every unit vector e ,

$$\mathbb{E}[J(\square_m, p'_e, q'_e; \mathbf{a}')] \leq K3^{-\alpha_0 m},$$

where the primes denote the corresponding quantities in the rescaled environment. Since also $\tilde{\Theta}'_0 \leq 1 + \delta \leq 2$, (5.8) gives $\Theta'_m \leq 2$ for every m . Combining this with the special-vector identity (5.75) yields

$$\Theta'_m - 1 \leq CK3^{-\alpha_0 m}.$$

On the other hand, the same monotonicity gives the uniform smallness

$$\Theta'_m - 1 \leq \delta \leq \frac{1}{4} \quad \forall m \in \mathbb{N}.$$

After decreasing $\alpha > 0$, depending only on the constants above, the last two estimates imply

$$\Theta'_m \leq 1 + 3^{-\alpha m} \quad \forall m \in \mathbb{N}.$$

Undoing the rescaling, $\Theta'_m = \Theta_{N_0+m}$, and therefore

$$\Theta_{N_0+m} \leq 1 + 3^{-\alpha m} \quad \forall m \in \mathbb{N},$$

which is (5.7). □

5.7 Quenched minimal scales

In this section we derive quenched estimates on the coarse-grained quantities. The stochastic input is a strengthened unit-scale ellipticity assumption. We write it in a form which includes both finite stretched-exponential moments and the uniformly elliptic endpoint.

(P5) *Quantitative coarse-grained ellipticity on the unit cube.* There exist exponents $\sigma \in (0, \infty]$, $s_1, s_2 \in (0, 1)$ and a constant $\hat{\Theta}_0$ satisfying

$$s_1 + s_2 < 1$$

such that

$$\bar{\sigma}_0^{-1} \Lambda_{s_1,1}(\square_0; \mathbf{a}) + \bar{\sigma}_0 \lambda_{s_2,1}^{-1}(\square_0; \mathbf{a}) = \mathcal{O}_{\Gamma_\sigma}(\hat{\Theta}_0). \quad (5.81)$$

When $\sigma = \infty$, this means the corresponding random variable is bounded by $\hat{\Theta}_0$ almost surely.

Here $\bar{\sigma}_0$ is defined in (5.2). The estimate (5.81) implies the qualitative assumption (P4) and, for a constant $C(d, s_1, s_2, \sigma) < \infty$ in the finite case,

$$\tilde{\Theta}_0 \leq C \hat{\Theta}_0^2. \quad (5.82)$$

It also gives, for every unit vector $e \in \mathbb{R}^{2d}$,

$$\mathbf{J}(\square_0, \bar{\mathbf{A}}_0^{-1/2} e, \bar{\mathbf{A}}_0^{1/2} e; \mathbf{a}) = \mathcal{O}_{\Gamma_\sigma}(C \hat{\Theta}_0^2),$$

with the same convention at $\sigma = \infty$. We denote $\bar{\mathbf{A}} := \lim_{m \rightarrow \infty} \bar{\mathbf{A}}_m$.

The first quenched estimate is the direct consequence of the concentration lemma and the annealed convergence theorem. It is the only place where the annealed algebraic exponent enters the proof.

Corollary 5.20. *Assume (P1), (P2), (P3), and (P5). There exists $C(d, s_1, s_2, \sigma) < \infty$ and an exponent $\alpha = \alpha(d, s_1, s_2) > 0$ such that, if we define*

$$N_0 := \lceil C \log^2(2 + \hat{\Theta}_0) \rceil, \quad (5.83)$$

then, for every $m, n \in \mathbb{N}$ with $n < m$ and every $e \in \mathbb{R}^{2d}$ with $|e| = 1$,

$$\mathbf{J}(\square_{N_0+m}, \bar{\mathbf{A}}^{-1/2} e, \bar{\mathbf{A}}^{1/2} e; \mathbf{a}) \leq 3^{-\alpha n} + \mathcal{O}_{\Gamma_{\sigma \wedge 2}}(C \hat{\Theta}_0^2 3^{-\frac{d}{2}(m-n)}). \quad (5.84)$$

Proof. We apply Lemma 4.27 with exponent $\sigma \wedge 2$. If $\sigma > 2$, the unit-scale Γ_σ bound is first downgraded to a Γ_2 bound. The deterministic center is controlled by Theorem 5.1; the passage from $\bar{\mathbf{A}}_0$ to $\bar{\mathbf{A}}$ is absorbed using the telescoping estimate for the annealed matrices, again a consequence of the annealed theorem. This gives

$$\begin{aligned} \mathbf{J}(\square_{N_0+m}, \bar{\mathbf{A}}^{-1/2}e, \bar{\mathbf{A}}^{1/2}e; \mathbf{a}) &\leq \mathbb{E}[\mathbf{J}(\square_{N_0+n}, \bar{\mathbf{A}}^{-1/2}e, \bar{\mathbf{A}}^{1/2}e; \mathbf{a})] \\ &\quad + \mathcal{O}_{\Gamma_{\sigma \wedge 2}}(C\hat{\Theta}_0^2 3^{-\frac{d}{2}(m-n)}) \\ &\leq 3^{-\alpha n} + \mathcal{O}_{\Gamma_{\sigma \wedge 2}}(C\hat{\Theta}_0^2 3^{-\frac{d}{2}(m-n)}). \end{aligned}$$

□

Theorem 5.21 (Quenched minimal scale). *Assume (P1), (P2), (P3), and (P5). Let $t \in (s_1 \vee s_2, 1]$ and define*

$$\eta := \begin{cases} \frac{\sigma(\sigma \wedge 2)\frac{d}{2}t}{\sigma t + (\sigma \wedge 2)(\frac{d}{2} - t)}, & \sigma < \infty, \\ d & \sigma = \infty. \end{cases} \quad (5.85)$$

Then there exist an exponent $\alpha(d, s_1, s_2) > 0$, a constant $C(d, s_1, s_2, t, \sigma) < \infty$ and a random variable $\mathcal{X} \geq 1$ satisfying

$$\mathcal{X} = \mathcal{O}_{\Gamma_\eta}(\exp(C \log^2(2 + \hat{\Theta}_0))) \quad (5.86)$$

such that, for every $m, n \in \mathbb{N}$ with $n < m$ and $3^m \geq \mathcal{X}$, and every $e \in \mathbb{R}^{2d}$ with $|e| \leq 1$,

$$3^{-t(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathbf{J}(z + \square_n, \bar{\mathbf{A}}^{-1/2}e, \bar{\mathbf{A}}^{1/2}e; \mathbf{a}) \leq \left(\frac{3^m}{\mathcal{X}}\right)^{-\alpha}. \quad (5.87)$$

Proof. We give the proof in the finite case; the endpoint is the same argument with the unit-scale random variables replaced by their almost-sure bounds. In the finite case, write $\tau := \sigma \wedge 2$. Let $a > 0$ denote the exponent α supplied by Corollary 5.20, and put $b = d/2$. Choose $\alpha > 0$ so small that

$$\alpha < t, \quad \alpha < a, \quad \alpha < b, \quad \alpha\left(1 + \frac{b}{a}\right) < b.$$

This choice depends only on the annealed parameters and not on σ .

For $m > n$, set

$$H_{m,n} := \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \max_{|e| \leq 1} \mathbf{J}(z + \square_n, \bar{\mathbf{A}}^{-1/2}e, \bar{\mathbf{A}}^{1/2}e; \mathbf{a}).$$

The maximum over the unit ball may be replaced, up to a dimensional constant, by a finite maximum over a fixed family of normalized vectors, since $e \mapsto \mathbf{J}(U, \bar{\mathbf{A}}^{-1/2}e, \bar{\mathbf{A}}^{1/2}e; \mathbf{a})$ is a nonnegative quadratic form. Stationarity, Corollary 5.20, and the maximum estimate (4.27) give, for $0 \leq \ell < n < m$ after the deterministic entry-scale shift,

$$H_{m,n} \leq C3^{-a\ell} + \mathcal{O}_{\Gamma_\tau}(C\hat{\Theta}_0^2(1+m-n)^{1/\tau}3^{-b(n-\ell)}). \quad (5.88)$$

We also use the crude estimate

$$H_{m,n} \leq \mathcal{O}_{\Gamma_\sigma}(C\hat{\Theta}_0^2(1+m)^{1/\sigma}), \quad (5.89)$$

which follows by subadditivity down to unit cubes and another finite maximum.

Let N be a candidate scale and let $\mathcal{B}_N$ be the event that there exist $m > n$ with $m \geq N$ such that

$$3^{-t(m-n)} H_{m,n} > 3^{-\alpha(m-N)}. \quad (5.90)$$

We estimate this event by splitting the pairs (m, n) . In the high range we use (5.88) and choose

$$\ell \simeq \left(a^{-1} (\alpha(m-N) - t(m-n)) \right)_+.$$

The deterministic term is then below one half of the right side of (5.90). The fluctuation term has a summable geometric gain; the inequalities imposed on α are exactly what make the coefficients of the free scales negative. In the complementary bottom range, including the finitely many scales below the entry scale, we use (5.89); here the discount $3^{-t(m-n)}$ supplies the decay. Balancing the localized Γ_τ branch with the crude Γ_σ branch gives the exponent (5.85). Thus, after increasing C ,

$$\mathbb{1}_{\mathcal{B}_N} = \mathcal{O}_{\Gamma_\eta} (\exp(C \log^2(2 + \hat{\Theta}_0)) 3^{-N}). \quad (5.91)$$

The polynomial factors in $m - n$ and the finite cardinalities are absorbed into the same geometric summation.

Define the minimal scale by the usual bad-scale construction,

$$\mathcal{X} := 1 \vee \sup \{ 3^{N+1} : \mathcal{B}_N \text{ occurs} \},$$

with the convention that the supremum of the empty set is 0. The tail bound (5.91) and the union bound over larger N imply (5.86). If $3^m \geq \mathcal{X}$, then the bad event $\mathcal{B}_m$ does not occur, and the definition of $\mathcal{B}_m$ gives (5.87), after absorbing the harmless rounding by changing C and decreasing α .

When $\sigma = \infty$, the bound in (5.81) is deterministic. The crude bottom branch is therefore deterministic up to the finite maximum over unit cubes, and the same bad-scale summation gives the endpoint exponent $\eta = d$. This proves the theorem in both cases. $\square$

We now turn the minimal-scale estimate into the state variables which appear in the deterministic homogenization theorem of Chapter 3. Write

$$\bar{\sigma} := \lim_{m \rightarrow \infty} \bar{\sigma}_m = \lim_{m \rightarrow \infty} \bar{\sigma}_{*,m}, \quad \bar{\mathbf{a}} := \bar{\sigma} \mathbf{I}_d.$$

The common limit exists by the annealed convergence theorem and is positive by the qualitative ellipticity inherited from (P5). For $t \in (0, 1]$, denote by $\eta_\sigma(t)$ the exponent in (5.85), with t in place of the parameter there. The map $t \mapsto \eta_\sigma(t)$ is nondecreasing: for $\sigma = \infty$ it is constant, and for $\sigma < \infty$ the denominator in (5.85) equals $(\sigma - (\sigma \wedge 2))t + (\sigma \wedge 2)\frac{d}{2}$, so that $\eta_\sigma(t)$ has the form $at/(bt + c)$ with $a, c > 0$ and $b \geq 0$.

Corollary 5.22 (Finite- q homogenization error above the minimal scale). *Assume (P1), (P2), (P3), and (P5). Let $q \in [1, \infty)$ and let τ, r satisfy*

$$s_1 \vee s_2 < \frac{\tau}{2}, \quad 0 < \tau \leq 1, \quad \frac{\tau}{2} < r.$$

Define

$$\eta_{\mathcal{E}} := \begin{cases} \eta_\sigma(\tau) \wedge \eta_\sigma(\tau/2), & \sigma < \infty, \\ d, & \sigma = \infty. \end{cases} \quad (5.92)$$

Then there exist $\alpha > 0$, independent of σ , a constant $C(d, s_1, s_2, \sigma, \tau, r, q) < \infty$, and a random variable $\mathcal{X} \geq 1$ satisfying

$$\mathcal{X} = \mathcal{O}_{\Gamma_{\eta_{\mathcal{E}}}} (\exp(C \log^2(2 + \hat{\Theta}_0))) \quad (5.93)$$

such that, almost surely, for every $m \in \mathbb{N}$ with $3^m \geq \mathcal{X}$,

$$\mathcal{E}_{r,\infty,q}(\square_m; \mathbf{a}, \bar{\mathbf{a}}) \leq C \left(\frac{3^m}{\mathcal{X}} \right)^{-\alpha/2}. \quad (5.94)$$

Moreover, for every $j \in \mathbb{N}$,

$$\mathcal{E}_{r,\infty,1}(\square_m, m-j; \mathbf{a}, \bar{\mathbf{a}}) \leq C 3^{rj} \left(\frac{3^m}{\mathcal{X}} \right)^{-\alpha/2}. \quad (5.95)$$

Proof. The estimate (5.87) gives a uniform bound on the scale-local normalized $\mathbf{J}$ quantities. The maximum over unit vectors in the definition of $\mathcal{E}$ may be replaced, up to a dimensional constant, by a finite maximum over a fixed set of normalized vectors, since the map in e is a nonnegative quadratic form. Summing the geometric series in the definition of $\mathcal{E}_{r,\infty,q}$ is possible because $r > \tau/2$, and gives (5.94). The localized bound (5.95) follows from (2.78). The scale $\mathcal{X}$ is the maximum of the two scales obtained by applying Theorem 5.21 with discount exponents τ and $\tau/2$, which explains the exponent (5.92). The maximum of two random variables with stretched-exponential tails at exponents $\eta_\sigma(\tau)$ and $\eta_\sigma(\tau/2)$ has Γ_{η_σ} integrability. In the case $\sigma = \infty$, the same argument uses the endpoint branch of Theorem 5.21, and the exponent is d . $\square$

Theorem 5.23 (Quenched homogenization above the minimal scale). *Assume (P1), (P2), (P3), and (P5). Let t, s, r_2 satisfy*

$$s_1 \vee s_2 < t, \quad 4t < s < 1, \quad 3t < r_2.$$

Then there exist $\alpha > 0$ ², independent of σ , a constant $C(d, s_1, s_2, \sigma, t, s, r_2) < \infty$, and a random variable $\mathcal{X} \geq 1$ satisfying

$$\mathcal{X} = \mathcal{O}_{\Gamma_{\eta_\sigma(t)}}(\exp(C \log^2(2 + \hat{\Theta}_0))) \quad (5.96)$$

with the following property. Almost surely, for every $m \in \mathbb{N}$ with $3^m \geq \mathcal{X}$, every $\mathbf{g} \in H^{r_2}(\square_m; \mathbb{R}^d)$, and every pair $u, v \in H^1(\square_m)$ satisfying

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ -\nabla \cdot \bar{\mathbf{a}} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u - v \in H_0^1(\square_m), \end{cases}$$

we have

$$\begin{aligned} & 3^{-sm} [\bar{\mathbf{a}}(\nabla u - \nabla v)]_{\dot{B}_{2,2}^{-s}(\square_m)} + 3^{-sm} [\mathbf{a} \nabla u - \bar{\mathbf{a}} \nabla v]_{\dot{B}_{2,2}^{-s}(\square_m)} \\ & \leq C \left(\frac{3^m}{\mathcal{X}} \right)^{-\alpha} \left(\bar{\sigma}^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + 3^{r_2 m} [\mathbf{g}]_{\dot{B}_{2,2}^{r_2}(\square_m)} \right). \end{aligned} \quad (5.97)$$

Proof. Define

$$\tau := 2t, \quad r := t + \frac{1}{2} \left(\frac{s}{2} \wedge \frac{2r_2}{3} \right),$$

so that $s_1 \vee s_2 < \frac{\tau}{2}$, $0 < \tau \leq 1$, and

$$\tau < r < \frac{s}{2} \wedge \frac{2r_2}{3}.$$

²In the Lean formalization, the exponent α is fixed before the parameters t , s , and r_2 , and the formalized statement carries the additional hypothesis $\alpha < t$, with the conclusion stated at exponent $\alpha/8$ as in the proof below. The manuscript statement needs no such hypothesis because the exponent supplied by Theorem 5.21 may be chosen smaller than $s_1 \vee s_2$, making the constraint automatic; the formalized version of that theorem does not record this choice.

Apply Corollary 5.22 with $q = 1$ and exponent r , and with $q = 2$ and exponent $r/2$. The second application is allowed because $\tau < r$. Taking the maximum of the two random scales gives a single scale $\mathcal{X}$ which is $\mathcal{O}_{\Gamma_{\eta_{\mathcal{E}}}}(\exp(C \log^2(2 + \hat{\Theta}_0)))$ with $\eta_{\mathcal{E}}$ as in (5.92). Since η_{σ} is nondecreasing, $\eta_{\mathcal{E}} = \eta_{\sigma}(\tau/2) = \eta_{\sigma}(t)$, which gives (5.96).

Suppose $3^m \geq \mathcal{X}$ and write

$$\delta := \left(\frac{3^m}{\mathcal{X}}\right)^{-\alpha/2} \leq 1,$$

with $\alpha > 0$ the smaller of the two exponents supplied by the corollary. The localized estimate (5.95) gives, for every $j \in \mathbb{N}$,

$$\mathcal{E}_{r,\infty,1}(\square_m, m - j; \mathbf{a}, \bar{\mathbf{a}}) \leq C 3^{rj} \delta,$$

while the full-cube estimate with $q = 2$ and Lemma 2.14 give

$$\max\{\bar{\sigma}^{-1} \Lambda_{r/2,2}(\square_m; \mathbf{a}), \bar{\sigma} \lambda_{r/2,2}^{-1}(\square_m; \mathbf{a})\} \leq C(1 + \delta^2) \leq C.$$

Consequently

$$\lambda_{r/2,2}^{-1/2}(\square_m; \mathbf{a}) \leq C \bar{\sigma}^{-1/2}, \quad \frac{\Lambda_{r/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{r/2,2}^{1/2}(\square_m; \mathbf{a})} \leq C.$$

We now apply Proposition 3.13 with r and r_2 in place of the exponents t and t_2 there, with $n = m - j$, and with $\mathbf{a}_0 = \bar{\mathbf{a}}$; the hypotheses hold because $r < \frac{s}{2}$ and $r < r_2$. Since $\bar{\mathbf{a}} = \bar{\sigma} \mathbf{I}_d$, the factors $|\bar{\mathbf{a}}|^{1/2} \bar{\sigma}^{-1/2}$ and $|\bar{\mathbf{a}}| \bar{\sigma}^{-1}$ are dimensional constants. Substituting the three displayed bounds into the right side of the deterministic estimate and absorbing the polynomial prefactors in s , r , and $1/2 - r$ into the constant, we obtain, for every $j \in \mathbb{N}$,

$$\begin{aligned} & 3^{-sm} [\bar{\mathbf{a}}(\nabla u - \nabla v)]_{\underline{B}_{2,2}^{-s}(\square_m)} + 3^{-sm} [\mathbf{a} \nabla u - \bar{\mathbf{a}} \nabla v]_{\underline{B}_{2,2}^{-s}(\square_m)} \\ & \leq C \bar{\sigma}^{1/2} 3^{rj} \delta \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C(3^{(r-r_2)j} + 3^{(3/2r-r_2)j} \delta) 3^{r_2 m} [\mathbf{g}]_{\underline{B}_{2,2}^{r_2}(\square_m)}. \end{aligned} \quad (5.98)$$

It remains to optimize over the localization depth. Choose

$$j := \left\lceil \frac{1}{2r} \log_3 \frac{1}{\delta} \right\rceil,$$

so that $\delta^{-1/2} \leq 3^{rj} \leq 3\delta^{-1/2}$. The first term in (5.98) is then at most $3C \bar{\sigma}^{1/2} \delta^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}$. For the second term, the inequality $r < \frac{2r_2}{3}$ gives $r_2 - r > \frac{r}{2}$, whence

$$3^{(r-r_2)j} \leq 3^{-\frac{r}{2}j} \leq \delta^{1/4}, \quad 3^{(3/2r-r_2)j} \delta \leq \delta.$$

Since $\delta \leq 1$, the right side of (5.98) is bounded by

$$C \delta^{1/4} \left(\bar{\sigma}^{1/2} \|\mathbf{s}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + 3^{r_2 m} [\mathbf{g}]_{\underline{B}_{2,2}^{r_2}(\square_m)} \right),$$

which is (5.97) after replacing α by $\alpha/8$. The endpoint $\sigma = \infty$ is included because the preceding corollary and the ellipticity bound use the endpoint minimal-scale estimate with stochastic exponent d . $\square$

Chapter 6

Large-scale regularity

Bibliography

- [1] S. Armstrong and T. Kuusi. Renormalization group and elliptic homogenization in high contrast. *Inventiones Mathematicae*, 242(3):895–1086, 2025.
- [2] E. Di Nezza, G. Palatucci, and E. Valdinoci. Hitchhiker’s guide to the fractional Sobolev spaces. *Bulletin des Sciences Mathématiques*, 136(5):521–573, 2012.
- [3] E. M. Stein. *Singular Integrals and Differentiability Properties of Functions*. Princeton University Press, Princeton, NJ, 1970.